

6 **Draft Standard for**  
7 **Local and Metropolitan Area Networks—**  
8 **Station and Media Access Control**  
9 **Connectivity Discovery**

10 Prepared by the

11 **Maintenance Task Group of IEEE 802.1**

12 Sponsor

13 **LAN/MAN Standards Committee of the IEEE Computer Society**

14 **This and the following cover pages are not part of the draft.** They provide revision and other information  
15 for IEEE 802.1 Working Group members and participants in the IEEE Standards Association ballot process,  
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1

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3 This draft, P802.1AB-2016-Rev/D2.0, has been prepared for the first Standards Association (SA) Ballot. The  
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7 The YANG modules specified by this standard are attached to the draft pdf as plain text (UTF-8) .yang files.

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<sup>1</sup> **Abstract:** A protocol and a set of managed objects that can be used for discovering the physical  
<sup>2</sup> topology from adjacent stations in IEEE 802<sup>®</sup> LANs are defined in this document.  
<sup>3</sup>

<sup>4</sup> IEEE 802.1AB<sup>™</sup>, link layer discovery protocol, management information base, topology discovery,  
<sup>5</sup> topology information

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7

## 1 Introduction

This introduction is not part of IEEE Std 802.1AB™-2016, IEEE Standard for Local and metropolitan area networks—Station and Media Access Control Connectivity Discovery.

2 This revision of IEEE Std 802.1AB incorporates the following amendments into the base text of the 2016  
3 revision:

4 — IEEE Std 802.1ABcu-2021

5 — IEEE Std 802.1ABdh-2021

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7 Revisions are anticipated within the next few years to clarify existing material, to correct possible errors, and  
8 to incorporate new related material. Information on the current revision state of this and other IEEE 802  
9 standards may be obtained from

10 Secretary, IEEE-SA Standards Board  
11 445 Hoes Lane  
12 Piscataway, NJ 08854-4141  
13 USA



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# 1 IEEE Standard for 2 Local and metropolitan area networks— 3 Station and Media Access Control 4 Connectivity Discovery

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6 *or environmental protection, or ensure against interference with or from other devices or networks.*  
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## 15 1. Overview

16 The Link Layer Discovery Protocol (LLDP) specified in this standard allows stations attached to an  
17 IEEE 802<sup>®</sup> LAN to advertise, to other stations attached to the same IEEE 802 LAN, the major capabilities  
18 provided by the system incorporating that station, the management address or addresses of the entity or  
19 entities that provide management of those capabilities, and the identification of the station’s point of  
20 attachment to the IEEE 802 LAN required by those management entity or entities. Basic LLDP provides a  
21 mechanism for advertising information that can fit within a single Protocol Data Unit (PDU), while  
22 Extended LLDP (XLLDP) allows advertisement of information spanning multiple PDUs.

23 The information distributed via this protocol is stored in an information repository (RP), which can support  
24 both a standard Management Information Base (MIB) for SNMP and standard data models defined in  
25 YANG. This enables the information to be accessed by a Network Management System (NMS) through  
26 compatible management protocols, such as SNMP or NETCONF.

### 27 1.1 Scope

28 The scope of this standard is to define a protocol and management elements, suitable for advertising  
29 information to stations attached to the same IEEE 802 LAN, for the purpose of populating physical topology  
30 and device discovery management information databases. The protocol facilitates the identification of  
31 stations connected by IEEE 802 LANs/MANs, their points of interconnection, and access points for  
32 management protocols.

<sup>1</sup> This standard defines a protocol that

- <sup>2</sup> a) Advertises connectivity and management information about the local station to adjacent stations on  
<sup>3</sup> the same IEEE 802 LAN.
- <sup>4</sup> b) Receives network management information from adjacent stations on the same IEEE 802 LAN.
- <sup>5</sup> c) Operates with all IEEE 802 access protocols and network media.
- <sup>6</sup> d) Establishes a network management information schema and object definitions that are suitable for  
<sup>7</sup> storing connection information about adjacent stations.
- <sup>8</sup> e) Can provide compatibility with the IETF PTOPO MIB (IETF RFC 2922 [B8]).<sup>1</sup>
- <sup>9</sup> f) Can provide compatibility with the IETF Data Model for Network Topologies (RFC 8345 [B15]).<sup>2</sup>

## <sup>10</sup> 1.2 Purpose

<sup>11</sup> This standard specifies the necessary protocol and management elements to

- <sup>12</sup> a) Facilitate multi-vendor inter-operability and the use of standard management tools to discover and  
<sup>13</sup> make available physical topology information for network management.
- <sup>14</sup> b) Make it possible for network management to discover certain configuration inconsistencies or  
<sup>15</sup> malfunctions that can result in impaired communication at higher layers.
- <sup>16</sup> c) Provide information to assist network management in making resource changes and/or  
<sup>17</sup> re-configurations that correct configuration inconsistencies or malfunctions identified in b) above.
- <sup>18</sup> d) Provide IETF MIB (IETF RFC 2922 [B8]) to describe a network's physical topology and associated  
<sup>19</sup> systems within that topology.
- <sup>20</sup> e) Provide YANG configuration and operational models supporting Station and Media Access Control  
<sup>21</sup> Connectivity Discovery.

---

<sup>1</sup> The numbers in brackets correspond to those in the bibliography in Annex D.

## 2. Normative references

The following referenced documents are indispensable for the application of this document (i.e., they must be understood and used, so each referenced document is cited in the text and its relationship to this document is explained). For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments or corrigenda) applies.

IEEE Std 802<sup>®</sup>, IEEE Standard for Local and Metropolitan Area Networks: Overview and Architecture.<sup>2, 3</sup>

IEEE Std 802.1AC<sup>™</sup>, IEEE Standard for Local and metropolitan area networks—Media Access Control (MAC) Service Definition.

IEEE Std 802.1AE<sup>™</sup>, IEEE Standard for Local and Metropolitan Area Networks—Media Access Control (MAC) Security.

IEEE Std 802.1AX<sup>™</sup>, IEEE Standard for Local and Metropolitan Area Networks—Link Aggregation.

IEEE Std 802.1Q<sup>™</sup>, IEEE Standards for Local and Metropolitan Area Networks: Bridges and Bridged Networks.

IEEE Std 802.1X<sup>™</sup>, IEEE Standard for Local and Metropolitan Area Networks—Port-Based Network Access Control.

IEEE Std 802.3<sup>™</sup>, IEEE Standard for Information technology—Telecommunications and information exchange between systems—Local and metropolitan area networks—Specific requirements—Part 3: Carrier Sense Multiple Access with Collision Detection (CSMA/CD) Access Method and Physical Layer Specifications.

IETF RFC 2863, The Interfaces Group MIB, June 2000.<sup>4</sup>

IETF RFC 3046, DHCP Relay Agent Information Option, January 2001.

IETF RFC 3232, Assigned Numbers: RFC 1700 is Replaced by an On-line Database, January 2002.<sup>5</sup>

IETF RFC 3410, Introduction and Applicability Statements for Internet Standard Management Framework, December 2002.

IETF RFC 3417, Transport Mappings for the Simple Network Management Protocol (SNMP), December 2002.

IETF RFC 3418, Management Information Base (MIB) for the Simple Network Management Protocol (SNMP), December 2002.

IETF RFC 3629, UTF-8, a transformation format of ISO 10646, November 2003.

IETF RFC 4502, Remote Network Monitoring Management Information Base Version 2, May 2006.

<sup>2</sup> IEEE and 802 are registered trademarks in the U.S. Patent & Trademark Office, owned by The Institute of Electrical and Electronics Engineers, Incorporated.

<sup>3</sup> IEEE publications are available from The Institute of Electrical and Electronics Engineers (<http://standards.ieee.org/>).

<sup>4</sup> IETF documents (i.e., RFCs) are available for download at <http://www.rfc-archive.org/>.

<sup>5</sup> The IETF RFC 3232 ianaAddressFamilyNumbers on-line database module is accessible at (<http://www.iana.org/>).

<sup>1</sup> IETF RFC 4639, Cable Device Management Information Base for Data-Over-Cable Service Interface  
<sup>2</sup> Specification (DOCSIS) Compliant Cable Modems and Cable Modem Termination Systems, December  
<sup>3</sup> 2006.

<sup>4</sup> IETF RFC 4789, Simple Network Management Protocol (SNMP) over IEEE 802 Networks, November  
<sup>5</sup> 2006.

<sup>6</sup> IETF RFC 6241, Network Configuration Protocol (NETCONF), June 2011. <sup>6</sup>

<sup>7</sup> IETF RFC 6933, Entity MIB (Version 4), May 2013. IETF RFC 6991, Common YANG Data Types, July  
<sup>8</sup> 2013.

<sup>9</sup> IETF RFC 6991, Common YANG Data Types, July 2013.

<sup>10</sup> IETF RFC 7950, The YANG 1.1 Data Modeling Language, August 2016.

<sup>11</sup> IETF RFC 8343, A YANG Data Model for Interface Management, March 2018.

<sup>12</sup> IETF RFC 8349, A YANG Data Model for Routing Management (NMDA Version), March 2018.

<sup>13</sup> ISO/IEC 8824-1 [ITU-T Rec. X.680 (2002)], Abstract Syntax Notation One (ASN.1): Specification of Basic  
<sup>14</sup> Notation. <sup>7</sup>

---

<sup>6</sup> IETF RFCs are available from the Internet Engineering Task Force (<https://www.rfc-archive.org/>).

<sup>7</sup> ASN.1 standards are available at <http://asn1.elibel.tm.fr/en/standards/index.htm#asn1>.



### 3. Definitions and numerical representation

#### 3.1 Definitions

For the purposes of this document, the following terms and definitions apply. The *IEEE Standards Dictionary Online* should be consulted for terms not defined in this clause.<sup>8</sup>

**alpha-numeric information:** Information that is encoded using the 8-bit Universal Character Set (UCS)/Unicode Transformation Format (UTF-8) octet sequence [IETF RFC 3629].

**chassis:** A physical component incorporating one or more IEEE 802<sup>®</sup> LAN stations and their associated application functionality.

**chassis identifier:** An administratively assigned name that identifies the particular chassis within the context of an administrative domain that comprises one or more networks.

**Extended Link Layer Discovery Protocol (XLLDP):** An LLDP extension used to retrieve the frames beyond the Normal LLDPDU of a multi-frame LLDP database.

**Extension LLDPDU (XPDU):** An LLDPDU containing an Extension Identifier TLV and not containing a Time to Live TLV or Extension Request TLV.

**Extension Request LLDPDU (XREQ):** An LLDPDU containing an Extension Request TLV and not containing a Time to Live TLV or Extension Identifier TLV.

**IEEE 802<sup>®</sup> LAN:** Local area network (LAN) technologies that provide a media access control (MAC) Service equivalent to the MAC Service defined in IEEE Std 802.1AC.

**IEEE 802<sup>®</sup> LAN station:** An IEEE 802-compatible entity that incorporates all the necessary mechanisms to participate in media access control of an IEEE 802 LAN, and that is at least capable of providing the MAC service plus the mandatory capabilities of the LLC.

NOTE 1—For example, routers and host computers are stations in a bridged network. Bridges, in addition to their relay function, also include station capabilities.<sup>9</sup>

**Link Layer Discovery Protocol (LLDP):** A media-independent protocol capable of running on all IEEE 802<sup>®</sup> LAN stations and to allow an LLDP agent to learn the connectivity and management information from adjacent stations.

**LLDP agent:** The protocol entity that implements LLDP for a particular MSAP associated with a Port.

**LLDP Information Repository (RP):** A management information repository which stores the information exchanged by LLDP in TLVs and can be modeled by SNMP MIBs, YANG, or other data models.

**MAC service access point (MSAP):** The access point for MAC services provided to the LLC sublayer.

**MSAP identifier:** The identifier of a MAC service access point.

NOTE 2—In this standard, the concatenation of the chassis identifier and the port identifier is used by LLDP as an MSAP identifier, to identify the port associated with an IEEE 802<sup>®</sup> LAN station.

<sup>8</sup> *IEEE Standards Dictionary Online* is available at: <http://dictionary.ieee.org>.

<sup>9</sup> Notes in text, tables, and figures are given for information only and do not contain requirements needed to implement the standard.

1 **management entity:** The protocol entity that implements a particular network management protocol and  
2 that provides access support to a information repository associated with the protocol and implemented on a  
3 host chassis.

4 **Management Information Base (MIB):** The instantiation of all MIB modules in a managed entity (e.g.,  
5 system or device).

6 **Management Information Base module (MIB module):** The specification or schema for a data base that  
7 can be populated with the information required to support a network management information system.

8 **Manifest LLDPDU:** A Normal LLDPDU containing a Manifest TLV.

9 **network:** An interconnected group of systems, each comprising one or more IEEE 802<sup>®</sup> LAN stations.

10 **Network Management System (NMS):** A management system that is capable of utilizing the information  
11 in a information repository.

12 **Normal LLDPDU:** An LLDPDU containing a Time To Live TLV and not containing an Extension Request  
13 TLV or an Extension Identifier TLV.

14 **object identifier (OID):** An identifier used to name an object. Structurally, an OID consists of a node in a  
15 hierarchically-assigned namespace, formally defined in ISO/IEC 8824-1, Abstract Syntax Notation 1  
16 (ASN.1). OIDs are used in this standard to identify MIB modules and the objects they contain.

17 **physical network topology:** The identification of systems, of IEEE 802<sup>®</sup> LAN stations that compose each  
18 system, and of the IEEE 802 LAN stations that attach to the same IEEE 802 LAN.

19 **port:** The entity in a chassis/system to support an MSAP. A port incorporates one and only one MSAP and  
20 identifies the collection of manageable entities that provide the MAC Service at the MSAP.

21 **port identifier:** An administratively assigned name that identifies the particular port within the context of a  
22 system, where the identification is convenient, local to the system, and persistent for the system's use and  
23 management (whereas the MAC address that globally identifies the MSAP can not be).

24 **Scope MAC Address:** The MAC address used as the destination address in Normal LLDPDUs, which  
25 determines the limits of propagation in the network.

26 **shutdown LLDPDU:** A Normal LLDPDU with a TTL value of zero.

27 **system:** A managed collection of hardware and software components incorporating one or more chassis,  
28 stations and ports.

29 **station only:** A non-forwarding IEEE 802<sup>®</sup> LAN station such as a user workstation, network file server, or  
30 print server.

31 **type, length, value (TLV):** A short, variable length encoding of an information element consisting of  
32 sequential type, length, and value fields where the type field identifies the type of information, the length  
33 field indicates the length of the information field in octets, and the value field contains the information,  
34 itself.

35 **YANG:** IETF defined data modeling language, published as IETF RFC 7950.

36 **YANG model:** One or more YANG modules used to configure and monitor the managed element or system.

1 **YANG module:** The description of the data model used to configure and monitor the managed element or  
2 system. A YANG module defines a hierarchy of nodes that can be used for NETCONF-based (see IETF  
3 RFC 7803 [B12]) and RESTCONF-based (see IETF RFC 8040 [B13]) operations.

## 4 3.2 Numerical representation

5 Decimal, hexadecimal, and binary numbers are used within this document. For clarity, decimal numbers are  
6 generally used to represent counts, hexadecimal numbers are used to represent addresses, and binary  
7 numbers are used to describe bit patterns within binary fields.

8 Decimal numbers are represented in their usual 0, 1, 2, ... format. Hexadecimal numbers are represented by  
9 a string of one or more hexadecimal (0–9, A–F) digits followed by the subscript 16, except in C-code  
10 contexts, where they are written as `0x123EF2`, etc. Binary numbers are represented by a string of one or  
11 more binary (0,1) digits, followed by the subscript 2. Thus the decimal number “26” may also be represented  
12 as “1A<sub>16</sub>” or “11010<sub>2</sub>”.

13 MAC addresses, EtherType values, and OUI/EUI values are represented as strings of 8-bit hexadecimal  
14 numbers separated by hyphens and without a subscript, as, for example, “01-80-C2-00-00-15” or  
15 “AA-55-11”, using the hexadecimal representation defined in IEEE Std 802.

## 4. Acronyms and abbreviations

|        |                                                    |
|--------|----------------------------------------------------|
| AP     | Access Point                                       |
| BSSID  | basic service set identification                   |
| DA     | Destination Address                                |
| DS     | Distribution System                                |
| EPD    | Ethertype Protocol Discrimination                  |
| EUI    | Extended Unique Identifier                         |
| FCS    | Frame Check Sequence                               |
| IANA   | Internet Assigned Numbers Authority                |
| ID     | Identifier                                         |
| IETF   | Internet Engineering Task Force                    |
| KaY    | Media access control Security Key Agreement Entity |
| LLC    | logical link control (sublayer)                    |
| LLDP   | Link Layer Discovery Protocol                      |
| LLDPDU | Link Layer Discovery Protocol data unit            |
| LSAP   | link service access point                          |
| MAC    | media access control (sublayer)                    |
| MAU    | medium attachment unit                             |
| MDI    | media dependent interface                          |
| MIB    | Management Information Base (module)               |
| MSAP   | Media access control service access point          |
| MSDU   | Media access control service data unit             |
| NMS    | Network Management System                          |
| OID    | object identifier                                  |
| OUI    | organizationally unique identifier                 |
| PD     | Powered Device                                     |
| PHY    | physical (sublayer)                                |

|    |                  |                                                                              |
|----|------------------|------------------------------------------------------------------------------|
| 1  | PMD              | physical media dependent (sublayer)                                          |
| 2  | PSE              | Power Sourcing Equipment                                                     |
| 3  | PVID             | port virtual local area network identifier                                   |
| 4  | PPVID            | port and protocol virtual local area network identifier                      |
| 5  | PTOPO            | the name of the Internet Engineering Task Force physical topology Management |
| 6  |                  | Information Base                                                             |
| 7  | RFC              | Request for comments                                                         |
| 8  | RP               | LLDP Information Repository                                                  |
| 9  | SA               | Source Address                                                               |
| 10 | SecY             | Media access control Security Entity                                         |
| 11 | SNAP             | Subnetwork Access Protocol                                                   |
| 12 | SNMP             | Simple Network Management Protocol                                           |
| 13 | STP              | Spanning Tree Protocol                                                       |
| 14 | TPMR             | Two-port Media access control relay                                          |
| 15 | TLV              | type, length, value                                                          |
| 16 | TTL              | time to live (value)                                                         |
| 17 | UCS              | Universal Character Set                                                      |
| 18 | UML <sup>®</sup> | Unified Modeling Language™ <sup>10</sup>                                     |
| 19 | UTF-8            | 8-bit Universal Character Set/Unicode Transformation Format                  |
| 20 | VID              | virtual local area network identifier                                        |
| 21 | XLLDP            | Extended Link Layer Discovery Protocol                                       |
| 22 | XPDU             | Extension LLDPDU                                                             |
| 23 | XREQ             | Extension Request LLDPDU                                                     |
| 24 |                  |                                                                              |

---

<sup>10</sup> UML<sup>®</sup> is a registered trademark of the Object Management Group, Inc., and Unified Modeling Language™ is a trademark of the Object Management Group, Inc.

## 1 5. Conformance

2 This clause specifies the mandatory and optional capabilities provided by conformant implementations of  
3 this standard.

### 4 5.1 Terminology

5 For consistency with IEEE and existing IEEE 802.1™ standards terminology, requirements placed upon  
6 conformant implementations of this standard are expressed using the following terminology:

- 7 a) *Shall* is used for mandatory requirements.
- 8 b) *May* is used to describe implementation or administrative choices (“may” means “is permitted to,”  
9 and hence, “may” and “may not” mean precisely the same thing).
- 10 c) *Should* is used for recommended choices (the behaviors described by “should” and “should not” are  
11 both permissible but not equally desirable choices).

12 The PICS proforma (see Annex A) reflects the occurrences of the words *shall*, *may*, and *should* within the  
13 standard.

14 The standard avoids needless repetition and apparent duplication of its formal requirements by using *is*, *is*  
15 *not*, *are*, and *are not* for definitions and the logical consequences of conformant behavior. Behavior that is  
16 permitted but is neither always required nor directly controlled by an implementor or administrator, or  
17 whose conformance requirement is detailed elsewhere, is described by *can*. Behavior that never occurs in a  
18 conformant implementation or system of conformant implementations is described by *can not*. The word  
19 *allow* is used as a replacement for the phrase “support the ability for,” and the word *capability* means “is  
20 able to, or can be configured to.”

### 21 5.2 Protocol Implementation Conformance Statement (PICS)

22 The supplier of an implementation that is claimed to conform to this standard shall complete a copy of the  
23 PICS proforma provided in Annex A and shall provide the information necessary to identify both the  
24 supplier and the implementation.

### 25 5.3 Required capabilities

26 A system for which conformance to this standard is claimed shall, for all ports for which support is claimed,  
27 include the following capabilities:

- 28 a) If port access is controlled by IEEE Std 802.1X,<sup>11</sup> LLDP exchanges shall be supported through the  
29 controlled port, as specified in Clause 6 of IEEE Std 802.1X-2020.
- 30 b) The destination and source addressing shall conform to 7.1 and 7.2.
- 31 c) EtherType encapsulation shall conform to 7.3.
- 32 d) LLDPDU recognition and reception shall conform to the operation of the receive state machine as  
33 defined in 9.2.10.
- 34 e) LLDPDU encapsulation shall conform to the specifications in 8.2.
- 35 f) The basic TLV format capability shall be implemented as defined in 8.4.
- 36 g) The basic management set of TLVs shall be implemented as defined in 8.5.
- 37 h) The Organizationally Specific TLV format capability shall be implemented as defined in 8.7.

<sup>11</sup> Information on references can be found in RFC 3629.

- 1 i) The protocol shall conform to the specifications for all Clause 9 subclauses indicated in Table 9-1  
2 for the particular operating mode (transmit only, receive only, or transmit and receive) being  
3 implemented.
- 4 j) If receipt of LLDPDUs is supported, for every set of TLVs (the basic management set, extended  
5 LLDP set, and any organizationally specific sets) supported, support shall be implemented for  
6 receipt of every TLV defined in the set.
- 7 k) If transmission of LLDPDUs is supported, for every set of TLVs (the basic management set,  
8 extended LLDP set, and any organizationally specific sets) supported, support shall be implemented  
9 for transmission of every TLV defined in the set.
- 10 l) If transmission of LLDPDUs is supported, for every set of TLVs (the basic management set,  
11 extended LLDP set, and any organizationally specific sets) supported, a capability shall be  
12 implemented for users to determine which optional TLVs are transmitted in any particular  
13 LLDPDU.
- 14 m) If SNMP is supported, then
  - 15 1) The system shall conform to the LLDP management specifications in Clause 11 and shall  
16 implement the sections of version 2 of the basic LLDP MIB indicated in Table 11-1 for the  
17 operating mode being implemented.
  - 18 2) The system shall support at least one of the transport mappings defined by IETF RFC 3417 or  
19 IETF RFC 4789.
- 20 n) If neither SNMP nor YANG are supported, the system shall provide storage and retrieval capability  
21 equivalent to the functionality specified in 10.1 for the operating mode being implemented.
- 22 o) If YANG is supported, then the system shall conform to the LLDP management specifications in 12  
23 and shall implement the sections of the LLDP YANG module for the operating mode being  
24 implemented.
- 25 p) If the Extended Link Layer Discovery Protocol is supported, then
  - 26 1) The extended LLDP set of TLVs shall be implemented as defined in 8.6.
  - 27 2) The Extension Request and Extension LLDPDU formats (8.2) and addressing (7.1.2) shall be  
28 implemented.
  - 29 3) The extended receive state machine (Figure 9-3) shall be implemented.

## 30 5.4 Optional capabilities

31 A system for which conformance to this standard is claimed may, for each port for which support is claimed,  
32 support the following capabilities:

- 33 a) If port access is controlled by IEEE Std 802.1X, LLDP exchanges may be supported through the  
34 uncontrolled port, as specified in Clause 6 of IEEE Std 802.1X-2020.

## 6. Principles of operation

LLDP is a link layer protocol that allows an IEEE 802 LAN station to advertise the capabilities and current status of the system associated with an MSAP. The MSAP provides the MAC service to an LLC Entity, and that LLC Entity provides an LSAP to an LLDP agent that transmits and receives information to and from the LLDP agents of other stations attached to the same LAN. The information distributed and received in each LLDPDU is stored in one or more Management Information Bases (MIBs) or YANG modules. Figure 6-1 illustrates the LLDP agent and its relationship to its LLC Entity and MSAP, and to additional MIBs or YANG modules designed by the IETF, IEEE 802, and others.

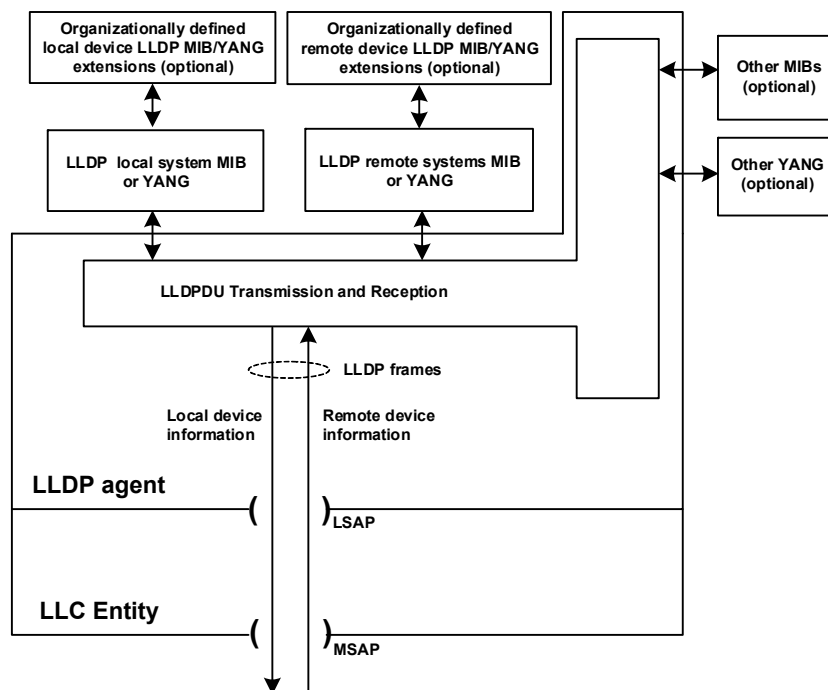


Figure 6-1—LLDP agent and its relationship to its LLC entity

This clause describes general principles of LLDP and Extended LLDP (XLLDP) operation. The following clauses specify transmission, reception, and addressing of LLPDUs (Clause 7); LLDPDU formats (Clause 8); operation of each LLDP agent in detail including state machines (Clause 9); management of LLDP (Clause 10); and the MIB module (Clause 11) or YANG module (Clause 12) used for LLDP management.

NOTE 1—For the purposes of this standard, the terms “LLC” and “LLC entity” include the service provided by the operation of entities that support protocol discrimination using an EtherType; i.e., protocol discrimination based on the Type interpretation of the Length/Type field as specified in IEEE Std 802.3, or the use of the SNAP encapsulation of EtherTypes as described in IEEE Std 802. Hence, “LSAP” in the above description can refer to the use of IEEE Std 802.2 LLC addresses, or an EtherType, as a means of higher layer protocol identification.

NOTE 2—The LLC Entity also provides distinct LSAPs to other higher layer protocol entities (such as those responsible for the Spanning Tree Protocol and IP). In a bridge, the LLC Entity associated with each Bridge Port is modeled as being directly connected to the attached LAN, as discussed in 8.13.9 of IEEE Std 802.1Q-2022, so the operation of LLDP is unaffected by the spanning tree Port State.



## 6.1 Transmission and reception

The information fields in each LLDP frame are contained in a Link Layer Discovery Protocol Data Unit (LLDPDU) as a sequence of variable length information elements, that each include type, length, and value fields (known as TLVs), where

- Type identifies what kind of information is being sent.
- Length indicates the length of the information string in octets.
- Value is the actual information that needs to be sent (for example, a binary bit map or an alpha-numeric string that can contain one or more fields).

Each LLDPDU begins with a mandatory sequence of three TLVs, as specified in 8.2, and can contain zero or more additional optional TLVs as selected by network management. The first three TLVs are:

- 1) A Chassis ID TLV.
- 2) A Port ID TLV.
- 3) A Time To Live TLV or, if XLLDP is supported, an Extension Request TLV or an Extension Identifier TLV.

The first two TLVs contain the chassis ID and the port ID values. These are concatenated to form a logical MSAP identifier. The combination of the MSAP identifier and the Scope MAC Address identify the LLDP agent/port that is the source of the information in the TLVs. Both the chassis ID and port ID values can be defined in a number of convenient forms. Once selected, however, the chassis ID/port ID value combination remains the same as long as the particular port remains operable (8.5.2, 8.5.3, 9.2.4.1).

The third TLV differentiates between three types of LLDPDUs:

- a) A Normal LLDPDU, where the third TLV is a Time To Live TLV that tells the receiving LLDP agent how long all information pertaining to this LLDPDU's MSAP identifier is valid so that all of the associated information can later be automatically discarded by the receiving LLDP agent if the sender fails to update it in a timely manner. It is useful to further differentiate two special cases of a Normal LLDPDU:
  - 1) A shutdown LLDPDU, where the third TLV is a Time To Live TLV with a TTL value of zero indicating that any information pertaining to this LLDPDU's MSAP identifier is to be discarded immediately. A TTL value of zero can be used, for example, to signal that the sending port has initiated a port shutdown procedure.
  - 2) A Manifest LLDPDU, where the third TLV is a Time To Live TLV with a non-zero TTL value and one of the subsequent TLVs is a Manifest TLV.
- b) An Extension Request LLDPDU (XREQ), where the third TLV is an Extension Request TLV that the recipient of a Normal LLDPDU containing a Manifest TLV uses to request one or more of the Extension LLDPDUs described in the Manifest TLV. The Extension Request LLDPDU does not contain a Time to Live TLV.
- c) An Extension LLDPDU (XPDU), where the third TLV is an Extension Identifier TLV. The Extension LLDPDU does not contain a Time to Live TLV.

NOTE —The receive state machine (in 9.2.10 and in prior versions of this standard) specifies that an implementation not supporting XLLDP will discard any received LLDPDUs not containing a TTL TLV as the third TLV (i.e., will discard any received Extension Request or Extension LLDPDUs). The state machine further specifies that only implementations supporting XLLDP will process a Manifest TLV and subsequently transmit Extension Request LLDPDUs, and that only implementations supporting XLLDP will process Extension Request TLVs and subsequently transmit Extension LLDPDUs.

An optional End Of LLDPDU TLV marks the end of the LLDPDU.

1 The format for the LLDPDU is defined in 8.2. The TLV categories and the basic TLV format are defined in  
2 8.3 and 8.4. The specific format and field contents for the Chassis ID TLV are defined in 8.5.2; for the  
3 Port ID TLV, in 8.5.3; for the Time To Live TLV in 8.5.4; and for the End Of LLDPDU TLV, in 8.5.1.

## 4 **6.2 LLDP operational modes**

5 The information advertised by LLDP is transferred in a single direction. An LLDP agent can periodically  
6 advertise information about the capabilities and current status of the system associated with its MSAP  
7 identifier. The LLDP agent can also collect information about the capabilities and current status of the  
8 system associated with a remote MSAP identifier. LLDP does not itself contain a mechanism for the LLDP  
9 agent to solicit specific information from other LLDP agents beyond what is being advertised by those  
10 LLDP agents, nor does it provide a specific means of confirming the receipt of information. Extended LLDP  
11 allows an LLDP agent collecting information to request detailed information (via an Extension Request  
12 LLDPDU) that is only advertised in a summary form (in a Manifest TLV), however this does not change the  
13 inherent LLDP property that the choice of information provided is at the discretion of the advertising LLDP  
14 agent.

15 NOTE—The LLDP protocol is designed to advertise information useful for discovering pertinent information about a  
16 remote port and to populate topology MIBs or YANG modules. It is not intended to act as a configuration protocol for  
17 remote systems, nor as a mechanism to signal control information between ports. During the operation of LLDP, it may  
18 be possible to discover configuration inconsistencies between systems on the same IEEE 802 LAN. LLDP does not  
19 provide a mechanism to resolve those inconsistencies. Rather, it provides a means to report discovered information to  
20 higher layer management entities.

21 LLDP allows the advertising function and collecting function of the LLDP agent to be separately enabled,  
22 making it possible to configure an implementation so the local LLDP agent can either advertise only or  
23 collect only, or so the local LLDP agent can advertise and collect LLDP information.

## 24 **6.3 LLDP information categories**

25 The following basic management TLV set (this set is required in all LLDP implementations) is defined by  
26 this standard and can be used to describe the system and/or to assist in the detection of configuration  
27 inconsistencies associated with the MSAP identifier:

- 28 a) Port Description TLV.
- 29 b) System Name TLV.
- 30 c) System Description TLV.
- 31 d) System Capabilities TLV (indicates both the system's capabilities and its current primary network  
32 function, such as end station, bridge, router).
- 33 e) Management Address TLV.

34 Table 8-1 includes a list of the optional TLVs in the basic management set and provides subclause references  
35 for their specific definitions.

36 Organizationally Specific TLVs can be defined by either the professional organizations or the individual  
37 vendors that are involved with the particular functionality being implemented within a system. The basic  
38 format and procedures for defining Organizationally Specific TLVs are provided in 8.7.

## 1 6.4 TLV selection

2 Information for constructing the various TLVs to be sent is stored in the LLDP local system information  
3 repository (RP). The selection of which particular TLVs to send is under control of network management.  
4 Information received from remote LLDP agents is stored in the LLDP remote systems information  
5 repository.

## 6 6.5 Transmission principles

7 A transmission cycle can be initiated either by the expiration of a transmit countdown timing counter or by a  
8 change in the value of one or more of the information elements (managed objects) associated with the local  
9 system. When a transmit cycle is initiated, the LLDP management entity extracts the managed objects from  
10 the LLDP local system information repository and formats this information into TLVs. The TLVs are  
11 inserted into an LLDPDU that is passed to the LLDP transmit module. When Extended LLDP is supported,  
12 some of the TLVs are inserted into one or more Extension LLDPDUs that are passed to the LLDP transmit  
13 module when requested by the receipt of a Extension Request LLDPDU. The LLDP transmit module  
14 prepends addressing parameters to the LLDPDU as defined in 8.2, 8.3, and 8.4. The LLDPDU and TLV  
15 formats are defined in Clause 8. The LLDP transmit state machine is described in 9.2.1 and 9.2.9.

16 NOTE 1—Because a transmission cycle can be initiated whenever a change occurs within the LLDP local system  
17 information repository, it is possible that a series of successive changes over a short period of time could trigger a  
18 number of LLDP frames to be sent, each reporting only a single change. LLDP utilizes a transmission delay timer that  
19 can be set by network management to limit the number of LLDP transmission cycles in a given time period.

20 NOTE 2—Under normal circumstances, the information in the receiving LLDP agent's remote systems information  
21 repository is refreshed periodically to avoid being discarded due to ageing. To prevent the receiving LLDP agent's  
22 remote systems information repository being aged out because a refresh frame has been lost in transmission, the sending  
23 LLDP agent typically sets the TTL value so that several refresh cycles can occur before the received repository  
24 information ages out.

25 LLDP can disclose information about a device and network that could be considered sensitive in certain  
26 environments. Implementers are encouraged to provide controls that take into account the link protection  
27 characteristics when including information in an LLDP message. Different information can be included if an  
28 LLDP message is being sent on the uncontrolled port, the controlled port, or a MACsec protected  
29 connectivity association.

## 30 6.6 Reception principles

31 The LLDP receive module uses the services of the LLC entity to recognize that the incoming  
32 MA\_UNITDATA.indication contains the correct combination of destination address and MSDU header  
33 values to identify it as resulting from a received LLDPDU. The LLDPDU recognition procedures are  
34 defined in 7.4 and 9.2.8.7.

### 35 6.6.1 LLDPDU and TLV error handling

36 The LLDPDU is checked to verify that it contains the correct sequence of three mandatory TLVs at the  
37 beginning of the frame, as specified in 8.2, and then each optional TLV is validated in succession.  
38 LLDPDUs that contain detectable errors in the first three mandatory TLVs are discarded. Optional TLVs that  
39 contain detectable errors are discarded [see 9.2.8.7.2 c)]. TLVs that are not recognized, but that also contain  
40 no basic format errors, are assumed to be valid and are stored for possible later retrieval by network  
41 management (see 9.2.8.7.1 and 9.2.8.5). If the End Of LLDPDU TLV is present, any octets that follow it are  
42 discarded.

1 TLVs in which the information string length field contains a value that is greater than the sum of the lengths  
2 of the fields within the information string are not discarded.

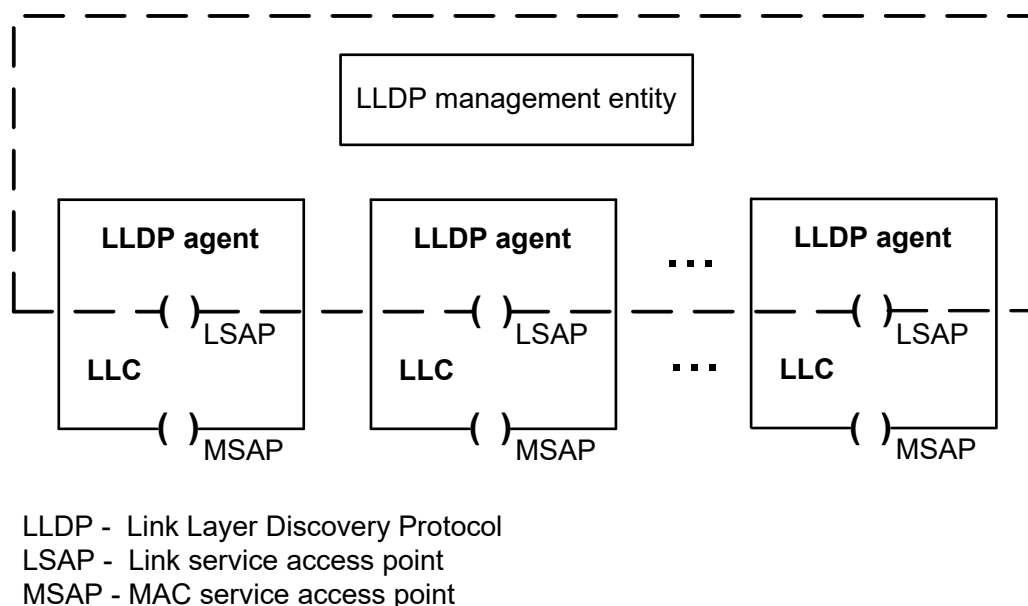
3 NOTE—This approach allows later versions of a TLV to define additional fields at the end of the information string;  
4 implementations based on an earlier version of the TLV can therefore continue to process the fields that were defined in  
5 that version and ignore any new fields added at the end of the information string in later versions. Any information  
6 contained in such new fields is not stored in the information repository and made available to network management  
7 protocols via the standard MIB or YANG modules.

## 8 6.6.2 LLDP remote systems information repository update

9 The LLDP remote systems information repository is updated after all TLVs have been validated. LLDP  
10 remote system information repository update procedures are defined in 9.2.8.9, 9.2.8.5, and 9.2.8.4. The  
11 LLDP receive state machine is described in 9.2.1 and 9.2.10.

## 12 6.7 Systems with multiple LLDP Agents

13 Each LLDP agent advertises a single set of information in the various TLVs it encodes in each transmission  
14 cycle, and is associated with the MSAP that supports the LLC entity that the LLDP agent uses to transmit  
15 and receive. Each LLDP agent uses its LSAP directly, without the use of any additional multiplexing or  
16 addressing above the LSAP to support the use of that LSAP by multiple LLDP agents; and each LLC entity  
17 provides service to one and only one protocol entity at each of its LSAPs that it supports, using the service  
18 provided by a single MSAP. It follows that each LLDP agent makes use of a unique MSAP, and that the  
19 LLDP agent can be uniquely identified by the receiving agent using the MSAP's identifier and the MAC  
20 address that determines the LLDP agent's address scope, as specified in 6.1. A single LLDP management  
21 entity can support the operation of multiple LLDP agents within the same system. Figure 6-2 illustrates the  
22 relationship between the LLDP agents, LLC Entities, MSAPs, and the LLDP management entity.



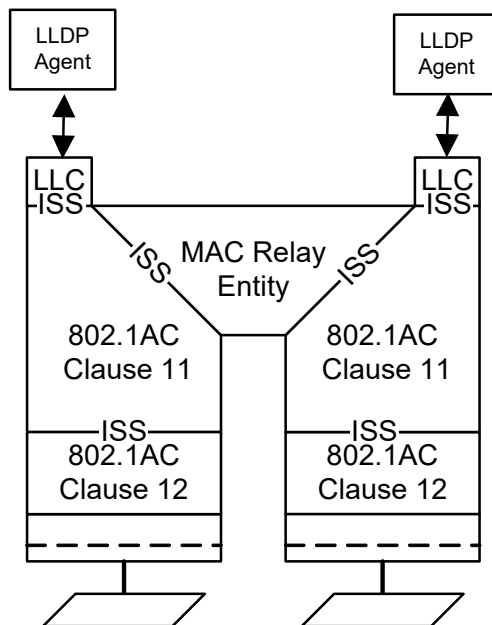
**Figure 6-2—Relationship between LLDP agents, LLC Entities, MSAPs, and the LLDP management entity**

23 A given system can require the use of multiple LLDP agents because it naturally comprises multiple  
24 MSAPs, or because it needs to be able to transmit different information (different sets of TLVs) to different  
25 sets of peers. These sets of peers can be determined by the way that the MAC service is supported within the

1 system (see IEEE Std 802.1AC for a description of the connectionless connectivity that characterizes an  
2 IEEE 802 LAN). A MAC Bridge (see IEEE Std 802.1Q), for example, attaches to multiple LANs, each  
3 providing the MAC service at a distinct MSAP. A different example is a station that uses port-based network  
4 access control (IEEE Std 802.1X) to provide both a Controlled Port and an Uncontrolled Port, i.e., two  
5 distinct MSAPs with potentially different connectivity, for transmission and reception to and from a single  
6 LAN. Similarly, different destination addresses can be associated with different sets of potential recipients.  
7 Table 7-1 identifies group MAC addresses that can be used for LLDPDU transmission, and 7.1 describes the  
8 different transmission scopes associated with each address. LLDP can also be used in conjunction with  
9 individual MAC addresses, and with other group MAC addresses. Distinct LLDP agents, and hence separate  
10 and distinct protocol state machines, local and remote information repository tables, and MSAPs are  
11 maintained for each LLDPDU destination address, even if the use of two or more MSAPs can result in  
12 transmission and reception on the same LAN.

13 NOTE—For a given MAC address that is used as a destination address in received LLDPDUs, there is the possibility for  
14 a station to receive LLDPDUs from multiple senders. All LLDPDUs that carry that destination MAC address are  
15 received by a single LLDP agent, via a single MSAP; however, as the LLDPDU carries information that identifies the  
16 source of the LLDPDU, information carried in LLDPDUs from different senders is able to be recorded independently in  
17 the remote systems information repository.

18 Figure 6-3 illustrates the use of LLDP in a MAC Bridge, with an LLDP agent for each of the two ports  
19 shown. Figure 6-4 shows the use of LLDP in an end system with port-based network access control  
20 (IEEE Std 802.1X) supported by MACsec (IEEE Std 802.1AE); an LLDP agent is supported by the  
21 Controlled Port and an additional LLDP agent may be supported by the Uncontrolled Port. Figure 6-5 shows  
22 LLDP in a MAC Bridge that uses port-based network access control on both ports.



**Figure 6-3—LLDP in a MAC Bridge**

23 LLDP frames sent on an unprotected link may be modified by an attacker. If an implementation is going to  
24 take action based upon a received LLDP attribute, it is advisable to take into account whether the message  
25 was received on a protected link, and allow for the system to be configured such that only information  
26 received in protected messages is acted upon.

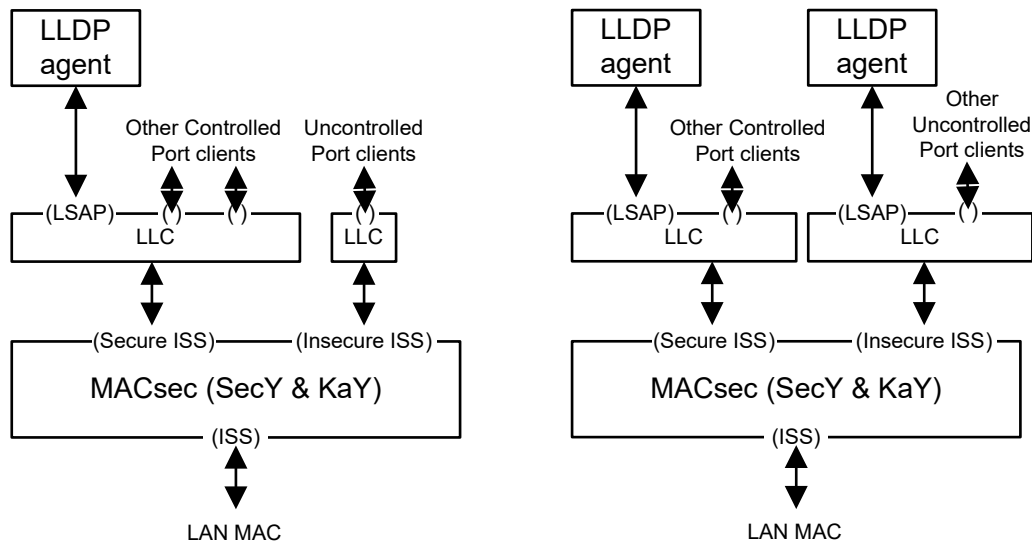


Figure 6-4—LLDP in an end system with port-based network access control

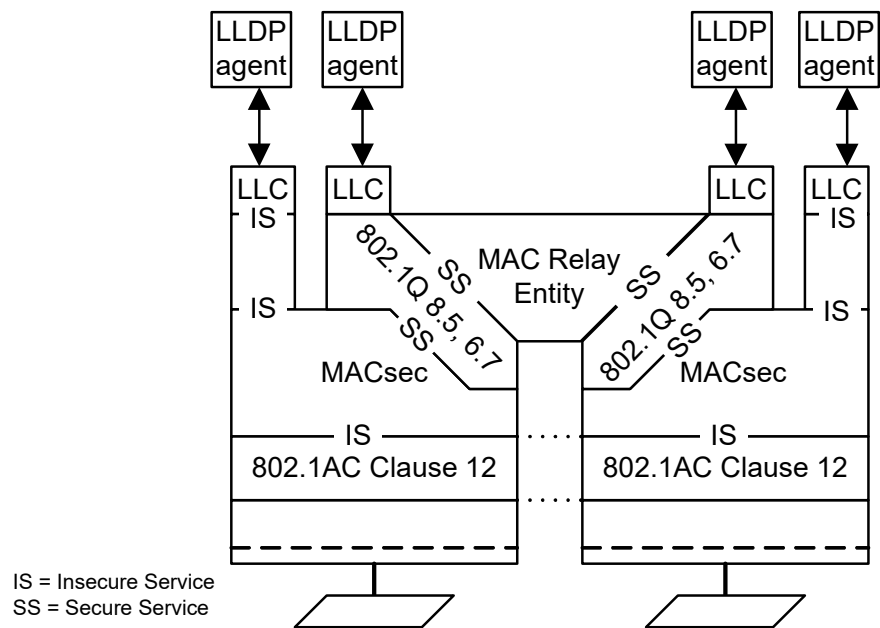
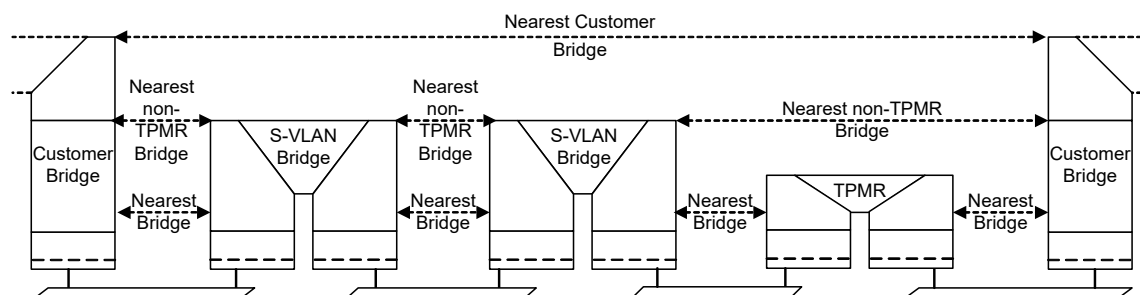


Figure 6-5—LLDP in a MAC Bridge that uses port-based network access control on both ports

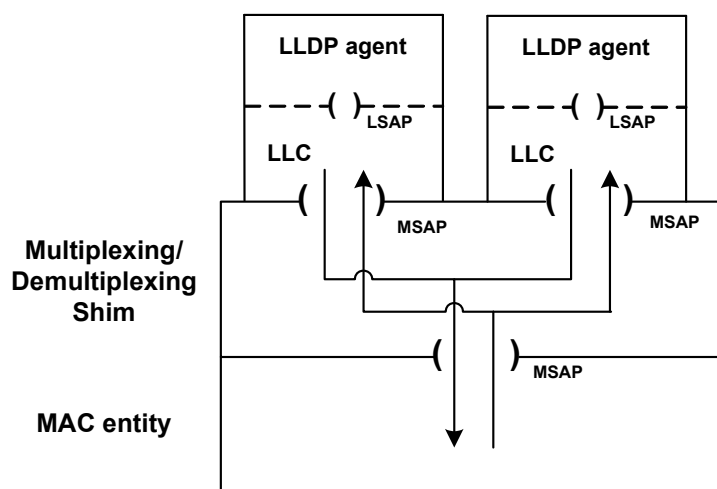
<sup>1</sup> The group addresses identified in Table 7-1 are taken from the set specified as reserved addresses by  
<sup>2</sup> bridging standards. Frames (including those conveying LLDPDUs) that use these destination addresses are  
<sup>3</sup> filtered or not by the various types of bridge components, as specified by Table 8-1 through Table 8-3 of  
<sup>4</sup> IEEE Std 802.1Q-2022. The choice of a particular address thus allows a given agent to limit the propagation

1 of LLDPDUs to an individual LAN, or to allow them a wider scope. Figure 6-6 illustrates the various scopes  
2 achievable with the three destination group MAC addresses identified in Table 7-1, their use is specified  
3 further in 7.1.



**Figure 6-6—Scope of group MAC addresses**

4 More than one LLDP agent, each using a different address scope, can be instantiated for a given system port  
5 by adding a simple shim that provides the necessary distinct MSAPs by multiplexing and demultiplexing  
6 between those MSAPs and a common MSAP for the port, as illustrated in Figure 6-7. Each MSAP shown in



**Figure 6-7—Multiplexing and demultiplexing using shims**

7 Figure 6-7 is allowed to share the same MSAP Identifier and individual MAC address. The LLDP agents are  
8 differentiated by address scope used, i.e., by the MAC address used as a Destination Address in frames  
9 containing a Normal LLDPDU.

## 10 6.8 LLDP and Link Aggregation

11 An LLDP agent can be configured on an IEEE 802.1AX™ Aggregated Port and/or on any number of the  
12 physical ports belonging to the Aggregation. Some TLVs (e.g., the Energy-Efficient Ethernet (EEE) TLV  
13 designated in subclause 79.3.5 of IEEE Std. 802.3-2022) only make sense when transmitted from an LLDP  
14 agent configured on a physical port, and can cause incorrect operation if configured on an Aggregation.  
15 Other TLVs (e.g., the DCBX TLVs defined in 38.4 of IEEE Std 802.1Q-2022) make sense only when  
16 transmitted from an Aggregated Port, because they carry information applicable to the simulated single

1 interface represented by the Aggregation. Not all standards that define LLDP TLVs are careful to state  
2 whether each TLV is applicable to an Aggregation, to a physical port, or to both. Only LLDP TLVs that are  
3 appropriate for transmission on a given Port should be transmitted on that Port.

4 An LLDP agent configured on an Aggregated Port, or on one of the physical ports of an Aggregation,  
5 transmits the Link Aggregation TLV, and processes received LLDPDUs, as specified in F of 802.1AX-2020.  
6 An LLDP agent configured on a physical port of an Aggregation uses the LLDP Parser/Multiplexer as  
7 specified in 6.1.3 of IEEE Std 802.1AX-2020 for the transmission and reception of LLDPDUs containing  
8 TLVs that are specific to that physical port. Since a port not running Link Aggregation is equivalent to an  
9 Aggregation with a single physical port, any LLDP agent can transmit the Link Aggregation TLV and  
10 process received LLDPDUs as specified in 802.1AX.

11 NOTE—The consideration of aggregated vs. Physical ports is similar to the distinction among destination MAC  
12 addresses discussed in 7.1. In both cases, LLDP TLVs that carry information specific to a physical link should be used  
13 only in the LLDP agent with the shortest reach.

## 14 6.9 Extended LLDP (XLLDP)

15 Extended LLDP allows an LLDP agent to advertise more TLVs than would fit within a single LLDPDU by  
16 transmitting the additional TLVs in one or more Extension LLDPDUs. The selection of TLVs to be included  
17 in LLDPDUs is controlled by management, but the mapping of TLVs to the initial Normal LLDPDU or to an  
18 Extension LLDPDU is implementation dependent (10.2.2). When a transmission cycle is initiated, a unique  
19 description of each Extension LLDPDU is encoded in a Manifest TLV, which is then included in the Normal  
20 LLDPDU transmitted at the beginning of the cycle. A Normal LLDPDU containing a Manifest TLV is  
21 referred to as a Manifest LLDPDU. Changes to any of the TLVs in an Extension LLDPDU changes the  
22 description included in the Manifest TLV, and thus triggers a new transmission cycle.

23 An implementation not supporting XLLDP that receives a Manifest TLV treats it as an unrecognized TLV  
24 and only stores the TLVs that were included in the Normal LLDPDU. An implementation supporting  
25 XLLDP that receives a Manifest TLV then requests the transmission of any new or modified Extension  
26 LLDPDUs by sending an Extension Request LLDPDU to the LLDP agent sourcing the Manifest LLDPDU.  
27 One or more Extension LLDPDUs can be requested in a single Extension Request LLDPDU. A new  
28 Extension Request LLDPDU can be transmitted when all Extension LLDPDUs previously requested have  
29 been received. This allows the requesting LLDP agent to pace the rate of information transfer. A timer on the  
30 request allows an Extension Request LLDPDU to be retried if it appears that either the request or a response  
31 has been lost.

32 Extension Request LLDPDUs and Extension LLDPDUs are transmitted with an individual MAC address as  
33 the destination address. The destination address in an Extension Request LLDPDU is taken from the  
34 contents of the Return MAC Address field of the received Manifest TLV. The requested Extension  
35 LLDPDUs are transmitted with a destination address taken from the contents the Return MAC Address field  
36 of the Extension Request TLV.



## 7. LLDPDU transmission, reception, and addressing

This standard is intended to be compatible with all IEEE 802 MACs.

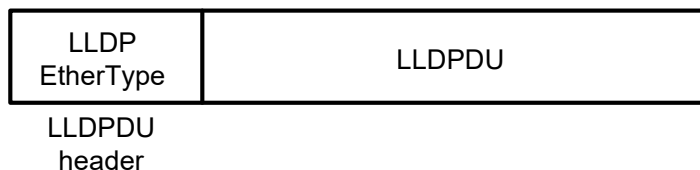
LLDP uses the service provided by the LLDP/LSAP and LLC to transmit and receive LLDPDUs. Each LLDPDU is transmitted as a single MAC service request by an LLC entity that uses a single instance of the MAC Service provided at an MSAP. Each incoming LLDP frame is received at the MSAP by the LLC entity as a MAC service indication.

NOTE—For the purposes of this standard, the terms “LLC” and “LLC entity” include the service provided by the operation of entities that support protocol discrimination using an EtherType, i.e., protocol discrimination based on the Type interpretation of the Length/Type field as specified in IEEE Std 802.3, or the use of the SNAP encapsulation of EtherTypes as described in IEEE Std 802. Hence, “LSAP” in the above description can refer to the use of ISO/IEC 8802-2 [B15] LLC addresses, or an EtherType, as a means of higher layer protocol identification.

The parameters of each service request and service indication comprise

- a) Destination address
- b) Source address
- c) EtherType
- d) LLDPDU

The LLDP EtherType, used to identify the LLDP protocol, is prepended to the LLDPDU as shown in Figure 7-1 to form the MSDU of the corresponding MAC service request.



**Figure 7-1—MSDU format**

The values of the parameters used by LLDP, and their encoding by the LLC entity that supports the LLDP LSAP, are specified in the following subclauses.

### 7.1 Destination address

#### 7.1.1 Destination address for Normal LLDPDUs

The MAC address used as the destination address in Normal LLDPDUs is referred to as the Scope MAC Address because it determines the limits of propagation of the LLDPDU within a network. A set of standard group MAC addresses that can be used by LLDP as the Scope MAC Address is specified in Table 7-1. These addresses are in the range of IEEE Std 802.1Q C-VLAN and MAC Bridge component Reserved addresses (see Table 8-1 in IEEE Std 802.1Q-2022), the S-VLAN component Reserved addresses (see Table 8-2 in IEEE Std 802.1Q-2022), and the TPMR component Reserved addresses (see Table 8-3 in IEEE Std 802.1Q-2022). The choice of address used determines the scope of propagation of LLDPDUs within a bridged LAN, as follows:

- a) The **nearest bridge** group MAC address is an address that no conformant Two-Port MAC Relay (TPMR) component, S-VLAN component, C-VLAN component, or MAC Bridge component can forward. LLDPDUs transmitted using this destination address can therefore travel no further than those stations that can be reached via a single individual LAN from the originating station.

1 LLDPDUs received on this address therefore represent information about stations that are attached  
2 to the same individual LAN segment as the recipient station.

3 NOTE 1—This address was selected in order to make it possible to transmit an LLDP frame containing information  
4 specific to a single individual LAN, and for that information not to be propagated further than the extent of that  
5 individual LAN, to avoid the meaning of that information being misinterpreted. For example, a TLV containing the  
6 auto-negotiation status of an IEEE 802.3™ LAN would be open to misinterpretation if it were to be propagated via a  
7 Bridge onto a second IEEE 802.3 LAN, and would be meaningless if propagated onto a LAN based on a different MAC  
8 technology. This is the “LLDP\_Multicast address” that was used in IEEE Std 802.1AB-2005.

9 b) The **nearest non-TPMR bridge** group MAC address is an address that no conformant C-VLAN  
10 component, S-VLAN component, or MAC Bridge can forward. However, this address is relayed by  
11 TPMR components. Therefore, LLDPDUs received on this address represent information about  
12 stations that are not separated from the recipient station by any intervening C-VLAN component,  
13 S-VLAN component, or MAC Bridge. There may, however, be one or more TPMR components in  
14 the path between the originating and receiving stations.

15 NOTE 2—This address was selected in order to make it possible to communicate information between adjacent bridges  
16 and for that communication to be transparent to the presence or absence of TPMRs in the transmission path. This address  
17 is primarily intended for use within provider bridged networks.

18 c) The **nearest Customer Bridge** group MAC address is an address that no conformant C-VLAN  
19 component or MAC Bridge forwards. However, this address is relayed by TPMR components and  
20 S-VLAN components. Therefore, LLDPDUs received on this address represent information about  
21 stations that are not separated from the recipient station by any intervening C-VLAN components  
22 (e.g., a C-VLAN Bridge or Provider Edge Bridge). There may, however, be TPMR components  
23 and/or S-VLAN components in the path between the originating and receiving stations.

24 NOTE 3—This address was selected in order to make it possible to communicate information between adjacent  
25 Customer Bridges and for that communication to be transparent to the presence or absence of TPMRs or S-VLAN  
26 components in the communication path. The scope of this address is the same as that of a customer-to-customer  
27 MACSec connection.

**Table 7-1—Group MAC addresses used by LLDP**

| Name                           | Value             | Purpose                                                                                                            |
|--------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------|
| <i>Nearest bridge</i>          | 01-80-C2-00-00-0E | Propagation constrained to a single physical link; stopped by all types of bridge                                  |
| <i>Nearest non-TPMR bridge</i> | 01-80-C2-00-00-03 | Propagation constrained by all bridges other than TPMRs; intended for use within provider bridged networks         |
| <i>Nearest Customer Bridge</i> | 01-80-C2-00-00-00 | Propagation constrained by customer bridges; this gives the same coverage as a customer-customer MACSec connection |

28 NOTE 4—The LSAP or EtherType field is necessary to distinguish between different protocols conveyed in frames with  
29 the same group or individual destination address. Prior to revision of this standard to allow multiple address scopes, the  
30 destination address 01-80-C2-00-00-00 was only explicitly specified for Spanning Tree Protocol BPDUs. It is therefore  
31 possible that some implementations recognize only a frame’s destination address before applying priority handling  
32 resources intended to guard against BPDUs loss. Since LLDP rate limits LLDPDU transmission, the additional  
33 consumption of these resources by LLDPDUs is unlikely to pose a problem for robust implementations.

34 It is assumed in the foregoing definitions of the scope of these group MAC addresses that an end station  
35 attached to an individual LAN is a potential recipient of any of these addresses, and therefore such end  
36 stations fall within the scope of these addresses as well as the identified bridge components.

1 In addition to determining the scope of transmission, the support of more than one destination MAC address  
2 allows different sets of TLVs to be supported over the different transmission scopes.

3 In addition to the prescribed support for standard group MAC addresses shown in Table 7-1,  
4 implementations of LLDP may support the following as the Scope MAC Address:

- 5 d) Any group MAC address.
- 6 e) Any individual MAC address.

7 Support for the use of each of these destination addresses, for both transmission and reception of Normal  
8 LLDPDUs, is either mandatory, recommended, permitted, or not permitted, according to the type of system  
9 in which LLDP is implemented, as shown in Table 7-2.

**Table 7-2—Support for the Scope MAC Address in different systems**

| Address                            | C-VLAN Bridge | S-VLAN Bridge | TPMR Bridge   | End station |
|------------------------------------|---------------|---------------|---------------|-------------|
| <i>Nearest bridge</i>              | Mandatory     | Mandatory     | Mandatory     | Mandatory   |
| <i>Nearest non-TPMR bridge</i>     | Mandatory     | Mandatory     | Not permitted | Recommended |
| <i>Nearest Customer Bridge</i>     | Mandatory     | Not permitted | Not permitted | Recommended |
| <i>Any other group MAC address</i> | Permitted     | Permitted     | Permitted     | Permitted   |
| <i>Any individual MAC address</i>  | Permitted     | Permitted     | Permitted     | Permitted   |

10 NOTE 5 —Where the implementation supports the use of individual MAC addresses, there is a need for some means  
11 whereby the sender can ascertain what address to use, and what scope is associated with that address; how this is  
12 achieved is outside the scope of this standard.

13 NOTE 6—The option of using an individual MAC addresses supports the use of LLDP in IEEE Std 802.11™ [B1] and  
14 other wireless LANs, where it is desirable to target specific TLV content at specific logical Ports. In particular, the use of  
15 an individual MAC address allows an access point to direct LLDP frames at an individual station with which it is  
16 associated.

17 NOTE 7—If an individual MAC address is specified as the Scope MAC Address, it is used as the destination for  
18 transmitted LLDPDUs and also used to validate received LLDPDUs. The primary purpose of using an individual  
19 address as the Scope MAC Address is that an LLDP agent configured as “transmit only” sends Normal LLDPDUs to a  
20 specific receiving LLDP agent. It is also allowable for an LLDP agent configured as “receive only” to use an individual  
21 address (typically the individual MAC address of the LLDP agent’s MSAP) so that only information from an LLDP  
22 agent configured as “transmit only” using this same address is collected. Using an individual MAC address as the  
23 destination address for an LLDP agent configured to “transmit and receive” is allowed, although it would not typically  
24 be useful.

25 NOTE 8—The destination MAC address used by a given LLDP agent defines only the scope of transmission and the  
26 intended recipient(s) of the LLDPDUs; it plays no part in protocol identification. In particular, the group MAC addresses  
27 identified in Table 7-1 are not used exclusively by LLDP; other protocols that require to use a similar transmission scope  
28 are free to use the same addresses.

## 29 7.1.2 Destination address for Extension Request and Extension LLDPDUs

30 The destination address in an Extension Request LLDPDU or an Extension LLDPDU is an individual  
31 address. Extension Request LLDPDUs are sent in response to the receipt of a Manifest LLDPDU, and use  
32 the contents of the Return MAC Address field of the Manifest TLV as the destination address of an  
33 Extension Request LLDPDU. Likewise, Extension LLDPDUs are sent in response to the receipt of an  
34 Extension Request LLDPDU, and use the contents of the Return MAC Address field of the Extension  
35 Request TLV as the destination address of the Extension LLDPDU.

1 **7.2 Source address**

2 The source address shall be the individual MAC address of the MSAP supporting the LLDP agent.

3 **7.3 EtherType use and encoding**

4 The EtherType used to identify the LLDP protocol shall be the LLDP EtherType specified in Table 7-3.

**Table 7-3—LLDP EtherType**

| Name           | Value |
|----------------|-------|
| LLDP EtherType | 88-CC |

5 Where the LLC entity uses an MSAP that is supported by a specific media access control method (for  
6 example, IEEE Std 802.3) or a media access control independent entity (for example, IEEE Std 802.1AE)  
7 that directly supports encoding of EtherTypes, the LLC entity shall encode the LLDP EtherType as the two  
8 octet LLDPDU header in the MSDU of the corresponding MAC service request.

9 Where the LLC entity uses an MSAP that is supported by a specific media access control method that does  
10 not directly support Ethertype encoding (for example, IEEE Std 802.11 [B1]), the encoding shall be as  
11 specified in IEEE Std 802.1AC.

12 NOTE—Annex C provides example LLDP transmission frame formats for both direct-encoded and SNAP-encoded  
13 LLDP EtherType encoding methods.

14 **7.4 LLDPDU reception**

15 The LLDPDU shall be delivered to the LLDP receive module if, and only if

- 16 a) The destination MAC address is equal to the individual MAC address of the MSAP supporting the  
17 LLDP agent, or the destination MAC address is equal to the Scope MAC Address associated with  
18 the corresponding LLDP agent; and

19 NOTE—The destination MAC address can be one of the addresses in Table 7-1, or any other valid MAC address—see  
20 7.1.

- 21 b) The EtherType is equal to the value shown in Table 7-3.

## 1 8. LLDPDU and TLV formats

### 2 8.1 LLDPDU bit and octet ordering conventions

3 All LLDPDUs contain an integral number of octets. The octets in an LLDPDU are numbered starting from 1  
4 and increasing in the order they are put into the LLDPDU. The bits in an octet are numbered from 1 to 8,  
5 where bit 1 is the low-order bit.

6 When consecutive bits within an octet are used to represent a binary number, the highest bit number has the  
7 most significant value. When consecutive octets are used to represent a binary number, the lower octet  
8 number has the most significant value. All TLVs respect these bit and octet ordering conventions.

9 When the encoding of a field or a number of fields is represented using a diagram

- 10 a) Octets are shown with the lowest numbered octet nearest the left of the page, the octet numbering  
11 increasing from left to right.  
12 b) Within an octet, bits are shown with bit 8 to the left and bit 1 to the right.

### 13 8.2 LLDPDU format

14 The LLDPDU contains the following ordered sequence of three mandatory TLVs followed by zero or more  
15 optional TLVs as shown in Figure 8-1. An End of LLDPDU TLV may be present as the last TLV in the  
16 LLDPDU.

- 17 a) Three mandatory TLVs are included at the beginning of each LLDPDU and are in the order shown.  
18 1) Chassis ID TLV  
19 2) Port ID TLV  
20 3) Time To Live TLV or, if XLLDP is supported, Extension Request TLV or Extension Identifier  
21 TLV  
22 b) Optional TLVs as selected by network management (may be inserted in any order).

23 NOTE 1—"Optional" in the sense that they are not required for LLDP operation; however, their presence could be  
24 required by other system elements that use LLDP.

- 25 c) If the End Of LLDPDU TLV is present, it is the last TLV in the LLDPDU.

1

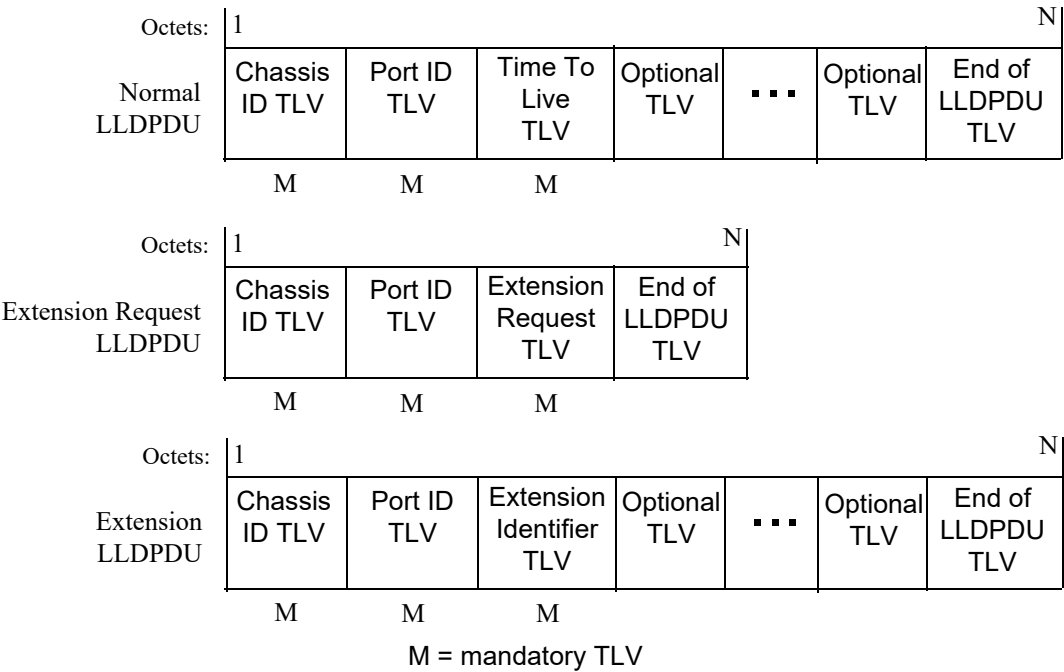


Figure 8-1—LLDPDU formats

2 The maximum length of the LLDPDU is less than or equal to the maximum information field length allowed  
3 by the particular transmission rate and protocol. In IEEE 802.3 MACs, for example, the maximum LLDPDU  
4 length is the maximum data field length for the basic, untagged MAC frame (1500 octets).

5 NOTE 2—Implementations can restrict the maximum length of the LLDPDU to a value smaller than what would be  
6 allowed by the underlying MAC technology. For example, an end station or bridge implementing Time-Sensitive  
7 Networking (TSN) features could restrict the maximum length to limit delay variation when transmitting frames in time  
8 sensitive streams.

9 NOTE 3—There is no defined minimum length of an LLDPDU, other than that implied by the requirement that  
10 conformant implementations support the mandatory TLVs specified in Table 8-1.

11 **8.3 TLV categories**

12 The general format of LLDP TLVs is defined in 8.4. The TLVs are grouped into three categories as follows:

- 13 a) A set of TLVs that are considered to be basic to the management of network stations. Support of the  
14 basic management set is required capability of all LLDP implementations. Each TLV in this  
15 category is identified by a unique TLV type value that indicates the particular kind of information  
16 contained in the TLV. The specific definitions and usage rules for the basic management set TLVs  
17 are defined in 8.5.
- 18 b) A set of TLVs that support Extended LLDP (XLLDP) operation. Support of the extended LLDP set  
19 is a required capability of all LLDP implementations supporting XLLDP. Each TLV in this category  
20 is identified by a unique TLV type value that indicates the particular kind of information contained  
21 in the TLV. The specific definitions and usage rules for the extended LLDP set TLVs are defined in  
22 8.6.
- 23 c) Organizationally specific extension sets of TLVs that are defined by standards groups such as  
24 IEEE 802.1 and IEEE 802.3 and others to enhance management of network stations that are

- 1
- operating with particular media and/or protocols. The format of Organizationally Specific TLVs, along with field definitions and usage rules common to all Organizationally Specific TLVs, are defined in 8.7.
- 2
- 3
- 4
- 1) TLVs in this category are identified by a common TLV type value that indicates the TLV as belonging to the set of Organizationally Specific TLVs.
- 5
- 6
- 2) Each organization is identified by its organizationally unique identifier (OUI), consisting of either an Organizationally Unique Identifier (OUI) or a Company ID (CID) <sup>12</sup>.
- 7
- 8
- 3) Organizationally Specific TLV subtype values indicate the kind of information contained in the TLV.
- 9

10

8.4 Basic TLV format

11

Figure 8-2 shows the basic TLV format.

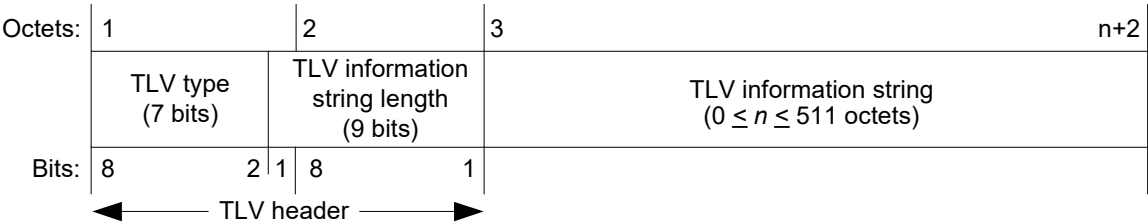


Figure 8-2—Basic TLV format

12

The TLV type field occupies the seven most significant bits of the first octet of the TLV format. The least

13

significant bit in the first octet of the TLV format is the most significant bit of the TLV information string

14

length field.

15

8.4.1 TLV type

16

The TLV type field is seven bits long and identifies the specific TLV. Two classes of TLVs are defined:

- 17
- a) Mandatory TLVs that are included in all LLDPDUs.
- 18
- b) Optional TLVs that may be included in LLDPDUs.

19

Table 8-1 lists the currently defined TLVs, their identifying TLV type values, and whether they are

20

mandatory or optional for inclusion in any particular LLDPDU.

21

8.4.2 TLV information string length

22

The TLV information string length field contain the length of the information string, in octets.

23

8.4.3 TLV information string

24

The information string

- 25
- a) May be fixed or variable length.
- 26
- b) May include one or more information fields with associated subtype identifiers and field length
- 27
- designators as in, for example, the Management Address TLV (see 8.5.9).

<sup>12</sup> Organizationally Unique Identifier and Company ID assignments are administered by the IEEE Registration Authority, and Organizationally Unique Identifier assignments are included in MAC Address-Large (MA-L) assignments; see <http://standards.ieee.org/regauth>.

**Table 8-1—TLV type values**

| TLV type <sup>a</sup> | TLV name                            | Usage in LLDPDU | Reference |
|-----------------------|-------------------------------------|-----------------|-----------|
| 0                     | End Of LLDPDU                       | Optional        | 8.5.1     |
| 1                     | Chassis ID                          | Mandatory       | 8.5.2     |
| 2                     | Port ID                             | Mandatory       | 8.5.3     |
| 3                     | Time To Live                        | Mandatory       | 8.5.4     |
| 4                     | Port Description                    | Optional        | 8.5.5     |
| 5                     | System Name                         | Optional        | 8.5.6     |
| 6                     | System Description                  | Optional        | 8.5.7     |
| 7                     | System Capabilities                 | Optional        | 8.5.8     |
| 8                     | Management Address                  | Optional        | 8.5.9     |
| 9                     | Manifest                            | Optional        | 8.6.1     |
| 10                    | Extension Request                   | Optional        | 8.6.2     |
| 11                    | Extension Identifier                | Optional        | 8.6.3     |
| 12–126                | Reserved for future standardization | —               | —         |
| 127                   | Organizationally Specific TLVs      | Optional        | —         |

<sup>a</sup>TLVs with type values 0–8 are members of the basic management set. TLVs with type values 9–11 are members of the extended LLDP set.

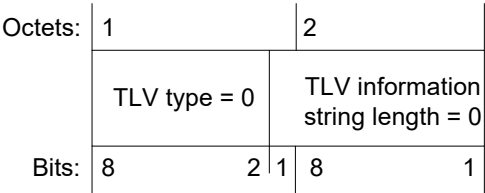
- 1 c) May contain either binary or alpha-numeric information that is instance specific for the particular
- 2 TLV type and/or subtype.
- 3 1) Bit 1 in binary bit maps is the least significant bit in the field.
- 4 2) The first octet of an alpha-numeric field is the most significant octet.
- 5 3) Alpha-numeric information is encoded in UTF-8 [IETF RFC 3629].



1 **8.5 Basic management TLV set formats and definitions**

2 **8.5.1 End Of LLDPDU TLV**

3 The End Of LLDPDU TLV is a 2-octet, all-zero TLV that is used to mark the end of the TLV sequence in  
4 LLDPDUs. The format for this TLV is shown in Figure 8-3.



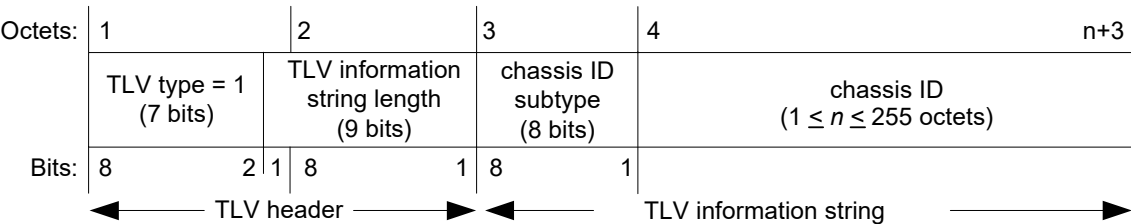
**Figure 8-3—End Of LLDPDU TLV format**

5 NOTE—Some IEEE 802 MACs require the data field in a frame to contain a minimum number of octets. For example,  
6 the IEEE 802.3 MAC adds pad octets to complete a minimum length data field if the user’s data is less than the  
7 minimum required length. Since pad octets are unspecified, an End Of LLDPDU TLV can be used to prevent non-zero  
8 pad octets from being interpreted by the receiving LLDP agent as another TLV.

9 **8.5.2 Chassis ID TLV**

10 The Chassis ID TLV is a mandatory TLV that identifies the chassis containing the IEEE 802 LAN station  
11 associated with the MSAP Identifier associated with the LLDP agent containing the LLDP local system  
12 information repository that is the source of the information in TLVs. There are several ways in which a  
13 chassis may be identified and a chassis ID subtype is used to indicate the type of component being  
14 referenced by the chassis ID field. Each LLDPDU contain one, and only one, Chassis ID TLV and the  
15 chassis ID field value remains constant for all LLDPDUs while the MAC status parameter for the Port  
16 remains operational.

17 The Chassis ID TLV is the first TLV in the LLDPDU. Its format is shown in Figure 8-4.



**Figure 8-4—Chassis ID TLV format**

18 **8.5.2.1 TLV information string length**

19 The TLV information string length field indicates the exact length, in octets, of the (chassis ID subtype +  
20 chassis ID) fields.

21 **8.5.2.2 chassis ID subtype**

22 The chassis ID subtype field contains an integer value indicating the basis for the chassis ID entity that is  
23 listed in the chassis ID field. The defined chassis ID subtypes and their preferred use order are listed in Table  
24 8-2.

Table 8-2—chassis ID subtype enumeration

| ID subtype | ID basis          | Reference                                                                                       |
|------------|-------------------|-------------------------------------------------------------------------------------------------|
| 0          | Reserved          | —                                                                                               |
| 1          | Chassis component | EntPhysicalAlias when entPhysClass has a value of ‘chassis(3)’ (IETF RFC 6933)                  |
| 2          | Interface alias   | IfAlias (IETF RFC 2863)                                                                         |
| 3          | Port component    | EntPhysicalAlias when entPhysicalClass has a value ‘port(10)’ or ‘backplane(4)’ (IETF RFC 6933) |
| 4          | MAC address       | MAC address (IEEE Std 802)                                                                      |
| 5          | Network address   | networkAddress <sup>a</sup>                                                                     |
| 6          | Interface name    | ifName (IETF RFC 2863)                                                                          |
| 7          | Locally assigned  | local <sup>b</sup>                                                                              |
| 8–255      | Reserved          | —                                                                                               |

<sup>a</sup>networkAddress is an octet string that identifies a particular network address family and an associated network address that are encoded in network octet order. An IP address, for example, would be encoded with the first octet containing the IANA Address Family Numbers enumeration value for the specific address type and octets 2 through *n* containing the address value (for example, the encoding for C0-00-02-0A would indicate the IPv4 address 192.0.2.10).

<sup>b</sup>local is an alpha-numeric string and is locally assigned.

1 8.5.2.3 chassis ID

2 The chassis ID field contains an octet string indicating the specific identifier for the particular chassis in this  
3 system. Because chassis ID and port ID values are concatenated to form the local MSAP identifier, the value  
4 chosen from Table 8-2 for the chassis ID is non-null.

5 8.5.2.4 Chassis ID TLV usage rules

6 An LLDPDU contains exactly one Chassis ID TLV.

7 8.5.3 Port ID TLV

8 The Port ID TLV is a mandatory TLV that identifies the port component of the MSAP identifier associated  
9 with the MSAP Identifier associated with the LLDP agent containing the LLDP local system information  
10 repository that is the source of the information in TLVs. As with the chassis, there are several ways in which  
11 a port is identified. A port ID subtype is used to indicate how the port is being referenced in the port ID field.  
12 Each LLDPDU contain one, and only one, Port ID TLV. The port ID value remain constant for all  
13 LLDPDUs while the transmitting port remains operational.

14 The chosen port is identified in an unambiguous manner (for example, backplane number, management  
15 entity, MAC address).

16 The Port ID TLV is the second TLV in the LLDPDU. Its format is shown in Figure 8-5.

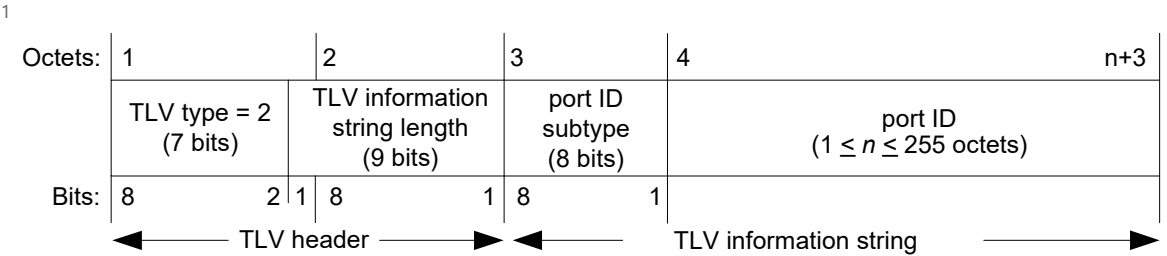


Figure 8-5—Port ID TLV format

2 8.5.3.1 TLV information string length

3 The TLV information string length field indicates the length, in octets, of the (port ID subtype + port ID)  
4 fields.

5 8.5.3.2 port ID subtype

6 The port ID subtype field contains an integer value indicating the basis for the identifier that is listed in the  
7 port ID field. The defined port ID subtypes and their preferred use order are listed in Table 8-3.

Table 8-3—port ID subtype enumeration

| ID subtype | ID basis         | References                                                                                      |
|------------|------------------|-------------------------------------------------------------------------------------------------|
| 0          | Reserved         | —                                                                                               |
| 1          | Interface alias  | ifAlias (IETF RFC 2863)                                                                         |
| 2          | Port component   | entPhysicalAlias when entPhysicalClass has a value 'port(10)' or 'backplane(4)' (IETF RFC 6933) |
| 3          | MAC address      | MAC address (IEEE Std 802)                                                                      |
| 4          | Network address  | networkAddress <sup>a</sup>                                                                     |
| 5          | Interface name   | ifName (IETF RFC 2863)                                                                          |
| 6          | Agent circuit ID | agent circuit ID (IETF RFC 3046)                                                                |
| 7          | Locally assigned | local <sup>b</sup>                                                                              |
| 8–255      | Reserved         | —                                                                                               |

<sup>a</sup>networkAddress is an octet string that identifies a particular network address family and an associated network address that are encoded in network octet order. An IP address, for example, would be encoded with the first octet containing the IANA Address Family Numbers enumeration value for the specific address type and octets 2 through *n* containing the address value (for example, the encoding for C0-00-02-0A would indicate the IP version 4 address 192.0.2.10).

<sup>b</sup>local is an alpha-numeric string and is locally assigned.

8 8.5.3.3 port ID

9 The port ID field is an alpha-numeric string that contains the specific identifier for the port from which this  
10 LLDPDU was transmitted. Because chassis ID and port ID values are concatenated to form the local MSAP  
11 identifier, the value chosen from Table 8-3 for the port ID is non-null.

12 NOTE—The port ID can contain alphanumeric data or binary data, depending upon the value of the port ID subtype.

### 8.5.3.4 Port ID TLV usage rules

2 An LLDPDU contains exactly one Port ID TLV.

### 8.5.4 Time To Live TLV

4 The Time To Live TLV indicates the number of seconds that the recipient LLDP agent is to regard the  
5 information associated with this MSAP identifier to be valid.

- 6 a) When the TTL field is non-zero the receiving LLDP agent is notified to completely replace all  
7 information associated with this MSAP identifier with the information in the received LLDPDU.
- 8 b) When the TTL field is set to zero, the receiving LLDP agent is notified to delete all system  
9 information associated with the LLDP agent/port. This TLV may be used, for example, to signal that  
10 the sending port has initiated a port shutdown procedure.

11 The Time To Live TLV is mandatory for all Normal LLDPDUs, and is the third TLV in the LLDPDU. Its  
12 format is shown in Figure 8-6.

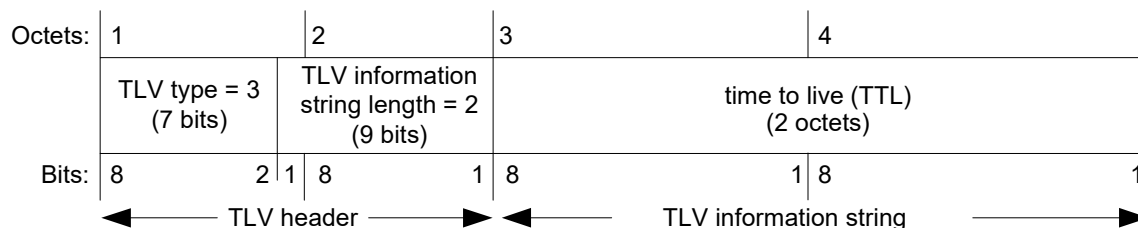


Figure 8-6—Time To Live TLV format

#### 8.5.4.1 time to live (TTL)

14 The TTL field contains an integer value in the range  $0 \leq t \leq 65535$  seconds and is set to the computed value  
15 of txTTL at the time the LLDPDU is constructed (see 9.2.5.22).

#### 8.5.4.2 Time To Live TLV usage rules

17 A Normal LLDPDU contains exactly one Time To Live TLV as the third TLV, and does not contain more  
18 than one Time To Live TLV. A Time To Live TLV is not contained in an Extension LLDPDU or an  
19 Extension Request LLDPDU.

### 8.5.5 Port Description TLV

21 The Port Description TLV allows network management to advertise the IEEE 802 LAN station's port  
22 description. The format for this TLV is shown in Figure 8-7.

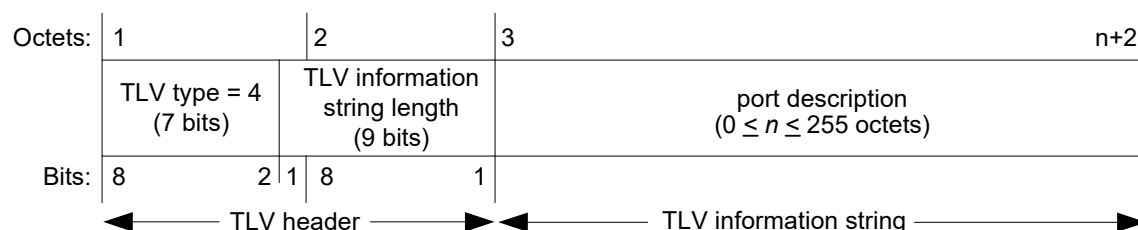


Figure 8-7—Port Description TLV format

1 **8.5.5.1 TLV information string length**

2 The TLV information string length field contains the exact length, in octets, of the port description field.

3 **8.5.5.2 port description**

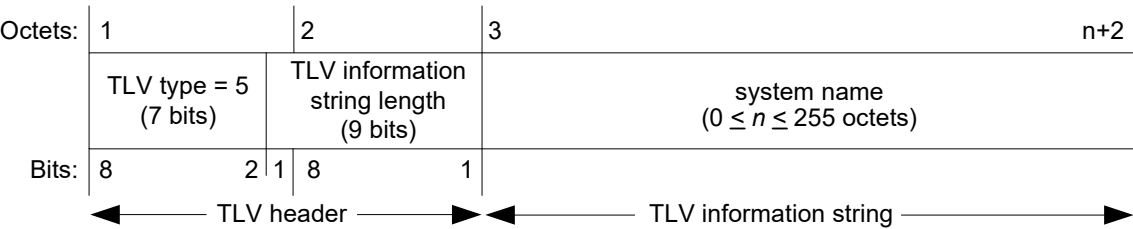
4 The port description field contains an alpha-numeric string that indicates the port’s description. If  
5 IETF RFC 2863 is implemented, the ifDescr object should be used for this field.

6 **8.5.5.3 Port Description TLV usage rules**

7 An LLDPDU should not contain more than one Port Description TLV.

8 **8.5.6 System Name TLV**

9 The System Name TLV allows network management to advertise the system’s assigned name. The format  
10 for this TLV is shown in Figure 8-8.



**Figure 8-8—System Name TLV format**

11 **8.5.6.1 TLV information string length**

12 The TLV information string length field contains the exact length, in octets, of the system name.

13 **8.5.6.2 system name**

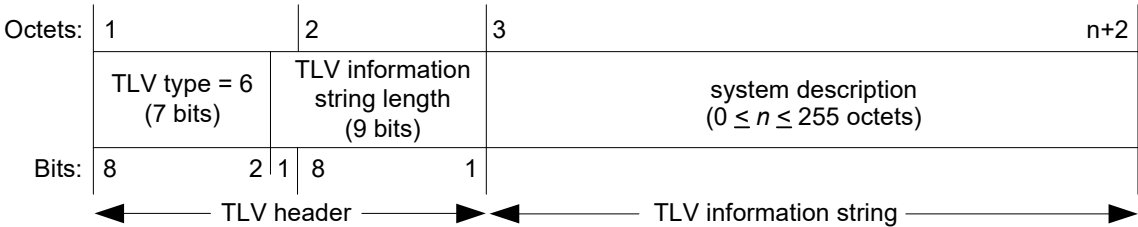
14 The system name field contains an alpha-numeric string that indicates the system’s administratively  
15 assigned name. The system name should be the system’s fully qualified domain name. If implementations  
16 support IETF RFC 3418, the sysName object should be used for this field.

17 **8.5.6.3 System Name TLV usage rules**

18 An LLDPDU does not contain more than one System Name TLV.

1 **8.5.7 System Description TLV**

2 The System Description TLV allows network management to advertise the system’s description. The format  
3 for this TLV is shown in Figure 8-9.



**Figure 8-9—System Description TLV format**

4 **8.5.7.1 TLV information string length**

5 The TLV information string length field indicates the exact length, in octets, of the system description.

6 **8.5.7.2 system description**

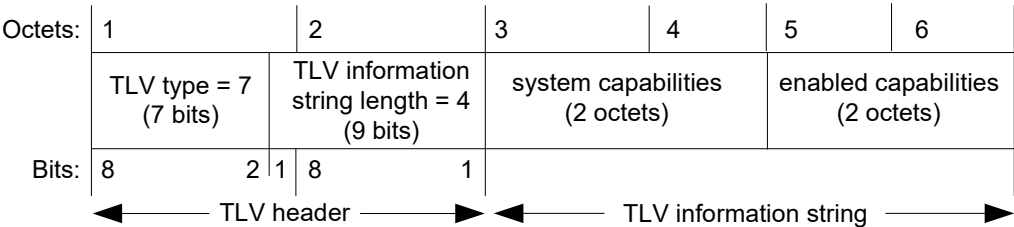
7 The system description field contains an alpha-numeric string that is the textual description of the network  
8 entity. The system description should include the full name and version identification of the system’s  
9 hardware type, software operating system, and networking software. If implementations support IETF RFC  
10 3418, the sysDescr object should be used for this field.

11 **8.5.7.3 System Description TLV usage rules**

12 An LLDPDU does not contain more than one System Description TLV.

13 **8.5.8 System Capabilities TLV**

14 The System Capabilities TLV is an optional TLV that identifies the primary function(s) of the system and  
15 whether or not these primary functions are enabled. Figure 8-10 shows the format of this TLV.



**Figure 8-10—System Capabilities TLV format**

16 **8.5.8.1 system capabilities**

17 The system capabilities field contains a bit-map of the capabilities that define the primary function(s) of the  
18 system. The bit positions for each function and the associated information repository or standard that are  
19 likely, but not guaranteed, to be supported are listed in Table 8-4. A binary one in the associated bit indicates  
20 the existence of that capability. Individual systems may indicate more than one implemented functional  
21 capability (for example, both a bridge and router capability).

**Table 8-4—System capabilities**

| Bit   | Capability                   | Reference                             |
|-------|------------------------------|---------------------------------------|
| 1     | Other                        | —                                     |
| 2     | Repeater                     | IETF RFC 2108 [B6]                    |
| 3     | MAC Bridge component         | IEEE Std 802.1Q                       |
| 4     | 802.11 Access Point (AP)     | IEEE Std 802.11 MIB                   |
| 5     | Router                       | IETF RFC 1812 [B5]                    |
| 6     | Telephone                    | IETF RFC 4293 [B9]                    |
| 7     | DOCSIS cable device          | IETF RFC 4639 and IETF RFC 4546 [B11] |
| 8     | Station Only <sup>a</sup>    | IETF RFC 4293 [B9]                    |
| 9     | C-VLAN component             | IEEE Std 802.1Q                       |
| 10    | S-VLAN component             | IEEE Std 802.1Q                       |
| 11    | Two-port MAC Relay component | IEEE Std 802.1Q                       |
| 12–16 | reserved                     | —                                     |

<sup>a</sup>The Station Only capability is intended for devices that implement only an end station capability, and for which none of the other capabilities in the table apply. Bit 8 should therefore not be set in conjunction with any other bits.

#### 1 8.5.8.2 enabled capabilities

2 The enabled capabilities field contains a bit map of the primary functions listed in Table 8-4. A binary one in  
3 a bit position indicates that the function associated with that bit is currently enabled.

#### 4 8.5.8.3 System Capabilities TLV usage rules

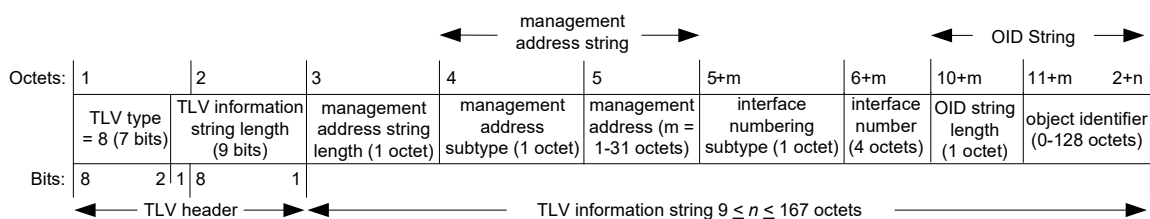
5 An LLDPDU does not contain more than one System Capabilities TLV.

6 If the system capabilities field does not indicate the existence of a capability that the enabled capabilities  
7 field indicates is enabled, the TLV is interpreted as containing an error and is discarded.

#### 8 8.5.9 Management Address TLV

9 The Management Address TLV identifies an address associated with the local LLDP agent that may be used  
10 to reach higher layer entities to assist discovery by network management. The TLV also provides room for  
11 the inclusion of both the system interface number and an object identifier (OID) that are associated with this  
12 management address, if either or both are known. Figure 8-11 shows the Management Address TLV format.

1



**Figure 8-11—Management Address TLV format**

## 2 8.5.9.1 TLV information string length

3 The TLV information string length field contains the length, in octets, of all the fields in the TLV  
4 information string.

## 5 8.5.9.2 management address string length

6 The management address string length field contains the length, in octets, of the (management address  
7 subtype + management address) fields.

## 8 8.5.9.3 management address subtype

9 The management address subtype field contains an integer value indicating the type of address that is listed  
10 in the management address field. Enumeration for this field is contained in the `ianaAddressFamilyNumbers`  
11 module of the IETF RFC 3232 on-line database that is accessible through a web page (<http://www.iana.org>).  
12 The management address subtype is contained in the first octet of the management address string.

13 NOTE—As a consequence of the limitation on the size of the management address field (a maximum of 31 octets), some  
14 options for the management address subtype (such as long DNS names) may be impractical.

## 15 8.5.9.4 management address

16 The management address field contains an octet string indicating the particular management address  
17 associated with this TLV.

- 18 a) The returned address should be the most appropriate for management use, typically a layer 3 address  
19 such as the IPv4 address, 192.0.2.10 (see also Table 8-2, footnote a).
- 20 b) If no management address is available, the return address should be the MAC address for the station  
21 or port. The `ianaAddressFamilyNumber` for a MAC address is all802 (value 6).

## 22 8.5.9.5 interface numbering subtype

23 The interface numbering subtype field contains an integer value indicating the numbering method used for  
24 defining the interface number. The following three values are currently defined:

- 25 1) Unknown
- 26 2) `ifIndex`
- 27 3) `system port number`

## 28 8.5.9.6 interface number

29 The interface number field contains the assigned number within the system that identifies the specific  
30 interface associated with this management address. If the value of the interface subtype is unknown, this  
31 field is set to zero.



#### 1 8.5.9.7 object identifier (OID) string length

2 The object identifier string length field contains the length, in octets, of the OID. A value of zero in this field  
3 indicates that the OID field is not provided.

#### 4 8.5.9.8 object identifier

5 The object identifier field contains an OID that identifies the type of hardware component or protocol entity  
6 associated with the indicated management address. The OID is the value portion of the ASN.1 encoding of  
7 the object identifier, encoded according to ASN.1 basic encoding rules [ISO/IEC 8824-1]. If no OID is  
8 available, this field is not provided.

9 NOTE—The interface number and OID are included in this TLV to assist NMS discovery by indicating Enterprise  
10 Specific or other starting points for the search, such as the Interface or Entity MIB (IETF RFC 6933).

#### 11 8.5.9.9 Management Address TLV usage rules

12 Management Address TLVs are subject to the following:

- 13 a) At least one Management Address TLV should be included in every LLDPDU.
  - 14 b) Since there are typically a number of different addresses associated with a MSAP identifier, an  
15 individual LLDPDU may contain more than one Management Address TLV.
  - 16 c) When Management Address TLV(s) are included in an LLDPDU, the included address(es) should  
17 be the address(es) offering the best management capability.
  - 18 d) If more than one Management Address TLV is included in an LLDPDU, each management address  
19 is different from the management address in any other management address TLV in the LLDPDU.
  - 20 e) If an OID is included in the TLV, it is reachable by the management address.
  - 21 f) In a properly formed Management Address TLV, the TLV information string length is equal to:  
22 (management address string length) + (OID string length) + 7.
- 23 If the TLV information string length in a received Management Address TLV is incorrect, then it is  
24 ignored and processing of that LLDPDU is terminated.

## 8.6 Extended LLDP set TLV formats and definitions

### 8.6.1 Manifest TLV

The Manifest TLV allows an LLDP agent supporting XLLDP to advertise more TLVs than fits in a single LLDPDU. The Manifest TLV contains a list of Extension PDUs that convey the additional TLVs. The format for this TLV is shown in Figure 8-12.

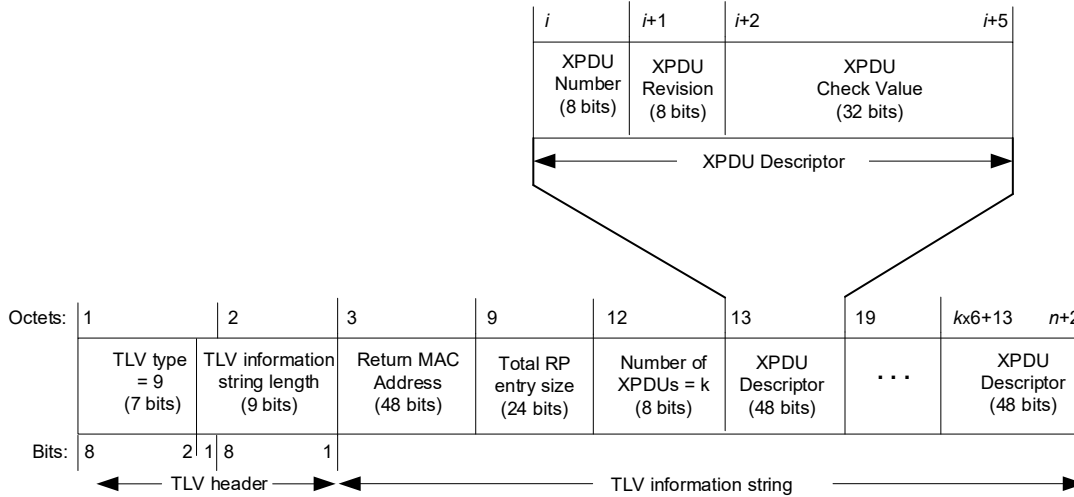


Figure 8-12—Manifest TLV format

#### 8.6.1.1 TLV information string length

The TLV information string length field contains the number of octets in the TLV information string.

#### 8.6.1.2 Return MAC Address

The individual MAC Address of the MSAP supporting the LLDP agent, to be used as the destination address for Extension Request LLDPDUs generated in response to this Manifest TLV.

#### 8.6.1.3 Total information repository (RP) entry size

The total information repository entry size is the number of octets in all TLVs associated with this MSAP Identifier (regardless of the number of XPDUs used to convey those TLVs). It is the sum of the information string length fields in all TLVs, plus two times the number of TLVs, and includes the TLVs in the initial Normal LLDPDU. This allows the LLDP remote systems information repository manager to determine whether it expects to have space to store the information prior to requesting any XPDUs.

#### 8.6.1.4 Number of XPDUs

The number of XPDUs field contains the count of the XPDU Descriptors in this TLV. Since the maximum size of the TLV information string is 511 octets, the minimum number of XPDU Descriptors is 1 and the maximum number is 83. When the number of XPDU Descriptors is  $k$ , the TLV information string length is at least  $k \times 6 + 10$ , but it need not be exactly  $k \times 6 + 10$ . This allows an implementation to use a fixed length TLV information string with a variable number of XPDU Descriptors.

1 **8.6.1.5 XPDU Descriptor**

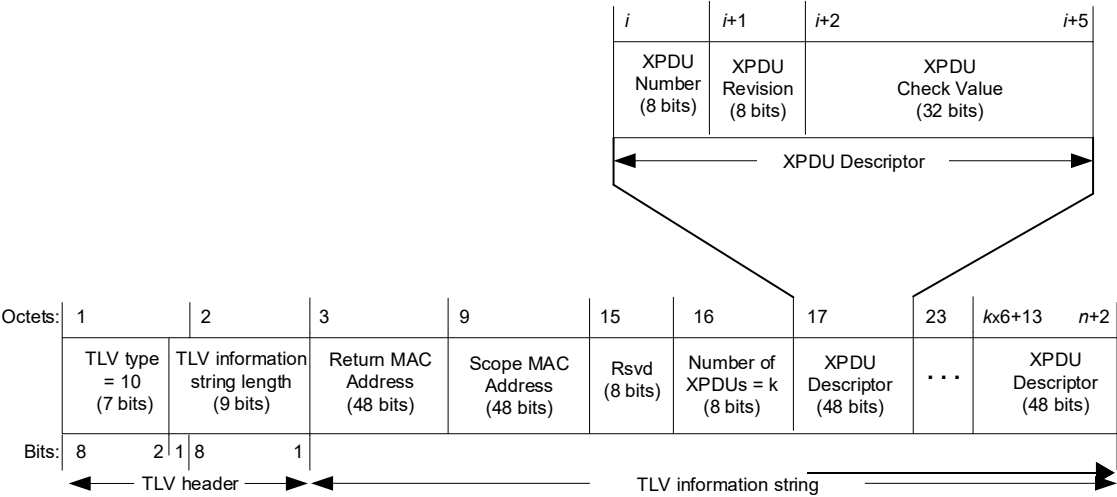
2 An XPDU Descriptor comprises an 8-bit XPDU Number, an 8-bit XPDU Revision, and a 32-bit XPDU  
3 Check Value. The XPDU Number uniquely identifies each XPDU in a manifest. The XPDU Revision is  
4 assigned a random value at initialization, and is incremented (modulo 256) each time there is a change to  
5 one or more TLVs in the XPDU. The XPDU Check Value is the low order 32 bits of an MD5 hash calculated  
6 across all octets of the XPDU.

7 **8.6.1.6 Manifest TLV usage rules**

8 An LLDPDU does not contain more than one Manifest TLV. Each XPDU Descriptor associated with an  
9 MSAP identifier and Scope MAC Address have a unique XPDU Number.

10 **8.6.2 Extension Request TLV**

11 The Extension Request TLV allows an LLDP agent supporting XLLDP to request the neighbor to transmit  
12 specific Extension PDUs. The format for this TLV is shown in Figure 8-13.



**Figure 8-13—Extension Request TLV format**

13 **8.6.2.1 TLV information string length**

14 The TLV information string length field contains the number of octets in the TLV information string.

15 **8.6.2.2 Return MAC Address**

16 The individual MAC address of the MSAP supporting the LLDP agent, to be used as the destination address  
17 for the requested Extension LLDPDUs.

18 **8.6.2.3 Scope MAC Address**

19 The MAC Address used by this LLDP agent as the destination address for Normal LLDPDUs.

20 **8.6.2.4 Rsvd**

21 The rsvd field is transmitted as zero and ignored on receipt.

1 **8.6.2.5 Number of XPDUs**

2 The number of XPDUs field contains the count of the XPDU Descriptors in this TLV. Since the maximum  
3 size of the TLV information string is 511 octets, the minimum number of XPDU Descriptors is 1 and the  
4 maximum number is 82. When the number of XPDU Descriptors is  $k$ , the TLV information string length is  
5 at least  $k \times 6 + 14$ , but it need not be exactly  $k \times 6 + 14$ . This allows an implementation to use a fixed length  
6 TLV information string with a variable number of XPDU Descriptors.

7 **8.6.2.6 XPDU Descriptor**

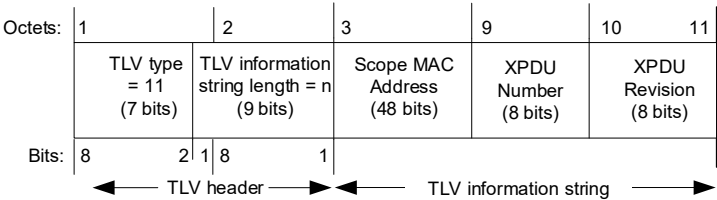
8 An XPDU Descriptor comprises an 8-bit XPDU Number, an 8-bit XPDU Revision, and a 32-bit XPDU  
9 Check, as specified in 8.6.1.5.

10 **8.6.2.7 Extension Request TLV usage rules**

11 An Extension Request LLDPDU contain an Extension Request TLV as the third TLV, and not contain more  
12 than one Extension Request TLV. An Extension Request TLV not be contained in a Normal LLDPDU or an  
13 Extension LLDPDU. Each XPDU Descriptor associated with a MSAP identifier and Scope MAC Address  
14 have a unique XPDU Number.

15 **8.6.3 Extension Identifier TLV**

16 The Extension Identifier TLV identifies an XPDU containing one or more TLVs. The format for this TLV is  
17 shown in Figure 8-11c.



**Figure 8-14—Extension Identifier TLV format**

18 **8.6.3.1 TLV information string length**

19 The TLV information string length field contain the number of octets in the TLV information string.

20 **8.6.3.2 Scope MAC Address**

21 The MAC Address used by this LLDP agent as the destination address for Normal LLDPDUs.

22 **8.6.3.3 XPDU Number**

23 The XPDU Number portion of the XPDU Descriptor specified in 8.6.1.5.

24 **8.6.3.4 XPDU Revision**

25 The XPDU Revision portion of the XPDU Descriptor specified in 8.6.1.5.

1 **8.6.3.5 Extension Identifier TLV usage rules**

2 An Extension LLDPDU contains an Extension Identifier TLV as the third TLV, and does not contain more  
3 than one Extension Identifier TLV. An Extension Identifier TLV is not contained in a Normal LLDPDU or  
4 an Extension Request LLDPDU.

5 **8.7 Organizationally Specific TLVs**

6 This TLV category is provided to allow different organizations, such as IEEE 802.1, IEEE 802.3, IETF, as  
7 well as individual software and equipment vendors, to define TLVs that advertise information to remote  
8 entities attached to the same media, subject to the following restrictions:

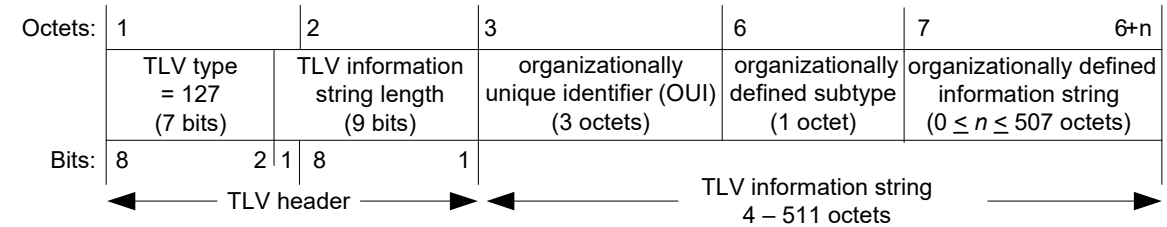
- 9 a) Information transmitted in an Organizationally Specific TLV is intended to be a one way  
10 advertisement.
- 11 b) Information transmitted in an Organizationally Specific TLV is independent from information in a  
12 TLV received from a remote port.
- 13 c) Information transmitted in one Organizationally Specific TLV is not concatenated with information  
14 transmitted in another TLV on the same media in order to provide a means for sending messages that  
15 are larger than would fit within a single TLV.
- 16 d) Information received in an Organizationally Specific TLV is not explicitly forwarded to other ports  
17 in the system.
- 18 e) Organizationally Specific TLVs conform to 8.4 and 8.7.1.

19 Each set of Organizationally Specific TLVs includes associated LLDP MIB or YANG extensions and the  
20 associated TLV selection management variables and MIB/YANG to TLV cross reference tables. Systems  
21 that implement LLDP and that also support standard protocols for which Organizationally Specific TLV  
22 extension sets have been defined support all TLVs and the LLDP MIB or YANG extensions defined for that  
23 particular TLV set.

24 Annex D of IEEE Std 802.1Q-2022 and Clause 79 of IEEE Std 802.3-2022 contain Organizationally  
25 Specific TLVs for IEEE 802.1 standards and IEEE Std 802.3-2022, respectively, along with their associated  
26 LLDP MIB or YANG extensions and LLDP MIB or YANG to TLV cross reference tables. Organizations  
27 wishing to define TLVs for their use should use these annexes as examples.

28 **8.7.1 Organizationally Specific TLV format**

29 The format for Organizationally Specific TLVs is shown in Figure 8-15.



**Figure 8-15—Format for Organizationally Specific TLVs**

30 The following subclauses indicate how the individual fields are defined.

31 **8.7.1.1 TLV type**

32 The same Organizationally Specific TLV type value, 127, is used for all organizationally defined TLVs.

### 1 8.7.1.2 TLV information string length

2 The TLV information string length field contains the length of the information string, in octets.

### 3 8.7.1.3 organizationally unique identifier (OUI)

4 The organizationally unique identifier field contains the organization's OUI (see Clause 8 of  
5 IEEE Std 802-2024.

### 6 8.7.1.4 organizationally defined subtype

7 The organizationally defined subtype field contains a unique subtype value assigned by the defining  
8 organization.

9 NOTE—Defining organizations are responsible for maintaining listings of organizationally defined subtypes in order to  
10 assure uniqueness.

### 11 8.7.1.5 organizationally defined information string

12 The actual format of the organizationally defined information string field is organizationally specific and can

- 13 a) Contain either binary or alpha-numeric information that is instance specific for the particular TLV  
14 type and/or subtype. Alpha-numeric information should be encoded in UTF-8 (IETF RFC 3629).
- 15 b) Include one or more information fields with their associated field-type identifiers and field length  
16 designators similar to those in the Management Address TLV (see 8.5.9).

### 17 8.7.2 Organizationally Specific TLV usage rules

18 Organizations defining their own Organizationally Specific TLVs should include a subclause that defines  
19 any specific usage rules and/or specific conditions that affects how the receiving LLDP agent treats the TLV.

20 The Organizationally Specific TLV usage rules should include the following:

- 21 a) The number of Organizationally Specific TLVs that may be contained in an LLDPDU and any  
22 additional information field subtypes that would identify differences between two TLVs with the  
23 same OUI and organizationally defined subtype.
- 24 b) Any error conditions that are specific to the particular Organizationally Specific TLV and the action  
25 that is taken for each defined error condition.

## 9. LLDP agent operation

This clause describes the operation of the protocol from the point of view of a single LLDP agent (see Clause 6); i.e., it describes the operation of the protocol for the transmission and reception of LLDPDUs on a single Port. Where the use of more than one Scope MAC Address (7.1.1) is supported on a given Port, a separate instance of the LLDP agent is responsible for the operation of the protocol for each Scope MAC Address that is supported.

### 9.1 Overview

Each LLDP agent is responsible for causing the following tasks to be performed:

- a) Maintaining current information in the LLDP local system information repository (RP).
- b) Extracting and formatting LLDP local system information repository information for transmission to the remote port LLDP agent(s) at regular intervals or whenever there is a change in the system condition or status.
- c) Recognizing and processing received LLDP frames.
- d) Maintaining current values in the LLDP remote system information repository.
- e) Using the procedure somethingChangedLocal() and the procedure somethingChangedRemote() to notify the PTOPO MIB manager and information repository managers of other optional MIBs and YANG modules whenever a value or status change has occurred in one or more objects in the LLDP local system LLDP remote systems information repository.

NOTE 1—A consequence of the support of multiple MAC addresses on a Port is that there are multiple copies of the LLDP local system information repository and remote system information repository, one copy for each address supported.

LLDP allows implementations that support three different operating modes: transmit only, receive only, or both transmit and receive. An LLDP agent shall conform to the specifications of each of the state machines indicated in Table 9-1 for the operating mode that it supports.

**Table 9-1—Subclause/operating mode applicability**

| Subclause number | Subclause title              | Transmit only                   | Receive only | Transmit and receive |
|------------------|------------------------------|---------------------------------|--------------|----------------------|
| 9.2.10           | Receive state machine        | Mandatory if XLLDP is supported | Mandatory    | Mandatory            |
| 9.2.9            | Transmit state machine       | Mandatory                       | —            | Mandatory            |
| 9.2.11           | Transmit timer state machine | Mandatory                       | —            | Mandatory            |

NOTE 2—When XLLDP is supported and the operating mode is transmit only, the receive state machine is only enabled to recognize and respond to Extension Request LLDPDUs. Information advertised by remote LLDP agents is not stored in the LLDP remote systems information repository.

#### 9.1.1 Frame transmission

An LLDPDU transmission cycle consists of the transmission of a Normal LLDPDU, potentially including a Manifest TLV if the implementation supports XLLDP, and the subsequent transmission of one or more Extension LLDPDUs upon receipt of Extension Request LLDPDUs. For an implementation not supporting XLLDP, the entire transmission cycle consists of the transmission of a single Normal LLDPDU.

1 The LLDP local system information repository contains the information needed for constructing individual  
2 TLVs and includes a capability that allows network managers to select the optional TLVs to be included in  
3 the Normal LLDPDU and, if XLLDP is supported, in one or more Extension LLDPDUs (see 10.2.2).

#### 4 9.1.1.1 Normal LLDPDUs

5 The transmission of Normal LLDPDUs is performed by the transmit state machine, and the timing of  
6 transmission cycles is controlled by the transmit timer state machine. Transmissions occur as a result of a  
7 number of different local events, as follows:

- 8 a) A regular background transmission cycle occurs as a result of a transmission timer expiring. This  
9 prevents the receiving system from timing out the information received, as a result of the “time to  
10 live” associated with the last received LLDPDU expiring.
- 11 b) If a *new neighbor* is recognized (i.e., an LLDPDU is received by the receive state machine from a  
12 hitherto unknown remote LLDP agent), then a default of four LLDPDU transmission cycles are  
13 performed at shorter time intervals to quickly update the new neighbor with current information  
14 about its own neighbors. This rapid transmission behavior is suppressed if the remote systems  
15 information repository is not able to accommodate the new neighbor information without removing  
16 current information relating to an older neighbor, in order to prevent excessive PDU transmissions  
17 resulting from “churn” effects.
- 18 c) If a condition (status or value) change occurs in one or more objects in the LLDP local system  
19 information repository, then a transmission cycle is initiated immediately, in order to communicate  
20 the local changes as quickly as possible to any neighboring systems. The transmit timer state  
21 machine operates a simple credit-based transmission strategy that allows a number of transmission  
22 cycles in quick succession (the maximum number being determined by the maximum credit value),  
23 thus allowing rapid updating of information in situations where the local state is changing quickly.  
24 However, once the LLDP agent’s transmission credit is exhausted, the rate of transmission is cut to  
25 an average of one transmission cycle per second.
- 26 d) The overall timing of regular transmission cycles is governed by a system “tick” that operates on a  
27 nominal timebase of 1 s. However, in shared media LANs, in order to avoid bunching of  
28 transmissions, the intervals between ticks include a random jitter component so that, while the  
29 average time interval between ticks is 1 s, the interval between adjacent pairs of ticks can vary (see  
30 9.2.2).

31 In order to reduce synchronization effects, it is recommended that transmission cycle intervals for different  
32 LLDP agents on the same multi-port implementation are staggered.

33 When the transmit state machine (9.2.9) initiates a transmission cycle, the LLDP information repository  
34 manager extracts the selected information from the LLDP local system information repository and  
35 constructs a Normal LLDPDU containing the following:

- 36 a) The mandatory TLVs as specified in 8.2:
  - 37 1) Chassis ID TLV (see 8.5.2).
  - 38 2) Port ID TLV (see 8.5.3).
  - 39 3) Time To Live TLV, with the TTL value set equal to txTTL (see 8.5.4).
- 40 b) A Manifest TLV (see 8.6.1), if XLLDP is supported and any of the selected information is to be  
41 included in an Extension LLDPDU. The Manifest TLV includes a descriptor for each Extension  
42 LLDPDU, even if the information selected to be included in that Extension LLDPDU has not  
43 changed.
- 44 c) Additional optional TLVs, from the basic management set or from one or more organizationally  
45 specific sets, as allowed by LLDPDU length restrictions and as selected in the LLDP local system  
46 information repository by network management (see Table 8-1).
- 47 d) Optionally, an End Of LLDPDU TLV.



Optional TLVs from the basic management set, such as the Management Address TLV, with different TLV type and subtype combinations as well as optional TLVs with different organizationally unique identifier and organizationally defined subtype combinations from one or more organizationally specific sets can be included within the same LLDPDU.

#### 9.1.1.2 Shutdown LLDPDUs

A special procedure exists for the case in which a LLDP agent knows an associated port is about to become non-operational (for example, the adminStatus for the port is transitioning to ‘disabled’). In the event a port, currently configured with LLDP frame transmission enabled, either becomes disabled for LLDP activity, or the interface is administratively disabled, the transmit state machine attempts to send a final LLDP shutdown LLDPDU with:

- a) The Chassis ID TLV.
- b) The Port ID TLV.
- c) The Time To Live TLV with the TTL field set to zero.
- d) Optionally, an End Of LLDPDU TLV.

The shutdown LLDPDU does not include any other optional TLVs and, if possible, should be transmitted before the interface is disabled.

NOTE—There is an inherent race condition between an interface knowing it is going down and its ability to send “one more frame.” If possible, the actual transition of the port to disabled is delayed until after this frame is transmitted. In the event where adminStatus is transitioning to the disabled state and the LLDP agent is shutting down, this shutdown procedure should be executed for all local ports.

#### 9.1.1.3 Extension Request LLDPDUs

Extension Request LLDPDUs are transmitted in response to the reception of a Manifest LLDPDU. The transmission is controlled by the extended receive state machine. When XLLDP is supported, and a Manifest LLDPDU or an Extension LLDPDU is received, and the Manifest TLV includes XPDU Descriptors that do not match the XPDU Descriptors in the LLDP remote systems information repository entry for the MSAP Identifier received in the LLDPDU, the LLDP information repository manager constructs an Extension Request LLDPDU containing the following:

- a) The mandatory TLVs as specified in 8.2:
  - 1) Chassis ID TLV (8.5.2).
  - 2) Port ID TLV (8.5.3).
  - 3) Extension Request TLV with a list of one or more XPDU descriptors (8.6.2).
- b) Optionally, an End Of LLDPDU TLV.

#### 9.1.1.4 Extension LLDPDUs

Extension LLDPDUs are transmitted in response to the reception of an Extension Request LLDPDU. The transmission is controlled by the receive state machine. When XLLDP is supported, and an Extension Request LLDPDU is received, then for each XPDU Number and XPDU Revision pair in the Extension Request TLV that matches an XPDU Number and XPDU Revision in the LLDP local system information repository, the LLDP information repository manager extracts the selected information from the LLDP local system information repository and constructs an Extension LLDPDU containing the following:

- a) The mandatory TLVs as specified in 8.2:
  - 1) Chassis ID TLV (8.5.2).
  - 2) Port ID TLV (8.5.3).
  - 3) Extension Identifier TLV with the requested XPDU Number and XPDU Revision pair (8.6.3).

- 1 b) Additional optional TLVs, from the basic management set or from one or more organizationally
- 2 specific sets, as allowed by LLDPDU length restrictions and as selected in the LLDP local system
- 3 information repository by network management (Table 8-1).
- 4 c) Optionally, an End Of LLDPDU TLV.

## 5 9.1.2 Frame reception

6 LLDP frame reception consists of four phases: frame recognition, frame validation, TLV collection, and  
7 LLDP remote systems information repository updating.

### 8 9.1.2.1 Frame recognition

9 Frame recognition is performed at the LLDP/LSAP (see Figure 6-2), where the destination address and  
10 EtherType values determine whether the frame is an LLDP frame destined for this LLDP agent or not.

### 11 9.1.2.2 Frame validation

12 Frames that are recognized as LLDP frames are then validated to determine whether they are properly  
13 constructed and contain the correct set of mandatory TLVs. Frames that do not pass the validation criteria  
14 are discarded.

### 15 9.1.2.3 TLV collection

16 All received TLVs are validated by checking for format errors, and invalid TLVs are ignored. If the TLVs  
17 received in the Normal LLDPDU did not include a Manifest TLV (or the implementation does not support  
18 XLLDP and does not recognize the Manifest TLV), then all validated TLVs are passed to the LLDP remote  
19 systems information repository immediately. If the implementation supports XLLDP and the Normal  
20 LLDPDU includes a Manifest TLV, then this TLV is used to determine which Extension LLDPDUs need to  
21 be received before a complete set of TLVs can be passed to the LLDP remote systems information  
22 repository.

23 An implementation supporting XLLDP compares the XPDU descriptors in the Manifest TLV to those  
24 received in a previous Manifest TLV to determine which XPDU (identified by the XPDU Number) contain  
25 new information. If no Manifest TLV has been received previously, then all of the XPDU described in the  
26 Manifest TLV contain new information. If there are no XPDU that have new information, then the validated  
27 TLVs received in the Normal LLDPDU are passed to the LLDP remote systems information repository.  
28 Otherwise these TLVs are retained until the XPDU with new information have been received.

29 If there are XPDU that have new information, then an Extension Request LLDPDU, requesting one or more  
30 of the XPDU with new information, is transmitted to the LLDP agent that sourced the Normal LLDPDU. A  
31 timer is started when an Extension Request LLDPDU is transmitted, and the request is retried once if the  
32 timer expires before all requested XPDU have been received. When all requested XPDU have been  
33 received, another Extension Request LLDPDU is transmitted requesting one or more of the remaining  
34 XPDU with new information. This process repeats until all XPDU with new information have been  
35 received. If a new Manifest LLDPDU is received before all XPDU have been received, any received  
36 XPDU that have the same Revision and Checksum in the new Manifest TLV are retained, and any received  
37 XPDU that have different Revision and Checksum are discarded. The extended receive state machine then  
38 continues to issue Extension Requests for any necessary XPDU.

39 Received frames containing Extension LLDPDU are recognized and validated following the same steps as  
40 frames containing Normal LLDPDU. Once validated, the XPDU check value is calculated. This check  
41 value, together with the XPDU Number and XPDU Revision from the Extension Identifier TLV, is  
42 compared to the XPDU descriptors in the Manifest TLV. If there is a matching descriptor then all valid TLVs

received in this Extension LLDPDU are retained. When all XPDU with new information have been received, all of the retained TLVs are passed to the LLDP remote systems information repository.

### 9.1.2.4 LLDP remote systems information repository updating

The collected TLVs are used to create, modify, or delete an entry in the LLDP remote systems information repository. The relevant entry in the LLDP remote systems information repository is determined by the originating MSAP identifier formed from the contents of the Chassis ID and Port ID TLVs. The LLDP remote systems information repository is updated as follows:

- a) If the TTL value in the Time To Live TLV is greater than zero and an entry does not exist in the LLDP remote systems information repository for the originating MSAP identifier and there is sufficient space (or sufficient space can be created by the steps in 9.2) for the new information, then a new entry in the information repository is created.
- b) If the TTL value in the Time To Live TLV is greater than zero and an entry already exists in the remote systems information repository for the originating MSAP identifier and there is sufficient space (or sufficient space can be created by the steps in 9.2) for the new information, then that entry is modified as follows:
  - 1) The TLVs received in the Normal LLDPDU are used to replace any information elements in the existing entry in the information repository that were previously received in a Normal LLDPDU. Note that if the Normal LLDPDU did not contain a Manifest TLV (or the implementation does not support XLLDP and does not recognize the Manifest TLV), then the new information contained in the LLDPDU is used to replace the whole of the existing entry in the information repository; i.e., if there are information elements in the existing entry for which there are corresponding elements in the received LLDPDU, then those elements are updated using the information received; any other information elements in the existing entry in the information repository are deleted.
  - 2) The TLVs received in an Extension LLDPDU are used to replace any information elements in the existing entry in the information repository that were previously received in an Extension LLDPDU with the same XPDU Number.
  - 3) Any information elements in the existing entry in the information repository that were previously received in an Extension LLDPDU with an XPDU Number that is not included in the received Manifest TLV are deleted.
- c) If the TTL value in the Time To Live TLV is greater than zero and an entry exists in the LLDP remote systems information repository for the originating MSAP identifier and there is insufficient space for the new information, then the entry is deleted from the information repository.
- d) If the TTL value in the Time To Live TLV is equal to zero and an entry exists in the LLDP remote systems information repository for the originating MSAP identifier, then the entry is deleted from the information repository.

When an entry in the LLDP remote systems information repository is created or modified (even if the modification is to replace all information elements with the same information), the rxInfoTTL timer associated with that entry is set to the TTL value from the Time To Live TLV. This determines how long the information associated with that MSAP identifier and destination MAC address is to be stored in the LLDP remote systems information repository before being aged out and deleted. The ageing out of old data purges the information repository of data that was originated by systems that are no longer neighbors as a result of topology changes, system failure, or system inactivation.

### 1 9.1.3 Too many neighbors

2 The amount of space needed in the LLDP remote systems information repository to accommodate the  
3 creation of several new information repository structures is beyond the scope of this standard. It may not  
4 always be possible to accommodate another new neighbor in implementations with limited memory. There  
5 are several possibilities for handling this case, including but not limited to the following examples:

- 6 a) Ignore and not process the new neighbor's information.
- 7 b) Delete the information from the oldest neighbor(s) until there is sufficient memory available to store  
8 the new neighbor's information.
- 9 c) Arbitrarily delete neighbors until there is sufficient memory available to store the new neighbor's  
10 information.

11 The method of handling the case where a new entry in the remote systems information repository can not be  
12 created is beyond the scope of this standard, however it is necessary for the implementation to keep track of  
13 this case by properly updating the variable `tooManyNeighbors` and the `tooManyNeighborsTimer` (9.2.5.16,  
14 9.2.2). The variable `tooManyNeighbors` identifies when there is insufficient space in the LLDP remote  
15 systems information repository to store information from all active neighbors. The `tooManyNeighborsTimer`  
16 indicates the minimum time that this condition exists.

17 Further detail on the handling of the too many neighbors condition can be found in 9.2.8.5.1.

### 18 9.1.4 LLDP remote systems `rxInfoTTL` timer expiration

19 If the `rxInfoTTL` timer (9.2.2.1) associated with an MSAP identifier expires, all information associated with  
20 that MSAP identifier is deleted and the procedure `somethingChangedRemote()` notifies the managers of the  
21 PTOPO MIB and other optional MIB or YANG modules (see Figure 11-1) that something has changed in the  
22 LLDP remote systems information repository associated with that MSAP identifier.

### 23 9.1.5 LLDP local port/connection failure

24 If the local port or the connection to the remote system fails unexpectedly before a shutdown frame is  
25 received, the LLDP management entity does not delete objects in the LLDP remote systems information  
26 repository pertaining to information received from any MSAP identifier through that local port until the port  
27 is re-initialized or the associated `rxInfoTTL` timer (9.2.2.1) expires.

## 28 9.2 State machines

29 The operation of the protocol is represented by the following three state machines and associated timers to  
30 indicate when local system managed object values are to be transmitted to refresh the values in remote  
31 LLDP agent's LLDP remote systems information repository and when values in the local LLDP agent's  
32 remote systems information repository have become invalid:

- 33 a) Transmit state machine (9.2.9).
- 34 b) Receive state machine (9.2.10).
- 35 c) Transmit timer state machine (9.2.11).

### 36 9.2.1 Notational conventions used in state diagrams

37 State diagrams are used to represent the operation of a function as a group of connected, mutually exclusive  
38 states. Only one state of a function can be active at any given time.

1 Each state is represented in the state diagram as a rectangular box, divided into two parts by a horizontal  
2 line. The upper part contains the state identifier, written in uppercase letters. The lower part contains any  
3 procedures that are executed on entry to the state.

4 All permissible transitions between states are represented by arrows, the arrowhead denoting the direction of  
5 the possible transition. Labels attached to arrows denote the condition(s) that shall be met in order for the  
6 transition to take place. A transition that is global in nature (i.e., a transition that occurs from any of the  
7 possible states if the condition attached to the arrow is met) is denoted by an open arrow; i.e., no specific  
8 state is identified as the origin of the transition.

9 On entry to a state, the procedures defined for the state (if any) are executed exactly once, in the order that  
10 they appear on the page. Each action is deemed to be atomic; i.e., execution of a procedure completes before  
11 the next sequential procedure starts to execute. No procedures execute outside of a state block. On  
12 completion of all of the procedures within a state, all exit conditions for the state (including all conditions  
13 associated with global transitions) are evaluated continuously until such a time as one of the conditions is  
14 met. All exit conditions are regarded as Boolean expressions that evaluate to TRUE or FALSE; if a condition  
15 evaluates to TRUE, then the condition is met. When the condition associated with a global transition is met,  
16 it supersedes all other exit conditions, including UCT. The label UCT denotes an unconditional transition  
17 (i.e., UCT always evaluates to TRUE). The label ELSE denotes a transition that occurs if none of the other  
18 conditions for transitions from the state are met (i.e., ELSE evaluates to TRUE if all other possible exit  
19 conditions from the state evaluate to FALSE).

20 A variable that is set to a particular value in a state block retains this value until a subsequent state block  
21 executes a procedure that modifies the value, or until the value is modified by an external event or timer tick.

22 Where it is necessary to segment a state machine description across more than one diagram, a transition  
23 between two states that appear on different diagrams is represented by an exit arrow drawn with dashed  
24 lines, plus a reference to the diagram that contains the destination state. Similarly, dashed arrows and a  
25 dashed state box are used on the destination diagram to show the transition to the destination state. In a state  
26 machine that has been segmented in this way, any global transitions that can cause entry to states defined in  
27 one of the diagrams are deemed to be potential exit conditions for all of the states of the state machine,  
28 regardless of which diagram the state boxes appear in.

29 Should a conflict exist between the interpretation of a state diagram and either the corresponding global  
30 transition tables or the textual description associated with the state machine, the state diagram takes  
31 precedence.

32 The interpretation of the special symbols and operators used in the state diagrams is defined in Table 9-2;  
33 these symbols and operators are derived from the notation of the “C” programming language.

**Table 9-2—State machine symbols**

| Symbol | Interpretation                                                                                                                                                                                        |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ( )    | Used to force the precedence of operators in Boolean expressions and to delimit the argument(s) of actions within state boxes.                                                                        |
| ;      | Used as a terminating delimiter for actions within state boxes. Where a state box contains multiple actions, the order of execution follows the normal English language conventions for reading text. |

**Table 9-2—State machine symbols (*continued*)**

| Symbol                         | Interpretation                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| =                              | Assignment action. The value of the expression to the right of the operator is assigned to the variable to the left of the operator. Where this operator is used to define multiple assignments, e.g.,<br>a = b = X<br>the action causes the value of the expression following the right-most assignment operator to be assigned to all of the variables that appear to the left of the right-most assignment operator. |
| !                              | Logical NOT operator.                                                                                                                                                                                                                                                                                                                                                                                                   |
| &&                             | Logical AND operator.                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                | Logical OR operator.                                                                                                                                                                                                                                                                                                                                                                                                    |
| if...then...                   | Conditional action. If the Boolean expression following the <b>if</b> evaluates to TRUE, then the action following the <b>then</b> is executed.                                                                                                                                                                                                                                                                         |
| {statement 1, ... statement N} | Compound statement. Braces are used to group statements that are executed together as if they were a single statement.                                                                                                                                                                                                                                                                                                  |
| !=                             | Inequality. Evaluates to TRUE if the expression to the left of the operator is not equal in value to the expression to the right.                                                                                                                                                                                                                                                                                       |
| ==                             | Equality. Evaluates to TRUE if the expression to the left of the operator is equal in value to the expression to the right.                                                                                                                                                                                                                                                                                             |
| <                              | Less than. Evaluates to TRUE if the value of the expression to the left of the operator is less than the value of the expression to the right.                                                                                                                                                                                                                                                                          |
| <=                             | Less than or equal to. Evaluates to TRUE if the value of the expression to the left of the operator is either less than or equal to the value of the expression to the right.                                                                                                                                                                                                                                           |
| >                              | Greater than. Evaluates to TRUE if the value of the expression to the left of the operator is greater than the value of the expression to the right.                                                                                                                                                                                                                                                                    |
| >=                             | Greater than or equal to. Evaluates to TRUE if the value of the expression to the left of the operator is either greater than or equal to the value of the expression to the right.                                                                                                                                                                                                                                     |
| +                              | Arithmetic addition operator.                                                                                                                                                                                                                                                                                                                                                                                           |
| –                              | Arithmetic subtraction operator.                                                                                                                                                                                                                                                                                                                                                                                        |

## 9.2.2 Timers

A set of timers is used by the LLDP state machines; these operate as countdown timers (i.e., they expire when their value reaches zero). These timers

- a) Have a resolution of one tick.
- b) Have an integer value  $n$ , where  $0 < n < 65535$  tick intervals.
- c) Are started by loading an initial integer value.
- d) Are decremented once per timer tick as long as  $n > 0$ .
- e) Represent the remaining time in the period.

For Ports connected to point-to-point LANs, the interval between timer ticks is 1 s. For Ports connected to shared media LANs, the interval between any pair of timer ticks is 0.8 s, plus a “jitter” component that is a random or pseudo-random value between 0.0 s and 0.4 s. Hence, the average interval between timer ticks is 1 s; however, the interval between any pair of ticks varies randomly. When a timer tick occurs, all instances of the variable txTick (9.2.5.22) for the Port are set TRUE.

<sup>1</sup> NOTE—The random component of the tick time used in shared media LANs is intended to minimize the possibility of  
<sup>2</sup> systems connected to the same LAN becoming synchronized in their transmission timings.

### <sup>3</sup> **9.2.2.1 rxInfoTTL**

<sup>4</sup> An instance of this timer exists for each entry in the LLDP remote systems information repository that has  
<sup>5</sup> been created by this instance of the receive state machine for a given MSAP identifier. The timer value  
<sup>6</sup> indicates the number of seconds remaining until the information in all the objects in the LLDP remote  
<sup>7</sup> systems information repository associated with the MSAP identifier is no longer valid and needs to be  
<sup>8</sup> deleted.

### <sup>9</sup> **9.2.2.2 tooManyNeighborsTimer**

<sup>10</sup> This timer indicates the number of seconds remaining during which it is known that an LLDP neighbor  
<sup>11</sup> exists that may not be stored in the remote systems information repository.

### <sup>12</sup> **9.2.2.3 txTTR**

<sup>13</sup> An instance of this timer exists for each Agent. The txTTR timer is used to determine the next LLDPDU  
<sup>14</sup> transmission is due; it is initialized by the transmit timer state machine, with the value msgTxInterval  
<sup>15</sup> (9.2.5.6) if txFast (9.2.5.19) is equal to zero, or with the value msgFastTx (9.2.5.4) if txFast is greater than  
<sup>16</sup> zero.

### <sup>17</sup> **9.2.2.4 txShutdownWhile**

<sup>18</sup> An instance of this timer exists for each Agent. The txShutdownWhile timer indicates the number of  
<sup>19</sup> seconds remaining until LLDP re-initialization can occur.

### <sup>20</sup> **9.2.2.5 rxTimer**

<sup>21</sup> An instance of this timer exists for each extended receive state machine to impose a time limit on each  
<sup>22</sup> extension request, and on the overall TLV collection process.

## <sup>23</sup> **9.2.3 Per-System variables**

<sup>24</sup> An instance of each of these variables exists per system; they are available for use by all of the state  
<sup>25</sup> machines of all of the LLDP agents in the system.

### <sup>26</sup> **9.2.3.1 BEGIN**

<sup>27</sup> This Boolean variable is controlled by the system initialization process. A value of TRUE causes all state  
<sup>28</sup> machines to continuously execute their initial state. A value of FALSE allows all state machines to perform  
<sup>29</sup> transitions out of their initial state, in accordance with the relevant state machine definitions.

## <sup>30</sup> **9.2.4 Per-Port variables**

<sup>31</sup> An instance of each of these variables exists per Port; they are available to all of the state machines  
<sup>32</sup> associated with a Port.

### <sup>33</sup> **9.2.4.1 portEnabled**

<sup>34</sup> This variable is externally controlled. Its value reflects the operational state of the MAC service supporting  
<sup>35</sup> the port. Its value is TRUE if the MAC service supporting the Port is in an operable condition; otherwise, it  
<sup>36</sup> is FALSE.

## 1 9.2.5 Per-Agent variables

2 An instance of each of these variables exists per LLDP agent; they are available to all of the state machines  
3 associated with an LLDP agent.

### 4 9.2.5.1 adminStatus

5 This variable indicates whether or not the LLDP agent is enabled. The defined values for this variable are as  
6 follows:

- 7 a) enabledRxTx: The LLDP agent is enabled for reception and transmission of LLDPDUs. This is the  
8 recommended default value.
- 9 b) enabledTxOnly: The LLDP agent is enabled for transmission of LLDPDUs only.
- 10 c) enabledRxOnly: The LLDP agent is enabled for reception of LLDPDUs only.
- 11 d) disabled: The LLDP agent is disabled for both reception and transmission.

### 12 9.2.5.2 localChange

13 This variable is set TRUE by the somethingChangedLocal() procedure (see 9.2.8.8) when there is a change  
14 in the value of the LLDP local system information repository associated with this LLDP agent. The variable  
15 is set FALSE by the operation of the transmit timer state machine (see 9.2.11).

### 16 9.2.5.3 Scope MAC Address

17 The MAC address associated with this instance of the LLDP agent that defines the scope of propagation of  
18 Normal LLDPDUs in the network. It is used as the destination MAC address when the LLDP agent  
19 transmits Normal LLDPDUs, and the destination MAC address that, when present in a received LLDPDU,  
20 causes the LLDPDU to be passed to this LLDP agent instance for processing.

### 21 9.2.5.4 msgFastTx

22 This variable defines the time interval in timer ticks between transmissions during fast transmission periods  
23 (i.e., txFast is non-zero). The recommended default value of msgFastTx is 1; this value can be changed by  
24 management to any value in the range 1 through 3600.

### 25 9.2.5.5 msgTxHold

26 This variable is used, as a multiplier of msgTxInterval, to determine the value of txTTL that is carried in  
27 LLDP frames transmitted by the LLDP agent. The recommended default value of msgTxHold is 4; this  
28 value can be changed by management to any value in the range 2 through 10.

### 29 9.2.5.6 msgTxInterval

30 This variable defines the time interval in timer ticks between transmissions during normal transmission  
31 periods (i.e., txFast is zero). The recommended default value for msgTxInterval is 30 s; this value can be  
32 changed by management to any value in the range 1 through 3600.

### 33 9.2.5.7 newNeighbor

34 This variable is set TRUE by the rxProcessFrame() procedure (see 9.2.8.7) when information is received  
35 from a new neighbor. It is set FALSE by the Transmit timer state machine (9.2.11).

36 NOTE—The newNeighbor variable is used to force a more rapid regular transmission rate when a new neighbor  
37 appears, to ensure that the new neighbor also receives new information from the receiving system.



### 1 9.2.5.8 rcvFrame

2 This variable indicates that an LLDP frame has been recognized by the LLDP LSAP function and is ready to  
3 be processed by the LLDP receive state machine. If both of the following conditions are TRUE, the variable  
4 rcvFrame is set to TRUE and the frame is made available to the receive state machine for validation and  
5 processing:

- 6 a) The destination address value is the Scope MAC Address associated with this specific LLDP agent  
7 or the individual MAC address of the MSAP supporting the LLDP agent.
- 8 b) The EtherType value carried by the frame is the LLDP EtherType defined in Table 7-3.

9 NOTE—The model that is assumed for reception is that incoming LLDPDUs are presented to all LLDP agents  
10 associated with the reception Port, and if the Scope MAC Address is not one that is associated with a given LLDP agent,  
11 that LLDPDU is ignored by that LLDP agent. In practice, at most one LLDP agent processes any received LLDPDU, as  
12 there is only one LLDP agent associated with any given Scope MAC Address on a given reception Port.

### 13 9.2.5.9 reinitDelay

14 This parameter indicates the amount of delay from when adminStatus becomes ‘disabled’ until  
15 re-initialization is attempted. The recommended default value for reinitDelay is 2 s.

### 16 9.2.5.10 remoteChanges

17 A Boolean indication of whether the status/value of one or more selected objects in the LLDP remote  
18 systems information repository has changed or that all objects associated with a particular MSAP identifier  
19 have been deleted. This variable takes the value (*rxInfoAge OR (rxChanges && (rxTTL !=0))*)

### 20 9.2.5.11 rxChanges

21 This variable indicates that the incoming LLDPDU has been received with different TLV values from those  
22 currently in the LLDP remote systems information repository associated with the LLDPDU’s MSAP  
23 identifier.

### 24 9.2.5.12 rxInfoAge

25 This variable is set TRUE when a timer expiry event occurs for the rxInfoTTL timing counter associated  
26 with a particular MSAP identifier in the LLDP remote systems information repository (i.e., the counter  
27 transitions from non-zero to zero). It is set FALSE by the operation of the Receive state machine.

### 28 9.2.5.13 rxTTL

29 This variable indicates the time to live value associated with all TLVs received in the current frame.

### 30 9.2.5.14 rxType

31 This variable indicates the type of the most recently received LLDP frame. It is set by the rxProcessFrame()  
32 procedure to one of the following values:

- 33 a) xreq: The LLDP agent supports XLLDP and identifies the frame as containing a request for  
34 extended PDUs.
- 35 b) xpdu: The LLDP agent supports XLLDP and identifies the frame as containing an extended PDU.
- 36 c) shutdown: The frame contains a shutdown LLDPDU (which has an rxTTL value equal to zero).
- 37 d) manifest: The LLDP agent supports XLLDP and the frame is a Normal LLDPDU that has a rxTTL  
38 value greater than zero and contains a Manifest TLV.

- 1 e) normal: The frame contains a Normal LLDPDU that has an rxTTL value greater than zero, and
- 2 either the LLDP agent does not support XLLDP or the frame does not contain a Manifest TLV.
- 3 f) invalid: The frame is improperly formatted for a valid LLDPDU, or the frame contains an extended
- 4 PDU or a request for extended PDUs but the LLDP agent does not support XLLDP.

#### 5 **9.2.5.15 sentManifest**

6 This Boolean variable is set to TRUE when the transmit state machine generates a Manifest LLDPDU, and  
7 is set to FALSE when the transmit state machine generates a Normal LLDPDU that does not contain a  
8 Manifest TLV. It is used to enable the receive state machine to recognize Extension Request LLDPDUs and  
9 respond with Extension LLDPDUs.

#### 10 **9.2.5.16 tooManyNeighbors**

11 This Boolean variable indicates that there is insufficient space in the LLDP remote systems information  
12 repository to store information from all connected active remote ports (see 9.2.8.5.1).

#### 13 **9.2.5.17 txCredit**

14 The number of consecutive LLDPDUs that can be transmitted at any time. This variable is incremented by  
15 the txAddCredit() procedure, up to the maximum value specified by the txCreditMax variable (see 9.2.5.18),  
16 and is decremented whenever an LLDPDU is transmitted by the transmit state machine.

#### 17 **9.2.5.18 txCreditMax**

18 The maximum value of txCredit. The recommended default value is 5; this value can be changed by  
19 management to any value in the range 1 through 10.

#### 20 **9.2.5.19 txFast**

21 This variable is used as a down counter to count the number of transmissions to be made during a fast  
22 transmission period. Fast transmission periods are initiated when a new neighbor is detected, and cause  
23 LLDPDU transmissions to take place at shorter time intervals than during normal (steady state) operation of  
24 the protocol. The fast transmission period ensures that more than one LLDPDU is transmitted when a new  
25 neighbor is detected. The first transmission is immediate; the subsequent transmissions occur at intervals  
26 determined by msgFastTx. The number of transmissions is determined by the value of txFastInit (see  
27 9.2.5.20).

#### 28 **9.2.5.20 txFastInit**

29 This variable is used as the initial value for the txFast variable (see 9.2.5.19). This value determines the  
30 number of LLDPDUs that are transmitted during a fast transmission period. The recommended default value  
31 of txFastInit is 4; this value can be changed by management to any value in the range 1 through 8.

#### 32 **9.2.5.21 txNow**

33 This variable is used to signal to the transmit state machine that a transmission is required. It is set TRUE by  
34 the transmit timer state machine, and set FALSE by the transmit state machine when the transmission has  
35 been initiated.

#### 36 **9.2.5.22 txTick**

37 This variable is set TRUE by the system timer tick at an average interval of one second (see 9.2.2). It is set  
38 FALSE by the operation of the transmit timer state machine.

### 1 9.2.5.23 txTTL

2 This variable indicates the time remaining before information in the outgoing LLDPDU is no longer valid.  
3 The TTL field in the Time To Live TLV is set to txTTL during LLDPDU construction (see 9.1.2). The value  
4 of txTTL depends on the following:

- 5 a) During normal operation, txTTL is set to whichever is the smaller of the values represented by  
6 Equation (1) and Equation (2):

$$7 \quad (\text{msgTxInterval} \times \text{msgTxHold}) + 1 \quad (1)$$

$$8 \quad 65535 \quad (2)$$

9 where msgTxInterval is defined in 9.2.5.6 and msgTxHold is defined in 9.2.5.5.

- 10 b) If adminStatus is transitioning to 'disabled' or portEnabled is transitioning to FALSE, txTTL shall  
11 be set to zero.

### 12 9.2.6 Per-Neighbor variables

13 An instance of each of these variables exists per neighbor (identified by an MSAP) for each LLDP agent;  
14 they are available to all of the state machines associated with an LLDP agent.

#### 15 9.2.6.1 createExtRx

16 This Boolean is used to initialize a per-neighbor extended receive state machine. It is set to TRUE when an  
17 extended receive state machine is created in response to the receipt of a Manifest LLDPDU from a neighbor  
18 for which an extended receive state machine does not already exist. It is set to FALSE by the extended  
19 receive state machine following initialization.

#### 20 9.2.6.2 rxExtended

21 This Boolean is used to pass control between the per-agent receive state machine and the per-neighbor  
22 extended receive state machine. It is set to TRUE by the receive state machine when an LLDPDU containing  
23 a Manifest TLV or an Extension Identifier TLV is received, and is set to FALSE by the extended receive  
24 state machine.

#### 25 9.2.6.3 manifestComplete

26 This Boolean indicates to the receive state machine whether the extended receive state machine has received  
27 all TLVs necessary to update the information for this MSAP in the LLDP remote systems information  
28 repository.

#### 29 9.2.6.4 xreqComplete

30 This Boolean is used within the extended receive state machine to indicate that all XPDUs requested for this  
31 MSAP have been received.

#### 32 9.2.6.5 rxTick

33 This Boolean is set TRUE by the system timer tick at an average interval of one second (see 9.2.2). It is set  
34 FALSE by the operation of the extended receive state machine.

#### 1 **9.2.6.6 rxTTLExpired**

2 This Boolean is used to indicate that the extended receive state machine has been unable to collect all the  
3 TLVs within a time interval greater than the received rxTTL value. It is set to TRUE when the rxTimer is  
4 decremented to zero and either all Extension LLDPDUs for the most recently transmitted Extension Request  
5 LLDPDU have been received, or that Extension Request LLDPDU was a retry of a previous request.

#### 6 **9.2.7 Per-Agent statistical counters**

7 An instance of each of these statistical counters exists per LLDP agent; they are used to count significant  
8 events in the transmit and receive state machines and are made available to the management functions  
9 described in Clause 10 and Clause 11. All counters are maintained as unsigned integer values, four octets in  
10 length, and are initialized to zero only on system initialization (i.e., when the BEGIN system variable is set  
11 TRUE).

#### 12 **9.2.7.1 statsAgeoutsTotal**

13 This counter provides a count of the times that a neighbor's information has been deleted from the LLDP  
14 remote systems information repository because the rxInfoTTL timer associated with the neighbor's MSAP  
15 has expired.

#### 16 **9.2.7.2 statsFramesDiscardedTotal**

17 This counter provides a count of all LLDPDUs received and then discarded for any of the following reasons:

- 18 a) One or more of the three mandatory TLVs at the beginning of the LLDPDU is missing, out of order,  
19 or contains an out of range information string length.
- 20 b) There is insufficient space in the remote systems information repository to store the LLDPDU.

#### 21 **9.2.7.3 statsFramesInErrorsTotal**

22 This counter provides a count of all LLDPDUs received with one or more detectable errors.

#### 23 **9.2.7.4 statsFramesInTotal**

24 This counter provides a count of all LLDP frames received.

#### 25 **9.2.7.5 statsFramesOutTotal**

26 This counter provides a count of all LLDP frames transmitted by this instance of the transmit state machine.  
27 The counter shall be maintained as an unsigned integer value, four octets in length.

#### 28 **9.2.7.6 statsTLVsDiscardedTotal**

29 This counter provides a count of all TLVs received and then discarded for any reason.

#### 30 **9.2.7.7 statsTLVsUnrecognizedTotal**

31 This counter provides a count of all TLVs received on the port that are not recognized by the LLDP local  
32 agent.

## 1 9.2.7.8 lldpduLengthErrors

2 A count of the number of LLDPDU length restriction errors detected by the rpConstrInfoLLDPDU()  
3 procedure (9.2.8.2).

## 4 9.2.8 State machine procedures

### 5 9.2.8.1 dec (variable-name)

6 If the state machine variable *variable-name* has a non-zero value, this procedure decrements the value of the  
7 variable by 1.

### 8 9.2.8.2 rpConstrInfoLLDPDU()

9 The rpConstrInfoLLDPDU() procedure constructs an information LLDPDU, structured according to the  
10 specification in 8.2, the associated basic TLV formats defined in 8.4, plus any optional Organizationally  
11 Specific TLVs as specified and their associated individual organizationally defined formats (see 8.7). The  
12 information LLDPDU contains the following:

- 13 a) The set of mandatory TLVs as specified in 8.2, with the value of txTTL set in accordance with the  
14 definition in 9.2.5.23.
- 15 b) Additional optional TLVs, from the basic management set or from one or more organizationally  
16 specific sets, as allowed by LLDPDU length restrictions and as selected in the LLDP local system  
17 information repository by network management (see Table 8-1).
- 18 c) Optionally, an End Of LLDPDU TLV.

19 If the set of TLVs that is selected in the LLDP local system information repository by network management  
20 would result in an LLDPDU that violates LLDPDU length restrictions and XLLDP is supported, each TLV  
21 selected for transmission is mapped to the Normal LLDPDU or to an Extension LLDPDU, as defined in  
22 10.2.2.

23 If the set of TLVs that is selected in the LLDP local system information repository by network management  
24 would result in an LLDPDU that violates LLDPDU length restrictions and XLLDP is not supported, then  
25 the variable lldpduLengthErrors (9.2.8.2) is incremented by 1, and an LLDPDU is sent containing the  
26 mandatory TLVs plus as many of the optional TLVs in the set as will fit in the remaining LLDPDU length.h.

### 27 9.2.8.3 rpConstrShutdownLLDPDU()

28 The rpConstrShutdownLLDPDU() procedure constructs a shutdown LLDPDU according to the LLDPDU  
29 and the associated TLV formats specified in 8.2 and 8.5. The shutdown LLDPDU contains only the  
30 following items:

- 31 a) The Chassis ID and Port ID TLVs.
- 32 b) The Time To Live TLV with the TTL field set to zero.
- 33 c) Optionally, an End Of LLDPDU TLV.

### 34 9.2.8.4 rpDeleteObjects()

35 The rpDeleteObjects() procedure deletes all information in the LLDP remote systems information repository  
36 associated with a given MSAP identifier (and destroys the associated extended receive state machine if one  
37 exists) if an LLDPDU is received with an rxTTL value of zero (see 9.2.8.7.1), the extended receive state  
38 machine sets rxTTLExpired to TRUE, or the timing counter rxInfoTTL expires (see 9.2.2.1).

### 9.2.8.5 rpUpdateObjects()

The rpUpdateObjects() procedure updates the information repository objects corresponding to the TLVs contained in the received Normal LLDPDU, and all TLVs contained in any received Extension LLDPDUs, for the LLDP remote system if information repository space is available, as follows:

- a) Compare the MSAP identifier in the current LLDPDU with the MSAP identifiers in the LLDP remote systems information repository:
  - 1) If a match is found, but no difference exists between the information in the TLVs just received and the information in the LLDP remote systems information repository, then set the control variable rxChanges to FALSE, and set the timing counter rxInfoTTL associated with the MSAP identifier to rxTTL.
  - 2) If a match is found and sufficient space is not available in the LLDP remote systems information repository for the updated information in the TLVs just received, then follow the tooManyNeighbors procedure (9.2.8.5.1) to determine whether to discard the TLVs from a new neighbor or to delete information from an existing neighbor that is already in the LLDP remote systems information repository.
  - 3) If a match is found, and sufficient space is available (or has been made available) in the LLDP remote systems information repository for the updated information in the TLVs just received, then:
    - i) Replace all information associated with the MSAP identifier in the LLDP remote systems information repository that was previously received in a Normal LLDPDU with the information in the just received Normal LLDPDU.
    - ii) For each received Extension LLDPDU associated with the MSAP identifier, if any, replace all information in the LLDP remote systems information repository that was previously received in an Extension LLDPDU of the same XPDU Number with the information in the just received Extension LLDPDU. If a received Extension LLDPDU has an XPDU Number that has not been previously received, add the new information to the LLDP remote system information repository entry.
    - iii) Set the control variable rxChanges to TRUE.
    - iv) Set the timing counter rxInfoTTL associated with the MSAP identifier to rxTTL.
  - 4) If no match is found and sufficient space is not available in the LLDP remote systems information repository, then follow the tooManyNeighbors procedure (9.2.8.5.1) to determine whether to discard the TLVs from a new neighbor or to delete information from an existing neighbor that is already in the LLDP remote systems information repository.
  - 5) If no match is found and sufficient space is available (or has been made available), create a new information repository structure to receive information associated with the new MSAP identifier, set these information repository objects to the values indicated in their respective TLVs, set the control variable rxChanges to TRUE, and set the timing counter rxInfoTTL associated with the MSAP identifier to rxTTL.

If an incoming TLV is not recognized by the receiving LLDP agent, the TLV is stored in the LLDP remote systems information repository as follows:

- b) If the TLV type value is in the range of the reserved TLV types in Table 8-1, the TLV can be from a later version of the basic management set and is stored according to the basic TLV format shown in Figure 8-2. These TLVs are indexed by their TLV type.
- c) If the TLV type value is 127, the TLV is an Organizationally Specific TLV and is stored according to the basic format for Organizationally Specific TLVs shown in Figure 8-15. These TLVs are indexed by their organizationally unique identifier and organizationally defined TLV subtype.

### 9.2.8.5.1 Too many neighbors

When there is insufficient space in the LLDP remote systems information repository to store information from a new neighbor something needs to be discarded. Either the received LLDPDU is discarded or existing information within the current remote systems information repository is discarded in order to make space for the new information received in the LLDPDU. If the space required to store the information from the new neighbor is larger than the storage space available to the remote systems information repository then the received LLDPDU shall be discarded rather than discarding existing information in the remote systems information repository. The procedure is as follows:

- a) Set the flag variable `tooManyNeighbors` to TRUE.
- b) If the information selected to be discarded is the information in the current LLDPDU:
  - 1) Set the `tooManyNeighborsTimer` as specified in Equation (3).

$$\text{tooManyNeighborsTimer} = \max(\text{tooManyNeighborsTimer}, \text{rxTTL}) \quad (3)$$

where `rxTTL` is the TTL value in the current LLDPDU

- 2) Discard the current LLDPDU and increment the `statsFramesDiscardedTotal` counter.
- 3) Set the control variable `rxChanges` to FALSE. Wait for the next LLDPDU.
- c) If the information selected to be discarded is currently in the LLDP remote systems information repository:
  - 1) From the data in the remote information repository, select a neighbor.
  - 2) Delete all information associated with the selected neighbor's MSAP identifier from the LLDP remote systems information repository, and destroy the associated extended receive state machine if one exists.
  - 3) Set the `tooManyNeighborsTimer` as specified in Equation (4).

$$\text{tooManyNeighborsTimer} = \max(\text{tooManyNeighborsTimer}, \text{rxInfoTTL}) \quad (4)$$

where `rxInfoTTL` = the selected neighbor's TTL value

- 4) If sufficient space exists to store the information received in the current LLDPDU in the LLDP remote systems information repository, set control variable `rxChanges` to TRUE.
- 5) If sufficient space still does not exist to store the information received in the current LLDPDU in the LLDP remote systems information repository, perform steps 1–4 again.

The variable `tooManyNeighbors` is automatically set to FALSE whenever the `tooManyNeighborsTimer` expires.

NOTE—The `tooManyNeighborsTimer` is not precise, as it is only cleared (and the `tooManyNeighbors` variable set FALSE) by timer expiration, and not by the removal of the too many neighbors condition.

### 9.2.8.6 rxInitializeLLDP()

The `rxInitializeLLDP()` procedure initializes the LLDP receive module. The `rxInitializeLLDP()` procedure waits till `portEnabled` is equal to TRUE and after the variable `portEnabled` is equal to TRUE, the LLDP receive module is initialized or re-initialized for frame reception. During this process, the local LLDP agent performs the following tasks:

- a) The variable `tooManyNeighbors` is set to FALSE.
- b) All information in the remote systems information repository associated with this LLDP agent is deleted.

## 1 9.2.8.7 rxProcessFrame()

2 The rxProcessFrame() procedure:

- 3 a) Validates the LLDPDU, sets the value of rxType, and validate the TLVs contained in the LLDPDU  
4 as specified in 9.2.8.7.1.  
5

### 6 9.2.8.7.1 LLDPDU validation

7 The receive module processes each incoming LLDPDU as it is received. The statsFramesInTotal counter for  
8 the port is incremented and the LLDPDU is checked to verify the presence of the three mandatory TLVs at  
9 the beginning of the LLDPDU as defined in 8.2.

- 10 a) The first TLV is extracted:
- 11 1) If the extracted TLV type value does not match the Chassis ID TLV type:
- 12 i) The LLDPDU is discarded.
- 13 ii) The statsFramesDiscardedTotal and statsFramesInErrorsTotal counters are both  
14 incremented.
- 15 iii) The rxType value is set equal to invalid.
- 16 iv) The procedure rxProcessFrame() is terminated.
- 17 2) If the extracted TLV type value matches the Chassis ID TLV type, and the chassis ID TLV  
18 information string length is not within the range  $2 \leq n \leq 256$ :
- 19 i) The LLDPDU is discarded.
- 20 ii) The statsFramesDiscardedTotal and statsFramesInErrorsTotal counters are both  
21 incremented.
- 22 iii) The rxType value is set equal to invalid.
- 23 iv) The procedure rxProcessFrame() is terminated.
- 24 3) If the extracted TLV type value matches the Chassis ID TLV type, and the chassis ID TLV  
25 information string length is within the range  $2 \leq n \leq 256$ , the chassis ID value is extracted to  
26 become the first part of the MSAP identifier.
- 27 b) The second TLV is extracted:
- 28 1) If the extracted TLV type value does not match the Port ID TLV type:
- 29 i) The LLDPDU is discarded.
- 30 ii) The statsFramesDiscardedTotal and statsFramesInErrorsTotal counters are both  
31 incremented.
- 32 iii) The rxType value is set equal to invalid.
- 33 iv) The procedure rxProcessFrame() is terminated.
- 34 2) If the extracted TLV type value matches the Port ID TLV type, and the port ID TLV  
35 information string length is not within the range  $2 \leq n \leq 256$ :
- 36 i) The LLDPDU is discarded.
- 37 ii) The statsFramesDiscardedTotal and statsFramesInErrorsTotal counters are both  
38 incremented.
- 39 iii) The rxType value is set equal to invalid.
- 40 iv) The procedure rxProcessFrame() is terminated.
- 41 3) If the extracted TLV type value matches the Port ID TLV type, and the port ID TLV  
42 information string length is within the range  $2 \leq n \leq 256$ , the port ID value is extracted and  
43 appended to the chassis ID value to complete construction of the MSAP identifier.
- 44 c) The third TLV is extracted:



- 1) If the extracted TLV type value does not match the Time To Live, Extension Request, or Extension TLV type, the LLDPDU is invalid:
  - i) The LLDPDU is discarded.
  - ii) The statsFramesDiscardedTotal and statsFramesInErrorsTotal counters are both incremented.
  - iii) The rxType value is set equal to invalid.
  - iv) The procedure rxProcessFrame() is terminated.
- 2) If the extracted TLV type value matches the Time To Live or Extension TLV type, and the adminStatus variable is enableTxOnly, the LLDPDU is discarded because the receive state machine is only enabled to receive Extension Request LLDPDUs.
  - i) The LLDPDU is discarded.
  - ii) The rxType value is set equal to invalid.
  - iii) The procedure rxProcessFrame() is terminated.
- 3) If the extracted TLV type value matches the Time To Live TLV type, and the Time To Live TLV information string length is less than 2:
  - i) The LLDPDU is discarded.
  - ii) The statsFramesDiscardedTotal and statsFramesInErrorsTotal counters are both incremented.
  - iii) The rxType value is set equal to invalid.
  - iv) The procedure rxProcessFrame() is terminated.
- 4) If the extracted TLV type value matches the Time To Live TLV type, and the Time To Live TLV information string length is greater than or equal to 2:
  - i) The first two octets of the TLV information string is extracted and rxTTL is set to this value.
  - ii) If rxTTL equals zero, the rxType value is set equal to shutdown and the procedure rxProcessFrame() is terminated.
  - iii) If rxTTL is non-zero, the rxType value is set to normal.
- 5) If the extracted TLV type value matches the Extension Request TLV type, XLLDP is supported, and the received MSAP identifier and Scope MAC Address match the MSAP and Scope MAC Address of this LLDP agent:
  - i) The rxType value is set equal to xreq.
  - ii) The procedure rxProcessFrame() is terminated.
- 6) If the extracted TLV type value matches the Extension Request TLV type, and either XLLDP is not supported or the received MSAP identifier and Scope MAC Address combination does not match the MSAP and Scope MAC Address combination of this LLDP agent:
  - i) The LLDPDU is discarded.
  - ii) The statsFramesDiscardedTotal and statsFramesInErrorsTotal counters are both incremented.
  - iii) The rxType value is set equal to invalid.
  - iv) The procedure rxProcessFrame() is terminated.
- 7) If the extracted TLV type value matches the Extension TLV type, XLLDP is supported, and an extended receive state machine exists for the neighbor identified by the received MSAP and Scope MAC Address:
  - i) The rxType value is set equal to xpdu.
- 8) If the extracted TLV type value matches the Extension TLV type, and either XLLDP is not supported or an extended receive state machine does not exist for the neighbor identified by the received MSAP and Scope MAC Address:

- 1 i) The LLDPDU is discarded.
- 2 ii) The statsFramesDiscardedTotal and statsFramesInErrorsTotal counters are both  
3 incremented.
- 4 iii) The rxType value is set equal to invalid.
- 5 iv) The procedure rxProcessFrame() is terminated.

6 Each of the remaining TLV information elements are decoded in succession as required by their particular  
7 TLV format definitions:

- 8 d) If the TLV type value equals 0, the TLV is the End Of LLDPDU TLV, and the procedure  
9 rxProcessFrame() is terminated.
- 10 e) If the TLV\_type\_value is less than 127 and is recognized by the LLDP implementation, it is  
11 validated according to the general rules for all TLVs defined in 9.2.8.7.2 as well as any specific rules  
12 defined for the particular TLVs defined in 8.5.
- 13 f) If the TLV\_type\_value is less than 127 and is not recognized by the LLDP implementation, it may  
14 be a basic TLV from a later LLDP version. The statsTLVsUnrecognizedTotal counter is  
15 incremented, and the TLV is assumed to be validated.
- 16 g) If the implementation supports XLLDP and the TLV type value matches the Manifest TLV type:
  - 17 1) Calculate the additional required space in the LLDP remote system information repository to  
18 store new or updated neighbor information:
    - 19 i) If the MSAP identifier in the current LLDPDU does not match an MSAP identifier in the  
20 LLDP remote systems information repository, the total information repository size is  
21 extracted from the Manifest TLV and used as the additional required space.
    - 22 ii) If the MSAP identifier in the current LLDPDU matches an MSAP identifier in the LLDP  
23 remote system, the additional required space is the difference between the total  
24 information repository entry size as extracted from the Manifest TLV and the current size  
25 of the data associated with the MSAP identifier in the LLDP remote system information  
26 repository.
  - 27 2) If there is insufficient space in the LLDP remote systems information repository to store the  
28 neighbor information (based on the additional required space calculated in 1 above), and the  
29 policy is to discard the current LLDPDU rather than delete information for other MSAPs or if  
30 there would be insufficient space after deleting all possible information for other MSAPs from  
31 the remote system information repository then:
    - 32 i) The flag variable tooManyNeighbors is set to TRUE.
    - 33 ii) If the rxTTL value in the LLDPDU is greater than the value of the  
34 tooManyNeighborsTimer, the tooManyNeighborsTimer is set to the rxTTL value.
    - 35 iii) The statsFrameDiscardedTotal counter is incremented.
    - 36 iv) The rxType value is changed to invalid.
  - 37 3) If there is sufficient space in the LLDP remote systems information repository or the policy  
38 allows deleting information from other MSAPs rather than discarding the current LLDPDU  
39 and sufficient space can be created (base on the space calculated in 1 above), then:
    - 40 i) The rxType value is changed to manifest.
    - 41 ii) If an extended receive state machine does not already exist for the MSAP identifier in the  
42 current LLDPDU, an extended receive state machine for that MSAP identifier is created  
43 and its createExtRx variable is set to TRUE.
- 44 h) If the TLV type value is 127, the TLV is an Organizationally Specific TLV:
  - 45 1) If the TLV's organizationally unique identifier and organizationally defined subtype are  
46 recognized, the TLV is validated according to the general rules for all TLVs defined in 9.2.8.7.2  
47 as well as the general rules for Organizationally Specific TLVs defined in 9.2.8.7.3 plus any  
48 specific rules defined for the particular TLV.

- 1        2) If the TLV's organizationally unique identifier and/or organizationally defined subtype are not  
2            recognized, the statsTLVsUnrecognizedTotal counter is incremented, and the TLV is assumed  
3            to be validated.
- 4        i) If the end of the LLDPDU has been reached the procedure rxProcessFrame() is terminated.

#### 5 **9.2.8.7.2 General validation rules for all TLVs**

6 The value in the TLV information string length field is the value used to validate the TLV and to indicate the  
7 location of the next TLV in the LLDPDU.

8 All TLVs are subject to the general validation rules listed in this subclause as well as any specific usage rules  
9 defined for the particular TLV (for example, see systems capabilities TLV usage rules in 8.5.8.3):

- 10    a) If the LLDPDU contains more than one Chassis ID TLV, Port ID TLV, or Time To Live TLV:
  - 11        1) The LLDPDU is discarded.
  - 12        2) The statsFramesDiscardedTotal and statsFramesInErrorsTotal counters are both incremented.
  - 13        3) The rxType value is set equal to invalid.
  - 14        4) The procedure rxProcessFrame() is terminated.
- 15    b) If the implementation supports XLLDP and the LLDPDU contains more than one Manifest TLV,  
16        Extension Request TLV, or Extension Identifier TLV:
  - 17        1) The LLDPDU is discarded.
  - 18        2) The statsFramesDiscardedTotal and statsFramesInErrorsTotal counters are both incremented.
  - 19        3) The rxType value is set equal to invalid.
  - 20        4) The procedure rxProcessFrame() is terminated.
- 21    c) If the LLDPDU contains more than one out of the 3 TLV types Time to Live TLV, Extension  
22        Request TLV and Extension TLV:
  - 23        1) The LLDPDU is discarded.
  - 24        2) The statsFramesDiscardedTotal and statsFramesInErrorsTotal counters are both incremented.
  - 25        3) The rxType value is set equal to invalid.
  - 26        4) The procedure rxProcessFrame() is terminated.
- 27    d) If the LLDPDU contains both a Manifest TLV and an Extension TLV:
  - 28        1) The LLDPDU is discarded.
  - 29        2) The statsFramesDiscardedTotal and statsFramesInErrorsTotal counters are both incremented.
  - 30        3) The rxType value is set equal to invalid.
  - 31        4) The procedure rxProcessFrame() is terminated.
- 32    e) If the TLV information string length value is not greater than or equal to the sum of the lengths of all  
33        fields contained in the TLV information string:
  - 34        1) The LLDPDU is discarded.
  - 35        2) The statsFramesDiscardedTotal and statsFramesInErrorsTotal counters are both incremented.
  - 36        3) The rxType value is set equal to invalid.
  - 37        4) The procedure rxProcessFrame() is terminated.
- 38    f) If any TLV contains an error condition specified for that particular TLV:
  - 39        1) The TLV is discarded.
  - 40        2) The statsTLVsDiscardedTotal counter is incremented and if the statsFramesInErrors has not  
41            been previously updated for this LLDPDU then the statsFramesInErrorsTotal counter is  
42            incremented.
  - 43        3) The location of the next TLV is based on the TLV information string length value of the current  
44            TLV.
- 45    g) With the exception of the TTL TLV for which specific rules apply (see 9.2.8.7.1), if the length of  
46        any field of a TLV is outside the length range specified for that particular TLV (for example, the

- 1 TLV information string length is greater than the maximum permitted length for the TLV  
2 information string as specified for that TLV):
- 3 1) The TLV is discarded.
  - 4 2) The statsTLVsDiscardedTotal counter is incremented and if the statsFramesInErrors has not  
5 been previously updated for this LLDPDU then the statsFramesInErrorsTotal counter is  
6 incremented.
  - 7 3) The location of the next TLV is based on the TLV information string length value of the current  
8 TLV.
- 9 h) If any TLV extends past the physical end of the frame:
- 10 1) The TLV is discarded.
  - 11 2) The statsTLVsDiscardedTotal counter is incremented and if the statsFramesInErrors has not  
12 been previously updated for this LLDPDU then the statsFramesInErrorsTotal counter is  
13 incremented.
- 14 i) Any information following an End Of LLDPDU TLV is ignored.

15 NOTE—Usage rules for individual TLVs allow some TLVs to appear more than once in an LLDPDU. Duplicate TLVs  
16 result in any one of the values being placed in the information repository, can cause the discard stats to increment, and  
17 can cause the change marker for the information repository entry to change if any of the TLV copies change the value  
18 even if the value finally recorded is unchanged. The only thing guaranteed is that the information repository value is set  
19 to one (unspecified) of the TLV values, and if that value is different to what was previously in the information repository  
20 then the change marker is set.

### 21 9.2.8.7.3 General validation rules for all Organizationally Specific TLVs

22 If an Organizationally Specific TLV is not recognized by the receiving LLDP agent, the content of the TLV  
23 can be ignored but the TLV is stored in the LLDP remote systems information repository for possible  
24 retrieval by network management using the basic Organizationally Specific TLV format (see 8.7) and the  
25 StatsTLVsUnrecognizedTotal counter is incremented. If more than one unrecognized Organizationally  
26 Specific TLV is received with the same OUI and organizationally defined subtype, but with identifiable  
27 differences in the organizationally defined information strings, all copies are assigned a temporary  
28 identification index and stored.

### 29 9.2.8.8 somethingChangedLocal()

30 This procedure is called to indicate that the status/value of one or more of the selected objects in the LLDP  
31 local system information repository has changed, and that there is therefore a need to transmit new  
32 information. It is used to notify the managers of the PTOPO and other optional MIBs or YANG modules that  
33 something has changed in the LLDP local systems information repository.

34 This procedure also sets the value of the variable localChange (see 9.2.5.2) to signal the need for the LLDP  
35 agent(s) to transmit the new information, and copies the value of the variable txFastInit (see 9.2.5.20) to the  
36 variable txFast (see 9.2.5.19) to ensure that the new values will be seen by the neighbors on the port.

37 It is assumed that the system provides the ability for any processes that need to receive such indications to  
38 register with this procedure so that appropriate signals can be received by those processes when the  
39 procedure is called.

#### 1 **9.2.8.9 somethingChangedRemote()**

2 This procedure is called after all the information associated with the particular MSAP identifier has been  
3 updated in the LLDP remote systems information repository. The procedure call serves as an indication to  
4 the managers of the PTOPO and other optional MIBs or YANG modules that one or more of the following  
5 conditions is true:

- 6 a) That the status/value of one or more objects in the LLDP remote systems information repository  
7 associated with the particular MSAP identifier has changed.
- 8 b) That an incoming LLDPDU contained an MSAP identifier requiring creation of a new information  
9 repository structure in the LLDP remote systems information repository to receive the information  
10 in that LLDPDU.
- 11 c) That information in the LLDP remote systems information repository associated with the MSAP  
12 identifier has been deleted.

13 It is assumed that the system provides the ability for any processes that need to receive such indications to  
14 register with this procedure so that appropriate signals can be received by those processes when the  
15 procedure is called.

#### 16 **9.2.8.10 txAddCredit()**

17 This procedure increments the value of the txCredit variable (9.2.5.17) by one if the current value of the  
18 txCredit variable is less than the value of txCreditMax (9.2.5.18).

#### 19 **9.2.8.11 txFrame()**

20 The txFrame() procedure makes use of the services of LLC to transmit the LLDPDU constructed by either  
21 the rpConstrInfoLLDPDU(), rpConstrShutdownLLDPDU(), constructXreqPDU(), or rxProcessXreq()  
22 procedures (9.2.8.2, 9.2.8.3, 9.2.8.16, 9.2.8.18). For Normal LLDPDUs, the destination MAC address used  
23 is the MAC address associated with the instance of the LLDP agent (see 7.1.1 and 9.2.5.3). For Extension  
24 Request and Extension LLDPDUs, the destination MAC address used is the Return MAC Address in the  
25 received Manifest TLV or Extension Request TLV respectively (see 8.6.1.2 and 8.6.2.2). The source MAC  
26 address used is the individual MAC address of the transmitting port. The EtherType used is the LLDP  
27 EtherType identified in Table 7-3.

28 If the frame conveys a Manifest LLDPDU, the sentManifest variable is set to TRUE; otherwise the  
29 sentManifest variable is set to FALSE.

30 After the frame has been transmitted, the statsFramesOutTotal counter (9.2.7.5) is incremented.

#### 31 **9.2.8.12 txInitializeLLDP()**

32 The txInitializeLLDP() procedure initializes the LLDP transmit module. It performs the following tasks:

- 33 a) If applicable, either the non-volatile configuration for the LLDP local system information repository  
34 are retrieved or the appropriate default values are assigned to all LLDP configuration variables.
- 35 b) The internal (implementation specific) data structures are initialized with appropriate local physical  
36 topology information.
- 37 c) The setManifest variable is set to FALSE.
- 38 d) System default values are set for the following timing parameters:
  - 39 1) reinitDelay (9.2.5.9)
  - 40 2) msgTxHold (9.2.5.5)
  - 41 3) msgTxInterval (9.2.5.6)

1           4) msgFastTx (9.2.5.4)

2 NOTE—It is recommended to introduce a degree of stagger into starting the LLDP local agent initialization in  
3 multi-port implementations so that the timing of LLDP frame transmission cycles can be distributed among system  
4 ports, and distributed among supported MAC addresses on each port. The method of accomplishing the stagger is an  
5 implementation issue and is beyond the scope of this standard.

6 **9.2.8.13 rxProcessManifest()**

7 The rxProcessManifest() procedure performs the following tasks:

- 8 a) Form the MSAP identifier from the chassis ID and Port ID of the received LLDPDU.
- 9 b) All TLVs received in the Manifest LLDPDU and passed from the receive state machine are retained  
10 in temporary storage associated with the MSAP identifier calculated in step a), replacing any TLVs  
11 retained from a previous Manifest LLDPDU.
- 12 c) If there is no current entry in the LLDP remote systems information repository for the MSAP  
13 identifier calculated in step a), or if the current entry does not contain a Manifest TLV, all XPDU  
14 Descriptors in the new Manifest TLV are assigned a status of “update”.
- 15 d) If there is a current manifest in the LLDP remote systems information repository for the MSAP  
16 identifier calculated in step a), each XPDU Descriptor in the new manifest is compared to the XPDU  
17 Descriptor in the current manifest. Any XPDU Descriptors in the new manifest that match an XPDU  
18 Descriptor in the current manifest are assigned a status of “current”. Any XPDU Descriptors in the  
19 new manifest that do not have a matching XPDU Descriptor in the current manifest are assigned a  
20 status of “update”.
- 21 e) If there are any XPDU in temporary storage associated with this MSAP identifier that match an  
22 XPDU Descriptor with a status of “update” in the new manifest, those XPDU are retained and the  
23 status of the corresponding XPDU Descriptors is changed to “new”. Any XPDU in temporary  
24 storage associated with the MSAP identifier calculated in step a) that do not match an XPDU  
25 descriptor received in the Manifest message with a status of update are discarded.

26 NOTE—Usage rules for the Manifest TLV specify that each XPDU Descriptor has a unique XPDU Number. There is no  
27 requirement for the LLDP agent receiving a Manifest TLV to check for duplicate XPDU Numbers, and the response of  
28 the receiving LLDP agent to duplicated XPDU Numbers is unspecified and implementation dependent.

29 **9.2.8.14 rxProcessXPDU()**

30 The rxProcessXPDU() procedure performs the following tasks:

- 31 a) The statsFramesInTotal counter for the port is incremented.
- 32 b) The 32-bit XPDU Check value of the XPDU is calculated. If the XPDU Number, Sequence, and  
33 Check do not match an XPDU Descriptor in the new manifest, the statsFramesDiscardedTotal and  
34 statsFramesInErrorsTotal counters are both incremented and the procedure rxProcessXPDU() is  
35 terminated.
- 36 c) If the XPDU Number, Sequence, and Check match an XPDU Descriptor in the new manifest, the  
37 status of that XPDU Descriptor is changed to “new”.
- 38 d) Validates the TLVs contained in the LLDPDU as defined in 9.2.8.7.
- 39 e) Retains all validated TLVs in temporary local storage associated with this XPDU Descriptor.
- 40 f) If there are any XPDU Descriptors in the new manifest with a status of “requested” or “retried”, the  
41 variable xreqComplete is set to FALSE. Otherwise the variable xreqComplete is set to TRUE.

42 **9.2.8.15 checkManifest()**

43 The checkManifest() procedure reviews the status of all XPDU Descriptors in the new manifest. If the status  
44 of one or more XPDU Descriptors is “update”, “requested”, or “retried”, the manifestComplete variable is  
45 set to FALSE. Otherwise the manifestComplete variable is set to TRUE and rxTimer is set to rxTTL.

### 1 **9.2.8.16 constructXreqPDU()**

2 The constructXreqPDU() procedure performs the following tasks:

- 3 a) If there are any XPDU Descriptors with a status of “requested” in the new manifest, these XPDU  
4 Descriptors are selected to be included in the Extension Request TLV, the status of these XPDU  
5 Descriptors is changed to “retried”, and rxTimer is set to rxTTL.
- 6 b) If there are no XPDU Descriptors with a status of “requested” or “retried” in the new manifest, one  
7 or more XPDU Descriptors with a status of “update” are selected to be included in the request. The  
8 status of these XPDU Descriptors is changed to “requested”, and rxTimer is set to the value of  
9 rxTTL divided by 32 and rounded up to the nearest integer.
- 10 c) If any XPDU have been selected in steps a) or b) then construct an Extension Request LLDPDU  
11 (8.2) including an Extension Request TLV, structured according to the specification in 8.6.2,  
12 containing the selected XPDU Descriptors..

13 The LLDP agent generating the Extension Request LLDPDU can pace the rate at which Extension  
14 LLDPDUs arrive by adjusting the number of XPDU Descriptors in a single Extension Request TLV and the  
15 timing of the Extension Request LLDPDU transmission. This pacing is implementation dependent and  
16 beyond the scope of this standard. An LLDP agent only transmits an Extension Request LLDPDU when it is  
17 prepared (e.g., has sufficient buffer resources) to receive all requested Extension LLDPDUs.

### 18 **9.2.8.17 checkRxTimer()**

19 The checkRxTimer() procedure reviews the status of all XPDU Descriptors in the new manifest. If the value  
20 of rxTimer is zero and there are any XPDU Descriptors with a status of “requested”, then this is a timeout for  
21 an Extension Request LLDPDU and the rxTTLExpired variable is set to FALSE. If the value of rxTimer is  
22 zero and there are no XPDU Descriptors with a status “requested”, then this is a timeout for the TLV  
23 collection process and rxTTLExpired is set to TRUE.

### 24 **9.2.8.18 rxProcessXREQ()**

25 If the MSAP identifier and Scope MAC Address in the received Extension Request LLDPDU match this  
26 LLDP agent, then for each XPDU Descriptor in the Extension Request TLV that matches one of the XPDU  
27 Descriptors in the most recent Manifest TLV created by the LLDP manager and transmitted in a Normal  
28 LLDPDU, an Extension LLDPDU (8.2) is constructed, including an Extension Identifier TLV structured  
29 according to the specification in 8.6.3, and the other TLVs selected by the LLDP manager, and transmits the  
30 frame using the txFrame() procedure.

31 NOTE—When the Extension Request TLV includes more than one XPDU Descriptor, the rate at which Extension  
32 LLDPDUs are transmitted is at the discretion of the transmitting agent and is implementation dependent.

## 9.2.9 Transmit state machine

The transmit state machine for an individual port and destination MAC address (see 7.1) shall implement the function specified by the state diagram contained in Figure 9-1 and the attendant definitions contained in 9.2.3 through 9.2.8.

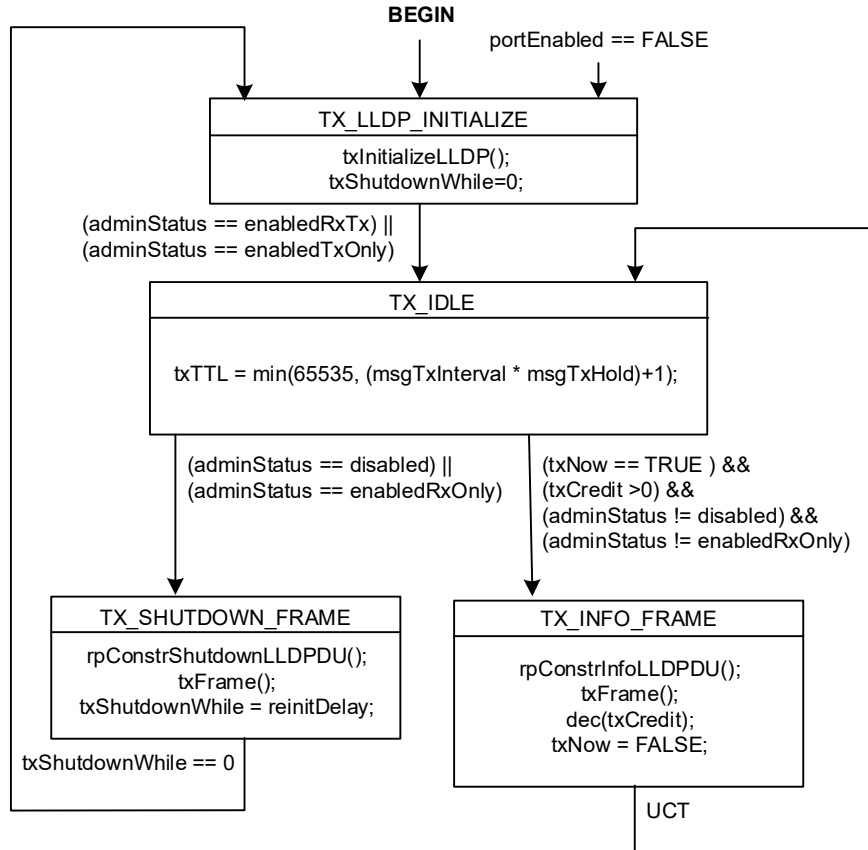


Figure 9-1—Transmit state machine

## 9.2.10 Receive state machine

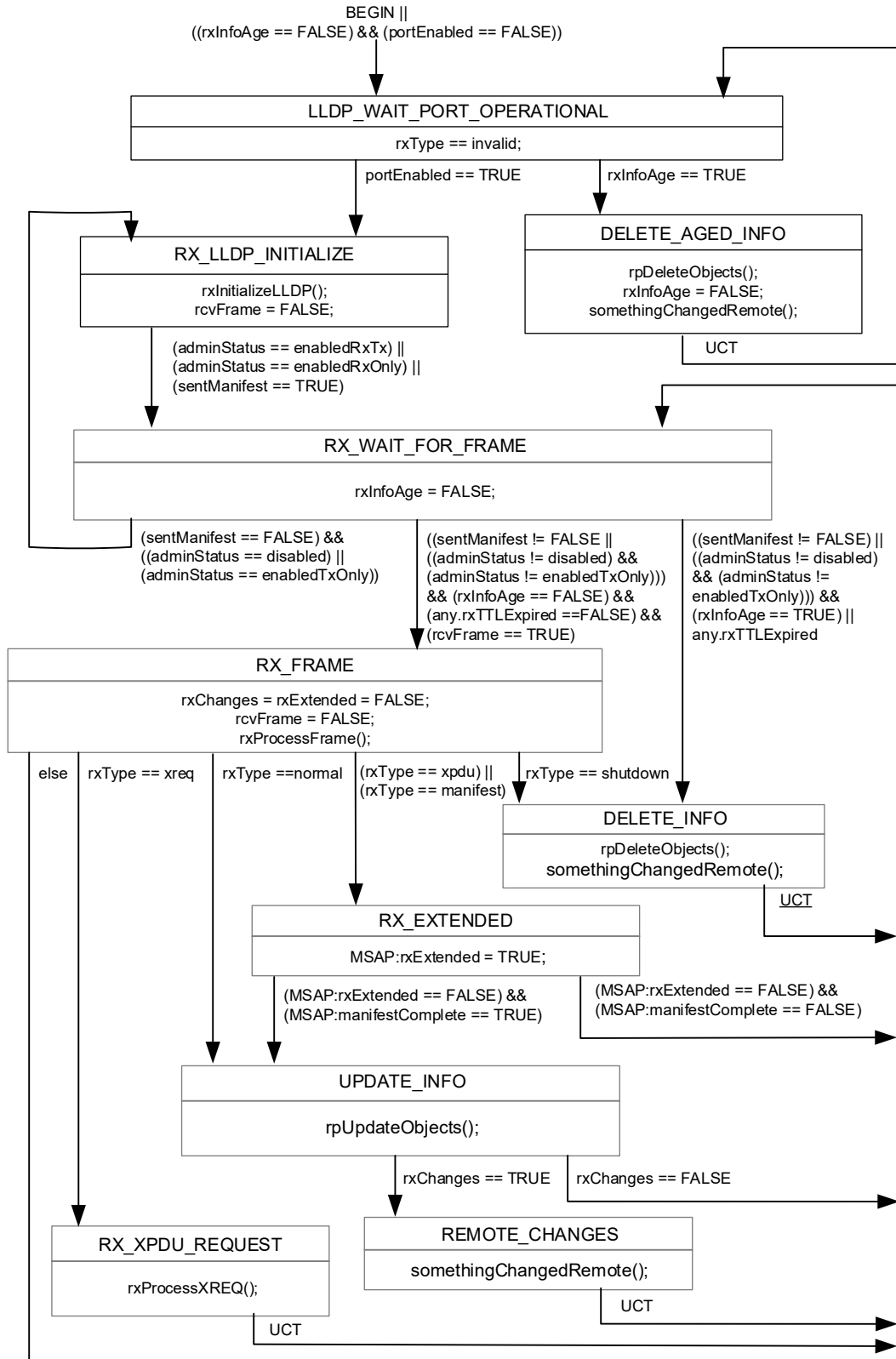
The receive state machine for an individual port and destination MAC address (see 7.1) shall implement the function specified by the state diagram contained in Figure 9-2 and the attendant definitions contained in 9.2.2 through 9.2.8. Additionally, when an implementation supports Extended LLDP and a Manifest TLV is received in an LLDPDU, an extended receive state machine for the received chassis ID and port ID shall implement the function specified by the state diagram contained in Figure 9-3 and the attendant definitions contained in 9.2.2 through 9.2.8.

Because there can be multiple extended receive state machines, the receive state machine uses a notational convention to reference per-neighbor variables (9.2.6) associated with an extended receive state machine. The variable is prefaced by “MSAP.” to indicate a variable associated with the specific MSAP Identifier contained in the frame currently being processed by the receive state machine. The variable is prefaced with “any.” to indicate the logical OR of the variable from all extended state machines.

An extended receive state machine associated with a neighbor is dynamically instantiated, if it does not already exist, when a Normal LLDPDU is received that contains a Manifest TLV. It is destroyed, and any



<sup>1</sup> TLVs held in temporary storage associated with this state machine are discarded, when a Normal LLDPDU  
<sup>2</sup> is received that does not contain a Manifest TLV, or by the assertion of BEGIN.



1

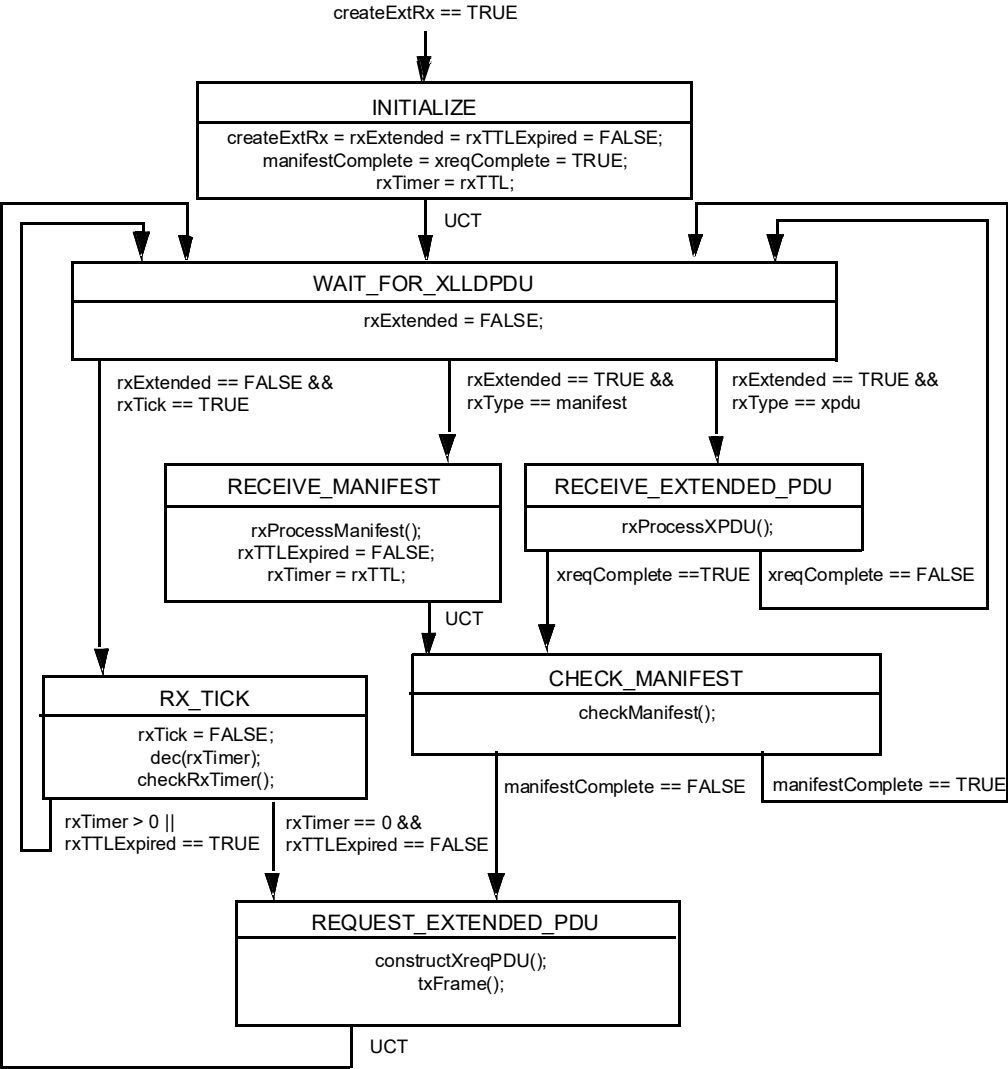


Figure 9-3—Extension receive state machine

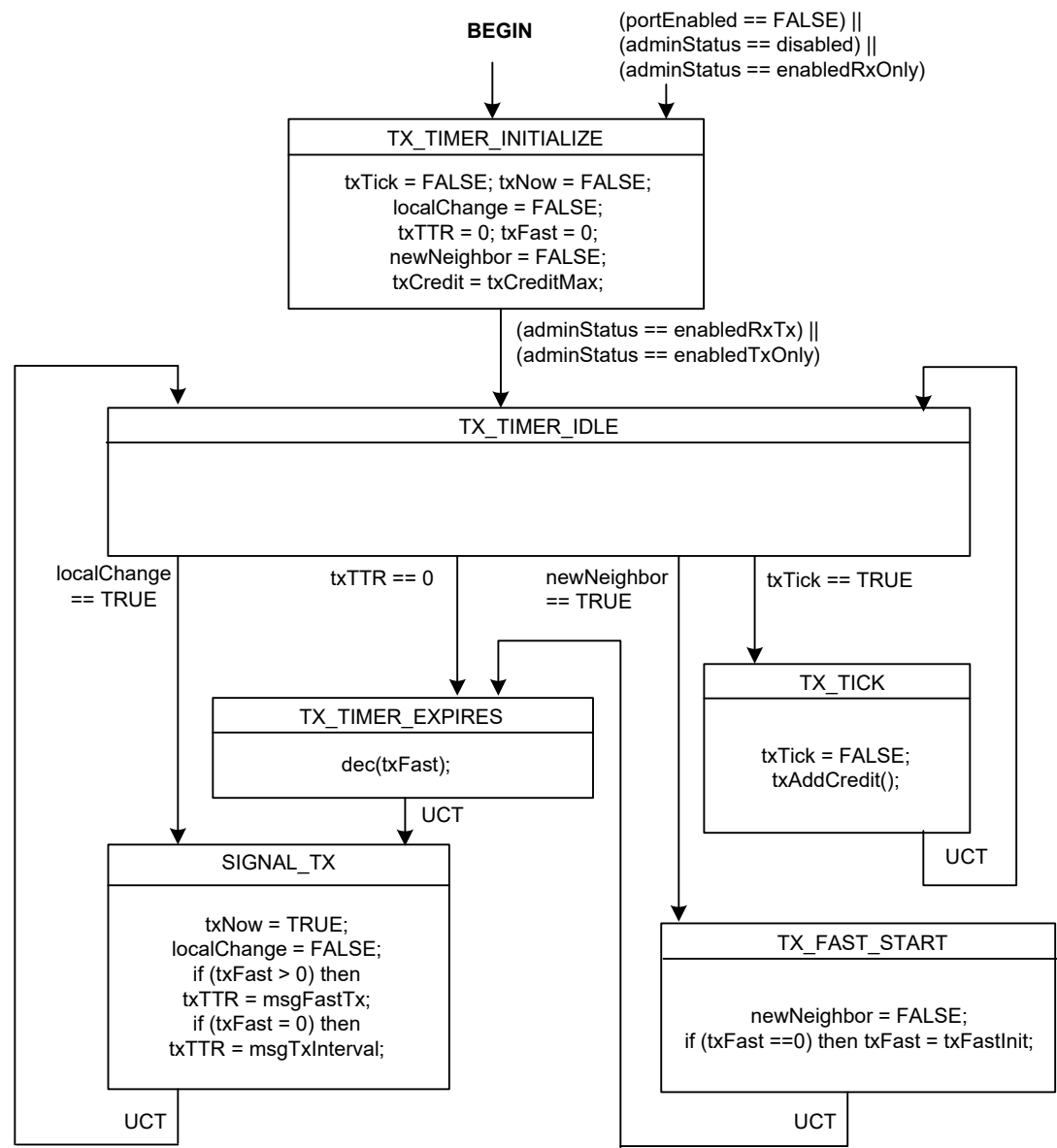
2

3

4

1 **9.2.11 Transmit timer state machine**

2 The transmit timer state machine for an individual port and destination MAC address (see 7.1) shall  
3 implement the function specified by the state diagram contained in Figure 9-4 and the attendant definitions  
4 contained in 9.2.3 through 9.2.8.



**Figure 9-4—Transmit timer state machine**

## 1 10. LLDP management

2 This clause defines the set of managed objects and their functionality that allow administrative configuration  
3 and monitoring of LLDP operation.

### 4 10.1 Data storage and retrieval

5 LLDP agents need to have a place to store both information about the local system and information they  
6 have received about remote systems. Data storage and retrieval capability to support the functionality  
7 defined in 7, 8, 9, and 10 shall be provided. No particular implementation is implied.

### 8 10.2 The LLDP management entity's responsibilities

9 The LLDP management entity has the following responsibilities:

- 10 a) Starting the transmit and receive state machines.
- 11 b) Providing a means for the network manager to select TLVs to be included in outgoing LLDPDUs.
- 12 c) Extracting the necessary information from the LLDP local system information repository and  
13 constructing the individual TLVs selected for insertion into an outgoing LLDPDU.
- 14 d) Prepending appropriate addressing and LLDP EtherType to the LLDPDU before submitting the  
15 MA\_UNITDATA.request for frame transmission.
- 16 e) Updating the LLDP remote systems information repository and monitoring for information  
17 repository object adds, deletions and value changes.
- 18 f) Maintaining operational statistics.

#### 19 10.2.1 Protocol initialization management

20 Protocol initialization consists of the following:

- 21 a) Retrieving non-volatile configuration values for the LLDP local system information repository.
- 22 b) Assigning default or management assigned values to LLDP configuration variables.
- 23 c) Loading physical topology information into their associated LLDP local system information  
24 repository objects.

#### 25 10.2.2 TLV selection management

26 TLV selection management consists of providing the network manager with the means to select which  
27 specific TLVs are enabled for inclusion in an LLDPDU. The following LLDP variables cross reference to  
28 LLDP local systems configuration information repository tables indicating which specific TLVs are enabled  
29 for the particular port(s) on the system. The specific port(s) through which each TLV is enabled for  
30 transmission may be set (or reset) by the network manager.

- 31 a) **mibBasicTLVsTxEnable:** This variable lists the single-instance-use basic management TLVs, each  
32 with a bit map indicating the system ports through which the referenced TLV is enabled for  
33 transmission.
- 34 b) **mibMgmtAddrInstanceTxEnable:** This variable lists the different management addresses (by  
35 subtype) that are defined for the system, each with a bit map indicating the system ports through  
36 which the particular management address/subtype TLV is enabled for transmission.

37 NOTE—Implementers of new TLVs should be aware that provision needs to be made to allow similar network  
38 management selection of new TLVs and that doing so requires linkage to the LLDP information repository, regardless of  
39 whether or not the TLV is part of the basic management set or part of an Organizationally Specific TLV set.

1 When XLLDP is not supported, the LLDP agent can only transmit the TLVs that fit within a single Normal  
2 LLDPU. If more TLVs are selected for inclusion in the LLDPU than fit, some TLVs are not transmitted.  
3 Which TLVs are not transmitted is implementation dependent and not determined by this standard.

4 When XLLDP is supported, each TLV selected for transmission is mapped to the Normal LLDPU or to an  
5 Extension LLDPU. Each Extension LLDPU is identified by an XPDU Number. The mapping of TLVs to  
6 XPDU is implementation dependent. The LLDP management entity maintains a list of which TLVs are  
7 mapped to which XPDU Number, and uses this list to create a Manifest TLV for inclusion in a Manifest  
8 LLDPU at the start of a transmission cycle. The addition, deletion, or modification of any TLV mapped to  
9 Extension LLDPU since the last Manifest LLDPU was transmitted cause the XPDU Revision for that  
10 Extension LLDPU to be incremented prior to creating a new Manifest TLV. Maintaining a consistent  
11 mapping of TLVs to Extension LLDPU whenever possible reduces the number of Extension LLDPU  
12 that need to be requested and transmitted during a transmission cycle.

### 13 10.2.3 Transmission management

14 Transmission management consists of the following:

- 15 a) Determining when a new transmission cycle is required, as signaled by the  
16 somethingChangedLocal() procedure (see 9.2.8.8), or when an Extension LLDPU has been  
17 requested by the rxProcessXREQ() procedure (see 9.2.8.18).
- 18 b) Extracting the appropriate LLDP local system information repository information for the three  
19 mandatory TLVs and inserting them at the beginning of the LLDPU in proper format order.
- 20 c) Creating a Manifest TLV for inclusion in the Normal LLDPU that starts a transmission cycle if  
21 XLLDP is supported and there are TLVs that are mapped to one or more Extension LLDPU.
- 22 d) Extracting the appropriate LLDP local system information repository information for the selected  
23 optional TLVs and inserting them into the LLDPU.
- 24 e) Optionally, appending an End Of LLDPU TLV after the last optional TLV in the LLDPU.
- 25 f) Maintaining length control during LLDPU construction so that the TLVs do not exceed the  
26 maximum length allowed for the LLDPU.
- 27 g) Submitting the frame for transmission.

### 28 10.2.4 Reception management

29 Reception management consists of the following:

- 30 a) Receiving and parsing the incoming LLDPU.
- 31 b) Validating, and checking the types of the first three TLVs to ensure that they are the required type  
32 and in LLDPU format order.
- 33 c) Validating optional TLVs and extracting information values.
- 34 d) Checking whether or not the current LLDPU represents a new MSAP identifier.
- 35 e) Monitoring whether or not there is sufficient space available in the LLDP remote systems  
36 information repository for all active neighbors.
- 37 f) Arbitrating which information is to be discarded in cases where insufficient space is available in the  
38 LLDP remote systems information repository for all active neighbors.
- 39 g) Checking whether or not the received information represents a status or value change to the existing  
40 information repository object.
- 41 h) Updating remote systems information repository objects as necessary.

42 When XLLDP is supported, the LLDP management entity associates each TLV that is stored in the remote  
43 systems information repository with the XPDU Number of the Extension LLDPU in which it was  
44 received. (For convenience an XPDU Number of zero can be used for TLVs received in a Normal LLDPU,

1 since no valid Extension LLDPDU has an XPDU Number equal to zero.) This association is necessary so  
2 that the appropriate TLVs are updated or deleted when a new Extension LLDPDU with that XPDU Number  
3 is received.

#### 4 **10.2.5 Performance management**

5 Performance management consists of the following:

- 6 a) Monitoring the LLDP local system information repository update activities for status or value  
7 changes to selected information repository objects and:
  - 8 1) Calling the somethingChangedLocal() procedure (see 9.2.8.8) whenever a status/value change  
9 occurs.
  - 10 2) Initiating a new transmit cycle if txCredit > 0.
- 11 b) Monitoring the txTTR timer for countdown expiration and initiating a new transmit cycle if txCredit  
12 > 0.
- 13 c) Monitoring the rxInfoTTL timer for countdown expiration and:
  - 14 1) Deleting the associated objects from the LLDP remote systems information repository  
15 whenever the timing counter expires.
  - 16 2) Performing the procedure somethingChangedRemote() associated with the MSAP identifier.
- 17 d) Monitoring rxTTL in the incoming LLDPDUs for shutdown indication and:
  - 18 1) Deleting the associated objects from the LLDP remote systems information repository  
19 whenever rxTTL = 0.
  - 20 2) Performing the procedure somethingChangedRemote() associated with the MSAP identifier.
- 21 e) Monitoring the “tooManyNeighbors” variable to determine whether an existing neighbor’s  
22 information needs to be deleted and:
  - 23 1) Deleting the objects associated with a selected MSAP identifier from the LLDP remote systems  
24 information repository whenever existing information is selected to be deleted.
  - 25 2) Performing the procedure somethingChangedRemote() associated with the MSAP identifier.
- 26 f) Monitoring the tooManyNeighborsTimer to determine when there is not sufficient space available to  
27 accommodate information from all active neighbors and setting the variable tooManyNeighbors  
28 associated with the port to FALSE when the tooManyNeighborsTimer expires.
- 29 g) Notifying the managers of the PTOPO MIB and other optional MIB and YANG modules (see Figure  
30 11-1) whenever the procedures somethingChangedLocal() or somethingChangedRemote() are  
31 performed.
- 32 h) Maintaining operational statistics regarding the LLDP frames sent and received.

#### 33 **10.3 Managed objects**

34 Managed objects model the semantics of management operations. Operations upon a managed object supply  
35 information concerning, or facilitate control over, the process or entity associated with that managed object.

36 Management of LLDP is described in terms of the managed resources that are associated with individual  
37 TLVs and that support frame transmission and reception.

#### 38 **10.4 Data types**

39 This subclause specifies the semantics of operations independent of their encoding in management protocol.  
40 The data types of the parameters of operations are defined for that specification.

1 The following data types are used:

- 2 a) Boolean.
- 3 b) Enumerated, for a collection of named values.
- 4 c) Unsigned, for all parameters specified as “number of” some quantity.
- 5 d) MAC address.
- 6 e) Time interval, an Unsigned value representing a positive integral number of seconds for all time out
- 7 parameters.
- 8 f) Counter, for all parameters specified as a “count” of some quantity (all counters increment and wrap
- 9 with a modulus of 2 to the power 32).

## 10 10.5 LLDP variables

11 LLDP managed objects fall into the following categories:

- 12 a) Status/control variables necessary for operation of the protocol.
- 13 b) Variables that accumulate operational statistics.
- 14 c) Variables that are required by the particular TLVs.

### 15 10.5.1 LLDP operational status and control

- 16 a) Global parameters and variables:
  - 17 1) **adminStatus**: The authority that controls whether or not a local LLDP agent is enabled for
  - 18 transmit and receive, transmit only, or receive only; or is disabled (9.2.5.1).
- 19 b) Transmit state machine parameters and variables:
  - 20 1) **msgTxHold**: A multiplier on msgTxInterval used to compute the TTL value of txTTL (9.2.5.5,
  - 21 9.2.5.23).
  - 22 2) **msgTxInterval**: The interval between successive transmit cycles in normal transmission mode
  - 23 (9.2.5.6).
  - 24 3) **reinitDelay**: The delay after adminStatus becomes ‘disabled’ before re-initialization is
  - 25 attempted (9.2.5.9).
  - 26 4) **txCreditMax**: The maximum value of transmission credit (9.2.5.18).
  - 27 5) **txFastInit**: The interval between successive LLDPDU transmissions in fast transmission mode
  - 28 (9.2.5.20).
- 29 c) Receive state machine parameters and variables:
  - 30 1) **remoteChanges**: An indication that the status/value of one or more selected objects in the
  - 31 LLDP remote systems information repository has changed or that all objects associated with a
  - 32 particular MSAP identifier have been deleted (9.2.5.10).
  - 33 2) **tooManyNeighbors**: An indication that there is insufficient space in the LLDP remote systems
  - 34 information repository to store information from all active neighbors (9.2.5.16).

### 35 10.5.2 LLDP operational statistics counters

36 The following counters provide operational statistics on a per port basis:

- 37 a) Transmission counters:
  - 38 1) **statsFramesOutTotal**: A count of all LLDP frames transmitted through the port (9.2.7.5).
  - 39 2) **statsLengthErrorsTotal**: A count of all LLDP length errors detected when constructing
  - 40 LLDPDU frames for transmission through the port (9.2.8.2).
- 41 b) Reception counters:
  - 42 1) **statsAgeoutsTotal**: A count of the times that a neighbor’s information is deleted from the
  - 43 LLDP remote systems information repository because of rxInfoTTL timer expiration (9.2.7.1).



- 1        2) **statsFramesDiscardedTotal:** A count of all LLDPDUs received and then discarded (9.2.7.2).
- 2        3) **statsFramesInErrorsTotal:** A count of all LLDPDUs received at the port with one or more
- 3        detectable errors (9.2.7.3).
- 4        4) **statsFramesInTotal:** A count of all LLDP frames received at the port (9.2.7.4).
- 5        5) **statsTlvsDiscardedTotal:** A count of all TLVs received at the port and discarded for any
- 6        reason. (9.2.7.6)
- 7        6) **statsTLVsUnrecognizedTotal:** This counter provides a count of all TLVs not recognized by
- 8        the receiving LLDP local agent (9.2.7.7).

### 9 **10.5.3 TLV required variables**

10 Variables in this category are defined by the requirements of the particular TLVs. They are maintained with  
11 reference to the local system in the LLDP local system information repository; and with reference to the  
12 remote system, in the LLDP remote systems information repository. TLV variables are outputs from the  
13 local LLDP agent and inputs to the remote LLDP agent.

14 Variables that pertain to basic set TLVs are listed in the following subclauses.

#### 15 **10.5.3.1 Chassis ID TLV objects**

- 16    a) **chassis ID subtype:** The type of identifier used for the chassis (see 8.5.2.2).
- 17    b) **chassis ID:** The identification assigned to the chassis containing the port (see 8.5.2.3).

#### 18 **10.5.3.2 Port ID TLV objects**

- 19    a) **port ID subtype:** The type of identifier used for the port (see 8.5.3.2).
- 20    b) **port ID:** The identification assigned to the port (see 8.5.3.3).

#### 21 **10.5.3.3 Port description TLV object**

- 22    a) **port description:** The port's description (see 8.5.5.2).

#### 23 **10.5.3.4 System name TLV object**

- 24    a) **system name:** The system's assigned name (see 8.5.6.2).

#### 25 **10.5.3.5 System description TLV object**

- 26    a) **system description:** The system's description (see 8.5.7.2).

#### 27 **10.5.3.6 System capabilities TLV objects**

- 28    a) **system capabilities:** The primary capabilities of the system (see 8.5.8.1).
- 29    b) **enabled capabilities:** The system's enabled capabilities (see 8.5.8.2).

#### 30 **10.5.3.7 Management address TLV objects**

- 31    a) **management address length:** The length of the management address (see 8.5.9.2).
- 32    b) **management address subtype:** The management address type (see 8.5.9.3).
- 33    c) **management address:** The management address (see 8.5.9.4).
- 34    d) **interface numbering subtype:** The interface type (see 8.5.9.5).
- 35    e) **interface number:** The interface number (see 8.5.9.6).

- 1 f) **OID length:** The object identifier length (see 8.5.9.7).
- 2 g) **OID:** The object identifier (see 8.5.9.8).

## 11. LLDP MIB definitions

This clause defines the LLDP basic MIB for use with SNMP in TCP/IP based internets. In particular, it defines objects for managing the operation of LLDP based on the specifications of 7, 8, 9, and 10.

### 11.1 Internet Standard Management Framework

LLDP MIBs are designed to operate in a manner consistent with the principles of the Internet Standard Management Framework, which describes the separation of a data modeling language (for example, SMIV2) from content-specific data models (for example, the LLDP remote systems MIB), and from messages and protocol operations used to manipulate the data (for example, SNMPv3).

For a detailed overview of the documents that describe the current Internet Standard Management Framework, please refer to section 7 of IETF RFC 3410.

Managed objects are accessed via a virtual information store, termed the Management Information Base or MIB. MIB objects are generally accessed through the Simple Network Management Protocol (SNMP). Objects in the MIB are defined using the mechanisms defined in the Structure of Management Information (SMI). This clause specifies a MIB module that is compliant to the SMIV2, which is described in IETF RFC 2578 (STD 58) [B2], IETF RFC 2579 (STD 58) [B3], and IETF RFC 2580 (STD 58) [B4].

### 11.2 Structure of the LLDP MIB

The LLDP MIB consists of two types of MIB modules, the mandatory basic MIB defined in this clause and from zero to  $n$  optional organizationally specific MIB extensions such as the IEEE 802.1 MIB in Annex D of IEEE Std 802.1Q-2022 and the IEEE 802.3 MIB in Clause 79 of IEEE Std 802.3-2022.

Each MIB module is divided into two major sections as shown in Figure 11-1 to allow selective MIB support for the particular operating mode (transmit only, receive only, or both transmit and receive) being implemented.

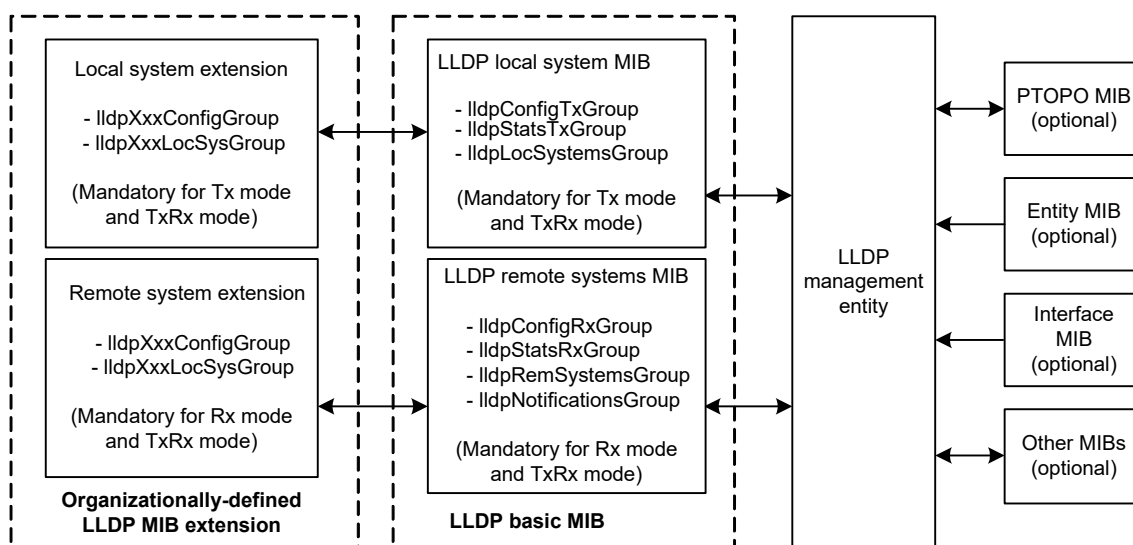


Figure 11-1—LLDP MIB block diagram

1 Table 11-1 summarizes the particular object groups that are required for each operating mode. The basic MIB  
2 shall comply with the MIB conformance section for the particular operating mode (Rx, Tx, or TxRx mode)  
3 being supported.

**Table 11-1—MIB object groups and operating mode applicability**

| MIB group                 | Rx mode        | Tx mode | TxRx mode |
|---------------------------|----------------|---------|-----------|
| lldpV2ConfigGroup         |                |         |           |
| lldpV2ConfigRxGroup       | M <sup>a</sup> | —       | M         |
| lldpV2ConfigTxGroup       | —              | M       | M         |
| lldpV2StatsRxGroup        | M              | —       | M         |
| lldpV2StatsTxGroup        | —              | M       | M         |
| lldpV2LocSysGroup         | —              | M       | M         |
| lldpV2RemSysGroup         | M              | —       | M         |
| lldpV2NotificationsGroup  | M              | —       | M         |
| ifGeneralInformationGroup | M              | M       | M         |

<sup>a</sup>M=Mandatory

4 Table 11-2 shows the structure of the MIB and the relationship of the MIB objects to the LLDP operational  
5 status/control variables, LLDP statistics variables, and TLV variables.

**Table 11-2—LLDP MIB structure and object cross reference**

| MIB table                       | MIB object                    | LLDP reference         |
|---------------------------------|-------------------------------|------------------------|
| <i>LLDP Configuration group</i> |                               |                        |
|                                 | lldpV2MessageTxInterval       | msgTxInterval, 9.2.5.6 |
|                                 | lldpV2MessageTxHoldMultiplier | msgTxHold, 9.2.5.5     |
|                                 | lldpV2ReinitDelay             | reinitDelay, 9.2.5.9   |
|                                 | lldpV2NotificationInterval    | msgTxInterval, 9.2.5.6 |
|                                 | lldpV2TxCreditMax             | txCreditMax, 9.2.5.18  |
|                                 | lldpV2MessageFastTx           | msgFastTx, 9.2.5.4     |
|                                 | lldpV2TxFastInit              | txFastInit, 9.2.5.20   |

**Table 11-2—LLDP MIB structure and object cross reference (continued)**

| MIB table                       | MIB object                           | LLDP reference         |
|---------------------------------|--------------------------------------|------------------------|
| lldpV2PortConfigTableV2         |                                      |                        |
|                                 | lldpV2PortConfigIfIndexV2            | (Table index)          |
|                                 | lldpV2PortConfigDestAddressIndexV2   | (Table index)          |
|                                 | lldpV2PortConfigAdminStatusV2        | adminStatus, 9.2.5.1   |
|                                 | lldpV2PortMessageTxInterval          | msgTxInterval, 9.2.5.6 |
|                                 | lldpV2PortMessageTxHoldMultiplier    | msgTxHold, 9.2.5.5     |
|                                 | lldpV2PortReinitDelay                | reinitDelay, 9.2.5.9   |
|                                 | lldpV2PortNotificationInterval       | msgTxInterval, 9.2.5.6 |
|                                 | lldpV2PortTxCreditMax                | txCreditMax, 9.2.5.18  |
|                                 | lldpV2PortMessageFastTx              | msgFastTx, 9.2.5.4     |
|                                 | lldpV2PortTxFastInit                 | txFastInit, 9.2.5.20   |
|                                 | lldpV2PortConfigNotificationEnableV2 | —                      |
|                                 | lldpV2PortConfigTLVsTxEnableV2       | 9.1.1.1                |
| lldpV2DestAddressTable          |                                      |                        |
|                                 | lldpV2AddressTableIndex              | (Table index)          |
|                                 | lldpV2DestMacAddress                 | (Table index)          |
| lldpV2ManAddrConfigTxPortsTable |                                      |                        |
|                                 | lldpV2ManAddrConfigIfIndex           | (Table index)          |
|                                 | lldpV2ManAddrConfigDestAddressIndex  | (Table index)          |
|                                 | lldpV2ManAddrConfigLocManAddrSubtype | 8.5.9.3 (Table index)  |
|                                 | lldpV2ManAddrConfigLocManAddr        | 8.5.9.4 (Table index)  |
|                                 | lldpV2ManAddrConfigTxEnable          | 9.1.1.1                |
|                                 | lldpV2ManAddrConfigRowStatus         | —                      |
| <i>LLDP Statistics group</i>    |                                      |                        |
|                                 | lldpV2StatsRemTablesLastChangeTime   | —                      |
|                                 | lldpV2StatsRemTablesInserts          | —                      |
|                                 | lldpV2StatsRemTablesDeletes          | —                      |
|                                 | lldpV2StatsRemTablesDrops            | —                      |
|                                 | lldpV2StatsRemTablesAgeouts          | —                      |

**Table 11-2—LLDP MIB structure and object cross reference (continued)**

| MIB table                      | MIB object                             | LLDP reference                                       |
|--------------------------------|----------------------------------------|------------------------------------------------------|
| lldpV2StatsTxPortTable         |                                        |                                                      |
|                                | lldpV2StatsTxIfIndex                   | (Table index)                                        |
|                                | lldpV2StatsTxDestMACAddress            | (Table index)                                        |
|                                | lldpV2StatsTxPortFramesTotal           | statsFramesOutTotal, 9.2.7.5                         |
|                                | lldpV2StatsTxLLDPDULengthErrors        | lldpduLengthErrors, 9.2.7.8                          |
| lldpV2StatsRxPortTable         |                                        |                                                      |
|                                | lldpV2StatsRxDestIfIndex               | (Table index)                                        |
|                                | lldpV2StatsRxDestMACAddress            | (Table index)                                        |
|                                | lldpV2StatsRxPortFramesDiscardedTotal  | statsFramesDiscardedTotal, 9.2.7.2                   |
|                                | lldpV2StatsRxPortFramesErrors          | statsFramesInErrorsTotal, 9.2.7.3                    |
|                                | lldpV2StatsRxPortFramesTotal           | statsFramesInTotal, 9.2.7.4                          |
|                                | lldpV2StatsRxPortTLVsDiscardedTotal    | statsTLVsDiscardedTotal, 9.2.7.6                     |
|                                | lldpV2StatsRxPortTLVsUnrecognizedTotal | statsTLVsUnrecognizedTotal, 9.2.7.7                  |
|                                | lldpV2StatsRxPortAgeoutsTotal          | statsAgeoutsTotal, 9.2.7.1                           |
| <i>Local System Data group</i> |                                        |                                                      |
|                                | lldpV2LocChassisIdSubtype              | chassis ID subtype, 8.5.2.2                          |
|                                | lldpV2LocChassisId                     | chassis ID, 8.5.2.3                                  |
|                                | lldpV2LocSysName                       | system name, 8.5.6.2                                 |
|                                | lldpV2LocSysDesc                       | system description, 8.5.7.2                          |
|                                | lldpV2LocSysCapSupported               | system capabilities, 8.5.8.1                         |
|                                | lldpV2LocSysCapEnabled                 | enabled capabilities, 8.5.8.2                        |
| lldpV2LocPortTable             |                                        |                                                      |
|                                | lldpV2LocPortIfIndex                   | (Table index)                                        |
|                                | lldpV2LocPortIdSubtype                 | port ID subtype, 8.5.3.2                             |
|                                | lldpV2LocPortId                        | port ID, 8.5.3.3                                     |
|                                | lldpV2LocPortDesc                      | port description, 8.5.5.2                            |
| lldpV2LocManAddrTable          |                                        |                                                      |
|                                | lldpV2LocManAddrSubtype                | management address subtype, 8.5.9.3<br>(Table index) |
|                                | lldpV2LocManAddr                       | management address, 8.5.9.4 (Table<br>index)         |
|                                | lldpV2LocManAddrLen                    | management address string length,<br>8.5.9.2         |
|                                | lldpV2LocManAddrIfSubtype              | interface numbering subtype, 8.5.9.5                 |

**Table 11-2—LLDP MIB structure and object cross reference (continued)**

| MIB table                        | MIB object                   | LLDP reference                                       |
|----------------------------------|------------------------------|------------------------------------------------------|
|                                  | lldpV2LocManAddrIfId         | interface number, 8.5.9.6                            |
|                                  | lldpV2LocManAddrOID          | object identifier, 8.5.9.8                           |
| <i>Remote Systems Data group</i> |                              |                                                      |
| lldpV2RemTable                   |                              |                                                      |
|                                  | lldpV2RemTimeMark            | (Table index)                                        |
|                                  | lldpV2RemLocalIfIndex        | (Table index)                                        |
|                                  | lldpV2RemLocalDestMACAddress | (Table index)                                        |
|                                  | lldpV2RemIndex               | (Table index)                                        |
|                                  | lldpV2RemChassisIdSubtype    | chassis ID subtype, 8.5.2.2                          |
|                                  | lldpV2RemChassisId           | chassis ID, 8.5.2.3                                  |
|                                  | lldpV2RemPortIdSubtype       | port ID sybtype, 8.5.3.2                             |
|                                  | lldpV2RemPortId              | port ID, 8.5.3.3                                     |
|                                  | lldpV2RemPortDesc            | port description, 8.5.5.2                            |
|                                  | lldpV2RemSysName             | system name, 8.5.6.2                                 |
|                                  | lldpV2RemSysDesc             | system description, 8.5.7.2                          |
|                                  | lldpV2RemSysCapSupported     | system capabilities, 8.5.8.1                         |
|                                  | lldpV2RemSysCapEnabled       | enabled capabilities, 8.5.8.2                        |
|                                  | lldpV2RemRemoteChanges       | remoteChanges, 9.2.5.10                              |
|                                  | lldpV2RemTooManyNeighbors    | tooManyNeighbors, 9.2.5.16                           |
| lldpV2RemManAddrTable            |                              | (Table index)                                        |
|                                  | lldpV2RemTimeMark            | (Table index)                                        |
|                                  | lldpV2RemLocalIfIndex        | (Table index)                                        |
|                                  | lldpV2RemLocalDestMACAddress | (Table index)                                        |
|                                  | lldpV2RemIndex               | (Table index)                                        |
|                                  | lldpV2RemManAddrSubtype      | management address subtype, 8.5.9.3<br>(Table index) |
|                                  | lldpV2RemManAddr             | management address, 8.5.9.4 (Table<br>index)         |
|                                  | lldpV2RemManAddrIfSubtype    | interface numbering subtype, 8.5.9.5                 |
|                                  | lldpV2RemManAddrIfId         | interface number, 8.5.9.6                            |
|                                  | lldpV2RemManAddrOID          | object identifier, 8.5.9.8                           |

**Table 11-2—LLDP MIB structure and object cross reference (continued)**

| MIB table                | MIB object                   | LLDP reference                                            |
|--------------------------|------------------------------|-----------------------------------------------------------|
| lldpV2RemUnknownTLVTable |                              |                                                           |
|                          | lldpV2RemTimeMark            | (Table index)                                             |
|                          | lldpV2RemLocalIfIndex        | (Table index)                                             |
|                          | lldpV2RemLocalDestMACAddress | (Table index)                                             |
|                          | lldpV2RemIndex               | (Table index)                                             |
|                          | lldpV2RemUnknownTLVType      | LLDPDU validation, 9.2.8.7.1 (Table index)                |
|                          | lldpV2RemUnknownTLVInfo      | LLDPDU validation, 9.2.8.7.1                              |
| lldpV2RemOrgDefInfoTable |                              |                                                           |
|                          | lldpV2RemTimeMark            | (Table index)                                             |
|                          | lldpV2RemLocalIfIndex        | (Table index)                                             |
|                          | lldpV2RemLocalDestMACAddress | (Table index)                                             |
|                          | lldpV2RemIndex               | (Table index)                                             |
|                          | lldpV2RemOrgDefInfoOUI       | organizationally unique identifier, 8.7.1.3 (Table index) |
|                          | lldpV2RemOrgDefInfoSubtype   | organizationally defined subtype, 8.7.1.4 (Table index)   |
|                          | lldpV2RemOrgDefInfoIndex     | (Table index)                                             |
|                          | lldpV2RemOrgDefInfo          | organizationally defined information, 8.7.1.5             |
| LLDP MIB Notifications   |                              |                                                           |
|                          | lldpV2RemTablesChange        |                                                           |

## 1 11.3 Relationship to other MIBs

2 This clause includes specifications for an LLDP Textual Conventions MIB module, an LLDP MIB module,  
3 and for IEEE 802.1 and IEEE 802.3 extension MIB modules that are compliant with the SMIV2 as defined in  
4 IETF RFC 2578 (STD 58) [B2], IETF RFC 2579 (STD 58) [B3], and IETF RFC 2580 (STD 58) [B4].

5 LLDP is designed to operate in conjunction with MIB modules defined by the IETF, IEEE 802, and others.  
6 The following subclauses discuss the relationship between LLDP and IETF MIB modules. LLDP agents  
7 automatically notify the managers of these MIB modules whenever there is a value or status change in an  
8 LLDP MIB object.

9 LLDP managed objects for the local system are stored in the LLDP local system MIB. Information received  
10 from a remote LLDP agent is stored in the local LLDP agent's LLDP remote system MIB.

11 NOTE—In this standard, managed objects for an LLDP MIB module are defined as they would be for an SNMP MIB.  
12 However, it is not required for LLDP implementations to support SNMP to store and retrieve system data. LLDP agents  
13 need to have a place to store both information about the local system and information they have received about remote  
14 systems. No particular implementation is implied.



### 11.3.1 Relationship to LLDP Version 1 MIB

The version 1 MIB module that was published in the initial 2005 publication of this standard has been superseded by the version 2 MIB module specified in 11.5.1, and support of the version 2 module is a requirement for conformance to the required or optional capabilities (Clause 5) in this revision of the standard. The version 2 MIB module reflects changes in indexation of the MIB objects that support the use of LLDP with multiple destination MAC addresses, as discussed in Clause 6.

In addition to the changes in indexation, the version 2 MIB module also supports changes to the management of the LLDP protocol that have resulted from changes to the LLDP protocol state machines. As these protocol changes affect only the timing and frequency of transmission of LLDPDUs, and not the content of the LLDPDUs, version 1 and version 2 implementations can successfully co-exist and interoperate.

From the perspective of a version 2 implementation, a version 1 neighbor behaves as a version 2 neighbor that is only using a single destination MAC address (the 01-80-C2-00-00-0E address specified for use in the IEEE 802.1AB-2005 version of the standard). From the perspective of a version 1 implementation, a version 2 neighbor that uses the 01-80-C2-00-00-0E destination MAC address behaves as a version 1 neighbor.

### 11.3.2 IETF Physical Topology MIB

The IETF Physical Topology (PTOPO) MIB (IETF RFC 2922 [B8]) allows a LLDP agent to expose learned physical topology information, using an IETF MIB. LLDP is intended to support the PTOPO MIB.

NOTE—The LLDP MIB module is a logical superset of the IETF PTOPO MIB; therefore, from a functional point of view, if the LLDP MIB module is implemented, it is likely not to be necessary to implement the PTOPO MIB defined in IETF RFC 2922 [B8]. However, for backward compatibility, this standard also supports the use of the PTOPO MIB.

### 11.3.3 IETF Entity MIB

The Entity MIB (IETF RFC 6933) allows the physical component inventory and hierarchy to be identified. Chassis IDs passed in the LLDPDU can identify entPhysicalTable entries. SNMP agents that implement the LLDP MIB should implement the entPhysicalAlias object from the Entity MIB version 2 or higher.

### 11.3.4 IETF Interfaces MIB

The Interfaces MIB (IETF RFC 2863) provides a standard mechanism for managing network ports. Port IDs passed in the LLDPDU can identify ifTable (or entPhysicalTable) entries. SNMP agents that implement the LLDP MIB shall also implement the ifTable and ifXTable for the ports that are represented in the Interfaces MIB.

## 11.4 Security considerations for LLDP base MIB module

There are a number of management objects defined in this MIB module with a MAX-ACCESS clause of read-write.<sup>13</sup> Such objects may be considered sensitive or vulnerable in some network environments. The support for SET operations in a non-secure environment without proper protection can have a negative effect on network operations.

- a) Setting the following objects to incorrect values can result in an excessive number of LLDP packets being sent by the LLDP agent:
  - 1) lldpV2MessageTxInterval, lldpV2PortMessageTxInterval
  - 2) lldpV2TxCreditMax, lldpV2PortTxCreditMax

<sup>13</sup> In IETF MIB definitions, the MAX-ACCESS clause defines the type of access that is allowed for particular data elements in the MIB. An explanation of the MAX-ACCESS mappings is given in section 7.3 of IETF RFC 2578 [B2].

- 1        3) lldpV2MessageFastTx, lldpV2PortMessageFastTx
- 2        4) lldpV2TxFastInit, lldpV2PortTxFastInit
- 3    b) Setting the object, lldpV2MessageTxHoldMultiplier or lldpV2PortMessageTxHoldMultiplier, to
- 4        incorrect values can cause the LLDP agent to transmit LLDPDUs with too-high TTL values, which
- 5        affect the expiration time of objects grouped under lldpV2RemoteSystemsData identifier.
- 6    c) Setting the object, lldpV2ReinitDelay or lldpV2PortReinitDelay, to too low a value can cause the
- 7        transmit state machine to attempt excessive re-initializations.
- 8    d) Setting incorrect bits in the object, lldpV2PortConfigTLVsTxEnableV2, can cause the LLDP agent
- 9        to transmit LLDPDUs with an undesired optional TLV sequence.
- 10   e) Setting incorrect bits in the object, lldpV2ConfigManAddrPortsTxEnable, can cause the LLDP
- 11        agent to advertise management addresses that were not meant to be disclosed and/or to omit
- 12        addresses that were desired.
- 13   f) Setting the following objects to incorrect values can result in improper operation of the MIB
- 14        notification process:
  - 15        1) lldpV2NotificationInterval
  - 16        2) lldpV2PortNotificationInterval
  - 17        3) lldpV2PortConfigNotificationEnableV2
- 18   g) Setting the object, lldpV2PortConfigAdminStatusV2, to the incorrect value can result in enabling a
- 19        non-desired operational mode.

20 The following readable objects in this MIB module may be considered to be sensitive or vulnerable in some  
21 network environments:

- 22   h) Objects that are associated with the transmit mode
  - 23        1) lldpV2LocChassisIdSubtype
  - 24        2) lldpV2LocChassisId
  - 25        3) lldpV2LocPortIdSubtype
  - 26        4) lldpV2LocPortId
  - 27        5) lldpV2LocPortDesc
  - 28        6) lldpV2LocSysName
  - 29        7) lldpV2LocSysDesc
  - 30        8) lldpV2LocSysCapSupported
  - 31        9) lldpV2LocSysCapEnabled
  - 32        10) lldpV2LocManAddrLen
  - 33        11) lldpV2LocManAddrIfSubtype
  - 34        12) lldpV2LocManAddrIfId
  - 35        13) lldpV2LocManAddrOID
- 36   i) Objects that are associated with the receive mode
  - 37        1) lldpV2NotificationInterval
  - 38        2) lldpV2PortConfigNotificationEnable
  - 39        3) lldpV2RemChassisIdSubtype
  - 40        4) lldpV2RemChassisId
  - 41        5) lldpV2RemPortIdSubtyp
  - 42        6) lldpV2RemPortId
  - 43        7) lldpV2RemPortDesc
  - 44        8) lldpV2RemSysName
  - 45        9) lldpV2RemSysDesc
  - 46        10) lldpV2RemSysCapSupported
  - 47        11) lldpV2RemSysCapEnabled
  - 48        12) lldpV2RemManAddrIfSubtype
  - 49        13) lldpV2RemManAddrIfId

- 1 14) lldpV2RemManAddrOID
- 2 15) lldpV2RemUnknownTLVInfo
- 3 16) lldpV2RemOrgDefInfo

4 This concern applies both to objects that describe the configuration of the local host, as well as for objects  
5 that describe information from the remote hosts, acquired via LLDP and displayed by the objects in this  
6 MIB module. It is thus important to control even GET and/or NOTIFY access to these objects and possibly  
7 to even encrypt the values of these objects when sending them over the network via SNMP.

8 SNMP versions prior to SNMPv3 did not include adequate security. Even if the network itself is secure (for  
9 example, by using IPSec), even then, there is no control as to who on the secure network is allowed to  
10 access and GET/SET (read/change/create/delete) the objects in this MIB module.

11 Implementers should consider the security features as provided by the SNMPv3 framework (see  
12 IETF RFC 3410, section 8), including full support for the SNMPv3 cryptographic mechanisms (for  
13 authentication and privacy).

14 Further, implementers should not deploy SNMP versions prior to SNMPv3. Instead, implementers should  
15 deploy SNMPv3 to enable cryptographic security. It is then a customer/operator responsibility to ensure that  
16 the SNMP entity giving access to an instance of this MIB module is properly configured to give access to the  
17 objects only to those principals (users) that have legitimate rights to indeed GET or SET  
18 (change/create/delete) them.

## 19 **11.5 LLDP MIB modules** <sup>14, 15</sup>

20 Two MIB modules are defined—a textual conventions MIB module (11.5.1) that contains the textual  
21 conventions used by the LLDP version 2 MIB module and by the version 2 extension MIB module in  
22 Annex D of IEEE Std 802.1Q-2022, and the LLDP version 2 MIB module itself (11.5.2). The textual  
23 conventions defined in the Textual-Convention MIB module are also available by extension MIB modules  
24 defined in other standards, such as the extension MIB modules defined in IEEE Std 802.1Q.

25

---

<sup>14</sup> *Copyright release for MIBs:* Users of this standard may freely reproduce the MIB contained in this subclause so that it can be used for its intended purpose.

<sup>15</sup> An ASCII version of this MIB module can be obtained by Web browser from the IEEE 802.1 Website at <http://www.ieee802.org/1/pages/MIBS.html>.

## 1 11.5.1 Definitions for the LLDP-V2-TC MIB module

2 In the following MIB module, should any discrepancy between the DESCRIPTION text and the  
3 corresponding definition in Clause 9 and Clause 10 occur, the definition in Clause 9 and Clause 10 shall take  
4 precedence.

```
5 LLDP-V2-TC-MIB DEFINITIONS ::= BEGIN
6 IMPORTS
7     MODULE-IDENTITY,
8     Unsigned32,
9     org
10    FROM SNMPv2-SMI
11    TEXTUAL-CONVENTION
12    FROM SNMPv2-TC;
13
14 lldpV2TcMIB MODULE-IDENTITY
15     LAST-UPDATED "202508150000Z" -- August 15, 2025
16 ORGANIZATION "IEEE 802.1 Working Group"
17     CONTACT-INFO
18         " WG-URL: http://www.ieee802.org/1/
19         WG-EMail: stds-802-1-L@ieee.org
20
21         Contact: IEEE 802.1 Working Group Chair
22         Postal: C/O IEEE 802.1 Working Group
23               IEEE Standards Association
24               445 Hoes Lane
25               Piscataway
26               NJ 08854
27               USA
28         E-mail: stds-802-1-L@ieee.org"
29 DESCRIPTION
30     "Textual conventions used throughout the IEEE Std 802.1AB
31     version 2 and later MIB modules.
32
33     Unless otherwise indicated, the references in this
34     MIB module are to IEEE 802.1AB-2026.
35
36     The TCs in this MIB are taken from the original LLDP-MIB,
37     LLDP-EXT-DOT1-MIB, and LLDP-EXT-DOT3-MIB published in
38     IEEE Std 802-1D-2004, with the addition of TCs to support
39     the management address table. They have been made available
40     as a separate TC MIB module to facilitate referencing from
41     other MIB modules.
42
43     Copyright (C) IEEE (2025). This version of this MIB module
44     is published as 11.5.1 of IEEE Std 802.1AB-2026;
45     see the standard itself for full legal notices."
46
47     REVISION "202508150000Z" -- August 15, 2025
48
49 DESCRIPTION
50     "Published as part of IEEE Std 802.1AB-2026.
51     This revision updated referencees."
52
53 REVISION "201603110000Z" -- March 11, 2016
54
55 DESCRIPTION
56     "Published as part of IEEE Std 802.1AB-2016.
57     This revision updated referencees."
```

```

1
2  REVISION "200906080000Z" -- June 08, 2009
3
4
5  DESCRIPTION
6      "Published as part of IEEE Std 802.1AB-2009 revision."
7
8  ::= { org ieee(111) standards-association-numbers-series-standards(2)
9      lan-man-stds(802) ieee802dot1(1) 1 12 }
10
11 --
12 -- Definition of the root OID arc for IEEE 802.1 MIBs
13 --
14
15 ieee802dot1mibs OBJECT IDENTIFIER
16     ::= { org ieee(111) standards-association-numbers-series-standards(2)
17         lan-man-stds(802) ieee802dot1(1) 1 }
18
19 --
20 -- *****
21 --
22 -- Textual Conventions
23 --
24 -- *****
25
26 LldpV2ChassisIdSubtype ::= TEXTUAL-CONVENTION
27     STATUS          current
28     DESCRIPTION
29         "This TC describes the source of a chassis identifier."
30
31         The enumeration 'chassisComponent(1)' represents a chassis
32         identifier based on the value of entPhysicalAlias object
33         (defined in IETF RFC 6933) for a chassis component (i.e.,
34         an entPhysicalClass value of 'chassis(3)').
35
36         The enumeration 'interfaceAlias(2)' represents a chassis
37         identifier based on the value of ifAlias object (defined in
38         IETF RFC 2863) for an interface on the containing chassis.
39
40         The enumeration 'portComponent(3)' represents a chassis
41         identifier based on the value of entPhysicalAlias object
42         (defined in IETF RFC 6933) for a port or backplane
43         component (i.e., entPhysicalClass value of 'port(10)' or
44         'backplane(4)'), within the containing chassis.
45
46         The enumeration 'macAddress(4)' represents a chassis
47         identifier based on the value of a unicast source address
48         (encoded in network byte order and IEEE 802.3 canonical bit
49         order), of a port on the containing chassis as defined in
50         IEEE Std 802.
51
52         The enumeration 'networkAddress(5)' represents a chassis
53         identifier based on a network address, associated with
54         a particular chassis. The encoded address is actually
55         composed of two fields. The first field is a single octet,
56         representing the IANA AddressFamilyNumbers value for the
57         specific address type, and the second field is the network
58         address value.
59

```

1           The enumeration 'interfaceName(6)' represents a chassis  
2           identifier based on the value of ifName object (defined in  
3           IETF RFC 2863) for an interface on the containing chassis.  
4  
5           The enumeration 'local(7)' represents a chassis identifier  
6           based on a locally defined value."  
7       SYNTAX   INTEGER {  
8                chassisComponent(1),  
9                interfaceAlias(2),  
10               portComponent(3),  
11               macAddress(4),  
12               networkAddress(5),  
13               interfaceName(6),  
14               local(7)  
15       }  
16  
17 LldpV2ChassisId ::= TEXTUAL-CONVENTION  
18     DISPLAY-HINT "1x:"  
19     STATUS       current  
20     DESCRIPTION  
21         "This TC describes the format of a chassis identifier string.  
22         Objects of this type are always used with an associated  
23         LldpChassisIdSubtype object, which identifies the format of  
24         the particular LldpChassisId object instance.  
25  
26         If the associated LldpChassisIdSubtype object has a value of  
27         'chassisComponent(1)', then the octet string identifies  
28         a particular instance of the entPhysicalAlias object  
29         (defined in IETF RFC 6933) for a chassis component (i.e.,  
30         an entPhysicalClass value of 'chassis(3)').  
31  
32         If the associated LldpChassisIdSubtype object has a value  
33         of 'interfaceAlias(2)', then the octet string identifies  
34         a particular instance of the ifAlias object (defined in  
35         IETF RFC 2863) for an interface on the containing chassis.  
36         If the particular ifAlias object does not contain any values,  
37         another chassis identifier type should be used.  
38  
39         If the associated LldpChassisIdSubtype object has a value  
40         of 'portComponent(3)', then the octet string identifies a  
41         particular instance of the entPhysicalAlias object (defined  
42         in IETF RFC 6933) for a port or backplane component within  
43         the containing chassis.  
44  
45         If the associated LldpChassisIdSubtype object has a value of  
46         'macAddress(4)', then this string identifies a particular  
47         unicast source address (encoded in network byte order and  
48         IEEE 802.3 canonical bit order), of a port on the containing  
49         chassis as defined in IEEE Std 802.  
50  
51         If the associated LldpChassisIdSubtype object has a value of  
52         'networkAddress(5)', then this string identifies a particular  
53         network address, encoded in network byte order, associated  
54         with one or more ports on the containing chassis. The first  
55         octet contains the IANA Address Family Numbers enumeration  
56         value for the specific address type, and octets 2 through  
57         N contain the network address value in network byte order.  
58  
59         If the associated LldpChassisIdSubtype object has a value

```
1         of 'interfaceName(6)', then the octet string identifies
2         a particular instance of the ifName object (defined in
3         IETF RFC 2863) for an interface on the containing chassis.
4         If the particular ifName object does not contain any values,
5         another chassis identifier type should be used.
6
7         If the associated LldpChassisIdSubtype object has a value of
8         'local(7)', then this string identifies a locally assigned
9         Chassis ID."
10    SYNTAX      OCTET STRING (SIZE (1..255))
11
12 LldpV2PortIdSubtype ::= TEXTUAL-CONVENTION
13     STATUS      current
14     DESCRIPTION
15         "This TC describes the source of a particular type of port
16         identifier used in the LLDP MIB.
17
18         The enumeration 'interfaceAlias(1)' represents a port
19         identifier based on the ifAlias MIB object, defined in IETF
20         RFC 2863.
21
22         The enumeration 'portComponent(2)' represents a port
23         identifier based on the value of entPhysicalAlias (defined in
24         IETF RFC 6933) for a port component (i.e., entPhysicalClass
25         value of 'port(10)'), within the containing chassis.
26
27         The enumeration 'macAddress(3)' represents a port identifier
28         based on a unicast source address (encoded in network
29         byte order and IEEE 802.3 canonical bit order), which has
30         been detected by the agent and associated with a particular
31         port (IEEE Std 802).
32
33         The enumeration 'networkAddress(4)' represents a port
34         identifier based on a network address, detected by the agent
35         and associated with a particular port.
36
37         The enumeration 'interfaceName(5)' represents a port
38         identifier based on the ifName MIB object, defined in IETF
39         RFC 2863.
40
41         The enumeration 'agentCircuitId(6)' represents a port
42         identifier based on the agent-local identifier of the circuit
43         (defined in IETF RFC 3046), detected by the agent and associated
44         with a particular port.
45
46         The enumeration 'local(7)' represents a port identifier
47         based on a value locally assigned."
48
49     SYNTAX      INTEGER {
50         interfaceAlias(1),
51         portComponent(2),
52         macAddress(3),
53         networkAddress(4),
54         interfaceName(5),
55         agentCircuitId(6),
56         local(7)
57     }
58
59 LldpV2PortId ::= TEXTUAL-CONVENTION
```

```
1  DISPLAY-HINT "1x:"
2  STATUS      current
3  DESCRIPTION
4      "This TC describes the format of a port identifier string.
5      Objects of this type are always used with an associated
6      LldpPortIdSubtype object, which identifies the format of the
7      particular LldpPortId object instance.
8
9      If the associated LldpPortIdSubtype object has a value of
10     'interfaceAlias(1)', then the octet string identifies a
11     particular instance of the ifAlias object (defined in IETF
12     RFC 2863). If the particular ifAlias object does not contain
13     any values, another port identifier type should be used.
14
15     If the associated LldpPortIdSubtype object has a value of
16     'portComponent(2)', then the octet string identifies a
17     particular instance of the entPhysicalAlias object (defined
18     in IETF RFC 6933) for a port or backplane component.
19
20     If the associated LldpPortIdSubtype object has a value of
21     'macAddress(3)', then this string identifies a particular
22     unicast source address (encoded in network byte order
23     and IEEE 802.3 canonical bit order) associated with the port
24     (IEEE Std 802).
25
26     If the associated LldpPortIdSubtype object has a value of
27     'networkAddress(4)', then this string identifies a network
28     address associated with the port. The first octet contains
29     the IANA AddressFamilyNumbers enumeration value for the
30     specific address type, and octets 2 through N contain the
31     networkAddress address value in network byte order.
32
33     If the associated LldpPortIdSubtype object has a value of
34     'interfaceName(5)', then the octet string identifies a
35     particular instance of the ifName object (defined in IETF
36     RFC 2863). If the particular ifName object does not contain
37     any values, another port identifier type should be used.
38
39     If the associated LldpPortIdSubtype object has a value of
40     'agentCircuitId(6)', then this string identifies a agent-local
41     identifier of the circuit (defined in IETF RFC 3046).
42
43     If the associated LldpPortIdSubtype object has a value of
44     'local(7)', then this string identifies a locally
45     assigned port ID."
46  SYNTAX      OCTET STRING (SIZE (1..255))
47
48 LldpV2ManAddrIfSubtype ::= TEXTUAL-CONVENTION
49  STATUS      current
50  DESCRIPTION
51      "This TC defines an enumeration value that identifies
52      the interface numbering method used for defining the
53      interface number associated with a management address.
54      An object with this syntax defines the format of an
55      interface number object.
56
57      The enumeration 'unknown(1)' represents the case where the
58      interface is not known. In this case, the corresponding
59      interface number is of zero length.
```



```
1
2     The enumeration 'ifIndex(2)' represents interface identifier
3     based on the ifIndex MIB object.
4
5     The enumeration 'systemPortNumber(3)' represents interface
6     identifier based on the system port numbering convention."
7 REFERENCE
8     "8.5.9.5"
9
10 SYNTAX  INTEGER {
11         unknown(1),
12         ifIndex(2),
13         systemPortNumber(3)
14     }
15
16 LldpV2ManAddress ::= TEXTUAL-CONVENTION
17     DISPLAY-HINT "1x:"
18     STATUS      current
19     DESCRIPTION
20         "The value of a management address associated with the LLDP
21         agent that may be used to reach higher layer entities to
22         assist discovery by network management.
23
24         It should be noted that appropriate security credentials,
25         such as SNMP engineId, may be required to access the LLDP
26         agent using a management address. These necessary credentials
27         should be known by the network management and the objects
28         associated with the credentials are not included in the
29         LLDP agent."
30 SYNTAX      OCTET STRING (SIZE (1..31))
31
32 LldpV2SystemCapabilitiesMap ::= TEXTUAL-CONVENTION
33     STATUS      current
34     DESCRIPTION
35         "This TC describes the system capabilities.
36
37         The bit 'other(0)' indicates that the system has capabilities
38         other than those listed below.
39
40         The bit 'repeater(1)' indicates that the system has repeater
41         capability.
42
43         The bit 'bridge(2)' indicates that the system has bridge
44         capability.
45
46         The bit 'wlanAccessPoint(3)' indicates that the system has
47         WLAN access point capability.
48
49         The bit 'router(4)' indicates that the system has router
50         capability.
51
52         The bit 'telephone(5)' indicates that the system has telephone
53         capability.
54
55         The bit 'docsisCableDevice(6)' indicates that the system has
56         DOCSIS Cable Device capability (IETF RFC 4639 & 2670).
57
58         The bit 'stationOnly(7)' indicates that the system has only
59         station capability and nothing else.
```

```
1
2      The bit 'cVLANComponent(8)' indicates that the system has
3      C-VLAN component functionality.
4
5      The bit 'sVLANComponent(8)' indicates that the system has
6      S-VLAN component functionality.
7
8      The bit 'twoPortMACRelay(10)' indicates that the system has
9      Two-port MAC Relay (TPMR) functionality."
10     SYNTAX BITS {
11         other(0),
12         repeater(1),
13         bridge(2),
14         wlanAccessPoint(3),
15         router(4),
16         telephone(5),
17         docsisCableDevice(6),
18         stationOnly(7),
19         cVLANComponent(8),
20         sVLANComponent(9),
21         twoPortMACRelay(10)
22     }
23
24
25
26 LldpV2DestAddressTableIndex ::= TEXTUAL-CONVENTION
27     DISPLAY-HINT "d"
28     STATUS current
29     DESCRIPTION
30         "An index value, used as the index to the table of destination
31         MAC addresses used both as the destination addresses on
32         transmitted LLDPDUs and on received LLDPDUs. This index value
33         is also used as a secondary index value in tables indexed
34         by fields of type ifIndex, in order to associate
35         a destination address with each row of the table."
36     SYNTAX Unsigned32(1..4096)
37
38 LldpV2PowerPortClass ::= TEXTUAL-CONVENTION
39     STATUS current
40     DESCRIPTION
41         "This TC describes the Power over Ethernet (PoE) port class."
42     SYNTAX INTEGER {
43         pClassPSE(1),
44         pClassPD(2)
45     }
46
47 END
48
```

## 1 11.5.2 LLDP MIB module - version 2

2 In the following MIB module, should any discrepancy between the DESCRIPTION text and the  
3 corresponding definition in Clause 9 and Clause 10 occur, the definition in Clause 9 and Clause 10 shall take  
4 precedence.

```

5 LLDP-V2-MIB DEFINITIONS ::= BEGIN
6 IMPORTS
7     MODULE-IDENTITY,
8     OBJECT-TYPE,
9     Unsigned32,
10    Counter32,
11    NOTIFICATION-TYPE
12        FROM SNMPv2-SMI
13    TimeStamp,
14    TruthValue,
15    MacAddress,
16    RowStatus
17        FROM SNMPv2-TC
18    SnmpAdminString
19        FROM SNMP-FRAMEWORK-MIB
20    MODULE-COMPLIANCE,
21    OBJECT-GROUP,
22    NOTIFICATION-GROUP
23        FROM SNMPv2-CONF
24    TimeFilter,
25    ZeroBasedCounter32
26        FROM RMON2-MIB
27    AddressFamilyNumbers
28        FROM IANA-ADDRESS-FAMILY-NUMBERS-MIB
29    ifGeneralInformationGroup,
30    InterfaceIndex
31        FROM IF-MIB
32    LldpV2ChassisIdSubtype,
33    LldpV2ChassisId,
34    LldpV2PortIdSubtype,
35    LldpV2PortId,
36    LldpV2ManAddrIfSubtype,
37    LldpV2ManAddress,
38    LldpV2SystemCapabilitiesMap,
39    LldpV2DestAddressTableIndex,
40    ieee802dot1mibs
41        FROM LLDP-V2-TC-MIB;
42
43 lldpV2MIB MODULE-IDENTITY
44     LAST-UPDATED "202508150000Z" -- August 15, 2025
45     ORGANIZATION "IEEE 802.1 Working Group"
46     CONTACT-INFO
47         "WG-URL: http://www.ieee802.org/1/
48         WG-EMail: stds-802-1-L@ieee.org
49
50         Contact: IEEE 802.1 Working Group Chair
51         Postal:  IEEE Standards Board
52                 445 Hoes Lane
53                 Piscataway, NJ 08854
54                 USA
55         E-mail:  stds-802-1-L@ieee.org"
56     DESCRIPTION
57         "Management Information Base module for LLDP configuration,
```

1           statistics, local system data and remote systems data  
2           components.  
3  
4           This MIB module supports the architecture described in  
5           Clause 6, where multiple LLDP agents can be associated with  
6           a single Port, each supporting transmission by means of a  
7           different MAC address.  
8  
9           Unless otherwise indicated, the references in this  
10          MIB module are to IEEE Std 802.1AB-2026.  
11  
12          Copyright (C) IEEE (2025). This version of this MIB module  
13          is published as 11.5.2 of IEEE Std 802.1AB-2026;  
14          see the standard itself for full legal notices."  
15  
16          REVISION "202508150000Z" -- August 15, 2025  
17  
18          DESCRIPTION  
19                  "Published as part of the 2026 revision of  
20                  IEEE Std 802.1AB.  
21                  This revision updates references to 802.1AB and  
22                  copyright dates."  
23  
24          REVISION "201603110000Z" -- March 11, 2016  
25  
26          DESCRIPTION  
27                  "Published as part of the 2016 revision of  
28                  IEEE Std 802.1AB.  
29                  This revision incorporated changes to the MIB to  
30                  address issues identified in maintenance item  
31                  0121 - see <http://www.ieee802.org/1/maint.html>.  
32                  The changes involved correcting the way that  
33                  lldpV2MessageTxHoldMultiplier is calculated, in  
34                  accordance with 9.2.5.23."  
35  
36  
37          REVISION "201502160000Z" -- February 16, 2015  
38  
39          DESCRIPTION  
40                  "Published as part of IEEE Std 802.1AB-2009 Cor-2.  
41                  This corrigendum incorporated changes to the MIB to  
42                  address issues identified in maintenance item  
43                  0121 - see <http://www.ieee802.org/1/maint.html>."  
44  
45          REVISION "200906080000Z" -- June 08, 2009  
46  
47          DESCRIPTION  
48                  "Published as part of IEEE Std 802.1AB-2009 revision.  
49                  This revision incorporated changes to the MIB to  
50                  support the use of LLDP with multiple destination MAC  
51                  addresses."  
52  
53          ::= { ieee802dot1mibs 13 }  
54  
55          lldpV2Notifications                   OBJECT IDENTIFIER ::= { lldpV2MIB 0 }  
56          lldpV2Objects                        OBJECT IDENTIFIER ::= { lldpV2MIB 1 }  
57          lldpV2Conformance                   OBJECT IDENTIFIER ::= { lldpV2MIB 2 }  
58  
59 --

```

1 -- LLDP MIB Objects
2 --
3
4 lldpV2Configuration          OBJECT IDENTIFIER ::= { lldpV2Objects 1 }
5 lldpV2Statistics             OBJECT IDENTIFIER ::= { lldpV2Objects 2 }
6 lldpV2LocalSystemData       OBJECT IDENTIFIER ::= { lldpV2Objects 3 }
7 lldpV2RemoteSystemsData     OBJECT IDENTIFIER ::= { lldpV2Objects 4 }
8 lldpV2Extensions            OBJECT IDENTIFIER ::= { lldpV2Objects 5 }
9
10 --
11 -- *****
12 --
13 --             L L D P       C O N F I G
14 --
15 -- *****
16 --
17
18 lldpV2MessageTxInterval OBJECT-TYPE
19     SYNTAX      Unsigned32(5..32768)
20     UNITS       "seconds"
21     MAX-ACCESS  read-write
22     STATUS      current
23     DESCRIPTION
24         "The interval at which LLDP frames are transmitted on
25         behalf of this LLDP agent.
26
27         The default value for lldpV2MessageTxInterval object is
28         30 seconds.
29
30         The value of this object is used as the initial value of
31         the lldpV2PortMessageTxInterval object on row creation in
32         the lldpV2PortConfigTableV2.
33
34         The value of this object is restored from non-volatile
35         storage after a re-initialization of the management system."
36     REFERENCE
37         "9.2.5.6"
38     DEFVAL      { 30 }
39     ::= { lldpV2Configuration 1 }
40
41 lldpV2MessageTxHoldMultiplier OBJECT-TYPE
42     SYNTAX      Unsigned32(2..10)
43     MAX-ACCESS  read-write
44     STATUS      current
45     DESCRIPTION
46         "The time to live value expressed as a multiple of the
47         lldpV2MessageTxInterval object. The actual time to live
48         value used in LLDP frames, transmitted on behalf of this
49         LLDP agent, can be expressed by the following formula:
50         TTL = min(65535,
51         (lldpV2MessageTxInterval*lldpV2MessageTxHoldMultiplier)+1)
52         For example, if the value of lldpV2MessageTxInterval is
53         '30', and the value of lldpV2MessageTxHoldMultiplier
54         is '4', then the value '121' is encoded in the
55         TTL field in the LLDP header.
56
57         The default value for lldpV2MessageTxHoldMultiplier
58         object is 4.
59

```

```

1          The value of this object is used as the initial value of
2          the lldpV2PortMessageTxHoldMultiplier object on row creation in
3          the lldpV2PortConfigTableV2.
4
5          The value of this object is restored from non-volatile
6          storage after a re-initialization of the management system."
7  REFERENCE
8      "9.2.5.5"
9  DEFVAL      { 4 }
10 ::= { lldpV2Configuration 2 }
11
12 lldpV2ReinitDelay OBJECT-TYPE
13     SYNTAX      Unsigned32(1..10)
14     UNITS       "seconds"
15     MAX-ACCESS  read-write
16     STATUS      current
17     DESCRIPTION
18         "The lldpV2ReinitDelay indicates the delay (in units of
19         seconds) from when lldpPortConfigAdminStatus object of a
20         particular port becomes 'disabled' until re-initialization
21         is attempted.
22
23         The default value for lldpV2ReinitDelay is 2 s.
24
25         The value of this object is used as the initial value of
26         the lldpV2PortReinitDelay object on row creation in
27         the lldpV2PortConfigTableV2.
28
29         The value of this object is restored from non-volatile
30         storage after a re-initialization of the management system."
31  REFERENCE
32      "9.2.5.9"
33  DEFVAL      { 2 }
34 ::= { lldpV2Configuration 3 }
35
36 lldpV2NotificationInterval OBJECT-TYPE
37     SYNTAX      Unsigned32(5..3600)
38     UNITS       "seconds"
39     MAX-ACCESS  read-write
40     STATUS      current
41     DESCRIPTION
42         "This object controls the interval between transmission of
43         LLDP notifications during normal transmission periods.
44
45         The value of this object is used as the initial value of
46         the lldpV2PortNotificationInterval object on row creation in
47         the lldpV2PortConfigTableV2.
48
49         The value of this object is restored from non-volatile
50         storage after a re-initialization of the management system."
51  DEFVAL { 30 }
52 ::= { lldpV2Configuration 4 }
53
54 lldpV2TxCreditMax OBJECT-TYPE
55     SYNTAX      Unsigned32(1..100)
56     UNITS       "PDUs"
57     MAX-ACCESS  read-write
58     STATUS      current
59     DESCRIPTION

```

1           "The maximum number of consecutive LLDPDUs that can be  
2           transmitted at any time.  
3  
4           The default value for lldpV2TxCreditMax object is 5 PDUs.  
5  
6           The value of this object is used as the initial value of  
7           the lldpV2PortTxCreditMax object on row creation in  
8           the lldpV2PortConfigTableV2.  
9  
10          The value of this object is restored from non-volatile  
11          storage after a re-initialization of the management system."  
12      REFERENCE  
13          "9.2.5.18"  
14      DEFVAL { 5 }  
15      ::= { lldpV2Configuration 5 }  
16  
17 lldpV2MessageFastTx OBJECT-TYPE  
18      SYNTAX Unsigned32(1..3600)  
19      UNITS "seconds"  
20      MAX-ACCESS read-write  
21      STATUS current  
22      DESCRIPTION  
23          "The interval at which LLDP frames are transmitted on  
24          behalf of this LLDP agent during fast transmission period  
25          (e.g., when a new neighbor is detected).  
26  
27          The default value for lldpV2MessageFastTx object is  
28          1 second.  
29  
30          The value of this object is used as the initial value of  
31          the lldpV2PortMessageFastTx object on row creation in  
32          the lldpV2PortConfigTableV2.  
33  
34          The value of this object is restored from non-volatile  
35          storage after a re-initialization of the management system."  
36      REFERENCE  
37          "9.2.5.4"  
38      DEFVAL { 1 }  
39      ::= { lldpV2Configuration 6 }  
40  
41 lldpV2TxFastInit OBJECT-TYPE  
42      SYNTAX Unsigned32(1..8)  
43      MAX-ACCESS read-write  
44      STATUS current  
45      DESCRIPTION  
46          "The initial value used to initialize the txFast variable  
47          which determines the number of transmissions that are  
48          made in fast transmission mode.  
49  
50          The default value for lldpV2TxFastInit object is 4.  
51  
52          The value of this object is used as the initial value of  
53          the lldpV2PortTxFastInit object on row creation in  
54          the lldpV2PortConfigTableV2.  
55  
56          The value of this object is restored from non-volatile  
57          storage after a re-initialization of the management system."  
58      REFERENCE  
59          "9.2.5.20"

```

1     DEFVAL { 4 }
2     ::= { lldpV2Configuration 7 }
3 --
4 -- lldpV2PortConfigTable: LLDP configuration indexed on a per port,
5 -- per destination address basis. The ifIndex, coupled with an
6 -- index into the lldpDestAddressTable, is used to index per port
7 -- per destination MAC address.
8 -- ***This table and its associated objects are now deprecated
9 -- and replaced by lldpV2PortConfigTableV2.***
10 --
11
12 lldpV2PortConfigTable    OBJECT-TYPE
13     SYNTAX                SEQUENCE OF LldpV2PortConfigEntry
14     MAX-ACCESS            not-accessible
15     STATUS                deprecated
16     DESCRIPTION
17         "The table that controls LLDP frame transmission on individual
18         ports that uses particular destination MAC addresses."
19     ::= { lldpV2Configuration 8 }
20
21 lldpV2PortConfigEntry    OBJECT-TYPE
22     SYNTAX                LldpV2PortConfigEntry
23     MAX-ACCESS            not-accessible
24     STATUS                deprecated
25     DESCRIPTION
26         "LLDP configuration information for a particular port and
27         destination MAC address.
28
29         This configuration parameter controls the transmission and
30         the reception of LLDP frames on those interface/address
31         combinations whose rows are created in this table.
32
33         Rows in this table can only be created for MAC addresses
34         that can validly be used in association with the type of
35         interface concerned, as defined by Table 7-2.
36
37         The contents of this table is persistent across
38         re-initializations or re-boots."
39     INDEX { lldpV2PortConfigIfIndex,
40             lldpV2PortConfigDestAddressIndex }
41     ::= { lldpV2PortConfigTable 1 }
42
43 lldpV2PortConfigEntry ::= SEQUENCE {
44     lldpV2PortConfigIfIndex      InterfaceIndex,
45     lldpV2PortConfigDestAddressIndex  LldpV2DestAddressTableIndex,
46     lldpV2PortConfigAdminStatus  INTEGER,
47     lldpV2PortConfigNotificationEnable  TruthValue,
48     lldpV2PortConfigTLVsTxEnable  BITS }
49
50 lldpV2PortConfigIfIndex    OBJECT-TYPE
51     SYNTAX                InterfaceIndex
52     MAX-ACCESS            not-accessible
53     STATUS                deprecated
54     DESCRIPTION
55         "The interface index value used to identify the port
56         associated with this entry. Its value is an index into
57         the interfaces MIB.
58
59         The value of this object is used as an index to the

```



```

1         lldpV2PortConfigTable."
2     ::= { lldpV2PortConfigEntry 1 }
3
4 lldpV2PortConfigDestAddressIndex OBJECT-TYPE
5     SYNTAX      LldpV2DestAddressTableIndex
6     MAX-ACCESS  not-accessible
7     STATUS      deprecated
8     DESCRIPTION
9         "The index value used to identify the destination
10        MAC address associated with this entry. Its value identifies
11        the row in the lldpV2DestAddressTable where the MAC address
12        can be found.
13
14        The value of this object is used as an index to the
15        lldpV2PortConfigTable."
16     ::= { lldpV2PortConfigEntry 2 }
17
18 lldpV2PortConfigAdminStatus OBJECT-TYPE
19     SYNTAX INTEGER {
20         txOnly(1),
21         rxOnly(2),
22         txAndRx(3),
23         disabled(4)
24     }
25     MAX-ACCESS  read-write
26     STATUS      deprecated
27     DESCRIPTION
28         "The administratively desired status of the local LLDP agent.
29
30         If the associated lldpV2PortConfigAdminStatus object is
31         set to a value of 'txOnly(1)', then LLDP agent transmits
32         LLDPframes on this port and it does not store any
33         information about the remote systems connected.
34
35         If the associated lldpV2PortConfigAdminStatus object is
36         set to a value of 'rxOnly(2)', then the LLDP agent
37         receives, but it does not transmit LLDP frames on this port.
38
39         If the associated lldpV2PortConfigAdminStatus object is set
40         to a value of 'txAndRx(3)', then the LLDP agent transmits
41         and receives LLDP frames on this port.
42
43         If the associated lldpV2PortConfigAdminStatus object is set
44         to a value of 'disabled(4)', then LLDP agent does not
45         transmit or receive LLDP frames on this port. If there is
46         remote systems information that is received on this port
47         and stored in other tables, before the port's
48         lldpV2PortConfigAdminStatus becomes disabled, then that
49         information is deleted."
50     REFERENCE
51         "9.2.5.1"
52     DEFVAL { txAndRx }
53     ::= { lldpV2PortConfigEntry 3 }
54
55 lldpV2PortConfigNotificationEnable OBJECT-TYPE
56     SYNTAX      TruthValue
57     MAX-ACCESS  read-write
58     STATUS      deprecated
59     DESCRIPTION

```

```

1          "The lldpV2PortConfigNotificationEnable controls, on a per
2          agent basis, whether or not notifications from the agent
3          are enabled. The value true(1) means that notifications are
4          enabled; the value false(2) means that they are not."
5  DEFVAL { false }
6  ::= { lldpV2PortConfigEntry 4 }
7
8  lldpV2PortConfigTLVsTxEnable OBJECT-TYPE
9      SYNTAX      BITS {
10         portDesc(0),
11         sysName(1),
12         sysDesc(2),
13         sysCap(3)
14     }
15     MAX-ACCESS   read-write
16     STATUS       deprecated
17     DESCRIPTION
18         "The lldpV2PortConfigTLVsTxEnable, defined as a bitmap,
19         includes the basic set of LLDP TLVs whose transmission is
20         allowed on the local LLDP agent by the network management.
21         Each bit in the bitmap corresponds to a TLV type associated
22         with a specific optional TLV.
23
24         It should be noted that the organizationally-specific TLVs
25         are excluded from the lldpV2PortConfigTLVsTxEnable bitmap.
26
27         LLDP Organization Specific Information Extension MIBs should
28         have similar configuration objects to control transmission
29         of their organizationally defined TLVs.
30
31         The bit 'portDesc(0)' indicates that LLDP agent should
32         transmit 'Port Description TLV'.
33
34         The bit 'sysName(1)' indicates that LLDP agent should transmit
35         'System Name TLV'.
36
37         The bit 'sysDesc(2)' indicates that LLDP agent should transmit
38         'System Description TLV'.
39
40         The bit 'sysCap(3)' indicates that LLDP agent should transmit
41         'System Capabilities TLV'.
42
43         There is no bit reserved for the management address TLV type
44         since transmission of management address TLVs are controlled
45         by another object, lldpV2ConfigManAddrTable.
46
47         The default value for lldpV2PortConfigTLVsTxEnable object is
48         empty set, which means no enumerated values are set.
49
50         The value of this object is restored from non-volatile
51         storage after a re-initialization of the management system."
52     REFERENCE
53         "9.1.2.1"
54     DEFVAL { { } }
55     ::= { lldpV2PortConfigEntry 5 }
56
57 --
58 -- lldpV2PortConfigTableV2: LLDP configuration indexed on a per port,
59 -- per destination address basis. The ifIndex, coupled with an

```

```

1-- index into the lldpDestAddressTable, is used to index per port
2-- per destination MAC address.
3--
4-- V2 extends the original table definition to include per-port
5-- per-MAC address parameters msgTxInterval, msgTxHold, reinitDelay,
6-- notificationInterval, txCreditMax, msgFastTx, and txFastInit.
7--
8
9 lldpV2PortConfigTableV2    OBJECT-TYPE
10  SYNTAX      SEQUENCE OF LldpV2PortConfigEntryV2
11  MAX-ACCESS  not-accessible
12  STATUS      current
13  DESCRIPTION
14      "The table that controls LLDP frame transmission on individual
15      ports and using particular destination MAC addresses."
16  ::= { lldpV2Configuration 11 }
17
18 lldpV2PortConfigEntryV2    OBJECT-TYPE
19  SYNTAX      LldpV2PortConfigEntryV2
20  MAX-ACCESS  not-accessible
21  STATUS      current
22  DESCRIPTION
23      "LLDP configuration information for a particular port and
24      destination MAC address.
25
26      This configuration parameter controls the transmission and
27      the reception of LLDP frames on those interface/address
28      combinations whose rows are created in this table.
29
30      Rows in this table can only be created for MAC addresses
31      that can validly be used in association with the type of
32      interface concerned, as defined by Table 7-2.
33
34      The contents of this table is persistent across
35      re-initializations or re-boots."
36  INDEX      { lldpV2PortConfigIfIndexV2,
37              lldpV2PortConfigDestAddressIndexV2 }
38  ::= { lldpV2PortConfigTableV2 1 }
39
40 LldpV2PortConfigEntryV2 ::= SEQUENCE {
41     lldpV2PortConfigIfIndexV2      InterfaceIndex,
42     lldpV2PortConfigDestAddressIndexV2 LldpV2DestAddressTableIndex,
43     lldpV2PortConfigAdminStatusV2  INTEGER,
44     lldpV2PortMessageTxInterval    Unsigned32,
45     lldpV2PortMessageTxHoldMultiplier Unsigned32,
46     lldpV2PortReinitDelay           Unsigned32,
47     lldpV2PortNotificationInterval Unsigned32,
48     lldpV2PortTxCreditMax           Unsigned32,
49     lldpV2PortMessageFastTx         Unsigned32,
50     lldpV2PortTxFastInit            Unsigned32,
51     lldpV2PortConfigNotificationEnableV2 TruthValue,
52     lldpV2PortConfigTLVsTxEnableV2 BITS }
53
54 lldpV2PortConfigIfIndexV2    OBJECT-TYPE
55  SYNTAX      InterfaceIndex
56  MAX-ACCESS  not-accessible
57  STATUS      current
58  DESCRIPTION
59      "The interface index value used to identify the port

```

```

1         associated with this entry. Its value is an index into
2         the interfaces MIB.
3
4         The value of this object is used as an index to the
5         lldpV2PortConfigTable."
6 ::= { lldpV2PortConfigEntryV2 1 }
7
8 lldpV2PortConfigDestAddressIndexV2 OBJECT-TYPE
9     SYNTAX      LldpV2DestAddressTableIndex
10    MAX-ACCESS   not-accessible
11    STATUS       current
12    DESCRIPTION
13        "The index value used to identify the destination
14        MAC address associated with this entry. Its value identifies
15        the row in the lldpV2DestAddressTable where the MAC address
16        can be found.
17
18        The value of this object is used as an index to the
19        lldpV2PortConfigTable."
20 ::= { lldpV2PortConfigEntryV2 2 }
21
22 lldpV2PortConfigAdminStatusV2 OBJECT-TYPE
23     SYNTAX INTEGER {
24         txOnly(1),
25         rxOnly(2),
26         txAndRx(3),
27         disabled(4)
28     }
29     MAX-ACCESS read-write
30     STATUS       current
31     DESCRIPTION
32         "The administratively desired status of the local LLDP agent.
33
34         If the associated lldpV2PortConfigAdminStatus object is
35         set to a value of 'txOnly(1)', then LLDP agent transmits
36         LLDPframes on this port and it does not store any
37         information about the remote systems connected.
38
39         If the associated lldpV2PortConfigAdminStatus object is
40         set to a value of 'rxOnly(2)', then the LLDP agent
41         receives, but it does not transmit, LLDP frames on this port.
42
43         If the associated lldpV2PortConfigAdminStatus object is set
44         to a value of 'txAndRx(3)', then the LLDP agent transmits
45         and receives LLDP frames on this port.
46
47         If the associated lldpV2PortConfigAdminStatus object is set
48         to a value of 'disabled(4)', then LLDP agent does not
49         transmit or receive LLDP frames on this port. If there is
50         remote systems information that is received on this port
51         and stored in other tables, before the port's
52         lldpV2PortConfigAdminStatus becomes disabled, then that
53         information is deleted."
54     REFERENCE
55         "9.2.5.1"
56     DEFVAL { txAndRx }
57 ::= { lldpV2PortConfigEntryV2 3 }
58
59 lldpV2PortMessageTxInterval OBJECT-TYPE

```

```

1  SYNTAX      Unsigned32(5..32768)
2  UNITS       "seconds"
3  MAX-ACCESS  read-write
4  STATUS      current
5  DESCRIPTION
6      "The interval at which LLDP frames are transmitted on
7      behalf of this LLDP agent.
8
9      This object takes its initial value from the
10     lldpV2MessageTxInterval object on table row creation.
11
12     The value of this object is restored from non-volatile
13     storage after a re-initialization of the management system."
14  REFERENCE
15     "9.2.5.6"
16  DEFVAL      { 30 }
17  ::= { lldpV2PortConfigEntryV2 4 }
18
19 lldpV2PortMessageTxHoldMultiplier OBJECT-TYPE
20  SYNTAX      Unsigned32(2..10)
21  MAX-ACCESS  read-write
22  STATUS      current
23  DESCRIPTION
24      "The time to live value expressed as a multiple of the
25      lldpV2MessageTxInterval object. The actual time to live
26      value used in LLDP frames, transmitted on behalf of this
27      LLDP agent, can be expressed by the following formula:
28      TTL = min(65535,
29      (lldpV2MessageTxInterval*lldpV2MessageTxHoldMultiplier)+1)
30      For example, if the value of lldpV2MessageTxInterval is
31      '30', and the value of lldpV2MessageTxHoldMultiplier
32      is '4', then the value '121' is encoded in the
33      TTL field in the LLDP header.
34
35      This object takes its initial value from the
36      lldpV2PortMessageTxHoldMultiplier object on table row creation.
37
38      The value of this object is restored from non-volatile
39      storage after a re-initialization of the management system."
40  REFERENCE
41     "9.2.5.5"
42  DEFVAL      { 4 }
43  ::= { lldpV2PortConfigEntryV2 5 }
44
45 lldpV2PortReinitDelay OBJECT-TYPE
46  SYNTAX      Unsigned32(1..10)
47  UNITS       "seconds"
48  MAX-ACCESS  read-write
49  STATUS      current
50  DESCRIPTION
51      "The lldpV2ReinitDelay indicates the delay (in units of
52      seconds) from when lldpPortConfigAdminStatus object of a
53      particular port becomes 'disabled' until re-initialization
54      is attempted.
55
56      This object takes its initial value from the
57      lldpV2PortReinitDelay object on table row creation.
58
59      The value of this object is restored from non-volatile

```

```

1         storage after a re-initialization of the management system."
2     REFERENCE
3         "9.2.5.9"
4     DEFVAL { 2 }
5     ::= { lldpV2PortConfigEntryV2 6 }
6
7 lldpV2PortNotificationInterval OBJECT-TYPE
8     SYNTAX      Unsigned32(5..3600)
9     UNITS       "seconds"
10    MAX-ACCESS  read-write
11    STATUS      current
12    DESCRIPTION
13        "This object controls the interval between transmission of
14        LLDP notifications during normal transmission periods.
15
16        This object takes its initial value from the
17        lldpV2PortNotificationInterval object on table row creation.
18
19        The value of this object is restored from non-volatile
20        storage after a re-initialization of the management system."
21    DEFVAL { 30 }
22    ::= { lldpV2PortConfigEntryV2 7 }
23
24 lldpV2PortTxCreditMax OBJECT-TYPE
25     SYNTAX Unsigned32(1..100)
26     UNITS "PDUs"
27     MAX-ACCESS read-write
28     STATUS current
29     DESCRIPTION
30         "The maximum number of consecutive LLDPDUs that can be
31         transmitted at any time.
32
33         This object takes its initial value from the
34         lldpV2PortTxCreditMax object on table row creation.
35
36         The value of this object is restored from non-volatile
37         storage after a re-initialization of the management system."
38     REFERENCE
39         "9.2.5.18"
40     DEFVAL { 5 }
41     ::= { lldpV2PortConfigEntryV2 8 }
42
43 lldpV2PortMessageFastTx OBJECT-TYPE
44     SYNTAX Unsigned32(1..3600)
45     UNITS "seconds"
46     MAX-ACCESS read-write
47     STATUS current
48     DESCRIPTION
49         "The interval at which LLDP frames are transmitted on
50         behalf of this LLDP agent during fast transmission period
51         (e.g., when a new neighbor is detected).
52
53         This object takes its initial value from the
54         lldpV2PortMessageFastTx object on table row creation.
55
56         The value of this object is restored from non-volatile
57         storage after a re-initialization of the management system."
58     REFERENCE
59         "9.2.5.4"

```

```

1  DEFVAL { 1 }
2  ::= { lldpV2PortConfigEntryV2 9 }
3
4  lldpV2PortTxFastInit OBJECT-TYPE
5      SYNTAX Unsigned32(1..8)
6      MAX-ACCESS read-write
7      STATUS current
8      DESCRIPTION
9          "The initial value used to initialize the txFast variable
10         which determines the number of transmissions that are
11         made in fast transmission mode.
12
13         This object takes its initial value from the
14         lldpV2PortTxFastInit object on table row creation.
15
16         The value of this object is restored from non-volatile
17         storage after a re-initialization of the management system."
18  REFERENCE
19      "9.2.5.20"
20  DEFVAL { 4 }
21  ::= { lldpV2PortConfigEntryV2 10 }
22
23
24  lldpV2PortConfigNotificationEnableV2 OBJECT-TYPE
25      SYNTAX      TruthValue
26      MAX-ACCESS  read-write
27      STATUS      current
28      DESCRIPTION
29          "The lldpV2PortConfigNotificationEnableV2 controls, on a per
30          agent basis, whether or not notifications from the agent
31          are enabled. The value true(1) means that notifications are
32          enabled; the value false(2) means that they are not."
33  DEFVAL { false }
34  ::= { lldpV2PortConfigEntryV2 11 }
35
36  lldpV2PortConfigTLVsTxEnableV2 OBJECT-TYPE
37      SYNTAX      BITS {
38          portDesc(0),
39          sysName(1),
40          sysDesc(2),
41          sysCap(3)
42      }
43      MAX-ACCESS  read-write
44      STATUS      current
45      DESCRIPTION
46          "The lldpV2PortConfigTLVsTxEnableV2, defined as a bitmap,
47          includes the basic set of LLDP TLVs whose transmission is
48          allowed on the local LLDP agent by the network management.
49          Each bit in the bitmap corresponds to a TLV type associated
50          with a specific optional TLV.
51
52          It should be noted that the organizationally-specific TLVs
53          are excluded from the lldpV2PortConfigTLVsTxEnable bitmap.
54
55          LLDP Organization Specific Information Extension MIBs should
56          have similar configuration objects to control transmission
57          of their organizationally defined TLVs.
58
59          The bit 'portDesc(0)' indicates that LLDP agent should

```

```

1          transmit 'Port Description TLV'.
2
3          The bit 'sysName(1)' indicates that LLDP agent should transmit
4          'System Name TLV'.
5
6          The bit 'sysDesc(2)' indicates that LLDP agent should transmit
7          'System Description TLV'.
8
9          The bit 'sysCap(3)' indicates that LLDP agent should transmit
10         'System Capabilities TLV'.
11
12         There is no bit reserved for the management address TLV type
13         since transmission of management address TLVs are controlled
14         by another object, lldpV2ConfigManAddrTable.
15
16         The default value for lldpV2PortConfigTLVsTxEnable object is
17         empty set, which means no enumerated values are set.
18
19         The value of this object is restored from non-volatile
20         storage after a re-initialization of the management system."
21 REFERENCE
22     "9.1.2.1"
23 DEFVAL  { { } }
24 ::= { lldpV2PortConfigEntryV2 12 }
25
26
27 --
28 -- lldpV2DestAddressTable: Destination MAC addresses used by LLDP
29 --
30
31 lldpV2DestAddressTable  OBJECT-TYPE
32     SYNTAX      SEQUENCE OF LldpV2DestAddressTableEntry
33     MAX-ACCESS  not-accessible
34     STATUS      current
35     DESCRIPTION
36         "The table that contains the set of MAC addresses used
37         by LLDP for transmission and reception of LLDPDUs."
38     ::= { lldpV2Configuration 9 }
39
40 lldpV2DestAddressTableEntry  OBJECT-TYPE
41     SYNTAX      LldpV2DestAddressTableEntry
42     MAX-ACCESS  not-accessible
43     STATUS      current
44     DESCRIPTION
45         "Destination MAC address information for LLDP.
46
47         This configuration parameter identifies a MAC address
48         corresponding to a LldpV2DestAddressTableIndex value.
49
50         Rows in this table are created as necessary, to support
51         MAC addresses needed by other tables in the MIB that
52         are indexed by MAC address.
53
54         A given row in this table cannot be deleted if the MAC
55         address table index value is in use in any other table
56         in the MIB.
57
58         The contents of this table are persistent across
59         re-initializations or re-boots."

```



```

1      INDEX { lldpV2AddressTableIndex }
2      ::= { lldpV2DestAddressTable 1 }
3
4  lldpV2DestAddressTableEntry ::= SEQUENCE {
5      lldpV2AddressTableIndex      LldpV2DestAddressTableIndex,
6      lldpV2DestMacAddress         MacAddress }
7
8  lldpV2AddressTableIndex OBJECT-TYPE
9      SYNTAX      LldpV2DestAddressTableIndex
10     MAX-ACCESS  not-accessible
11     STATUS      current
12     DESCRIPTION
13         "The index value used to identify the destination
14         MAC address associated with this entry.
15
16         The value of this object is used as an index to the
17         lldpV2DestAddressTable."
18     ::= { lldpV2DestAddressTableEntry 1 }
19
20 lldpV2DestMacAddress OBJECT-TYPE
21     SYNTAX      MacAddress
22     MAX-ACCESS  read-only
23     STATUS      current
24     DESCRIPTION
25         "The MAC address associated with this entry.
26
27         The octet string identifies an individual or a group
28         MAC address that is in use by LLDP as a destination
29         MAC address.
30
31         The MAC address is encoded in the octet string in
32         canonical format (see IEEE Std 802)."
```

--

```

36 -- lldpV2ManAddrConfigTxPortsTable : selection of management addresses
37 -- to be transmitted on a specified set of port/destination
38 -- address pairs.
39 --
40
41 lldpV2ManAddrConfigTxPortsTable OBJECT-TYPE
42     SYNTAX      SEQUENCE OF LldpV2ManAddrConfigTxPortsEntry
43     MAX-ACCESS  not-accessible
44     STATUS      current
45     DESCRIPTION
46         "The table that controls selection of LLDP management address
47         TLV instances to be transmitted on individual port/
48         destination address pairs."
49     ::= { lldpV2Configuration 10 }
50
51 lldpV2ManAddrConfigTxPortsEntry OBJECT-TYPE
52     SYNTAX      LldpV2ManAddrConfigTxPortsEntry
53     MAX-ACCESS  not-accessible
54     STATUS      current
55     DESCRIPTION
56         "LLDP configuration information that specifies the set
57         of port/destination address pairs on which the local
58         system management address instance is transmitted.
59
```

```

1      Each active lldpManAddrConfigTxPortsTableV2Entry is
2      restored from non-volatile storage and re-created
3      after a re-initialization of the management system."
4  INDEX {
5      lldpV2ManAddrConfigIfIndex,
6      lldpV2ManAddrConfigDestAddressIndex,
7      lldpV2ManAddrConfigLocManAddrSubtype,
8      lldpV2ManAddrConfigLocManAddr }
9  ::= { lldpV2ManAddrConfigTxPortsTable 1 }
10
11 lldpV2ManAddrConfigTxPortsEntry ::= SEQUENCE {
12     lldpV2ManAddrConfigIfIndex      InterfaceIndex,
13     lldpV2ManAddrConfigDestAddressIndex  LldpV2DestAddressTableIndex,
14     lldpV2ManAddrConfigLocManAddrSubtype  AddressFamilyNumbers,
15     lldpV2ManAddrConfigLocManAddr      LldpV2ManAddress,
16     lldpV2ManAddrConfigTxEnable      TruthValue,
17     lldpV2ManAddrConfigRowStatus      RowStatus
18 }
19
20 lldpV2ManAddrConfigIfIndex OBJECT-TYPE
21     SYNTAX      InterfaceIndex
22     MAX-ACCESS  not-accessible
23     STATUS      current
24     DESCRIPTION
25         "The interface index value used to identify the port
26         associated with this entry. Its value is an index into
27         the interfaces MIB.
28
29         The value of this object is used as an index to the
30         lldpV2PortConfigTable.
31
32         The value in this column of the table MUST match
33         the IfIndex value specified in the BridgePort table."
34     ::= { lldpV2ManAddrConfigTxPortsEntry 1 }
35
36 lldpV2ManAddrConfigDestAddressIndex OBJECT-TYPE
37     SYNTAX      LldpV2DestAddressTableIndex
38     MAX-ACCESS  not-accessible
39     STATUS      current
40     DESCRIPTION
41         "The index value used to identify the destination
42         MAC address associated with this entry. Its value identifies
43         the row in the lldpV2DestAddressTable where the MAC address
44         can be found.
45
46         The value of this object is used as an index to the
47         lldpV2PortConfigTable."
48     ::= { lldpV2ManAddrConfigTxPortsEntry 2 }
49
50
51 lldpV2ManAddrConfigLocManAddrSubtype OBJECT-TYPE
52     SYNTAX      AddressFamilyNumbers
53     MAX-ACCESS  not-accessible
54     STATUS      current
55     DESCRIPTION
56         "The type of management address identifier encoding used in
57         the associated 'lldpLocManagmentAddr' object.
58
59         It should be noted that only a subset of the possible

```

```

1         address encodings enumerated in AddressFamilyNumbers
2         are appropriate for use as a LLDP management
3         address, either because some are just not applicable or
4         because the maximum size of a LldpV2ManAddress octet string
5         would prevent the use of some address identifier encodings."
6     REFERENCE
7         "8.5.9.3"
8     ::= { lldpV2ManAddrConfigTxPortsEntry 3 }
9
10 lldpV2ManAddrConfigLocManAddr OBJECT-TYPE
11     SYNTAX      LldpV2ManAddress
12     MAX-ACCESS  not-accessible
13     STATUS      current
14     DESCRIPTION
15         "The string value used to identify the management address
16         component associated with the local system. The purpose of
17         this address is to contact the management entity."
18     REFERENCE
19         "8.5.9.4"
20     ::= { lldpV2ManAddrConfigTxPortsEntry 4 }
21
22
23
24 lldpV2ManAddrConfigTxEnable OBJECT-TYPE
25     SYNTAX      TruthValue
26     MAX-ACCESS  read-create
27     STATUS      current
28     DESCRIPTION
29         "A Boolean controlling the transmission of system
30         management address instance for the specified port,
31         destination, subtype, and MAN address used to index
32         this table. If set to the default value of false,
33         no transmission occurs. If set to true, the
34         appropriate information is transmitted out of the
35         port specified in the row's index."
36     REFERENCE
37         "9.1.2.1"
38     DEFVAL { false }      -- not transmitted
39     ::= { lldpV2ManAddrConfigTxPortsEntry 5 }
40
41 lldpV2ManAddrConfigRowStatus OBJECT-TYPE
42     SYNTAX      RowStatus
43     MAX-ACCESS  read-create
44     STATUS      current
45     DESCRIPTION
46         "Indicates the status of an entry in this table, and is used
47         to create/delete entries.
48         The corresponding instances of the following objects
49         must be set before this object can be made active(1):
50             lldpV2ManAddrConfigDestAddressIndex
51             lldpV2ManAddrConfigLocManAddrSubtype
52             lldpV2ManAddrConfigLocManAddr
53             lldpV2ManAddrConfigTxEnable
54
55         The corresponding instances of the following objects
56         may not be changed while this object is active(1):
57             lldpV2ManAddrConfigDestAddressIndex
58             lldpV2ManAddrConfigLocManAddrSubtype
59             lldpV2ManAddrConfigLocManAddr "
```

```

1 ::= { lldpV2ManAddrConfigTxPortsEntry 6 }
2
3 --
4 -- *****
5 --
6 --             L L D P       S T A T S
7 --
8 -- *****
9 --
10 -- LLDP Stats Group
11
12 lldpV2StatsRemTablesLastChangeTime OBJECT-TYPE
13     SYNTAX      TimeStamp
14     MAX-ACCESS  read-only
15     STATUS      current
16     DESCRIPTION
17         "The value of sysUpTime object (defined in IETF RFC 3418)
18         at the time an entry is created, modified, or deleted in the
19         in tables associated with the lldpV2RemoteSystemsData objects
20         and all LLDP extension objects associated with remote systems.
21
22         An NMS can use this object to reduce polling of the
23         lldpV2RemoteSystemsData objects."
24     ::= { lldpV2Statistics 1 }
25
26 lldpV2StatsRemTablesInserts OBJECT-TYPE
27     SYNTAX      ZeroBasedCounter32
28     UNITS       "table entries"
29     MAX-ACCESS  read-only
30     STATUS      current
31     DESCRIPTION
32         "The number of times the complete set of information
33         advertised by a particular MSAP has been inserted into tables
34         contained in lldpV2RemoteSystemsData and lldpV2Extensions objects.
35
36         The complete set of information received from a particular
37         MSAP should be inserted into related tables. If partial
38         information cannot be inserted for a reason such as lack
39         of resources, all of the complete set of information should
40         be removed.
41
42         This counter should be incremented only once after the
43         complete set of information is successfully recorded
44         in all related tables. Any failures during inserting
45         information set that result in deletion of previously
46         inserted information should not trigger any changes in
47         lldpV2StatsRemTablesInserts since the insert is not completed
48         yet or in lldpStatsRemTablesDeletes since the deletion
49         would only be a partial deletion. If the failure was the
50         result of lack of resources, the lldpStatsRemTablesDrops
51         counter should be incremented once."
52     ::= { lldpV2Statistics 2 }
53
54 lldpV2StatsRemTablesDeletes OBJECT-TYPE
55     SYNTAX      ZeroBasedCounter32
56     UNITS       "table entries"
57     MAX-ACCESS  read-only
58     STATUS      current
59

```

```

1      DESCRIPTION
2          "The number of times the complete set of information
3          advertised by a particular MSAP has been deleted from
4          tables contained in lldpV2RemoteSystemsData and lldpV2Extensions
5          objects.
6
7          This counter should be incremented only once when the
8          complete set of information is completely deleted from all
9          related tables. Partial deletions, such as deletion of
10         rows associated with a particular MSAP from some tables,
11         but not from all tables are not allowed, thus should not
12         change the value of this counter."
13     ::= { lldpV2Statistics 3 }
14
15 lldpV2StatsRemTablesDrops OBJECT-TYPE
16     SYNTAX      ZeroBasedCounter32
17     UNITS        "table entries"
18     MAX-ACCESS  read-only
19
20     STATUS      current
21     DESCRIPTION
22         "The number of times the complete set of information
23         advertised by a particular MSAP could not be entered into
24         tables contained in lldpV2RemoteSystemsData and lldpV2Extensions
25         objects because of insufficient resources."
26     ::= { lldpV2Statistics 4 }
27
28 lldpV2StatsRemTablesAgeouts OBJECT-TYPE
29     SYNTAX      ZeroBasedCounter32
30     UNITS        "table entries"
31     MAX-ACCESS  read-only
32     STATUS      current
33     DESCRIPTION
34         "The number of times the complete set of information
35         advertised by a particular MSAP has been deleted from tables
36         contained in lldpV2RemoteSystemsData and lldpV2Extensions objects
37         because the information timeliness interval has expired.
38
39         This counter should be incremented only once when the complete
40         set of information is completely invalidated (aged out)
41         from all related tables. Partial ageing, similar to deletion
42         case, is not allowed, and thus, should not change the value
43         of this counter."
44     ::= { lldpV2Statistics 5 }
45
46 --
47 -- TX statistics
48 -- Indexed by port (via ifIndex) and
49 -- destination MAC address.
50 --
51
52 lldpV2StatsTxPortTable OBJECT-TYPE
53     SYNTAX      SEQUENCE OF LldpV2StatsTxPortEntry
54     MAX-ACCESS  not-accessible
55     STATUS      current
56     DESCRIPTION
57         "A table containing LLDP transmission statistics for
58         individual port/destination address combinations.
59         Entries are not required to exist in

```

```

1           this table while the lldpPortConfigEntry object is equal to
2           'disabled(4)'."
3 ::= { lldpV2Statistics 6 }
4
5 lldpV2StatsTxPortEntry OBJECT-TYPE
6     SYNTAX      LldpV2StatsTxPortEntry
7     MAX-ACCESS  not-accessible
8     STATUS      current
9     DESCRIPTION
10        "LLDP frame transmission statistics for a particular port
11        and destination MAC address.
12        The port is contained in the same chassis as the
13        LLDP agent.
14
15        All counter values in a particular entry shall be
16        maintained on a continuing basis and shall not be deleted
17        upon expiration of rxInfoTTL timing counters in the LLDP
18        remote systems MIB of the receipt of a shutdown frame from
19        a remote LLDP agent.
20
21        All statistical counters associated with a particular
22        port on the local LLDP agent become frozen whenever the
23        adminStatus is disabled for the same port.
24
25        Rows in this table can only be created for MAC addresses
26        that can validly be used in association with the type of
27        interface concerned, as defined by Table 7-2."
28     INDEX { lldpV2StatsTxIfIndex,
29             lldpV2StatsTxDestMACAddress }
30 ::= { lldpV2StatsTxPortTable 1 }
31
32 lldpV2StatsTxPortEntry ::= SEQUENCE {
33     lldpV2StatsTxIfIndex      InterfaceIndex,
34     lldpV2StatsTxDestMACAddress LldpV2DestAddressTableIndex,
35     lldpV2StatsTxPortFramesTotal Counter32,
36     lldpV2StatsTxLLDPDULengthErrors Counter32 }
37
38 lldpV2StatsTxIfIndex OBJECT-TYPE
39     SYNTAX      InterfaceIndex
40     MAX-ACCESS  not-accessible
41     STATUS      current
42     DESCRIPTION
43        "The interface index value used to identify the port
44        associated with this entry. Its value is an index
45        into the interfaces MIB.
46
47        The value of this object is used as an index to the
48        lldpV2StatsTxPortTable."
49 ::= { lldpV2StatsTxPortEntry 1 }
50
51 lldpV2StatsTxDestMACAddress OBJECT-TYPE
52     SYNTAX      LldpV2DestAddressTableIndex
53     MAX-ACCESS  not-accessible
54     STATUS      current
55     DESCRIPTION
56        "The index value used to identify the destination
57        MAC address associated with this entry. Its value identifies
58        the row in the lldpV2DestAddressTable where the MAC address
59        can be found.

```

```

1
2         The value of this object is used as an index to the
3         lldpV2StatsTxPortTable."
4     ::= { lldpV2StatsTxPortEntry 2 }
5
6 lldpV2StatsTxPortFramesTotal OBJECT-TYPE
7     SYNTAX      Counter32
8     UNITS       "LLDP frames"
9     MAX-ACCESS  read-only
10    STATUS      current
11    DESCRIPTION
12        "The number of LLDP frames transmitted by this LLDP agent
13        on the indicated port to the destination MAC address
14        associated with this row of the table."
15    REFERENCE
16        "9.2.7.5"
17    ::= { lldpV2StatsTxPortEntry 3 }
18
19 lldpV2StatsTxLLDPDULengthErrors OBJECT-TYPE
20     SYNTAX      Counter32
21     UNITS       "LLDP frames"
22     MAX-ACCESS  read-only
23     STATUS      current
24     DESCRIPTION
25         "The number of LLDPDU Length Errors recorded for the Port."
26     REFERENCE
27         "9.2.7.8"
28     ::= { lldpV2StatsTxPortEntry 4 }
29
30 --
31 -- lldpV2StatsRxPortTable - RX statistics
32 -- This table is indexed by ifIndex and destination MAC address.
33 --
34
35 lldpV2StatsRxPortTable OBJECT-TYPE
36     SYNTAX      SEQUENCE OF LldpV2StatsRxPortEntry
37     MAX-ACCESS  not-accessible
38     STATUS      current
39     DESCRIPTION
40         "A table containing LLDP reception statistics for individual
41         ports and destination MAC addresses.
42         Entries are not required to exist in this table while
43         the lldpPortConfigEntry object is equal to 'disabled(4)'."
44     ::= { lldpV2Statistics 7 }
45
46 lldpV2StatsRxPortEntry OBJECT-TYPE
47     SYNTAX      LldpV2StatsRxPortEntry
48     MAX-ACCESS  not-accessible
49     STATUS      current
50     DESCRIPTION
51         "LLDP frame reception statistics for a particular port.
52         The port is contained in the same chassis as the
53         LLDP agent.
54
55         All counter values in a particular entry shall be
56         maintained on a continuing basis and shall not be deleted
57         upon expiration of rxInfoTTL timing counters in the LLDP
58         remote systems MIB of the receipt of a shutdown frame from
59         a remote LLDP agent.

```

```

1
2      All statistical counters associated with a particular
3      port on the local LLDP agent become frozen whenever the
4      adminStatus is disabled for the same port.
5
6      Rows in this table can only be created for MAC addresses
7      that can validly be used in association with the type of
8      interface concerned, as defined by Table 7-2.
9
10     The contents of this table is persistent across
11     re-initializations or re-boots."
12     INDEX { lldpV2StatsRxDestIfIndex,
13             lldpV2StatsRxDestMACAddress }
14     ::= { lldpV2StatsRxPortTable 1 }
15
16 LldpV2StatsRxPortEntry ::= SEQUENCE {
17     lldpV2StatsRxDestIfIndex      InterfaceIndex,
18     lldpV2StatsRxDestMACAddress    LldpV2DestAddressTableIndex,
19     lldpV2StatsRxPortFramesDiscardedTotal Counter32,
20     lldpV2StatsRxPortFramesErrors Counter32,
21     lldpV2StatsRxPortFramesTotal Counter32,
22     lldpV2StatsRxPortTLVsDiscardedTotal Counter32,
23     lldpV2StatsRxPortTLVsUnrecognizedTotal Counter32,
24     lldpV2StatsRxPortAgeoutsTotal ZeroBasedCounter32
25 }
26
27 lldpV2StatsRxDestIfIndex OBJECT-TYPE
28     SYNTAX      InterfaceIndex
29     MAX-ACCESS  not-accessible
30     STATUS      current
31     DESCRIPTION
32         "The interface index value used to identify the port
33         associated with this entry. Its value is an index
34         into the interfaces MIB.
35
36         The value of this object is used as an index to the
37         lldpStatsRxPortV2Table."
38     ::= { lldpV2StatsRxPortEntry 1 }
39
40 lldpV2StatsRxDestMACAddress OBJECT-TYPE
41     SYNTAX      LldpV2DestAddressTableIndex
42     MAX-ACCESS  not-accessible
43     STATUS      current
44     DESCRIPTION
45         "The index value used to identify the destination
46         MAC address associated with this entry. Its value identifies
47         the row in the lldpV2DestAddressTable where the MAC address
48         can be found.
49
50         The value of this object is used as an index to the
51         lldpStatsRxPortV2Table."
52     ::= { lldpV2StatsRxPortEntry 2 }
53
54
55 lldpV2StatsRxPortFramesDiscardedTotal OBJECT-TYPE
56     SYNTAX      Counter32
57     UNITS        "LLDP frames"
58     MAX-ACCESS  read-only
59     STATUS      current

```



```

1  DESCRIPTION
2      "The number of LLDP frames received by this LLDP agent on
3      the indicated port, and then discarded for any reason.
4      This counter can provide an indication that LLDP header
5      formatting problems may exist with the local LLDP agent in
6      the sending system or that LLDPDU validation problems may
7      exist with the local LLDP agent in the receiving system."
8  REFERENCE
9      "9.2.7.2"
10 ::= { lldpV2StatsRxPortEntry 3 }
11
12 lldpV2StatsRxPortFramesErrors OBJECT-TYPE
13     SYNTAX      Counter32
14     UNITS       "LLDP frames"
15     MAX-ACCESS  read-only
16     STATUS      current
17     DESCRIPTION
18         "The number of invalid LLDP frames received by this LLDP
19         agent on the indicated port, while this LLDP agent is enabled."
20     REFERENCE
21         "9.2.7.3"
22     ::= { lldpV2StatsRxPortEntry 4 }
23
24 lldpV2StatsRxPortFramesTotal OBJECT-TYPE
25     SYNTAX      Counter32
26     UNITS       "LLDP frames"
27     MAX-ACCESS  read-only
28     STATUS      current
29     DESCRIPTION
30         "The number of valid LLDP frames received by this LLDP agent
31         on the indicated port, while this LLDP agent is enabled."
32     REFERENCE
33         "9.2.7.4"
34     ::= { lldpV2StatsRxPortEntry 5 }
35
36 lldpV2StatsRxPortTLVsDiscardedTotal OBJECT-TYPE
37     SYNTAX      Counter32
38     UNITS       "TLVs"
39     MAX-ACCESS  read-only
40     STATUS      current
41     DESCRIPTION
42         "The number of LLDP TLVs discarded for any reason by this LLDP
43         agent on the indicated port."
44     REFERENCE
45         "9.2.7.6"
46     ::= { lldpV2StatsRxPortEntry 6 }
47
48 lldpV2StatsRxPortTLVsUnrecognizedTotal OBJECT-TYPE
49     SYNTAX      Counter32
50     UNITS       "TLVs"
51     MAX-ACCESS  read-only
52     STATUS      current
53     DESCRIPTION
54         "The number of LLDP TLVs received on the given port that
55         are not recognized by this LLDP agent on the indicated port.
56
57         An unrecognized TLV is referred to as the TLV whose type value
58         is in the range of reserved TLV types (000 1100 - 111 1110)
59         in Table 8-1 of IEEE Std 802.1AB. An unrecognized

```

```

1           TLV may be a basic management TLV from a later LLDP version."
2  REFERENCE
3      "9.2.7.7"
4      ::= { lldpV2StatsRxPortEntry 7 }
5
6  lldpV2StatsRxPortAgeoutsTotal  OBJECT-TYPE
7      SYNTAX      ZeroBasedCounter32
8      MAX-ACCESS  read-only
9      STATUS      current
10     DESCRIPTION
11         "The counter that represents the number of age-outs that
12         occurred on a given port. An age-out is the number of
13         times the complete set of information advertised by a
14         particular MSAP has been deleted from tables contained in
15         lldpV2RemoteSystemsData and lldpV2Extensions objects because
16         the information timeliness interval has expired.
17
18         This counter is similar to lldpV2StatsRemTablesAgeouts, except
19         that the counter is on a per port basis. This enables NMS to
20         poll tables associated with the lldpV2RemoteSystemsData objects
21         and all LLDP extension objects associated with remote systems
22         on the indicated port only.
23
24         This counter is set to zero during agent initialization
25         and its value should not be saved in non-volatile storage.
26
27         This counter is incremented only once when the
28         complete set of information is invalidated (aged out) from
29         all related tables on a particular port. Partial ageing
30         is not allowed."
31  REFERENCE
32      "9.2.7.1"
33      ::= { lldpV2StatsRxPortEntry 8 }
34
35
36  -- *****
37  --
38  --          L O C A L      S Y S T E M      D A T A
39  --
40  -- *****
41
42  lldpV2LocChassisIdSubtype  OBJECT-TYPE
43      SYNTAX      LldpV2ChassisIdSubtype
44      MAX-ACCESS  read-only
45      STATUS      current
46      DESCRIPTION
47          "The type of encoding used to identify the chassis
48          associated with the local system."
49  REFERENCE
50      "8.5.2.2"
51      ::= { lldpV2LocalSystemData 1 }
52
53  lldpV2LocChassisId  OBJECT-TYPE
54      SYNTAX      LldpV2ChassisId
55      MAX-ACCESS  read-only
56      STATUS      current
57      DESCRIPTION
58          "The string value used to identify the chassis component
59          associated with the local system."

```

```

1  REFERENCE
2      "8.5.2.3"
3  ::= { lldpV2LocalSystemData 2 }
4
5  lldpV2LocSysName OBJECT-TYPE
6      SYNTAX      SnmpAdminString (SIZE(0..255))
7      MAX-ACCESS  read-only
8      STATUS      current
9      DESCRIPTION
10         "The string value used to identify the system name of the
11         local system. If the local agent supports IETF RFC 3418,
12         lldpLocSysName object should have the same value as sysName
13         object."
14  REFERENCE
15      "8.5.6.2"
16  ::= { lldpV2LocalSystemData 3 }
17
18  lldpV2LocSysDesc OBJECT-TYPE
19      SYNTAX      SnmpAdminString (SIZE(0..255))
20      MAX-ACCESS  read-only
21      STATUS      current
22      DESCRIPTION
23         "The string value used to identify the system description
24         of the local system. If the local agent supports IETF RFC 3418,
25         lldpLocSysDesc object should have the same value as sysDesc
26         object."
27  REFERENCE
28      "8.5.7.2"
29  ::= { lldpV2LocalSystemData 4 }
30
31  lldpV2LocSysCapSupported OBJECT-TYPE
32      SYNTAX      LldpV2SystemCapabilitiesMap
33      MAX-ACCESS  read-only
34      STATUS      current
35      DESCRIPTION
36         "The bitmap value used to identify which system capabilities
37         are supported on the local system."
38  REFERENCE
39      "8.5.8.1"
40  ::= { lldpV2LocalSystemData 5 }
41
42  lldpV2LocSysCapEnabled OBJECT-TYPE
43      SYNTAX      LldpV2SystemCapabilitiesMap
44      MAX-ACCESS  read-only
45      STATUS      current
46      DESCRIPTION
47         "The bitmap value used to identify which system capabilities
48         are enabled on the local system."
49  REFERENCE
50      "8.5.8.2"
51  ::= { lldpV2LocalSystemData 6 }
52
53
54 --
55 -- lldpV2LocPortTable : Port specific Local system data
56 -- Indexed by ifIndex.
57 --
58
59 lldpV2LocPortTable OBJECT-TYPE

```

```

1  SYNTAX      SEQUENCE OF LldpV2LocPortEntry
2  MAX-ACCESS  not-accessible
3  STATUS      current
4  DESCRIPTION
5      "This table contains one row per port of information
6      associated with the local system known to this agent."
7  ::= { lldpV2LocalSystemData 7 }
8
9  lldpV2LocPortEntry OBJECT-TYPE
10 SYNTAX      LldpV2LocPortEntry
11 MAX-ACCESS  not-accessible
12 STATUS      current
13 DESCRIPTION
14     "Information about a particular port component.
15
16     Entries may be created and deleted in this table by the
17     agent.
18
19     Rows in this table can only be created for MAC addresses
20     that can validly be used in association with the type of
21     interface concerned, as defined by Table 7-2.
22
23     The contents of this table is persistent across
24     re-initializations or re-boots."
25 INDEX      { lldpV2LocPortIfIndex }
26 ::= { lldpV2LocPortTable 1 }
27
28 LldpV2LocPortEntry ::= SEQUENCE {
29     lldpV2LocPortIfIndex      InterfaceIndex,
30     lldpV2LocPortIdSubtype    LldpV2PortIdSubtype,
31     lldpV2LocPortId           LldpV2PortId,
32     lldpV2LocPortDesc         SnmpAdminString
33 }
34
35 lldpV2LocPortIfIndex OBJECT-TYPE
36 SYNTAX      InterfaceIndex
37 MAX-ACCESS  not-accessible
38 STATUS      current
39 DESCRIPTION
40     "The interface index value used to identify the port
41     associated with this entry. Its value is an index
42     into the interfaces MIB.
43
44     The value of this object is used as an index to the
45     lldpV2LocPortTable."
46 ::= { lldpV2LocPortEntry 1 }
47
48 lldpV2LocPortIdSubtype OBJECT-TYPE
49 SYNTAX      LldpV2PortIdSubtype
50 MAX-ACCESS  read-only
51 STATUS      current
52 DESCRIPTION
53     "The type of port identifier encoding used in the associated
54     'lldpLocPortId' object."
55 REFERENCE
56     "8.5.3.2"
57 ::= { lldpV2LocPortEntry 2 }
58
59 lldpV2LocPortId OBJECT-TYPE

```

```

1  SYNTAX      LldpV2PortId
2  MAX-ACCESS  read-only
3  STATUS      current
4  DESCRIPTION
5      "The string value used to identify the port component
6      associated with a given port in the local system."
7  REFERENCE
8      "8.5.3.3"
9  ::= { lldpV2LocPortEntry 3 }
10
11 lldpV2LocPortDesc OBJECT-TYPE
12     SYNTAX      SnmpAdminString (SIZE(0..255))
13     MAX-ACCESS  read-only
14     STATUS      current
15     DESCRIPTION
16         "The string value used to identify the IEEE 802 LAN station's port
17         description associated with the local system. If the local
18         agent supports IETF RFC 2863, lldpLocPortDesc object should
19         have the same value of ifDescr object."
20     REFERENCE
21         "8.5.5.2"
22     ::= { lldpV2LocPortEntry 4 }
23
24 --
25 -- lldpV2LocManAddrTable : Management addresses of the local system
26 --
27
28 lldpV2LocManAddrTable OBJECT-TYPE
29     SYNTAX      SEQUENCE OF LldpV2LocManAddrEntry
30     MAX-ACCESS  not-accessible
31     STATUS      current
32     DESCRIPTION
33         "This table contains management address information on the
34         local system known to this agent."
35     ::= { lldpV2LocalSystemData 8 }
36
37 lldpV2LocManAddrEntry OBJECT-TYPE
38     SYNTAX      LldpV2LocManAddrEntry
39     MAX-ACCESS  not-accessible
40     STATUS      current
41     DESCRIPTION
42         "Management address information about a particular chassis
43         component. There may be multiple management addresses
44         configured on the system identified by a particular
45         lldpLocChassisId. Each management address should have
46         distinct 'management address type' (lldpV2LocManAddrSubtype) and
47         'management address' (lldpLocManAddr.)
48
49         Entries may be created and deleted in this table by the
50         agent.
51
52         Since a variable length octetstring is used as an index
53         in a table, the address length is encoded as part of the OID
54         (as per IETF RFC 2578)."
```

```

55
56     INDEX      { lldpV2LocManAddrSubtype,
57                  lldpV2LocManAddr }
58     ::= { lldpV2LocManAddrTable 1 }
59

```

```

1 lldpV2LocManAddrEntry ::= SEQUENCE {
2     lldpV2LocManAddrSubtype    AddressFamilyNumbers,
3     lldpV2LocManAddr           LldpV2ManAddress,
4     lldpV2LocManAddrLen        Unsigned32,
5     lldpV2LocManAddrIfSubtype  LldpV2ManAddrIfSubtype,
6     lldpV2LocManAddrIfId       Unsigned32,
7     lldpV2LocManAddrOID        OBJECT IDENTIFIER
8 }
9
10 lldpV2LocManAddrSubtype OBJECT-TYPE
11     SYNTAX      AddressFamilyNumbers
12     MAX-ACCESS  not-accessible
13     STATUS      current
14     DESCRIPTION
15         "The type of management address identifier encoding used in
16         the associated 'lldpLocManagmentAddr' object.
17
18         It should be noted that only a subset of the possible
19         address encodings enumerated in AddressFamilyNumbers
20         are appropriate for use as a LLDP management
21         address, either because some are just not applicable or
22         because the maximum size of a LldpV2ManAddress octet string
23         would prevent the use of some address identifier encodings."
24     REFERENCE
25         "8.5.9.3"
26     ::= { lldpV2LocManAddrEntry 1 }
27
28 lldpV2LocManAddr OBJECT-TYPE
29     SYNTAX      LldpV2ManAddress
30     MAX-ACCESS  not-accessible
31     STATUS      current
32     DESCRIPTION
33         "The string value used to identify the management address
34         component associated with the local system. The purpose of
35         this address is to contact the management entity."
36     REFERENCE
37         "8.5.9.4"
38     ::= { lldpV2LocManAddrEntry 2 }
39
40 lldpV2LocManAddrLen OBJECT-TYPE
41     SYNTAX      Unsigned32
42     MAX-ACCESS  read-only
43     STATUS      current
44     DESCRIPTION
45         "The total length of the management address subtype and the
46         management address fields in LLDPDUs transmitted by the
47         local LLDP agent.
48
49         The management address length field is needed so that the
50         receiving systems that do not implement SNMP are not
51         required to implement an Internet Assigned Numbers
52         Authority (IANA) family numbers/address length equivalency
53         table in order to decode the management address."
54     REFERENCE
55         "8.5.9.2"
56     ::= { lldpV2LocManAddrEntry 3 }
57
58 lldpV2LocManAddrIfSubtype OBJECT-TYPE
59     SYNTAX      LldpV2ManAddrIfSubtype

```

```

1  MAX-ACCESS  read-only
2  STATUS      current
3  DESCRIPTION
4      "The enumeration value that identifies the interface numbering
5      method used for defining the interface number
6      (lldpV2LocManAddrIfId), associated with the local system."
7  REFERENCE
8      "8.5.9.5"
9  ::= { lldpV2LocManAddrEntry 4 }
10
11 lldpV2LocManAddrIfId  OBJECT-TYPE
12     SYNTAX      Unsigned32
13     MAX-ACCESS  read-only
14     STATUS      current
15     DESCRIPTION
16         "The integer value used to identify the interface number
17         regarding the management address component associated with
18         the local system."
19     REFERENCE
20         "8.5.9.6"
21     ::= { lldpV2LocManAddrEntry 5 }
22
23 lldpV2LocManAddrOID  OBJECT-TYPE
24     SYNTAX      OBJECT IDENTIFIER
25     MAX-ACCESS  read-only
26     STATUS      current
27     DESCRIPTION
28         "The OID value used to identify the type of hardware component
29         or protocol entity associated with the management address
30         advertised by the local system agent."
31     REFERENCE
32         "8.5.9.8"
33     ::= { lldpV2LocManAddrEntry 6 }
34
35
36 -- *****
37 --
38 --             R E M O T E       S Y S T E M S       D A T A
39 --
40 -- *****
41
42
43 --
44 -- lldpV2RemTable
45 -- Indexed by ifIndex and destination MAC address.
46 --
47
48 lldpV2RemTable  OBJECT-TYPE
49     SYNTAX      SEQUENCE OF LldpV2RemEntry
50     MAX-ACCESS  not-accessible
51     STATUS      current
52     DESCRIPTION
53         "This table contains one or more rows per physical network
54         connection known to this agent. The agent may wish to ensure
55         that only one lldpRemEntry is present for each local port
56         and destination MAC address,
57         or it may choose to maintain multiple lldpRemEntries for
58         the same local port and destination MAC address.
59

```

The following procedure may be used to retrieve remote systems information updates from an LLDP agent:

1. NMS polls all tables associated with remote systems and keeps a local copy of the information retrieved. NMS polls periodically the values of the following objects:
  - a. lldpV2StatsRemTablesInserts
  - b. lldpV2StatsRemTablesDeletes
  - c. lldpV2StatsRemTablesDrops
  - d. lldpV2StatsRemTablesAgeouts
  - e. lldpV2StatsRxPortAgeoutsTotal for all ports.
2. LLDP agent updates remote systems MIB objects, and sends out notifications to a list of notification destinations.
3. NMS receives the notifications and compares the new values of objects listed in step 1.

Periodically, NMS should poll the object lldpV2StatsRemTablesLastChangeTime to find out if anything has changed since the last poll. If something has changed, NMS polls the objects listed in step 1 to figure out what kind of changes occurred in the tables.

If value of lldpV2StatsRemTablesInserts has changed, then NMS walks all tables by employing TimeFilter with the last-pollled time value. This request returns new objects or objects whose values have been updated since the last poll.

If value of lldpV2StatsRemTablesAgeouts has changed, then NMS walks the lldpStatsRxPortAgeoutsTotal and compares the new values with previously recorded ones. For ports whose lldpStatsRxPortAgeoutsTotal value is greater than the recorded value, NMS can retrieve objects associated with those ports from table(s) without employing a TimeFilter (which is performed by specifying 0 for the TimeFilter).

lldpV2StatsRemTablesDeletes and lldpV2StatsRemTablesDrops objects are provided for informational purposes."

::= { lldpV2RemoteSystemsData 1 }

lldpV2RemEntry OBJECT-TYPE

SYNTAX LldpV2RemEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"Information about a particular physical network connection. Entries may be created and deleted in this table by the agent, if a physical topology discovery process is active.

Rows in this table can only be created for MAC addresses that can validly be used in association with the type of interface concerned, as defined by Table 7-2.

The contents of this table is persistent across



```

1         re-initializations or re-boots."
2     INDEX    {
3         lldpV2RemTimeMark,
4         lldpV2RemLocalIfIndex,
5         lldpV2RemLocalDestMACAddress,
6         lldpV2RemIndex
7     }
8     ::= { lldpV2RemTable 1 }
9
10 lldpV2RemEntry ::= SEQUENCE {
11     lldpV2RemTimeMark          TimeFilter,
12     lldpV2RemLocalIfIndex      InterfaceIndex,
13     lldpV2RemLocalDestMACAddress LldpV2DestAddressTableIndex,
14     lldpV2RemIndex            Unsigned32,
15     lldpV2RemChassisIdSubtype  LldpV2ChassisIdSubtype,
16     lldpV2RemChassisId        LldpV2ChassisId,
17     lldpV2RemPortIdSubtype     LldpV2PortIdSubtype,
18     lldpV2RemPortId           LldpV2PortId,
19     lldpV2RemPortDesc         SnmpAdminString,
20     lldpV2RemSysName          SnmpAdminString,
21     lldpV2RemSysDesc          SnmpAdminString,
22     lldpV2RemSysCapSupported  LldpV2SystemCapabilitiesMap,
23     lldpV2RemSysCapEnabled    LldpV2SystemCapabilitiesMap,
24     lldpV2RemRemoteChanges    TruthValue,
25     lldpV2RemTooManyNeighbors TruthValue
26 }
27
28 lldpV2RemTimeMark OBJECT-TYPE
29     SYNTAX      TimeFilter
30     MAX-ACCESS  not-accessible
31     STATUS      current
32     DESCRIPTION
33         "A TimeFilter for this entry. See the TimeFilter textual
34         convention in IETF RFC 4502 and
35         http://www.ietf.org/IESG/Implementations/RFC2021-Implementation.txt
36         to see how TimeFilter works."
37     REFERENCE
38         "IETF RFC 4502 section 6"
39     ::= { lldpV2RemEntry 1 }
40
41
42 lldpV2RemLocalIfIndex OBJECT-TYPE
43     SYNTAX      InterfaceIndex
44     MAX-ACCESS  not-accessible
45     STATUS      current
46     DESCRIPTION
47         "The interface index value used to identify the port
48         associated with this entry. Its value is an index
49         into the interfaces MIB
50
51         The value of this object is used as an index to the
52         lldpV2RemTable."
53     ::= { lldpV2RemEntry 2 }
54
55 lldpV2RemLocalDestMACAddress OBJECT-TYPE
56     SYNTAX      LldpV2DestAddressTableIndex
57     MAX-ACCESS  not-accessible
58     STATUS      current
59     DESCRIPTION

```

```

1          "The index value used to identify the destination
2          MAC address associated with this entry. Its value identifies
3          the row in the lldpV2DestAddressTable where the MAC address
4          can be found.
5
6          The value of this object is used as an index to the
7          lldpV2RemTable."
8 ::= { lldpV2RemEntry 3 }
9
10
11 lldpV2RemIndex OBJECT-TYPE
12     SYNTAX      Unsigned32(1..2147483647)
13     MAX-ACCESS  not-accessible
14     STATUS      current
15     DESCRIPTION
16         "This object represents an arbitrary local integer value used
17         by this agent to identify a particular connection instance,
18         unique only for the indicated remote system.
19
20         An agent is encouraged to assign monotonically increasing
21         index values to new entries, starting with one, after each
22         reboot. It is considered unlikely that the lldpRemIndex
23         can wrap between reboots."
24     ::= { lldpV2RemEntry 4 }
25
26 lldpV2RemChassisIdSubtype OBJECT-TYPE
27     SYNTAX      LldpV2ChassisIdSubtype
28     MAX-ACCESS  read-only
29     STATUS      current
30     DESCRIPTION
31         "The type of encoding used to identify the chassis associated
32         with the remote system."
33     REFERENCE
34         "8.5.2.2"
35     ::= { lldpV2RemEntry 5 }
36
37 lldpV2RemChassisId OBJECT-TYPE
38     SYNTAX      LldpV2ChassisId
39     MAX-ACCESS  read-only
40     STATUS      current
41     DESCRIPTION
42         "The string value used to identify the chassis component
43         associated with the remote system."
44     REFERENCE
45         "8.5.2.3"
46     ::= { lldpV2RemEntry 6 }
47
48 lldpV2RemPortIdSubtype OBJECT-TYPE
49     SYNTAX      LldpV2PortIdSubtype
50     MAX-ACCESS  read-only
51     STATUS      current
52     DESCRIPTION
53         "The type of port identifier encoding used in the associated
54         'lldpRemPortId' object."
55     REFERENCE
56         "8.5.3.2"
57     ::= { lldpV2RemEntry 7 }
58
59 lldpV2RemPortId OBJECT-TYPE

```

```

1  SYNTAX      LldpV2PortId
2  MAX-ACCESS  read-only
3  STATUS      current
4  DESCRIPTION
5      "The string value used to identify the port component
6      associated with the remote system."
7  REFERENCE
8      "8.5.3.3"
9  ::= { lldpV2RemEntry 8 }
10
11 lldpV2RemPortDesc  OBJECT-TYPE
12  SYNTAX      SnmpAdminString (SIZE(0..255))
13  MAX-ACCESS  read-only
14  STATUS      current
15  DESCRIPTION
16      "The string value used to identify the description of
17      the given port associated with the remote system."
18  REFERENCE
19      "8.5.5.2"
20  ::= { lldpV2RemEntry 9 }
21
22 lldpV2RemSysName   OBJECT-TYPE
23  SYNTAX      SnmpAdminString (SIZE(0..255))
24  MAX-ACCESS  read-only
25  STATUS      current
26  DESCRIPTION
27      "The string value used to identify the system name of the
28      remote system."
29  REFERENCE
30      "8.5.6.2"
31  ::= { lldpV2RemEntry 10 }
32
33 lldpV2RemSysDesc   OBJECT-TYPE
34  SYNTAX      SnmpAdminString (SIZE(0..255))
35  MAX-ACCESS  read-only
36  STATUS      current
37  DESCRIPTION
38      "The string value used to identify the system description
39      of the remote system."
40  REFERENCE
41      "8.5.7.2"
42  ::= { lldpV2RemEntry 11 }
43
44 lldpV2RemSysCapSupported  OBJECT-TYPE
45  SYNTAX      LldpV2SystemCapabilitiesMap
46  MAX-ACCESS  read-only
47  STATUS      current
48  DESCRIPTION
49      "The bitmap value used to identify which system capabilities
50      are supported on the remote system."
51  REFERENCE
52      "8.5.8.1"
53  ::= { lldpV2RemEntry 12 }
54
55 lldpV2RemSysCapEnabled  OBJECT-TYPE
56  SYNTAX      LldpV2SystemCapabilitiesMap
57  MAX-ACCESS  read-only
58  STATUS      current
59  DESCRIPTION

```

```

1         "The bitmap value used to identify which system capabilities
2         are enabled on the remote system."
3     REFERENCE
4         "8.5.8.2"
5     ::= { lldpV2RemEntry 13 }
6
7 lldpV2RemRemoteChanges    OBJECT-TYPE
8     SYNTAX      TruthValue
9     MAX-ACCESS   read-only
10    STATUS      current
11    DESCRIPTION
12        "Indicates that there are changes in the remote systems
13        MIB, as determined by the variable remoteChanges."
14    REFERENCE
15        "9.2.5.10"
16    ::= { lldpV2RemEntry 14 }
17
18 lldpV2RemTooManyNeighbors OBJECT-TYPE
19     SYNTAX      TruthValue
20     MAX-ACCESS   read-only
21     STATUS      current
22     DESCRIPTION
23         "Indicates that there are too many neighbors
24         as determined by the variable tooManyNeighbors."
25     REFERENCE
26         "9.2.5.16"
27     ::= { lldpV2RemEntry 15 }
28
29 --
30 -- lldpV2RemManAddrTable : Management addresses of the remote system
31 -- Version 2 includes additional index values for ifIndex and
32 -- destination MAC address.
33 --
34
35 lldpV2RemManAddrTable OBJECT-TYPE
36     SYNTAX      SEQUENCE OF LldpV2RemManAddrEntry
37     MAX-ACCESS   not-accessible
38     STATUS      current
39     DESCRIPTION
40         "This table contains one or more rows per management address
41         information on the remote system learned on a particular port
42         contained in the local chassis known to this agent."
43     ::= { lldpV2RemoteSystemsData 2 }
44
45 lldpV2RemManAddrEntry OBJECT-TYPE
46     SYNTAX      LldpV2RemManAddrEntry
47     MAX-ACCESS   not-accessible
48     STATUS      current
49     DESCRIPTION
50         "Management address information about a particular chassis
51         component. There may be multiple management addresses
52         configured on the remote system identified by a particular
53         lldpRemIndex whose information is received on
54         an interface of the local system and a given destination
55         MAC address. Each management
56         address should have distinct 'management address
57         type' (lldpRemManAddrSubtype) and 'management address'
58         (lldpRemManAddr).
59

```

```

1      Entries may be created and deleted in this table by the
2      agent.
3      Since a variable length octetstring is used as an index
4      in a table, the address length is encoded as part of the OID
5      (as per IETF RFC 2578).
6      INDEX { lldpV2RemTimeMark,
7              lldpV2RemLocalIfIndex,
8              lldpV2RemLocalDestMACAddress,
9              lldpV2RemIndex,
10             lldpV2RemManAddrSubtype,
11             lldpV2RemManAddr
12 }
13 ::= { lldpV2RemManAddrTable 1 }
14
15
16 LldpV2RemManAddrEntry ::= SEQUENCE {
17     lldpV2RemManAddrSubtype    AddressFamilyNumbers,
18     lldpV2RemManAddr           LldpV2ManAddress,
19     lldpV2RemManAddrIfSubtype  LldpV2ManAddrIfSubtype,
20     lldpV2RemManAddrIfId       Unsigned32,
21     lldpV2RemManAddrOID        OBJECT IDENTIFIER
22 }
23
24 lldpV2RemManAddrSubtype OBJECT-TYPE
25     SYNTAX      AddressFamilyNumbers
26     MAX-ACCESS  not-accessible
27     STATUS      current
28     DESCRIPTION
29         "The type of management address identifier encoding used in
30         the associated 'lldpRemManagmentAddr' object.
31
32         It should be noted that only a subset of the possible
33         address encodings enumerated in AddressFamilyNumbers
34         are appropriate for use as a LLDP management
35         address, either because some are just not applicable or
36         because the maximum size of a LldpV2ManAddress octet string
37         would prevent the use of some address identifier encodings."
38     REFERENCE
39         "8.5.9.3"
40     ::= { lldpV2RemManAddrEntry 1 }
41
42 lldpV2RemManAddr OBJECT-TYPE
43     SYNTAX      LldpV2ManAddress
44     MAX-ACCESS  not-accessible
45     STATUS      current
46     DESCRIPTION
47         "The string value used to identify the management address
48         component associated with the remote system. The purpose
49         of this address is to contact the management entity."
50     REFERENCE
51         "8.5.9.4"
52     ::= { lldpV2RemManAddrEntry 2 }
53
54 lldpV2RemManAddrIfSubtype OBJECT-TYPE
55     SYNTAX      LldpV2ManAddrIfSubtype
56     MAX-ACCESS  read-only
57     STATUS      current
58     DESCRIPTION
59         "The enumeration value that identifies the interface numbering

```

```

1         method used for defining the interface number, associated
2         with the remote system."
3     REFERENCE
4         "8.5.9.5"
5     ::= { lldpV2RemManAddrEntry 3 }
6
7 lldpV2RemManAddrIfId OBJECT-TYPE
8     SYNTAX      Unsigned32
9     MAX-ACCESS  read-only
10    STATUS      current
11    DESCRIPTION
12        "The integer value used to identify the interface number
13        regarding the management address component associated with
14        the remote system. The value depends upon the value of the
15        lldpV2RemManAddrIfSubtype for the table row."
16    REFERENCE
17        "8.5.9.6"
18    ::= { lldpV2RemManAddrEntry 4 }
19
20 lldpV2RemManAddrOID OBJECT-TYPE
21    SYNTAX      OBJECT IDENTIFIER
22    MAX-ACCESS  read-only
23    STATUS      current
24    DESCRIPTION
25        "The OID value used to identify the type of hardware component
26        or protocol entity associated with the management address
27        advertised by the remote system agent."
28    REFERENCE
29        "8.5.9.8"
30    ::= { lldpV2RemManAddrEntry 5 }
31
32
33 --
34 -- lldpV2RemUnknownTLVTable : Unrecognized TLV information
35 -- This version has additional indexes for
36 -- ifIndex and destination MAC address
37 --
38
39 lldpV2RemUnknownTLVTable OBJECT-TYPE
40     SYNTAX      SEQUENCE OF LldpV2RemUnknownTLVEntry
41     MAX-ACCESS  not-accessible
42     STATUS      current
43     DESCRIPTION
44         "This table contains information about an incoming TLV that
45         is not recognized by the receiving LLDP agent. The TLV may
46         be from a later version of the basic management set.
47
48         This table should only contain TLVs that are found in
49         a single LLDP frame. Entries in this table, associated
50         with an MAC service access point (MSAP, the access point
51         for MAC services provided to the LCC sublayer, defined
52         in IEEE Standards Dictionary Online, which is also
53         identified with a particular lldpRemLocalPortNum,
54         lldpRemIndex pair) are overwritten with most recently
55         received unrecognized TLV from the same MSAP, or they
56         naturally age out when the rxInfoTTL timer
57         (associated with the MSAP) expires."
58     REFERENCE
59         "9.2.8.7.1"

```

```

1      ::= { lldpV2RemoteSystemsData 3 }
2
3 lldpV2RemUnknownTLVEntry OBJECT-TYPE
4     SYNTAX      LldpV2RemUnknownTLVEntry
5     MAX-ACCESS  not-accessible
6     STATUS      current
7     DESCRIPTION
8         "Information about an unrecognized TLV received from a
9         physical network connection. Entries may be created and
10        deleted in this table by the agent, if a physical topology
11        discovery process is active."
12     INDEX      {
13         lldpV2RemTimeMark,
14         lldpV2RemLocalIfIndex,
15         lldpV2RemLocalDestMACAddress,
16         lldpV2RemIndex,
17         lldpV2RemUnknownTLVType
18     }
19     ::= { lldpV2RemUnknownTLVTable 1 }
20
21 LldpV2RemUnknownTLVEntry ::= SEQUENCE {
22     lldpV2RemUnknownTLVType      Unsigned32,
23     lldpV2RemUnknownTLVInfo      OCTET STRING
24 }
25
26 lldpV2RemUnknownTLVType OBJECT-TYPE
27     SYNTAX      Unsigned32(9..126)
28     MAX-ACCESS  not-accessible
29     STATUS      current
30     DESCRIPTION
31         "This object represents the value extracted from the type
32         field of the TLV."
33     REFERENCE
34         "9.2.8.7.1"
35     ::= { lldpV2RemUnknownTLVEntry 1 }
36
37 lldpV2RemUnknownTLVInfo OBJECT-TYPE
38     SYNTAX      OCTET STRING (SIZE(0..511))
39     MAX-ACCESS  read-only
40     STATUS      current
41     DESCRIPTION
42         "This object represents the value extracted from the value
43         field of the TLV."
44     REFERENCE
45         "9.2.8.7.1"
46     ::= { lldpV2RemUnknownTLVEntry 2 }
47
48 -----
49 -- Remote Systems Extension Table - Organizationally Defined Information
50 -----
51 --
52 -- lldpV2RemOrgDefInfoTable - indexed by ifIndex and destination
53 -- MAC address.
54 --
55
56 lldpV2RemOrgDefInfoTable OBJECT-TYPE
57     SYNTAX      SEQUENCE OF LldpV2RemOrgDefInfoEntry
58     MAX-ACCESS  not-accessible
59     STATUS      current

```

```

1  DESCRIPTION
2      "This table contains one or more rows per physical network
3      connection which advertises the organizationally defined
4      information.
5
6      Note that this table contains one or more rows of
7      organizationally defined information that is not recognized
8      by the local agent.
9
10     If the local system is capable of recognizing any
11     organizationally defined information, appropriate extension
12     MIBs from the organization should be used for information
13     retrieval."
14     ::= { lldpV2RemoteSystemsData 4 }
15
16 lldpV2RemOrgDefInfoEntry OBJECT-TYPE
17     SYNTAX      LldpV2RemOrgDefInfoEntry
18     MAX-ACCESS  not-accessible
19     STATUS      current
20     DESCRIPTION
21         "Information about the unrecognized organizationally
22         defined information advertised by the remote system.
23         The lldpRemTimeMark, lldpRemLocalPortNum, lldpRemIndex,
24         lldpRemOrgDefInfoOUI, lldpRemOrgDefInfoSubtype, and
25         lldpRemOrgDefInfoIndex are indexes to this table. If there is
26         an lldpRemOrgDefInfoEntry associated with a particular remote
27         system identified by the lldpRemLocalPortNum and lldpRemIndex,
28         then there is an lldpRemEntry associated with the same
29         instance (i.e., using same indexes.) When the lldpRemEntry
30         for the same index is removed from the lldpRemTable, the
31         associated lldpRemOrgDefInfoEntry is removed from
32         the lldpRemOrgDefInfoTable.
33
34         Entries may be created and deleted in this table by the
35         agent."
36     INDEX      { lldpV2RemTimeMark,
37                 lldpV2RemLocalIfIndex,
38                 lldpV2RemLocalDestMACAddress,
39                 lldpV2RemIndex,
40                 lldpV2RemOrgDefInfoOUI,
41                 lldpV2RemOrgDefInfoSubtype,
42                 lldpV2RemOrgDefInfoIndex }
43     ::= { lldpV2RemOrgDefInfoTable 1 }
44
45 LldpV2RemOrgDefInfoEntry ::= SEQUENCE {
46     lldpV2RemOrgDefInfoOUI      OCTET STRING,
47     lldpV2RemOrgDefInfoSubtype  Unsigned32,
48     lldpV2RemOrgDefInfoIndex    Unsigned32,
49     lldpV2RemOrgDefInfo         OCTET STRING
50 }
51
52 lldpV2RemOrgDefInfoOUI OBJECT-TYPE
53     SYNTAX      OCTET STRING (SIZE(3))
54     MAX-ACCESS  not-accessible
55     STATUS      current
56     DESCRIPTION
57         "The Organizationally Unique Identifier (OUI), as defined
58         in IEEE Std 802, is a 24 bit (three octets) globally
59         unique assigned number referenced by various standards,

```



```

1         of the information received from the remote system."
2     REFERENCE
3         "8.7.1.3"
4     ::= { lldpV2RemOrgDefInfoEntry 1 }
5
6 lldpV2RemOrgDefInfoSubtype OBJECT-TYPE
7     SYNTAX      Unsigned32(1..255)
8     MAX-ACCESS  not-accessible
9     STATUS      current
10    DESCRIPTION
11        "The integer value used to identify the subtype of the
12        organizationally defined information received from the
13        remote system.
14
15        The subtype value is required to identify different instances
16        of organizationally defined information that could not be
17        retrieved without a unique identifier that indicates the
18        particular type of information contained in the information
19        string."
20    REFERENCE
21        "8.7.1.4"
22    ::= { lldpV2RemOrgDefInfoEntry 2 }
23
24 lldpV2RemOrgDefInfoIndex OBJECT-TYPE
25     SYNTAX      Unsigned32(1..2147483647)
26     MAX-ACCESS  not-accessible
27     STATUS      current
28     DESCRIPTION
29        "This object represents an arbitrary local integer value
30        used by this agent to identify a particular unrecognized
31        organizationally defined information instance, unique only
32        for the lldpRemOrgDefInfoOUI and lldpRemOrgDefInfoSubtype
33        from the same remote system.
34
35        An agent is encouraged to assign monotonically increasing
36        index values to new entries, starting with one, after each
37        reboot. It is considered unlikely that the
38        lldpRemOrgDefInfoIndex can wrap between reboots."
39    ::= { lldpV2RemOrgDefInfoEntry 3 }
40
41 lldpV2RemOrgDefInfo OBJECT-TYPE
42     SYNTAX      OCTET STRING(SIZE(0..507))
43     MAX-ACCESS  read-only
44     STATUS      current
45     DESCRIPTION
46        "The string value used to identify the organizationally
47        defined information of the remote system. The encoding for
48        this object should be as defined for SnmpAdminString TC."
49    REFERENCE
50        "8.7.1.5"
51    ::= { lldpV2RemOrgDefInfoEntry 4 }
52
53
54 --
55 -- *****
56 --
57 --     L L D P   M I B   N O T I F I C A T I O N S
58 --
59 -- *****

```

```

1 --
2
3 lldpV2NotificationPrefix OBJECT IDENTIFIER ::= { lldpV2Notifications 0 }
4
5 lldpV2RemTablesChange NOTIFICATION-TYPE
6     OBJECTS {
7         lldpV2StatsRemTablesInserts,
8         lldpV2StatsRemTablesDeletes,
9         lldpV2StatsRemTablesDrops,
10        lldpV2StatsRemTablesAgeouts
11    }
12    STATUS      current
13    DESCRIPTION
14        "A lldpV2RemTablesChange notification is sent when the value
15        of lldpV2StatsRemTablesLastChangeTime changes. It can be
16        utilized by an NMS to trigger LLDP remote systems table
17        maintenance polls.
18
19        Note that transmission of lldpV2RemTablesChange
20        notifications are throttled by the agent, as specified by the
21        'lldpV2NotificationInterval' object."
22 ::= { lldpV2NotificationPrefix 1 }
23
24
25 --
26 -- *****
27 --
28 --             L L D P   M I B   C O N F O R M A N C E
29 --
30 -- *****
31 --
32
33 lldpV2Compliances OBJECT IDENTIFIER ::= { lldpV2Conformance 1 }
34 lldpV2Groups      OBJECT IDENTIFIER ::= { lldpV2Conformance 2 }
35
36 -- compliance statements
37
38 lldpV2TxRxCompliance MODULE-COMPLIANCE
39     --V2 to add ifGeneralInformationGroup
40     --and support re-indexed tables
41     STATUS      current
42     DESCRIPTION
43         "A compliance statement for all SNMP entities that
44         implement the LLDP MIB as either a transmitter or
45         a receiver of LLDPDUs.
46
47         This version defines compliance requirements for
48         V2 of the LLDP MIB module."
49     MODULE -- this module
50         MANDATORY-GROUPS { lldpV2ConfigGroup,
51                             ifGeneralInformationGroup
52         }
53
54     ::= { lldpV2Compliances 1 }
55
56 lldpV2TxCompliance MODULE-COMPLIANCE
57     --V2 requirements for transmitters of LLDPDUs
58     --and support re-indexed tables
59     STATUS      current

```

```

1  DESCRIPTION
2      "A compliance statement for SNMP entities that implement
3      the LLDP MIB and have the capability of transmitting
4      LLDP frames.
5
6      This version defines compliance requirements for
7      V2 of the LLDP MIB module."
8  MODULE  -- this module
9      MANDATORY-GROUPS { lldpV2ConfigTxGroup,
10                         lldpV2StatsTxGroup,
11                         lldpV2LocSysGroup
12                     }
13
14      ::= { lldpV2Compliances 2 }
15
16 lldpV2RxCompliance MODULE-COMPLIANCE
17     --V2 requirements for receivers of LLDPDUs
18     --and support re-indexed tables
19     STATUS current
20     DESCRIPTION
21         "The compliance statement for SNMP entities that implement
22         the LLDP MIB and have the capability of receiving
23         LLDP frames.
24
25         This version defines compliance requirements for
26         V2 of the LLDP MIB module."
27     MODULE  -- this module
28         MANDATORY-GROUPS { lldpV2ConfigRxGroup,
29                             lldpV2StatsRxGroup,
30                             lldpV2RemSysGroup,
31                             lldpV2NotificationsGroup
32                         }
33
34     ::= { lldpV2Compliances 3 }
35
36 -- MIB groupings
37
38
39 lldpV2ConfigGroup  OBJECT-GROUP
40     OBJECTS {
41         lldpV2PortConfigAdminStatusV2
42     }
43     STATUS current
44     DESCRIPTION
45         "The collection of objects that are used to configure the
46         LLDP implementation behavior."
47     ::= { lldpV2Groups 1 }
48
49
50 lldpV2ConfigRxGroup  OBJECT-GROUP
51     OBJECTS {
52         lldpV2NotificationInterval,
53         lldpV2PortConfigNotificationEnableV2
54     }
55     STATUS current
56     DESCRIPTION
57         "The collection of objects that are used to configure the
58         LLDP reception implementation behavior."
59     ::= { lldpV2Groups 2 }

```

```

1
2
3 lldpV2ConfigTxGroup      OBJECT-GROUP
4     OBJECTS {
5         lldpV2MessageTxInterval,
6         lldpV2MessageTxHoldMultiplier,
7         lldpV2ReinitDelay,
8         lldpV2PortConfigTLVsTxEnableV2,
9         lldpV2ManAddrConfigTxEnable,
10        lldpV2ManAddrConfigRowStatus,
11        lldpV2TxCreditMax,
12        lldpV2MessageFastTx,
13        lldpV2TxFastInit,
14        lldpV2DestMacAddress,
15        lldpV2PortMessageTxInterval,
16        lldpV2PortMessageTxHoldMultiplier,
17        lldpV2PortReinitDelay,
18        lldpV2PortNotificationInterval,
19        lldpV2PortTxCreditMax,
20        lldpV2PortMessageFastTx,
21        lldpV2PortTxFastInit
22    }
23    STATUS    current
24    DESCRIPTION
25        "The collection of objects that are used to configure the
26        LLDP transmission implementation behavior."
27    ::= { lldpV2Groups 3 }
28
29
30 lldpV2StatsRxGroup      OBJECT-GROUP
31     OBJECTS {
32         lldpV2StatsRemTablesLastChangeTime,
33         lldpV2StatsRemTablesInserts,
34         lldpV2StatsRemTablesDeletes,
35         lldpV2StatsRemTablesDrops,
36         lldpV2StatsRemTablesAgeouts,
37         lldpV2StatsRxPortFramesDiscardedTotal,
38         lldpV2StatsRxPortFramesErrors,
39         lldpV2StatsRxPortFramesTotal,
40         lldpV2StatsRxPortTLVsDiscardedTotal,
41         lldpV2StatsRxPortTLVsUnrecognizedTotal,
42         lldpV2StatsRxPortAgeoutsTotal
43     }
44    STATUS    current
45    DESCRIPTION
46        "The collection of objects that are used to represent LLDP
47        reception statistics."
48    ::= { lldpV2Groups 4 }
49
50
51 lldpV2StatsTxGroup      OBJECT-GROUP
52     OBJECTS {
53         lldpV2StatsTxPortFramesTotal,
54         lldpV2StatsTxLLDPDULengthErrors
55     }
56    STATUS    current
57    DESCRIPTION
58        "The collection of objects that are used to represent LLDP
59        transmission statistics."

```

```
1 ::= { lldpV2Groups 5 }
2
3
4 lldpV2LocSysGroup OBJECT-GROUP
5   OBJECTS {
6     lldpV2LocChassisIdSubtype,
7     lldpV2LocChassisId,
8     lldpV2LocPortIdSubtype,
9     lldpV2LocPortId,
10    lldpV2LocPortDesc,
11    lldpV2LocSysDesc,
12    lldpV2LocSysName,
13    lldpV2LocSysCapSupported,
14    lldpV2LocSysCapEnabled,
15    lldpV2LocManAddrLen,
16    lldpV2LocManAddrIfSubtype,
17    lldpV2LocManAddrIfId,
18    lldpV2LocManAddrOID
19  }
20  STATUS current
21  DESCRIPTION
22    "The collection of objects that are used to represent LLDP
23    Local System Information."
24  ::= { lldpV2Groups 6 }
25
26 lldpV2RemSysGroup OBJECT-GROUP
27   OBJECTS {
28     lldpV2RemChassisIdSubtype,
29     lldpV2RemChassisId,
30     lldpV2RemPortIdSubtype,
31     lldpV2RemPortId,
32     lldpV2RemPortDesc,
33     lldpV2RemSysName,
34     lldpV2RemSysDesc,
35     lldpV2RemSysCapSupported,
36     lldpV2RemSysCapEnabled,
37     lldpV2RemRemoteChanges,
38     lldpV2RemTooManyNeighbors,
39     lldpV2RemManAddrIfSubtype,
40     lldpV2RemManAddrIfId,
41     lldpV2RemManAddrOID,
42     lldpV2RemUnknownTLVInfo,
43     lldpV2RemOrgDefInfo
44  }
45  STATUS current
46  DESCRIPTION
47    "The collection of objects that are used to represent
48    LLDP Remote Systems Information. The objects represent the
49    information associated with the basic TLV set. Please note
50    that even if the agent does not implement some of the optional
51    TLVs, it shall recognize all the optional TLV information
52    that the remote system may advertise."
53  ::= { lldpV2Groups 7 }
54
55 lldpV2NotificationsGroup NOTIFICATION-GROUP
56   NOTIFICATIONS {
57     lldpV2RemTablesChange
58   }
59   STATUS current
```

```
1      DESCRIPTION
2          "The collection of notifications used to indicate LLDP MIB
3          data consistency and general status information."
4      ::= { lldpV2Groups 8 }
5
6 END
```

## 12. LLDP YANG definitions

This clause specifies YANG modules that provide control and status monitoring of systems and system components that implement functionality specified in this standard.

This clause:

- a) Introduces the YANG framework that governs the naming and hierarchy of configuration and operational data structures in the data models, and the modeling of network interfaces (12.1).
- b) Describes the information data model and its relationship to the operational processes and managed objects specified in the other clauses of this standard, and provides a Unified Modeling Language (UML) representation of each data model (12.2).
- c) Describes the structure of the data models, each of which comprises or makes use of one or more YANG modules (12.3).
- d) Includes a relationship description of other modules imported in YANG modules (12.4).
- e) Reviews security considerations applicable to each of the modules, with specific reference to data nodes in the YANG modules that compose the model (12.5).
- f) Includes each of the YANG modules and its data schema (12.6).

### 12.1 Internet Standard Management Framework

This YANG module uses the YANG 1.1 Data Modeling Language as specified in IETF RFC 7950.

The YANG framework applies hierarchy in the following areas:

- a) The uniform resource name (URN), as specified in IEEE Std 802.
- b) The YANG objects form a hierarchy of configuration and operational data structures that define the YANG model.

### 12.2 Information model for LLDP management

The YANG objects are based on the TLVs detailed in 8.5 and 8.7, and the agent operation variables detailed in 9.1 and 9.2. A UML-like representation of the management model is provided in the following subclauses.

The purpose of a UML-like<sup>16</sup> diagram is to express the model design on a single piece of paper. The structure of the UML-like representation shows the name of the object followed by a list of properties for the object. The properties indicate its type and accessibility. It should be noted that the UML-like representation is meant to express simplified semantics for the properties. It is not meant to provide the specific datatype as used to encode the object in either MIB or YANG. In the UML-like representation, a box with a white background represents information that comes from sources outside of the IEEE. A box with a gray background represents objects that are defined by this IEEE Standard.

The YANG hierarchical structure that incorporates the LLDP YANG modules supported by this standard is represented by Figure 12-1. In the figures in this clause, items that are shaded gray are described in this document, items with no background shading are defined elsewhere. The YANG data model is realized in two YANG modules. One module *ieee802-dot1ab-types* provides data types that are needed by the LLDP configuration and monitoring objects. The *ieee802-dot1ab-lldp* module provides the LLDP configuration

<sup>16</sup> A description of the UML-like diagrams used in this clause is provided at <https://1.ieee802.org/uml-like-diagrams>

1 and monitoring objects. The ability to augment the port container to support extension TLVs is also shown.

2 The LLDP capabilities are not only applicable to IEEE Std 802.1Q bridges, but also (for example) end

3 stations, and routers.

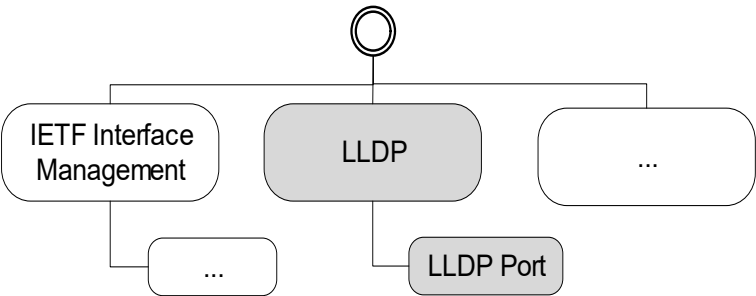


Figure 12-1—YANG root hierarchy with LLDP YANG modules

4 12.2.1 LLDP UML

5 The LLDP configuration and monitoring objects in Figure 12-2 show the objects that are applicable on a

6 per-agent or chassis associated with the LLDP agent.

7

| lldp                 |                               |  |                   |
|----------------------|-------------------------------|--|-------------------|
| uint32               | message-fast-tx;              |  | // (9.2.5.5) r-w  |
| uint32               | message-tx-hold-multiplier;   |  | // (9.2.5.6) r-w  |
| uint32               | message-tx-interval;          |  | // (9.2.5.7) r-w  |
| uint32               | reinit-delay;                 |  | // (9.2.5.10) r-w |
| uint32               | tx-credit-max;                |  | // (9.2.5.17) r-w |
| uint32               | tx-fast-init;                 |  | // (9.2.5.19) r-w |
| uint32               | notification-interval;        |  | // (9.2.5.7) r-w  |
| remote-statistics    |                               |  |                   |
| timestamp            | last-change-time;             |  | // r              |
| zero-based-counter32 | remote-inserts;               |  | // r              |
| zero-based-counter32 | remote-deletes;               |  | // r              |
| zero-based-counter32 | remote-drops;                 |  | // r              |
| zero-based-counter32 | remote-ageouts;               |  | // r              |
| local-system-data    |                               |  |                   |
| enum                 | chassis-id-subtype;           |  | // (8.5.2.2) r    |
| string               | chassis-id;                   |  | // (8.5.2.3) r    |
| string               | system-name;                  |  | // (8.5.6.2) r    |
| string               | system-description;           |  | // (8.5.7.2) r    |
| bits                 | system-capabilities-supported |  | // (8.5.8.1) r    |
| bits                 | system-capabilities-enabled   |  | // (8.5.8.2) r    |

Figure 12-2—LLDP configuration and monitoring objects



1 12.2.2 LLDP Port UML

2 The LLDP port configuration and monitoring objects in Figure 12-3 show the objects that are applicable to  
3 an LLDP port.

4

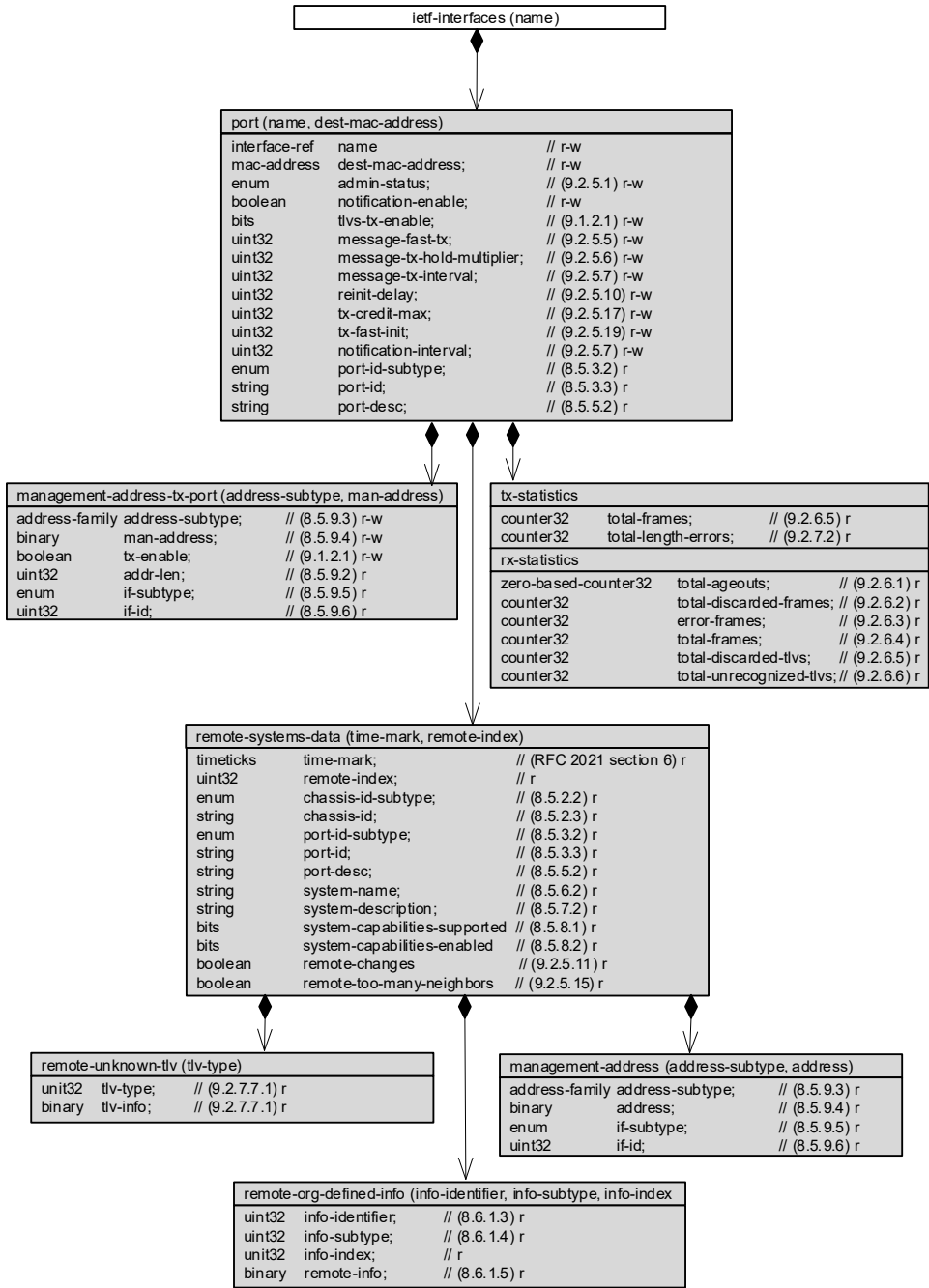


Figure 12-3—LLDP port configuration and monitoring objects

1 **12.3 Structure of the LLDP YANG model**

2 The IEEE YANG model specified in this standard is divided into two YANG modules. A summary of the  
3 modules contained in this clause is represented in Table 12-1.

**Table 12-1—Structure of the YANG modules**

| Module               | Subclause       | Notes                               |
|----------------------|-----------------|-------------------------------------|
| ieee802-dot1ab-types | Clause 12.6.2.1 | Type definitions used for LLDP YANG |
| ieee802-dot1ab-lldp  | Clause 12.6.2.2 | LLDP Generic Management             |

4 In the YANG module definitions below, if any discrepancy between the DESCRIPTION text and the  
5 corresponding definition in any other part of this standard occurs, the definitions outside this subclause take  
6 precedence.

### 1 12.3.1 Structure of the *ieee802-dot1ab-lldp* module

2 The *ieee802-dot1ab-lldp* YANG module is divided into several YANG branches (e.g., subtrees). A summary  
3 of the YANG subtrees associated with this module is presented in Table 12-2.

**Table 12-2—Structure of the *ieee802-dot1ab-lldp* module**

| Branches                   | References | Notes                                                                                  |
|----------------------------|------------|----------------------------------------------------------------------------------------|
| lldp                       |            | Link Layer Discovery Protocol (LLDP) configuration and operational information         |
| message-fast-tx            | 9.2.5.4    |                                                                                        |
| message-tx-hold-multiplier | 9.2.5.5    |                                                                                        |
| message-tx-interval        | 9.2.5.7    |                                                                                        |
| reinit-delay               | 9.2.5.9    |                                                                                        |
| tx-credit-max              | 9.2.5.18   |                                                                                        |
| tx-fast-init               | 9.2.5.20   |                                                                                        |
| notification-interval      | 9.2.5.6    |                                                                                        |
| remote-statistics          | 10.5.2     | LLDP remote operational statistics data                                                |
| local-system-data          | 10.5.1     | LLDP local system operational data                                                     |
| port                       | 9.2.4      | LLDP configuration information for a particular port                                   |
| management-address-tx-port | 8.5.9      | Set of ports on which the local system management address instance will be transmitted |
| tx-statistics              | 9.2.7      | LLDP frame transmission statistics for a particular port                               |
| rx-statistics              | 9.2.7      | LLDP frame reception statistics for a particular port                                  |
| remote-system-data         | 9.2.6      | Information about a particular physical network connection                             |

### 4 12.4 Relationship to other YANG modules

5 This clause describes how the *ieee802-dot1ab-lldp* YANG module is related to the YANG modules that are  
6 imported.

#### 7 12.4.1 IEEE LLDP Types Module

8 The *ieee802-dot1ab-types* module provides reusable types that are used by the *ieee802-dot1ab-lldp* module.

#### 9 12.4.2 IETF Routing Module

10 The *ietf-routing* YANG module (IETF RFC 8349) is a module defined by the IETF for management of a  
11 routing subsystem. This document only uses the address-family identity, which is used as the base identity  
12 for address families.

### 1 12.4.3 IETF YANG Types Module

2 The *ietf-yang-types* YANG module (IETF RFC 6991) contains a set of derived YANG types. This document  
3 leverages timestamp, timeticks, zero-based-counter32, and counter32.

### 4 12.4.4 IETF Interfaces YANG Module

5 The *ietf-interfaces* YANG module (IETF RFC 8343) contains a set of YANG definitions for managing  
6 network interfaces. This document models a port as an interface.

### 7 12.4.5 IEEE 802 Types Module

8 The *ieee802-types* module provides reusable types that are used in IEEE 802 standards.

9 The type for mac-addresses defined in *ieee802-types* has a pattern that allows upper and lower case letters.  
10 To avoid issues with string comparison, it is suggested to only use upper case for the letters in the  
11 hexadecimal numbers. Implementers using code comparing MAC addresses should note that there is still an  
12 issue with a difference between the IETF mac-address definition and the IEEE mac-address definition.

## 13 12.5 Security considerations

14 The YANG modules defined in this clause are designed to be accessed via a network configuration protocol  
15 (e.g., NETCONF protocol). In the case of NETCONF, the lowest NETCONF layer is the secure transport  
16 layer and the mandatory to implement secure transport is SSH. The NETCONF access control model  
17 provides the means to restrict access for particular NETCONF users to a pre-configured subset of all  
18 available NETCONF protocol operations and content.

19 It is the responsibility of a system's implementor and administrator to ensure that the protocol entities in the  
20 system that support NETCONF, and any other remote configuration protocols that make use of these YANG  
21 modules, are properly configured to allow access only to those users who have legitimate rights to read or  
22 write data nodes. This standard does not specify how the credentials of those users are to be stored or  
23 validated.

### 24 12.5.1 Security considerations of the *ieee802-dot1ab-lldp* YANG module

25 There are several management objects defined in the *ieee802-dot1ab-lldp* YANG module that are  
26 configurable (i.e., read-write) and/or operational (i.e., read-only). Such objects may be considered sensitive  
27 or vulnerable in some network environments. A network configuration protocol, such as NETCONF (IETF  
28 RFC 6241), can support protocol operations that can edit or delete YANG module configuration data  
29 (e.g., edit-config, delete-config, copy-config). If this is done in a non-secure environment without proper  
30 protection, then negative effects on the network operation are possible.

31 The following objects in the *ieee802-dot1ab-lldp* YANG module can be manipulated to interfere with the  
32 operation of LLDP. This could, for example, be used to return incorrect information about neighbor nodes,  
33 or cause a denial-of-service attack.

34       lldp/message-fast-tx  
35       lldp/message-tx-hold-multiplier  
36       lldp/message-tx-interval  
37       lldp/reinit-delay  
38       lldp/tx-credit-max  
39       lldp/tx-fast-init

1       lldp/notification-interval  
2       lldp/port/name  
3       lldp/port/dest-mac-address  
4       lldp/port/admin-status  
5       lldp/port/notification-enable  
6       lldp/port/tlvs-tx-enable  
7       lldp/port/message-fast-tx  
8       lldp/port/message-tx-hold-multiplier  
9       lldp/port/message-tx-interval  
10       lldp/port/reinit-delay  
11       lldp/port/tx-credit-max  
12       lldp/port/tx-fast-init  
13       lldp/port/notification-interval  
14       lldp/port/management-address-tx-port/address-subtype  
15       lldp/port/management-address-tx-port/man-address  
16       lldp/port/management-address-tx-port/tx-enable

17 Some of the readable data in this YANG module may be considered sensitive or vulnerable in some network  
18 environments. It is important to control all types of access (e.g., including NETCONF get and get-config  
19 operations) to these objects and possibly to even encrypt the values of these objects when sending them over  
20 the network. For example the system name and other information about the remote systems could provide  
21 information about the configuration and topology of the network and could be considered a privacy threat.

## 22 12.6 Definition of the YANG modules<sup>17, 18, 19</sup>

### 23 12.6.1 YANG schema definitions

24 A simplified graphical representation of the data model is used in this document. The meaning of the  
25 symbols in these diagrams is as follows:

- 26 — Brackets “[“ and “]” enclose list keys.
- 27 — Abbreviations before data node names: “rw” means configuration (read-write), and “ro” means state  
28 data (read-only).
- 29 — Symbols after data node names: “?” means an optional node, “!” means a presence container, and  
30 “\*” denotes a list and leaf-list.
- 31 — Parentheses enclose choice and case nodes, and case nodes are also marked with a colon (“:”).
- 32 — Ellipsis (“...”) stand for contents of subtrees that are not shown.

#### 33 12.6.1.1 YANG schema definition for *ieee802-dot1ab-lldp* YANG module

```
34 module: ieee802-dot1ab-lldp
35   +---rw lldp
36     +---rw message-fast-tx?          uint32
37     +---rw message-tx-hold-multiplier?  uint32
```

<sup>17</sup> *Copyright release for YANG modules:* Users of this standard may freely reproduce the YANG modules contained in this subclause so that they can be used for their intended purpose.

<sup>18</sup> The ieee802-types YANG module is common across various IEEE 802.1 standards and can be updated by any IEEE 802.1 standard that imports the ieee802-types YANG module.

<sup>19</sup> Consult the IEEE 802.1 website at <https://1.ieee802.org/yang-modules/> to determine and obtain the current revision of any IEEE 802.1 YANG module.

```

1      +---rw message-tx-interval?          uint32
2      +---rw reinit-delay?                 uint32
3      +---rw tx-credit-max?                uint32
4      +---rw tx-fast-init?                 uint32
5      +---rw notification-interval?        uint32
6      +---ro remote-statistics
7      |   +---ro last-change-time?         yang:timestamp
8      |   +---ro remote-inserts?           yang:zero-based-counter32
9      |   +---ro remote-deletes?           yang:zero-based-counter32
10     |   +---ro remote-drops?             yang:zero-based-counter32
11     |   +---ro remote-ageouts?           yang:zero-based-counter32
12     +---ro local-system-data
13     |   +---ro chassis-id-subtype?        ieee:chassis-id-subtype-type
14     |   +---ro chassis-id?               ieee:chassis-id-type
15     |   +---ro system-name?              string
16     |   +---ro system-description?        string
17     |   +---ro system-capabilities-supported? lldp-types:system-capabilities-map
18     |   +---ro system-capabilities-enabled? lldp-types:system-capabilities-map
19     +---rw port* [name dest-mac-address]
20         +---rw name                      if:interface-ref
21         +---rw dest-mac-address           ieee:mac-address
22         +---rw admin-status?              enumeration
23         +---rw notification-enable?        boolean
24         +---rw tlvs-tx-enable?            bits
25         +---rw message-fast-tx?           uint32
26         +---rw message-tx-hold-multiplier? uint32
27         +---rw message-tx-interval?        uint32
28         +---rw reinit-delay?              uint32
29         +---rw tx-credit-max?              uint32
30         +---rw tx-fast-init?              uint32
31         +---rw notification-interval?      uint32
32         +---rw management-address-tx-port* [address-subtype man-address]
33         |   +---rw address-subtype         identityref
34         |   +---rw man-address             lldp-types:man-addr-type
35         |   +---rw tx-enable?              boolean
36         |   +---ro addr-len?               uint32
37         |   +---ro if-subtype?             lldp-types:man-addr-if-subtype
38         |   +---ro if-id?                  uint32
39         +---ro port-id-subtype?            ieee:port-id-subtype-type
40         +---ro port-id?                    ieee:port-id-type
41         +---ro port-desc?                  string
42         +---ro tx-statistics
43         |   +---ro total-frames?           yang:counter32
44         |   +---ro total-length-errors?     yang:counter32
45         +---ro rx-statistics
46         |   +---ro total-ageouts?          yang:zero-based-counter32
47         |   +---ro total-discarded-frames? yang:counter32
48         |   +---ro error-frames?           yang:counter32
49         |   +---ro total-frames?           yang:counter32
50         |   +---ro total-discarded-tlvs?   yang:counter32
51         |   +---ro total-unrecognized-tlvs? yang:counter32
52         +---ro remote-systems-data* [time-mark remote-index]
53             +---ro time-mark               yang:timeticks
54             +---ro remote-index            uint32
55             +---ro remote-too-many-neighbors? boolean
56             +---ro remote-changes?         boolean
57             +---ro chassis-id-subtype?     ieee:chassis-id-subtype-type
58             +---ro chassis-id?             ieee:chassis-id-type
59             +---ro port-id-subtype?        ieee:port-id-subtype-type

```

```

1      +---ro port-id?                               ieee:port-id-type
2      +---ro port-desc?                             string
3      +---ro system-name?                           string
4      +---ro system-description?                     string
5      +---ro system-capabilities-supported?
6      | lldp-types:system-capabilities-map
7      +---ro system-capabilities-enabled?
8      | lldp-types:system-capabilities-map
9      +---ro management-address* [address-subtype address]
10     | +---ro address-subtype identityref
11     | +---ro address lldp-types:man-addr-type
12     | +---ro if-subtype? lldp-types:man-addr-if-subtype
13     | +---ro if-id? uint32
14     +---ro remote-unknown-tlv* [tlv-type]
15     | +---ro tlv-type uint32
16     | +---ro tlv-info? binary
17     +---ro remote-org-defined-info*
18         [info-identifier info-subtype info-index]
19         +---ro info-identifier uint32
20         +---ro info-subtype uint32
21         +---ro info-index uint32
22         +---ro remote-info? binary
23
24 notifications:
25     +---n remote-table-change
26         +---ro remote-insert? -> /lldp/remote-statistics/remote-inserts
27         +---ro remote-delete? -> /lldp/remote-statistics/remote-deletes
28         +---ro remote-drops? -> /lldp/remote-statistics/remote-drops
29         +---ro remote-ageouts? -> /lldp/remote-statistics/remote-ageouts
30

```

## 31 12.6.2 YANG data model definitions

### 32 12.6.2.1 Definition for the *ieee802-dot1ab-types* module

```

33 module ieee802-dot1ab-types {
34     yang-version "1.1";
35     namespace urn:ieee:std:802.1Q:yang:ieee802-dot1ab-types;
36     prefix lldp-types;
37     organization
38         "IEEE 802.1 Working Group";
39     contact
40         "WG-URL: http://ieee802.org/1/
41         WG-EMail: stds-802-1-1@ieee.org
42         Contact: IEEE 802.1 Working Group Chair
43         Postal: C/O IEEE 802.1 Working Group
44         IEEE Standards Association
45         445 Hoes Lane
46         Piscataway, NJ 08854
47         USA
48
49         E-mail: stds-802-1-chairs@ieee.org";
50     description
51         "Common types used within ieee802-dot1ab-lldp modules. Copyright(C) IEEE
52         (2025). All rights reserved. This version of this YANG module is part of
53         IEEE Std 802.1AB-2026; see the standard itself for full legal notices.";
54     revision 2025-09-29 {
55         description

```

```
1      "Published as part of IEEE Std 802.1AB-2026.
2
3      The following reference statement identifies each referenced IEEE
4      Standard as updated by applicable amendments.";
5  reference
6      "IEEE Std 802.1AB Station and Media Access Control Connectivity Discovery:
7      IEEE Std 802.1AB-2026.";
8  }
9  revision 2022-03-15 {
10     description
11         "Published as part of IEEE Std 802.1ABcu.";
12     reference
13         "IEEE Std 802.1AB-2016";
14 }
15 typedef man-addr-if-subtype {
16     type enumeration {
17         enum unknown {
18             value 1;
19             description
20                 "Interface is not known.";
21         }
22         enum port-ref {
23             value 2;
24             description
25                 "Interface based on the port-ref MIB object.";
26         }
27         enum system-port-number {
28             value 3;
29             description
30                 "Interface based on the system port number.";
31         }
32     }
33     description
34         "Management address interface subtype.";
35     reference
36         "8.5.9.3 of IEEE Std 802.1AB";
37 }
38 typedef man-addr-type {
39     type string {
40         pattern "[0-9A-F]{2}([0-9A-F]{2}){0,30}";
41     }
42     description
43         "Management address associated with the LLDP agent.";
44     reference
45         "8.5.9.4 of IEEE Std 802.1AB";
46 }
47 typedef system-capabilities-map {
48     type bits {
49         bit other {
50             position 0;
51             description
52                 "System has capabilities other than those listed below.";
53         }
54         bit repeater {
55             position 1;
56             description
57                 "System has repeater capability.";
58         }
59         bit bridge {
```



```
1      position 2;
2      description
3      "System has bridge capability.";
4  }
5  bit wlan-access-point {
6      position 3;
7      description
8      "System has WLAN access point capability.";
9  }
10 bit router {
11     position 4;
12     description
13     "System has router capability.";
14 }
15 bit telephone {
16     position 5;
17     description
18     "System has telephone capability.";
19 }
20 bit docsis-cable-device {
21     position 6;
22     description
23     "System has DOCSIS Cable Device capability.";
24     reference
25     "IETF RFC 4639";
26 }
27 bit station-only {
28     position 7;
29     description
30     "System has only station capability.";
31 }
32 bit cvlan-component {
33     position 8;
34     description
35     "System has C-VLAN component functionality.";
36 }
37 bit svlan-component {
38     position 9;
39     description
40     "System has S-VLAN component functionality.";
41 }
42 bit two-port-mac-relay {
43     position 10;
44     description
45     "System has Two-port MAC Relay (TPMR) functionality.";
46 }
47 }
48 description
49 "This describes system capabilities.";
50 reference
51 "8.5.8.1 of IEEE Std 802.1AB";
52 }
53 typedef port-list {
54     type binary {
55         length "0..512";
56     }
57     description
58     "Each octet within this value specifies a set of eight ports, with the
59     first octet specifying ports 1 through 8, the second octet specifying
```

```
1     ports 9 through 16, etc. Within each octet, the most significant bit
2     represents the lowest numbered port, and the least significant bit
3     represents the highest numbered port. Thus, each port of the system is
4     represented by a single bit within the value of this object. If that
5     bit has a value of '1' then that port is included in the set of ports;
6     the port is not included if its bit has a value of '0'.
7     reference
8     "IETF RFC 2674 section 5";
9 }
10 }
11
```

#### 12 12.6.2.2 Definition for *ieee802-dot1ab-lldp* YANG module

```
13 module ieee802-dot1ab-lldp {
14   yang-version 1.1;
15   namespace "urn:ieee:std:802.1AB:yang:ieee802-dot1ab-lldp";
16   prefix lldp;
17
18   import ieee802-dot1ab-types {
19     prefix lldp-types;
20   }
21   import ietf-routing {
22     prefix rt;
23   }
24   import ietf-yang-types {
25     prefix yang;
26   }
27   import ietf-interfaces {
28     prefix if;
29   }
30   import ieee802-types {
31     prefix ieee;
32   }
33
34   organization
35     "Institute of Electrical and Electronics Engineers";
36   contact
37     "WG-URL: http://ieee802.org/1/
38     WG-EMail: stds-802-1-1@ieee.org
39     Contact: IEEE 802.1 Working Group Chair
40     Postal: C/O IEEE 802.1 Working Group
41     IEEE Standards Association
42     445 Hoes Lane
43     Piscataway, NJ 08854
44     USA
45
46     E-mail: stds-802-1-chairs@ieee.org";
47   description
48     "Management Information Base module for LLDP configuration, statistics,
49     local system data, and remote systems data components. Copyright(C) IEEE
50     (2025). All rights reserved. This version of this YANG module is part of
51     IEEE Std 802.1AB-2026; see the standard itself for full legal notices.";
52
53   revision 2025-09-29 {
54     description
55       "Published as part of IEEE Std 802.1AB-2026.
56
```

```
1      The following reference statement identifies each referenced IEEE
2      Standard as updated by applicable amendments.";
3      reference
4      "IEEE Std 802.1AB Station and Media Access Control Connectivity Discovery:
5        IEEE Std 802.1AB-2026.";
6    }
7    revision 2022-03-15 {
8      description
9        "Published as part of IEEE Std 802.1ABcu.";
10     reference
11       "IEEE Std 802.1AB-2016";
12   }
13
14   grouping lldp-cfg {
15     description
16       "LLDP basic configuration group.";
17     leaf message-fast-tx {
18       type uint32 {
19         range "1..3600";
20       }
21       units "ticks";
22       default "1";
23       description
24         "Time interval in timer ticks between transmissions during fast
25         transmission periods (i.e., txFast is non-zero).";
26       reference
27         "9.1.1, 9.2.5.4 of IEEE Std 802.1AB";
28     }
29     leaf message-tx-hold-multiplier {
30       type uint32 {
31         range "2..10";
32       }
33       default "4";
34       description
35         "Multiplier of msg-tx-interval.";
36       reference
37         "9.2.5.5 of IEEE Std 802.1AB";
38     }
39     leaf message-tx-interval {
40       type uint32 {
41         range "1..3600";
42       }
43       units "ticks";
44       default "30";
45       description
46         "Time interval in timer ticks between transmissions during normal
47         transmission periods (i.e., txFast is zero).";
48       reference
49         "9.1.1, 9.2.5.6 of IEEE Std 802.1AB";
50     }
51     leaf reinit-delay {
52       type uint32 {
53         range "1..10";
54       }
55       units "second";
56       default "2";
57       description
58         "Amount of delay (in units of seconds) from when admin-status becomes
59         'disabled' until re-initialization is attempted.";
```

```
1     reference
2       "9.2.5.9 of IEEE Std 802.1AB";
3   }
4   leaf tx-credit-max {
5       type uint32 {
6           range "1..10";
7       }
8       default "5";
9       description
10        "The maximum number of consecutive LLDPDUs that can be transmitted at
11        any time.";
12        reference
13        "9.2.5.18 of IEEE Std 802.1AB";
14    }
15    leaf tx-fast-init {
16        type uint32 {
17            range "1..8";
18        }
19        default "4";
20        description
21        "Initial value for the fast transmitting LLDPDU.";
22        reference
23        "9.2.5.20 of IEEE Std 802.1AB";
24    }
25    leaf notification-interval {
26        type uint32 {
27            range "1..3600";
28        }
29        units "second";
30        default "30";
31        description
32        "Controls the transmission of LLDP notifications.";
33        reference
34        "9.2.5.6 of IEEE Std 802.1AB";
35    }
36 }
37
38 container lldp {
39     description
40     "Link Layer Discovery Protocol configuration and operational
41     information.";
42     uses lldp-cfg;
43     container remote-statistics {
44         config false;
45         description
46         "LLDP remote operational statistics data.";
47         leaf last-change-time {
48             type yang:timestamp;
49             description
50             "The value of sysUpTime object.";
51             reference
52             "11.5.1 of IEEE Std 802.1AB:
53             lldpV2StatsRemTablesLastChangeTime";
54         }
55         leaf remote-inserts {
56             type yang:zero-based-counter32;
57             units "table entries";
58             description
59             "The number of times the complete set of information advertised by
```

```
1         a particular MSAP has been inserted into tables contained in
2         remote-systems-data and lldpExtensions objects.";
3     reference
4         "11.5.1 of IEEE Std 802.1AB: lldpV2StatsRemTablesInserts";
5 }
6 leaf remote-deletes {
7     type yang:zero-based-counter32;
8     units "table entries";
9     description
10        "The number of times the complete set of information advertised by
11        a particular MSAP has been deleted from tables contained in
12        remote-systems-data and lldpExtensions objects.";
13    reference
14        "11.5.1 of IEEE Std 802.1AB: lldpV2StatsRemTablesDeletes";
15 }
16 leaf remote-drops {
17     type yang:zero-based-counter32;
18     units "table entries";
19     description
20        "The number of times the complete set of information advertised by
21        a particular MSAP could not be entered into tables contained in
22        remote-systems-data and lldpExtensions objects because of
23        insufficient resources.";
24    reference
25        "11.5.1 of IEEE Std 802.1AB: lldpV2StatsRemTablesDrops";
26 }
27 leaf remote-ageouts {
28     type yang:zero-based-counter32;
29     description
30        "The number of times the complete set of information advertised by
31        a particular MSAP has been deleted from tables contained in
32        remote-systems-data and lldpExtensions objects because the
33        information timeliness interval has expired.";
34    reference
35        "11.5.1 of IEEE Std 802.1AB: lldpV2StatsRemTablesAgeouts";
36 }
37 }
38 container local-system-data {
39     config false;
40     description
41        "LLDP local system operational data.";
42     leaf chassis-id-subtype {
43         type ieee:chassis-id-subtype-type;
44         description
45            "The type of encoding used to identify the chassis associated with
46            the local system.";
47         reference
48            "8.5.2.2 of IEEE Std 802.1AB";
49     }
50     leaf chassis-id {
51         type ieee:chassis-id-type;
52         description
53            "Chassis component associated with the local system.";
54         reference
55            "8.5.2.3 of IEEE Std 802.1AB";
56     }
57     leaf system-name {
58         type string {
59             length "0..255";
```

```

1      }
2      description
3          "System name of the local system.";
4      reference
5          "8.5.6.2 of IEEE Std 802.1AB";
6  }
7  leaf system-description {
8      type string {
9          length "0..255";
10     }
11     description
12         "System description of the local system.";
13     reference
14         "8.5.7.2 of IEEE Std 802.1AB";
15 }
16 leaf system-capabilities-supported {
17     type lldp-types:system-capabilities-map;
18     description
19         "System capabilities are supported on the local system.";
20     reference
21         "8.5.8.1 of IEEE Std 802.1AB";
22 }
23 leaf system-capabilities-enabled {
24     type lldp-types:system-capabilities-map;
25     description
26         "System capabilities that are enabled on the local system.";
27     reference
28         "8.5.8.2 of IEEE Std 802.1AB";
29 }
30 }
31 list port {
32     key "name dest-mac-address";
33     description
34         "LLDP configuration information for a particular port.";
35     leaf name {
36         type if:interface-ref;
37         description
38             "The port name used to identify the port component (contained in
39             the local chassis with the LLDP agent) associated with this entry.";
40     }
41     leaf dest-mac-address {
42         type ieee:mac-address;
43         description
44             "Destination MAC address. The ieee:mac-address type has a pattern
45             that allows upper and lower case letters. To avoid issues with
46             string compairson, it is suggested to only use upper case for the
47             letters in the hexadecimal numbers. Implementers using code
48             comparing MAC addresses should note that there is still an issue
49             with a difference between the IETF mac-address definition and the
50             IEEE mac-address definition.";
51     }
52     leaf admin-status {
53         type enumeration {
54             enum tx-only {
55                 value 1;
56                 description
57                     "Transmit LLDP frames only.";
58             }
59             enum rx-only {

```

```

1         value 2;
2         description
3             "Receive LLDP frames only.";
4     }
5     enum tx-and-rx {
6         value 3;
7         description
8             "Transmit and Receive LLDP frames.";
9     }
10    enum disabled {
11        value 4;
12        description
13            "Do Not Transmit or Receive LLDP frames.";
14    }
15 }
16 default "tx-and-rx";
17 description
18     "Administrative status of the local LLDP agent.";
19 reference
20     "9.2.5.1 of IEEE Std 802.1AB";
21 }
22 leaf notification-enable {
23     type boolean;
24     default "false";
25     description
26         "Notification status.";
27 }
28 leaf tlvs-tx-enable {
29     type bits {
30         bit port-desc {
31             position 0;
32             description
33                 "Transmit 'Port Description TLV'.";
34         }
35         bit sys-name {
36             position 1;
37             description
38                 "Transmit 'System Name TLV'.";
39         }
40         bit sys-desc {
41             position 2;
42             description
43                 "Transmit 'System Description TLV'.";
44         }
45         bit sys-cap {
46             position 3;
47             description
48                 "Transmit 'System Capabilities TLV'.";
49         }
50     }
51     description
52         "LLDP TLVs whose transmission is allowed on the local LLDP agent by
53         the network management.";
54     reference
55         "9.1.1.1 of IEEE Std 802.1AB";
56 }
57 uses lldp-cfg;
58 list management-address-tx-port {
59     key "address-subtype man-address";

```

```
1      description
2          "Set of ports (represented as a PortList) on which the local system
3          management address instance will be transmitted.";
4      leaf address-subtype {
5          type identityref {
6              base rt:address-family;
7          }
8          description
9              "Type of address.";
10         reference
11             "8.5.9.3 of IEEE Std 802.1AB";
12     }
13     leaf man-address {
14         type lldp-types:man-addr-type;
15         description
16             "Management address associated with this TLV.";
17         reference
18             "8.5.9.4 of IEEE Std 802.1AB";
19     }
20     leaf tx-enable {
21         type boolean;
22         default "false";
23         description
24             "Transmission enabled status.";
25         reference
26             "9.1.1.1 of IEEE Std 802.1AB";
27     }
28     leaf addr-len {
29         type uint32;
30         config false;
31         description
32             "Length of the management address subtype and the management
33             address fields in LLDPDUs transmitted by the local LLDP agent.";
34         reference
35             "8.5.9.2 of IEEE Std 802.1AB";
36     }
37     leaf if-subtype {
38         type lldp-types:man-addr-if-subtype;
39         config false;
40         description
41             "Interface numbering method used for defining the interface
42             number, associated with the local system.";
43         reference
44             "8.5.9.5 of IEEE Std 802.1AB";
45     }
46     leaf if-id {
47         type uint32;
48         config false;
49         description
50             "Interface number for the management address component associated
51             with the local system.";
52         reference
53             "8.5.9.6 of IEEE Std 802.1AB";
54     }
55 }
56 leaf port-id-subtype {
57     type ieee:port-id-subtype-type;
58     config false;
59     description
```



```
1         "Port identifier encoding used in the associated 'port-id' object.";
2     reference
3         "8.5.3.2 of IEEE Std 802.1AB";
4 }
5 leaf port-id {
6     type ieee:port-id-type;
7     config false;
8     description
9         "Port component associated with a given port in the local system.";
10    reference
11        "8.5.3.3 of IEEE Std 802.1AB";
12 }
13 leaf port-desc {
14     type string {
15         length "0..255";
16     }
17     config false;
18     description
19         "802 LAN station's port description associated with the local
20         system.";
21     reference
22         "8.5.5.2 of IEEE Std 802.1AB";
23 }
24 container tx-statistics {
25     config false;
26     description
27         "LLDP frame transmission statistics for a particular port.";
28     leaf total-frames {
29         type yang:counter32;
30         description
31             "A count of all LLDP frames transmitted through the port.";
32         reference
33             "9.2.7.5 of IEEE Std 802.1AB";
34     }
35     leaf total-length-errors {
36         type yang:counter32;
37         description
38             "A count of all LLDP length errors detected when constructing
39             LLDPDU frames for transmission through the port.";
40         reference
41             "9.2.7.8 of IEEE Std 802.1AB";
42     }
43 }
44 container rx-statistics {
45     config false;
46     description
47         "LLDP frame reception statistics for a particular port.";
48     leaf total-ageouts {
49         type yang:zero-based-counter32;
50         description
51             "A count of the times that a neighbor's information is deleted
52             because of rxInfoTTL timer expiration.";
53         reference
54             "9.2.7.1 of IEEE Std 802.1AB";
55     }
56     leaf total-discarded-frames {
57         type yang:counter32;
58         description
59             "A count of all LLDPDUs received and then discarded.";
```

```
1         reference
2         "9.2.7.2 of IEEE Std 802.1AB";
3     }
4     leaf error-frames {
5         type yang:counter32;
6         description
7             "A count of all LLDPDUs received at the port with one or more
8             detectable errors.";
9         reference
10        "9.2.7.3 of IEEE Std 802.1AB";
11    }
12    leaf total-frames {
13        type yang:counter32;
14        description
15            "A count of all LLDP frames received at the port.";
16        reference
17            "9.2.7.4 of IEEE Std 802.1AB";
18    }
19    leaf total-discarded-tlvs {
20        type yang:counter32;
21        description
22            "A count of all TLVs received at the port and discarded for any
23            reason.";
24        reference
25            "9.2.7.6 of IEEE Std 802.1AB";
26    }
27    leaf total-unrecognized-tlvs {
28        type yang:counter32;
29        description
30            "A count of all TLVs not recognized by the receiving LLDP local
31            agent.";
32        reference
33            "9.2.7.7 of IEEE Std 802.1AB";
34    }
35 }
36 list remote-systems-data {
37     key "time-mark remote-index";
38     config false;
39     description
40         "Information about a particular physical network connection.";
41     leaf time-mark {
42         type yang:timeticks;
43         description
44             "A TimeFilter for this entry.";
45         reference
46             "IETF RFC 2021 section 6";
47     }
48     leaf remote-index {
49         type uint32 {
50             range "1..2147483647";
51         }
52         description
53             "Represents an arbitrary local integer value used to identify a
54             remote system.";
55         reference
56             "11.5.1 of IEEE Std 802.1AB: lldpV2RemIndex";
57     }
58     leaf remote-too-many-neighbors {
59         type boolean;
```

```
1      description
2      "Indicates that there are too many neighbors as determined by the
3      variable tooManyNeighbors.";
4      reference
5      "9.2.5.16 of IEEE Std 802.1AB";
6  }
7  leaf remote-changes {
8      type boolean;
9      description
10     "Indicates that there are changes in the remote system's data, as
11     determined by the variable remoteChanges.";
12     reference
13     "9.2.5.10 of IEEE Std 802.1AB";
14 }
15 leaf chassis-id-subtype {
16     type ieee:chassis-id-subtype-type;
17     description
18     "Identify the chassis associated with the remote system.";
19     reference
20     "8.5.2.2 of IEEE Std 802.1AB";
21 }
22 leaf chassis-id {
23     type ieee:chassis-id-type;
24     description
25     "Identify the chassis component associated with the remote
26     system.";
27     reference
28     "8.5.2.3 of IEEE Std 802.1AB";
29 }
30 leaf port-id-subtype {
31     type ieee:port-id-subtype-type;
32     description
33     "The type of port identifier encoding used in the associated
34     'port-id' object.";
35     reference
36     "8.5.3.2 of IEEE Std 802.1AB";
37 }
38 leaf port-id {
39     type ieee:port-id-type;
40     description
41     "Port component associated with the remote system.";
42     reference
43     "8.5.3.3 of IEEE Std 802.1AB";
44 }
45 leaf port-desc {
46     type string {
47         length "0..255";
48     }
49     description
50     "Description of the given port associated with the remote system.";
51     reference
52     "8.5.5.2 of IEEE Std 802.1AB";
53 }
54 leaf system-name {
55     type string {
56         length "0..255";
57     }
58     description
59     "System name of the remote system.";
```

```
1         reference
2         "8.5.6.2 of IEEE Std 802.1AB";
3     }
4     leaf system-description {
5         type string {
6             length "0..255";
7         }
8         description
9         "System description of the remote system.";
10        reference
11        "8.5.7.2 of IEEE Std 802.1AB";
12    }
13    leaf system-capabilities-supported {
14        type lldp-types:system-capabilities-map;
15        description
16        "Capabilities that are supported on the remote system.";
17        reference
18        "8.5.8.1 of IEEE Std 802.1AB";
19    }
20    leaf system-capabilities-enabled {
21        type lldp-types:system-capabilities-map;
22        description
23        "System capabilities that are enabled on the remote system.";
24        reference
25        "8.5.8.2 of IEEE Std 802.1AB";
26    }
27    list management-address {
28        key "address-subtype address";
29        description
30        "Management address information about a particular chassis
31        component.";
32        leaf address-subtype {
33            type identityref {
34                base rt:address-family;
35            }
36            description
37            "Management address identifier encoding.";
38            reference
39            "8.5.9.3 of IEEE Std 802.1AB";
40        }
41        leaf address {
42            type lldp-types:man-addr-type;
43            description
44            "Management address component associated with the remote
45            system.";
46            reference
47            "8.5.9.4 of IEEE Std 802.1AB";
48        }
49        leaf if-subtype {
50            type lldp-types:man-addr-if-subtype;
51            description
52            "Interface numbering method used for defining the interface
53            number, associated with the remote system.";
54            reference
55            "8.5.9.5 of IEEE Std 802.1AB";
56        }
57        leaf if-id {
58            type uint32;
59            description
```

```
1         "Interface number regarding the management address component
2         associated with the remote system.";
3     reference
4         "8.5.9.6 of IEEE Std 802.1AB";
5     }
6 }
7 list remote-unknown-tlv {
8     key "tlv-type";
9     description
10        "Information about an unrecognized TLV received from a physical
11        network connection. Entries may be created and deleted in this
12        table by the agent, if a physical topology discovery process is
13        active.";
14     leaf tlv-type {
15         type uint32 {
16             range "9..126";
17         }
18         description
19             "Type of TLV.";
20         reference
21             "9.2.8.7.1 of IEEE Std 802.1AB";
22     }
23     leaf tlv-info {
24         type binary {
25             length "0..511";
26         }
27         description
28             "Value extracted from TLV.";
29         reference
30             "9.2.8.7.1 of IEEE Std 802.1AB";
31     }
32 }
33 list remote-org-defined-info {
34     key "info-identifier info-subtype info-index";
35     description
36        "Information about the unrecognized organizationally defined
37        information advertised by the remote system.";
38     leaf info-identifier {
39         type uint32 {
40             range "0..16777215";
41         }
42         description
43             "The Organizationally Unique Identifier (OUI) or Company ID
44             (CID).";
45         reference
46             "8.7.1.3 of IEEE Std 802.1AB";
47     }
48     leaf info-subtype {
49         type uint32 {
50             range "1..255";
51         }
52         description
53             "The subtype of the organizationally defined information
54             received from the remote system.";
55         reference
56             "8.7.1.4 of IEEE Std 802.1AB";
57     }
58     leaf info-index {
59         type uint32 {
```

```
1         range "1..2147483647";
2     }
3     description
4         "Arbitrary local integer value.";
5     }
6     leaf remote-info {
7         type binary {
8             length "0..507";
9         }
10        description
11            "The organizationally defined information of the remote system.";
12        reference
13            "8.7.1.5 of IEEE Std 802.1AB";
14    }
15 }
16 }
17 }
18 }
19
20 notification remote-table-change {
21     description
22         "A rem-table-change notification is sent when the value of
23         remote-table-last-change-time changes. It can be utilized by an NMS to
24         trigger LLDP remote systems table maintenance polls.";
25     leaf remote-insert {
26         type leafref {
27             path "/lldp:lldp/lldp:remote-statistics/lldp:remote-inserts";
28         }
29         description
30             "The number of times the complete set of information advertised by a
31             particular MSAP has been inserted into tables contained in
32             remote-systems-data and lldpExtensions objects.";
33         reference
34             "11.5.1 of IEEE Std 802.1AB: lldpV2StatsRemTablesInserts";
35     }
36     leaf remote-delete {
37         type leafref {
38             path "/lldp:lldp/lldp:remote-statistics/lldp:remote-deletes";
39         }
40         description
41             "The number of times the complete set of information advertised by a
42             particular MSAP has been deleted from tables contained in
43             remote-systems-data and lldpExtensions objects.";
44         reference
45             "11.5.1 of IEEE Std 802.1AB: lldpV2StatsRemTablesDeletes";
46     }
47     leaf remote-drops {
48         type leafref {
49             path "/lldp:lldp/lldp:remote-statistics/lldp:remote-drops";
50         }
51         description
52             "The number of times the complete set of information advertised by a
53             particular MSAP could not be entered into tables contained in
54             remote-systems-data and lldpExtensions objects because of
55             insufficient resources.";
56         reference
57             "11.5.1 of IEEE Std 802.1AB: lldpV2StatsRemTablesDrops";
58     }
59     leaf remote-ageouts {
```

```
1     type leafref {
2         path "/lldp:lldp/lldp:remote-statistics/lldp:remote-ageouts";
3     }
4     description
5         "The number of times the complete set of information advertised by a
6         particular MSAP has been deleted from tables contained in
7         remote-systems-data and lldpExtensions objects because the
8         information timeliness interval has expired.";
9     reference
10        "11.5.1 of IEEE Std 802.1AB: lldpV2StatsRemTablesAgeouts";
11 }
12 }
13 }
14
15
```

# **Annex A**

(normative)

## **PICS proforma<sup>20</sup>**

### **A.1 Introduction**

The supplier of a protocol implementation that is claimed to conform to this standard shall complete the following Protocol Implementation Conformance Statement (PICS) proforma.

A completed PICS proforma is the PICS for the implementation in question. The PICS is a statement of which capabilities and options of the protocol have been implemented. The PICS can have a number of uses, including use

- a) By the protocol implementers, as a checklist to reduce the risk of failure to conform to the standard through oversight.
- b) By the supplier and acquirer—or potential acquirer—of the implementation, as a detailed indication of the capabilities of the implementation, stated relative to the common basis for understanding provided by the standard PICS proforma.
- c) By the user—or potential user—of the implementation, as a basis for initially checking the possibility of interworking with another implementation (note that although interworking can never be guaranteed, failure to interwork can often be predicted from incompatible PICS).
- d) By a protocol tester, as the basis for selecting appropriate tests against which to assess the claim for conformance of the implementation.

### **A.2 Abbreviations and special symbols**

#### **A.2.1 Status symbols**

- |            |                                                                                                                |
|------------|----------------------------------------------------------------------------------------------------------------|
| M          | Mandatory                                                                                                      |
| O          | Optional                                                                                                       |
| <i>O.n</i> | Optional, but support of at least one of the group of options labeled by the same numeral <i>n</i> is required |
| X          | Prohibited                                                                                                     |
| pred:      | Conditional-item symbol, including predicate identification: See B.3.4                                         |
| ¬          | Logical negation, applied to a conditional item's predicate                                                    |

#### **A.2.2 General abbreviations**

- |      |                                               |
|------|-----------------------------------------------|
| N/A  | Not applicable                                |
| PICS | Protocol Implementation Conformance Statement |

<sup>20</sup> *Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.



## **1 A.3 Instructions for completing the PICS proforma**

### **2 A.3.1 General structure of the PICS proforma**

3 The first part of the PICS proforma, implementation identification and protocol summary, is to be completed  
4 as indicated with the information necessary to identify fully both the supplier and the implementation.

5 The main part of the PICS proforma is a fixed-format questionnaire, divided into several subclauses, each  
6 containing a number of individual items. Answers to the questionnaire items are to be provided in the  
7 right-most column, either by simply marking an answer to indicate a restricted choice (usually Yes or No), or  
8 by entering a value or a set or range of values. (Note that there are some items in which two or more choices  
9 from a set of possible answers can apply; all relevant choices are to be marked.)

10 Each item is identified by an item reference in the first column. The second column contains the question to  
11 be answered; the third column records the status of the item—whether support is mandatory, optional, or  
12 conditional; see also B.3.4. The fourth column contains the reference or references to the material that  
13 specifies the item in the main body of this standard, and the fifth column provides the space for the answers.

14 A supplier may also provide (or be required to provide) further information, categorized as either Additional  
15 Information or Exception Information. When present, each kind of further information is to be provided in a  
16 further subclause of items labeled  $A_i$  or  $X_i$ , respectively, for cross-referencing purposes, where  $i$  is any  
17 unambiguous identification for the item (e.g., simply a numeral). There are no other restrictions on its format  
18 and presentation.

19 A completed PICS proforma, including any Additional Information and Exception Information, is the  
20 Protocol Implementation Conformation Statement for the implementation in question.

21 NOTE—Where an implementation is capable of being configured in more than one way, a single PICS may be able to  
22 describe all such configurations. However, the supplier has the choice of providing more than one PICS, each covering  
23 some subset of the implementation's configuration capabilities, in case that makes for easier and clearer presentation of  
24 the information.

### **25 A.3.2 Additional information**

26 Items of Additional Information allow a supplier to provide further information intended to assist the  
27 interpretation of the PICS. It is not intended or expected that a large quantity will be supplied, and a PICS  
28 can be considered complete without any such information. Examples might be an outline of the ways in  
29 which a (single) implementation can be set up to operate in a variety of environments and configurations, or  
30 information about aspects of the implementation that are outside the scope of this standard but that have a  
31 bearing on the answers to some items.

32 References to items of Additional Information may be entered next to any answer in the questionnaire and  
33 may be included in items of Exception Information.

### **34 A.3.3 Exception information**

35 It may occasionally happen that a supplier will wish to answer an item with mandatory status (after any  
36 conditions have been applied) in a way that conflicts with the indicated requirement. No preprinted answer  
37 will be found in the Support column for this: instead, the supplier shall write the missing answer into the  
38 Support column, together with an  $X_i$  reference to an item of Exception Information, and shall provide the  
39 appropriate rationale in the Exception item.

- 1 An implementation for which an Exception item is required in this way does not conform to this standard.  
2 NOTE—A possible reason for the situation described above is that a defect in this standard has been reported, a  
3 correction for which is expected to change the requirement not met by the implementation.

#### 4 **A.3.4 Conditional status**

##### 5 **A.3.4.1 Conditional items**

6 The PICS proforma contains a number of conditional items. These are items for which both the applicability  
7 of the item itself, and its status if it does apply—mandatory or optional—are dependent upon whether or not  
8 certain other items are supported.

9 Where a group of items is subject to the same condition for applicability, a separate preliminary question  
10 about the condition appears at the head of the group, with an instruction to skip to a later point in the  
11 questionnaire if the “Not Applicable” answer is selected. Otherwise, individual conditional items are  
12 indicated by a conditional symbol in the Status column.

13 A conditional symbol is of the form “pred: S,” where pred is a predicate as described in A.3.4.2, and S is a  
14 status symbol, M or O.

15 If the value of the predicate is TRUE (see A.3.4.2), the conditional item is applicable, and its status is  
16 indicated by the status symbol following the predicate: the answer column is to be marked in the usual way.  
17 If the value of the predicate is FALSE, the “Not Applicable” (N/A) answer is to be marked.

##### 18 **A.3.4.2 Predicates**

19 A predicate is one of the following:

- 20 a) An item-reference for an item in the PICS proforma: The value of the predicate is TRUE if the item  
21 is marked as supported, and is FALSE otherwise.
- 22 b) A predicate-name, for a predicate defined as a Boolean expression constructed by combining  
23 item-references using the Boolean operator OR: The value of the predicate is TRUE if one or more  
24 of the items is marked as supported.
- 25 c) A predicate-name, for a predicate defined as a Boolean expression constructed by combining  
26 item-references using the Boolean operator AND: The value of the predicate is TRUE if all of the  
27 items are marked as supported.
- 28 d) The logical negation symbol “¬” prefixed to an item-reference or predicate-name: The value of the  
29 predicate is TRUE if the value of the predicate formed by omitting the “¬” symbol is FALSE, and  
30 vice versa.

31 Each item whose reference is used in a predicate or predicate definition, or in a preliminary question for  
32 grouped conditional items, is indicated by an asterisk<sup>21</sup> in the Item column.

33

34

35

#### 36 **PICS proforma for IEEE Std 802.1AB-2016**

<sup>21</sup> Asterisks are currently not used in this PICS.

### A.3.5 Implementation identification

|                                                                                                                                                                                                                                                                                                                                            |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Supplier</b>                                                                                                                                                                                                                                                                                                                            |  |
| Contact point for queries about the PICS                                                                                                                                                                                                                                                                                                   |  |
| Implementation Name(s) and Version(s)                                                                                                                                                                                                                                                                                                      |  |
| Other information necessary for full identification—e.g., name(s) and version(s) of machines and/or operating system names                                                                                                                                                                                                                 |  |
| <p>NOTE 1—Only the first three items are required for all implementations; other information may be completed as appropriate in meeting the requirement for full identification.</p> <p>NOTE 2—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).</p> |  |

### A.3.6 Protocol summary, IEEE Std 802.1AB-2016

|                                                                                                                                             |                                                                                                                                       |   |         |
|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|---|---------|
| <b>Identification of protocol specification</b>                                                                                             | IEEE Std 802.1AB-2016, IEEE Standard for Local and metropolitan area networks—Station and Media Access Control Connectivity Discovery |   |         |
| Identification of amendments and corrigenda to the PICS proforma that have been completed as part of the PICS                               | Amd.                                                                                                                                  | : | Corr. : |
|                                                                                                                                             | Amd.                                                                                                                                  | : | Corr. : |
| Have any Exception items been required? (See B.3.3: The answer Yes means that the implementation does not conform to IEEE Std 802.1AB-2016) | No [ ] Yes [ ]                                                                                                                        |   |         |

|                          |  |
|--------------------------|--|
| <b>Date of Statement</b> |  |
|--------------------------|--|

## A.4 Major capabilities and options

| Item        | Feature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Status                                                                                                                                        | References                                                                                                                                                                | Support                                                                                                                                                                                                                                                                                                                               |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cntrlport   | Are LLDP exchanges supported through a controlled port if port access is controlled by IEEE Std 802.1X?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | M                                                                                                                                             | 5.3, 6                                                                                                                                                                    | Yes [ ] N/A [ ]                                                                                                                                                                                                                                                                                                                       |
| uncntrlport | Are LLDP exchanges supported through the uncontrolled port if port access is controlled by IEEE Std 802.1X?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | O                                                                                                                                             | 5.4, 6                                                                                                                                                                    | Yes [ ] No [ ] N/A [ ]                                                                                                                                                                                                                                                                                                                |
| addr        | Are LLDP addressing and LLDP EtherType encoding for Normal LLDPDUs in conformance with the defined requirements?<br><br>All systems:<br>DA = Any group MAC address<br>DA = Any individual MAC address<br>SA = station MAC address<br>LLDP EtherType encoding<br><br>C-VLAN Bridge:<br>DA = Nearest bridge address<br>DA = Nearest non-TPMR bridge address<br>DA = Nearest customer bridge address<br><br>S-VLAN Bridge:<br>DA = Nearest bridge address<br>DA = Nearest non-TPMR bridge address<br>DA = Nearest customer bridge address<br><br>TPMR Bridge:<br>DA = Nearest bridge address<br>DA = Nearest non-TPMR bridge address<br>DA = Nearest customer bridge address<br><br>End station:<br>DA = Nearest bridge address<br>DA = Nearest non-TPMR bridge address<br>DA = Nearest customer bridge address | <br><br><br><br>O<br>O<br>M<br>M<br><br><br><br><br>M<br>M<br>M<br><br><br><br><br><br><br><br><br><br>M<br>X<br>X<br><br><br><br>M<br>O<br>O | <br><br><br><br>7.1<br>7.1<br>7.2<br>7.3<br><br><br>7.1<br>7.1<br>7.1<br><br>7.1<br>7.1<br>7.1<br><br>7.1<br>7.1<br>7.1<br><br>7.1<br>7.1<br>7.1<br><br>7.1<br>7.1<br>7.1 | <br><br><br><br>Yes [ ] No [ ]<br>Yes [ ] No [ ]<br>Yes [ ]<br>Yes [ ]<br><br>Yes [ ] N/A [ ]<br>Yes [ ] N/A [ ]<br>Yes [ ] N/A [ ]<br><br>Yes [ ] N/A [ ]<br>Yes [ ] N/A [ ]<br>No [ ] N/A [ ]<br><br>Yes [ ] N/A [ ]<br>No [ ] N/A [ ]<br>No [ ] N/A [ ]<br><br>Yes [ ] N/A [ ]<br>Yes [ ] No [ ] N/A [ ]<br>Yes [ ] No [ ] N/A [ ] |
| lldpdu      | Is the Normal LLDAPDU encapsulation in conformance with the TLV order specified by the Normal LLDAPDU format?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | M                                                                                                                                             | 7.3                                                                                                                                                                       | Yes [ ]                                                                                                                                                                                                                                                                                                                               |
| tlvfmt      | Is the basic TLV capability implemented?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | M                                                                                                                                             | 8.4                                                                                                                                                                       | Yes [ ]                                                                                                                                                                                                                                                                                                                               |
| basictlv    | Is each TLV in the basic management set implemented?<br><br>End Of LLDAPDU TLV<br>Chassis ID TLV<br>Port ID TLV<br>Time To Live TLV<br>Port Description TLV<br>System Name TLV<br>System Description TLV<br>System Capabilities TLV<br>Management Address TLV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <br><br>O<br>M<br>M<br>M<br>M<br>M<br>M<br>M<br>M<br>M                                                                                        | <br><br>8.5.1<br>8.5.2<br>8.5.3<br>8.5.4<br>8.5.5<br>8.5.6<br>8.5.2<br>8.5.8<br>8.5.9                                                                                     | <br><br>Yes [ ] No [ ]<br>Yes [ ]<br>Yes [ ]<br>Yes [ ]<br>Yes [ ]<br>Yes [ ]<br>Yes [ ]<br>Yes [ ]<br>Yes [ ]<br>Yes [ ]                                                                                                                                                                                                             |
| xltlvfmt    | Is the Organizationally Specific TLV capability implemented?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | M                                                                                                                                             | 8.7                                                                                                                                                                       | Yes [ ]                                                                                                                                                                                                                                                                                                                               |

## A.4 Major capabilities and options

| Item        | Feature                                                                                                                                                                                | Status                  | References | Support                |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|------------|------------------------|
|             | Which of the following operational modes are implemented?                                                                                                                              |                         |            |                        |
| optxrx      | Transmit and receive                                                                                                                                                                   | O.1                     | 6.1        | Yes [ ] No [ ]         |
| optx        | Transmit only                                                                                                                                                                          | O.1                     | 6.1        | Yes [ ] No [ ]         |
| oprxx       | Receive only                                                                                                                                                                           | O.1                     | 6.1        | Yes [ ] No [ ]         |
| txmode      | Is the transmit mode in conformance with all operational specifications indicated for the Tx mode in Table 9-1?                                                                        | optxrx<br>OR optx:<br>M | Clause 9   | Yes [ ] N/A [ ]        |
| rxmode      | Is the receive module in conformance with all operational specifications indicated for the Rx mode in Table 9-1?                                                                       | optxrx<br>OR oprx:<br>M | Clause 9   | Yes [ ] N/A [ ]        |
|             | Which type of data store/retrieval is implemented?                                                                                                                                     |                         |            |                        |
| mib         | SNMP MIB is supported                                                                                                                                                                  | O.2                     | 11.5, 5.3  | Yes [ ] No [ ]         |
| nomib       | SNMP MIB is not supported                                                                                                                                                              | O.2                     | 10.1, 5.3  | Yes [ ] No [ ]         |
| snmpmib     | Is the MIB module in conformance with the MIB sections indicated in Table 11-1 for the operating mode being implemented?                                                               | mib:M                   | 11.5, 5.3  | Yes [ ] N/A [ ]        |
| snmpsupport | Which of the transport mappings defined by IETF RFC 3417 or IETF RFC 4789 is used to support SNMP?                                                                                     |                         |            |                        |
|             | IETF RFC 3417                                                                                                                                                                          | mib:O.3                 | 5.3, 5.4   | Yes [ ] No [ ] N/A [ ] |
|             | IETF RFC 4789                                                                                                                                                                          | mib:O.3                 | 5.3, 5.4   | Yes [ ] No [ ] N/A [ ] |
| equivstor   | If the SNMP is not supported, is functionally equivalent storage and retrieval capability specified in Clause 8, Clause 9, Clause10 provided for the operating mode being implemented? | nomib:M                 | 10.1       | Yes [ ] N/A [ ]        |

## A.4 Major capabilities and options

| Item         | Feature                                                                                                                                                                                                                                                                                                                                                                                                                            | Status                                                             | References                                     | Support                                                                                                            |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| xlldp        | Is the Extended Link Layer Discovery Protocol supported?                                                                                                                                                                                                                                                                                                                                                                           | O                                                                  |                                                | Yes [ ] No [ ]                                                                                                     |
| xlldpdu      | Are the Extension and Extension Request LLDPDU encapsulation in conformance with the TLV order specified?                                                                                                                                                                                                                                                                                                                          | xlldp:M                                                            | 8.2                                            | Yes [ ] N/A [ ]                                                                                                    |
| xaddr        | Are LLDP addressing and LLDP EtherType encoding for Extension and Extension Request LLDPDUs in conformance with the defined requirements?<br><br><i>Extension Request LLDPDUs:</i><br>DA = Return MAC address from Manifest TLV<br>SA = station MAC address<br>LLDP EtherType encoding<br><br><i>Extension LLDPU:</i><br>DA = Return MAC address from Extension Request TLV<br>SA = station MAC address<br>LLDP EtherType encoding | xlldp:M<br>xlldp:M<br>xlldp:M<br><br>xlldp:M<br>xlldp:M<br>xlldp:M | 7.1.2<br>7.2<br>7.4<br><br>7.1.2<br>7.2<br>7.4 | Yes [ ] N/A [ ]<br>Yes [ ] N/A [ ]<br>Yes [ ] N/A [ ]<br><br>Yes [ ] N/A [ ]<br>Yes [ ] N/A [ ]<br>Yes [ ] N/A [ ] |
| yang         | Does the implementation support management operations using YANG modules?                                                                                                                                                                                                                                                                                                                                                          | O                                                                  | 12.6                                           | Yes[ ] No[ ]                                                                                                       |
| yang modules | Is the ieee802-dot1ab-lldp module supported?<br><br>Is the ieee802-dot1ab-types module supported?                                                                                                                                                                                                                                                                                                                                  | yang:M<br><br>yang:M                                               | 12.6.2.2<br><br>12.6.2.1                       | Yes[ ] N/A[ ]<br><br>Yes[ ] N/A[ ]                                                                                 |

## **1 Annex B**

### **2 (normative)**

### **3 PTOPO MIB update**

4 The PTOPO MIB should be updated according to the following rules:

- 5 a) If any objects in the LLDP remote systems MIB age out, the equivalent objects in the PTOPO MIB  
6 should be deleted.
- 7 b) If the TTL value in the Time To Live TLV is zero, then the port is being shutdown and the PTOPO  
8 MIB objects associated with the MSAP identifier should be deleted.
- 9 c) If the TTL field is non-zero, then the appropriate ptopoRemEntry is found or created, based on the  
10 data elements included in the LLDP frame. If the indicated entry is dynamic (i.e., ptopoConnIsStatic  
11 is FALSE), then the current sysUpTime value is stored in the ptopoConnLastVerifyTime field for the  
12 entry.
- 13 d) If a ptopoRemEntry was added then the ptopoConnTabInserts counter is incremented.
- 14 e) If any ptopoRemEntry was added or deleted, or if information other than the  
15 ptopoRemLastVerifyTime changed for any entry due to the processing of this LLDP frame, the  
16 ptopoLastChangeTime object is set with the current sysUpTime, and a ptopoConfigChange trap  
17 event is generated. (See the PTOPO MIB for information on ptopoConfigChange trap generation.)

1 **Annex C**

2 (informative)

3 **Example LLDP transmission frame formats**

4 The LLDP MAC frame format is based on the particular transmission protocol. The following example  
5 formats illustrate the indicated LLDP EtherType encoding method.

6 **C.1 Direct-encoded LLDP frame format**

7 The IEEE 802.3 LLDP frame format is illustrated in Figure C.1.

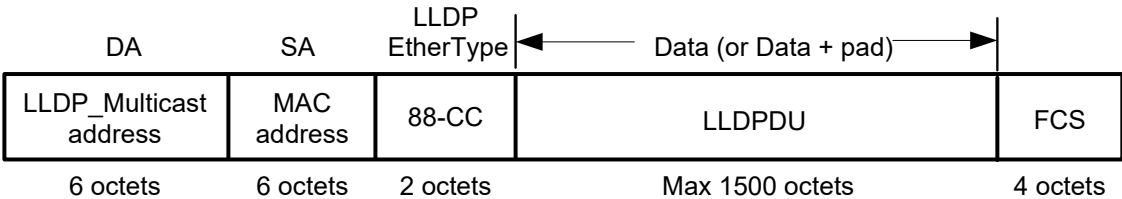


Figure C.1—IEEE 802.3 LLDP frame format

8 NOTE—The illustration shows the simplest form of an LLDP frame on an IEEE 802.3 medium; i.e., where the frame  
9 has had no IEEE 802.1Q tag header, or IEEE 802.1AE™ security tag, or any other form of encapsulation applied to it.

10 **C.2 SNAP-encoded LLDP frame format**

11 The IEEE 802.11™ frame format is illustrated in Figure C.2, for the case where the value of To DS and  
12 From DS in the Frame Control field are both zero.

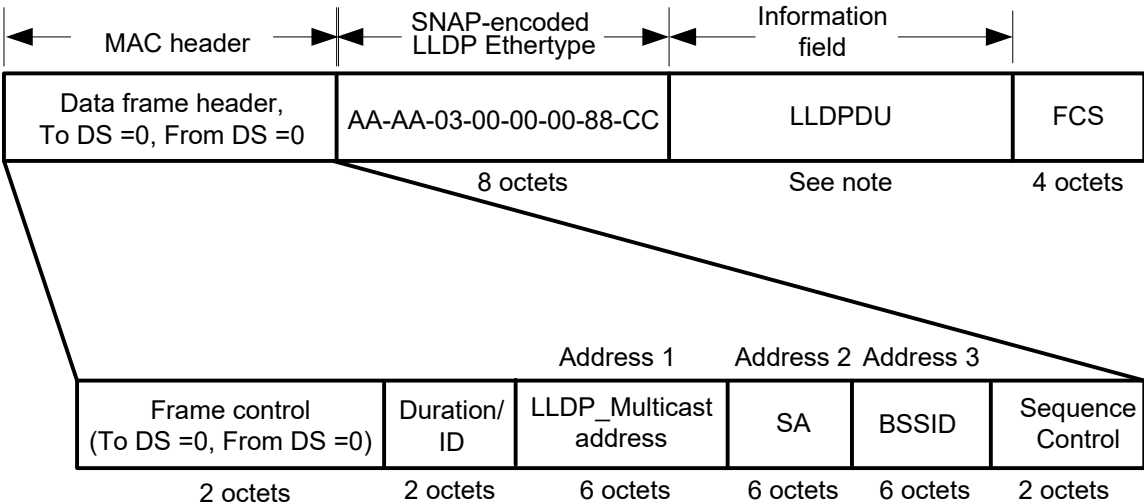


Figure C.2—IEEE 802.11 LLDP frame format

13 NOTE—The illustration shows the simplest form of an LLDP frame on an IEEE 802.11 medium; i.e., where the frame  
14 has had no IEEE 802.1Q tag header, or IEEE 802.1AE security tag, or any other form of encapsulation applied to it.



## <sup>1</sup> **Annex D**

<sup>2</sup> (informative)

## <sup>3</sup> **Bibliography**

<sup>4</sup> [B1] IEEE Std 802.11, Standard for Information technology—Telecommunications and information  
<sup>5</sup> exchange between systems—Local and metropolitan area networks—Specific requirements—Part 11:  
<sup>6</sup> Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) specifications.<sup>22</sup>

<sup>7</sup> [B2] IETF RFC 2578 (STD 58), Structure of Management Information for Version 2 of the Simple Network  
<sup>8</sup> Management Protocol (SNMPv2), McCloghrie, K., Perkins, D., Schoenwaelder, J., Case, J., Rose, M.,  
<sup>9</sup> Waldbusser, S., April 1999.<sup>23</sup>

<sup>10</sup> [B3] IETF RFC 2579 (STD 58), Textual Conventions for Version 2 of the Simple Network Management  
<sup>11</sup> Protocol (SNMPv2), McCloghrie, K., Perkins, D., Schoenwaelder, J., Case, J., Rose, M., Waldbusser, S.,  
<sup>12</sup> April 1999.

<sup>13</sup> [B4] IETF RFC 2580 (STD 58), Conformance Statements for SMIV2, McCloghrie, K., Perkins, D.,  
<sup>14</sup> Schoenwaelder, J., Case, J., Rose, M., Waldbusser, S., April 1999.

<sup>15</sup> [B5] IETF RFC 1812, Requirements for IP Version 4 Routers, June 1995.

<sup>16</sup> [B6] IETF RFC 2108, Definitions of Managed Objects for IEEE 802.3 Repeater Devices using SMIV2,  
<sup>17</sup> February 1997.

<sup>18</sup> [B7] IETF RFC 2670, Radio Frequency (RF) Interface Management Information Base for MCNS/DOCSIS  
<sup>19</sup> compliant RF interfaces, August 1999.

<sup>20</sup> [B8] IETF RFC 2922, Physical Topology MIB, Bierman, A., and Jones, K., November 1998.

<sup>21</sup> [B9] IETF RFC 4293, Management Information Base for the Internet Protocol (IP), April 2006.

<sup>22</sup> [B10] IETF RFC 4363, Definitions of Managed Objects for Bridges with Traffic Classes, Multicast  
<sup>23</sup> Filtering, and Virtual LAN Extensions, January 2006.

<sup>24</sup> [B11] IETF RFC 4546, Radio Frequency (RF) Interface Management Information Base for Data over Cable  
<sup>25</sup> Service Interface Specifications (DOCSIS) 2.0 Compliant RF Interfaces, June 2006.

<sup>26</sup> [B12] IETF RFC 7803, Changing the Registration Policy for the NETCONF Capability URNs Registry,  
<sup>27</sup> February 2016.

<sup>28</sup> [B13] IETF RFC 8040, RESTCONF Protocol, January 2017.

<sup>29</sup> [B14] IETF RFC 8345, A YANG Data Model for Network Topologies, March 2018.

<sup>30</sup> [B15] ISO/IEC 8802-2:1998, Standard for Information technology—Telecommunications and information  
<sup>31</sup> exchange between systems—Local and metropolitan area networks—Specific requirements—Part 2:  
<sup>32</sup> Logical link control.<sup>24</sup>

<sup>22</sup> IEEE publications are available from The Institute of Electrical and Electronic Engineers (<http://standards.ieee.org>).

<sup>23</sup> IETF RFCs are available from the Internet Engineering Task Force (<https://www.rfc-archive.org/>).

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<sup>24</sup> ISO/IEC publications are available from the International Organization for Standardization (<https://www.iso.org/>), the International Electrotechnical Commission (<https://www.iec.ch/>), and the American National Standards Institute (<https://www.ansi.org/>).